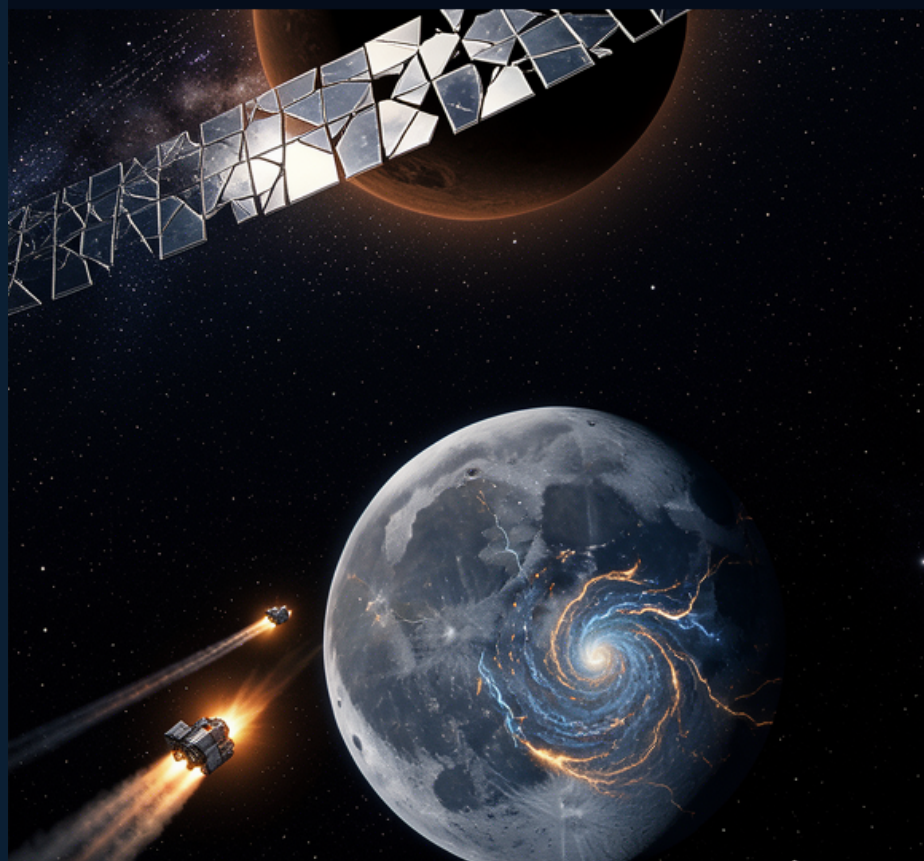


A SCIENCE FICTION NOVEL

THE SILENCE AFTER TOMORROW

*The future sent one final order.
Humanity chose what it could not know.*



TAPIO KINNUNEN

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Chapter 1 — The Warning That Arrived Alone

The forecast desk on *Aster Vale* smelled the way it always did at the start of shift: warm electronics, the resin tang of the cable insulation Tev was always patching somewhere deeper in the spine, and a faint breath of coolant that someone had learned to live with. Iven Mora liked the hour before traffic. They liked the way the crew compartments settled, the soft tick of the hull breathing under spin-gravity, and the fact that for ninety seconds the mirror deck belonged to nobody but them.

The day-clock on the bulkhead read 06:14 ship-local. The forecast window would open at 06:30, when Nox rose above the relay horizon and the lattice woke.

Iven's hands found the console by memory and pulled the overnight standing queue. Forty-three routine advisories queued for distribution. Twelve engineering improvements. Six epidemiological nudges. Two route corrections. One agricultural note about hydroponic nitrogen cycling on the trailing habitat *Hesper's Rest*. A whole city of small kindnesses, sent backward along the only constrained geodesic the universe had offered the species.

They had read the morning queue a thousand times. The rhythm had weight. Each advisory carried a timestamp in the future it had been sent from, a hash signature from a return mirror, a short body of advice, and the courteous signature of whoever on the far side had decided their descendants needed to hear it. Some were warm. Some were perfunctory. The hydroponic note had included a

weather aside that Iven had read three times and then sent down to the agronomy deck with a one-line addendum. That was the work. Hold the queue, scrub the noise, walk the lot to the bridge for transmission.

At 06:27 the lattice pinged awake.

Iven watched the inbound feed come up green. They had not yet noticed they were watching.

The first advisory arrived at 06:29:01, a routine engineering improvement on capacitor tolerance in the *Pell*-class tug fleet. The second followed it at 06:29:04. Then a third. Then a gap. Then a fourth advisory, and another, and the rhythm of the feed smoothed into what they had seen every morning for the seven years they had run this desk.

At 06:31 the feed stopped.

Iven counted eight seconds by habit, then by choice, watching the indicator stay amber. Twelve. Sixteen. The lattice still hummed; the return mirrors still drank power from the reactor down-spin and held their frame on Nox. A second advisory queue opened, called by the relay logic from a different mirror pair. That queue was also empty.

By 06:33 the deck officer on watch, a young cartographer named Pell Arent, had drifted in from the next compartment. Pell did not ask. They set a second cup of tea beside Iven's elbow and watched the screen.

"Both queues down," Iven said.

"Both."

"Send a maintenance ping."

Pell sent the ping. The maintenance channel answered within two seconds. The mirrors were healthy, the lattice was aligned, the courier corridor was open. There was simply nothing coming down it.

Iven pulled the long-baseline diagnostic. The relay tree reported nominal at every node from the wake-leading relay

station *Ossein's Lantern* to the trailing courier buoy *Hespar's Rest* kept in orbit around its frozen moon. The Wake itself, two hundred and fourteen habitats strung across eleven light-days of dark, was silent on the morning band. The other survey carriers and courier stations were silent too. The Commonwealth still breathed somewhere out there; the relay lattice was full of faint pings from convoy traffic, marriage announcements, and an unremarkable hull-dent report from a tanker called *Measured Rain*. What had stopped was the *advice*.

"Tell me again how long since we had a total stop," Iven said.

"Never," Pell said.

"Since the lattice came up."

"Since the lattice came up. There's no record."

Iven rested their hand on the console. The amber light painted their knuckles.

"Log it," they said. "Mark it as anomaly only. Do not file an event report until we know what we are looking at."

At 06:37 the door chime sounded. That alone was wrong; the forecast deck ran on a sealed rotation during the window, and nobody was supposed to knock until the queue had been walked.

Iven keyed the panel. The door unseated with the soft pneumatic sigh that meant someone had overridden it from the bridge.

Doctor Nadi Okoro stood in the frame. Okoro had not been on *Aster Vale* in three days. She had been aboard *Everspring* with her sleeper count, auditing the adaptive biosphere against the seedship's own updated forecast profile. She was wearing a civilian jacket and a face that meant she had been running.

"Skipper wants you on the bridge," Okoro said. "Now. Don't walk the queue."

"Why."

"Because there is no queue."

Okoro turned and went. Iven followed her up the central spine, past the cable runs Tev had been repairing earlier that morning, past the sealed hatch that led toward the mirror bay, and up two ladders to the bridge deck. The lift would have been faster; Okoro had chosen the ladders, which meant she wanted the time to think.

They did not speak on the way up. The spine was quiet.

On the bridge, the duty officer was standing rather than sitting. Captain Thessaly Bren stood near the commander's chair with her arms folded. Admiral Rhea Calder stood at the holo table with her back to the door. Iven registered the admiral's uniform before they registered anything else. Calder did not live on *Aster Vale*. Calder commanded the Wake Defense Fleet from a heavy cruiser two days out by courier burn.

When Calder turned, Iven saw that she had been carrying a single printed sheet.

"Close the hatch," Calder said.

The hatch sealed. Okoro crossed to the far side of the holo table and stood with her hands at her sides. Pell had not followed; the deck below them was empty.

"We received an authenticated return this morning," Calder said. "Single advisory. Highest legal priority. I want you to read it."

She handed Iven the sheet.

The header carried the standard return-mirror formatting: a timestamp stamped 53 years forward, a mirror-pair identifier, a hash signature, and an authentication chain that Iven knew at sight. They had read a thousand such headers. The signature chain was intact. The mirror that had carried it had spent one frame-dragging second of its operational budget to deliver

this single page. No advisory in living memory had consumed that much.

The body of the advisory was four paragraphs. Iven read the first three twice and the fourth once more. Then they read the first three again.

The advisory was addressed to all operational authorities of the Wake Commonwealth. It identified by name the seedship *Everspring*, its civilian registry of eighteen thousand four hundred sleepers, its projected Serein insertion corridor, and the timestamp at which it would reach operational status. The advisory named the Director of *Everspring*'s civilian mission. It named Doctor Nadi Okoro.

It ordered the destruction of *Everspring* before Serein insertion. It ordered the destruction to be carried out by the Wake Defense Fleet under standing forecast-compact authority. It ordered the destruction to be carried out in a manner that precluded civilian evacuation, lest the act become legible to whatever inhabited Serein's system.

There was a fifth page. Calder had folded it inside the sheet. Iven did not open it. They did not need to.

"There is an annex," Iven said.

"There is an annex," Calder said.

"It specifies the means."

"It specifies a list of acceptable means. It includes the timing window. It names this ship, this crew, and your name."

Okoro did not move. Bren had stopped breathing.

"You have read it," Calder said.

"I have read the first four paragraphs."

"You have not read the annex."

"No."

Calder set her hands flat on the holo table. Her fingers did not tremble. Iven noticed that, because they had expected them to.

"Walk me through what you are going to do next," Calder said.

Iven held the sheet against their thigh. The bridge was small enough that they could feel the heat of the admiral's hands reflected off the surface.

"I am going to copy the first four pages," Iven said. "I am going to walk the copy to the legal desk for filing. I am going to leave the annex in your custody."

"And the original."

"I am going to leave the original with you as the receiving authority."

"You are not going to read the annex."

"I have not read the annex."

"Why."

"Because I do not want my hands on a kill order for eighteen thousand civilians. Because if my hands are on it, I cannot pretend I am not responsible for what is done with it."

Calder studied them for three seconds. The amber light from the holo table made her face look older than Iven had remembered. The admiral had lost her brother at Hespar's Rest when a council delayed a verified forecast by ninety minutes to confirm it. Iven had been a junior cartographer then. They remembered the cable traffic. They remembered the silence afterward, which had lasted almost as long.

"Take the annex out of the sheet," Calder said. "Leave it on the table."

Iven opened the sheet. The annex was a single page, folded again. They set it on the holo table and let go of it.

"Thank you, Chief," Calder said. "Take the desk back. Do not transmit the advisory. Do not transmit anything. The lattice stays open, but you send nothing down it."

"Understood."

"If anything arrives, you bring it to me unread."

"Understood."

Iven turned and walked toward the hatch. Okoro caught their arm at the door. Her grip was cold. She had not moved from where she had stood since Calder turned. Her face still held the look of a woman who had just heard her own name read out of a future.

"Iven."

"Director."

Okoro opened her mouth and closed it again. Her hand tightened on Iven's sleeve and then let go.

"They sent my name," she said.

"They sent all our names," Iven said. "Including mine."

Okoro looked at the folded annex on the table. She did not reach for it.

"Yes," she said. "Including yours."

Iven went down the ladders, past the cable runs, and through the hatch onto the forecast deck. Pell was still there, alone, watching the empty queues.

Iven sat down at the desk and put their hands flat on the console. Their hands were shaking. They let them shake.

Iven made themselves wait one full minute before they stood up again. They poured the cold tea into the recycler, rinsed the cup, set it back on the warmer. They pulled up the legal-desk filing log to confirm the envelope had been timestamped. They opened the long-baseline diagnostic again and noted, for the second time, that every relay node in the Wake reported nominal except the inbound advisory stream. They wrote the time of the second observation on

the corner of a working slip and initialed it. The minute passed. Their hands stopped shaking.

They had thought, on the ladder down, that they would walk to the legal desk immediately and file the working copy before the admiral could send someone down to stop them. They had not. The admiral had not, either. Either Bren had cleared it with Calder, or Calder had decided the working copy was acceptable to leave in the drawer as long as the annex stayed with her, or both. Iven did not know which. They suspected they would learn within the hour.

"The bridge is sealed," Pell said quietly.

"Yes."

"Acknowledged."

Iven closed their eyes for six seconds. When they opened them, the screen still showed an empty feed. The lattice hummed. The mirror deck waited.

Iven took the printed sheet from the inner pocket where they had folded it on the way down and laid it on the console beside the keyboard. They had copied the first four pages in their head on the ladder. They copied them now, line by line, onto a working slip of forecast paper, in the careful block letters the legal desk used for filings of this weight. They dated the slip. They time-stamped it. They initialed the bottom.

They did not copy the fifth page.

They put the working slip into a sealed envelope marked LEGAL PRIORITY and walked it two compartments down to where the duty legal officer kept her station. The legal officer took the envelope without reading it and put it into the filing drawer. Iven did not stop to explain.

On the way back they passed Tev Arin in the cable junction. Tev was kneeling with his shoulders half inside a hatch, swearing softly about a connector that had lost its keying pin. He had a coil of patch cable over one shoulder

and a cup of something that was not tea balanced on a cross-strut beside him.

"Cable's fine," Tev reported without looking up. "Connector was loose. I tightened it."

"Good."

"You okay, chief?"

"No."

"Do you want me to come up."

"Yes."

Tev set his tools down, took the cup off the cross-strut, drank from it, set it back, and followed Iven up the ladder to the forecast deck. Pell looked up when they came through and moved over without being asked. The three of them sat in the empty room and watched the queues not move.

Iven found, in the next ten minutes, that they could do small useful things. They pulled the mirror-hour temperature log and confirmed the bay was still at operational cold. They opened the courier burn pad and saw Sena Vale's name on the next-day priority queue, with *Everspring* as her destination, and a maintenance block flagged open on her tug's number-two radiator. They closed the pad without printing it. They checked the long-baseline diagnostic one more time and saw the silence held across all eight relay pairs.

The amber indicators did not move. The lattice drank power and held its frame. Pell made tea and brought three cups and then, because none of them had eaten, set out the small foil of pressed grain and dried fruit she kept in her station drawer. Tev drank his cup, took a single piece of fruit, and went back to the cable junction to finish the patch, because somebody ought to be repairing something on this ship and it might as well be him.

At 06:51 the chime sounded again.

This time it was Tev who answered it, because Tev was nearest the panel and because Iven's hands had decided they were not ready to stand yet. The hatch opened on Captain Bren, who looked in, saw the three of them, and stepped through without greeting.

"The admiral is coming down," Bren said. "She wants the deck."

"She wants the deck," Iven repeated.

"She wants the mirror deck and she wants the legal desk. I'm bringing her the keys."

Bren set two physical key cards on the console. They were pale green with a black magnetic stripe; Iven had only seen one once before, on the day they had been promoted to chief cartographer. The key to the mirror bay opened the room that held the actual return-mirror hardware. The key to the legal desk opened the filing drawer where Iven had just placed their working copy.

"Iven," Bren said carefully. "You filed the working copy."

"Yes, Captain."

"You filed only the working copy."

"That is correct, Captain."

"Tell me that is what you did."

"That is what I did."

Bren looked at the envelope drawer and then at Iven.

"Good," she said. "Keep the deck tidy. The admiral will be here in six minutes."

Bren left. Tev closed the hatch behind her and stood with his back against it.

"Chief," Tev said.

"Yes."

"Are we at war."

"No."

"Are we about to be."

"I don't know."

"Are those the keys for the mirror bay."

"Yes."

Tev nodded once. He did not ask what Iven had filed, or what Iven had not filed, or why the admiral wanted both the bay and the legal drawer at the same moment. Tev did not need to ask. Tev had grown up on a courier station with a maintenance key on his carabiner, and he knew the difference between a key you carry because it is your job and a key you hand over because someone has just taken your job away.

"Six minutes," Tev said.

"Six minutes."

"Stay on the deck?"

"Yes."

"Stay with the hatch locked?"

"Yes."

Tev locked the hatch. The lock engaged with a sound that was not the soft pneumatic sigh of a normal override but a hard mechanical clunk that meant the deck would not open from the other side without a key. Tev pocketed the key and sat back down at the second station.

Pell, who had not spoken since Bren left, spoke now.

"Chief."

"Yes."

"There is something on the feed."

Iven turned to the console. The amber indicator had flickered once. A single advisory, arriving now, four minutes after the first one, twenty-one minutes after the lattice had

gone silent. The header was not in the standard return-mirror format. The authentication chain was intact. The mirror pair was different. The forward timestamp matched the first advisory to the second.

"Mark it," Iven said. "Do not open it. Mark it for the admiral."

Pell marked it. The indicator held steady. The deck waited. The relay pair kept its frame on Nox. The hatch stayed locked.

Six minutes was a long time when the room was empty and the only sound was the soft tick of the hull and the patient breath of a mirror bay that had just been handed over to a fleet that had come two days out by courier burn to claim it.

Chapter 2 — The Starless Sea

The courier burn was supposed to begin at 07:12 ship-local. Sena Vale climbed into the cockpit of her tug at 06:40 and sat with her gloves on her knees, watching the indicator lights along the spine settle into their pre-burn sequence. She had thirty-two minutes to think, which was a luxury. She used them the way she always used them, by not thinking at all.

Quiet Verdict was a *Pell*-class courier tug, twelve years old, with a port-side radiator that ran ten degrees hot in deep cruise and a thrust frame her last pilot had nicknamed Honest John because it told you when something was about to go wrong. Sena liked her. She had flown worse. She had flown better. Honest John was the tug she would choose if she were allowed one tug for the rest of her career, which she would not be allowed because courier pilots rotated by hull hours and the deck had a schedule.

The cockpit was small and warm. The board was old but it had been maintained by people who cared about it, and every light had a label and every label had been updated within the last quarter. Sena ran her hand down the column of switches by habit and felt each one click under her finger. The trick of a courier cockpit was that nothing surprised you. The trick of a courier career was that you made sure nothing surprised you.

The drill today was nominal. A short transfer burn to a waypoint buoy, a rendezvous with a tanker that needed a course correction, a long slow approach back into the *Aster*

Vale hangar on cold thrust. Twenty-three minutes from light to dock. Standard courier work, the kind that kept a pilot's reflexes honest and her hand on the throttle. The kind the deck scheduled to keep the wake-leading relay stations honest too, because a courier who could rendezvous on cold thrust at the end of a long burn was a courier you could trust to deliver something that mattered.

Sena did not know, when she sat down at 06:40, that the day was going to require her to deliver something that mattered. The deck had not told her. The lattice had not told her. The lattice, in fact, had stopped telling anyone anything an hour earlier, and the rumor was making its way through the crew compartments with the speed that rumors have on a small ship in the dark.

At 07:09 her board chimed. The chime meant the deck had updated the burn profile.

Sena pulled the file. It was not the drill profile.

The new profile was a four-burn courier run to *Everspring*, departure in eleven minutes, return open. The rendezvous tanker was no longer on the manifest. The deck note said, in the clipped prose of the operations desk, that the run had been added by senior medical authority to support a civilian equipment transfer. The note did not say whose authority. The note did not need to. Only one person on *Aster Vale* would reroute a courier tug to a seedship in the middle of a drill.

Sena closed the file and called the deck.

"Acknowledged," the deck said. "Stand by for clearance."

She did not stand by. She pulled up the maintenance log on her number-two radiator and saw that the open block was still open. The note beside it read, in Tev Arin's careful capitals: BEARING 3 WORN — RUN TO REPAIR NEXT PORT. Tev had written it three days ago. Tev had not fixed it. Tev did not usually leave things unfixed. The bearing would hold for the burn, but it would not hold for the burn

home if anything else went wrong, and a courier pilot who had to make a long approach to a seedship with a hot radiator was a courier pilot who was going to ask for help whether she wanted to or not.

The clearance came at 07:11. Sena took it and ran.

The first burn was clean. She felt Honest John settle into the long curve of the corridor and the tug asked nothing of her except patience. She watched the indicator readouts the way she watched them on every burn: not for anomalies, but for the small departures that meant something wanted her attention. None came.

At the waypoint buoy she matched rotation with the tanker and let the autopilot hold station. *Measured Rain* carried the hull dent from the report Iven's desk had logged that morning. Her captain sent a one-line greeting. Sena sent one back. They did their drill. The tanker began its slow arc away. Sena began her long approach to *Everspring*.

She let the autopilot handle the rendezvous. She had nothing to do for nineteen minutes, and she used eight of them to call down to the thermal bay and ask for Tev.

Tev answered on the third ring, which meant he had been elbow-deep in something and had to climb out.

"Vale."

"Arin. I'm looking at your note on bearing three."

"Which ship."

"My ship."

"Ah." Tev paused. "Yeah. I put it in the log three days back. I was going to do it next port."

"We have a four-burn run."

"Sena."

"A four-burn run to *Everspring*, Tev."

"Give me your coordinates. I'll meet you at the hangar with the kit."

"It is a four-burn run. We may not come back to the same hangar."

"Then I'll meet you at whatever hangar you land in. Bring the bearing on the cart. Ten minutes."

"Tev."

"Sena."

"I do not have time for a proper repair in a hangar. I have time for what you can do in an airlock."

"Then I will do what I can do in an airlock. Stand by."

Sena did not say thank you. Tev did not want her to. Tev wanted her to land.

She let the autopilot carry the long arc toward the rendezvous. She had nineteen minutes of free fall and used the first eight to confirm Tev was on his way.

She docked at *Everspring* at 08:31 and cycled the airlock with the seedship's medical officer watching through the inner port. The seedship smelled different from any ship Sena had been on: cool, vegetal, and faintly sweet. The adaptive biosphere breathed through the ventilation.

Tev was waiting for her on the gantry outside, breathing mask on, tool kit in one hand and the bearing assembly in the other. He did not ask how the run had been. He climbed into her airlock with her and began to work.

"You have a four-burn run," he said, by way of greeting.

"I have a four-burn run."

"And the order came from senior medical."

"Yes."

"And senior medical meant Doctor Okoro."

"Tev."

"I know, I know. I am not asking."

Tev worked in the airlock with his shoulders half out of the inner hatch. Sena held the bearing steady while he pressed the new race into place. The work took twenty-three minutes. Tev checked the alignment twice. He checked it a third time.

The bearing itself was a small thing. A ring of pressed carbide, a race of polished alloy, a sleeve of braided fiber that the manuals called the keeper. Honest John's number-two radiator mounted on a yoke that pivoted on three such bearings, and the yoke's job was to keep the radiator pointed at the cold side of the ship while the tug pitched and yawed through a long burn. A worn bearing meant the yoke could not keep the radiator pointed, and an unpointed radiator meant the tug bled heat through her skin instead of through her sails, which meant the tug ran hot and the tug ran short on propellant because a hot tug had to burn longer to make the same course correction.

The manuals said a worn bearing would hold for two more burns at the upper tolerance. The manuals also said that the upper tolerance assumed the pilot was gentle, and the manuals knew that courier pilots were not gentle, and the manuals had been written by people who understood that the upper tolerance was a number and not a promise.

Tev set the alignment to a conservative spec. He wrote the new alignment into the log on the bearing sleeve with a grease pencil, the way he always did, so the next engineer would not have to chase the history.

"You are welcome," he said.

"I did not say thank you."

"You did not."

"How much longer would the old bearing have held."

"Two burns. Maybe three, if you were gentle."

"I am never gentle."

"I know. I set the alignment to a conservative spec because of that."

Sena handed him the used bearing. He put it in the kit.

"Have you heard from the bridge about the forecast queue?" he asked, not looking up.

"No. Why."

"The lattice has been empty since 06:31. Nobody on this ship is talking about anything else."

Sena waited.

"A single authenticated advisory came in," Tev said. "I've filed it. The admiral is on the ship. The mirror bay is locked."

Sena looked at the sealed airlock indicator between her tug and *Everspring's* hull. The indicator was amber. She looked at the inner port, where the seedship's medical officer still stood. The officer looked back.

"Run your seals," Sena said quietly. "Check them again."

Tev ran them. They held. He did not ask why she had asked.

Sena cycled the airlock open and walked into *Everspring's* civilian section. Okoro was waiting for her in the small conference room between the medical bay and the adaptive biosphere. Okoro had changed into a deck uniform. She had not changed out of the look on her face from that morning.

"Sit down," Okoro said.

Sena sat.

"I am going to ask you to do something," Okoro said. "If you say no, I will find another way. If you say yes, I will not be able to tell you why until it is over. You can still say no."

Sena waited. The conference room was quiet. The seedship's adaptive biosphere breathed through the vents in slow, regular pulses.

Okoro set a small sealed data spindle on the table between them. The spindle was no larger than a finger. Its casing was medical white, with a green band that meant priority legal hold.

"This spindle is a legal filing," Okoro said. "It is addressed to the civilian registry office aboard *Everspring*. It authorizes the seedship's mission director to undertake any defensive maneuver the director judges necessary to preserve the lives of the sleepers, in the event that the ship's chain of command is compromised by an external order inconsistent with the welfare of those sleepers."

Sena did not touch the spindle.

"It is a standing filing," Okoro said. "Every seedship director has one. They are filed before launch and held in legal escrow. They are only to be opened if the ship receives an order from outside that conflicts with the welfare of the sleepers."

Sena looked at the spindle for a long time. The green band caught the conference room light and held it. Then she looked at Okoro.

"What do you need me to do."

"I need you to carry this spindle to the legal escrow desk in the civilian registry office. I cannot carry it myself. I am under observation."

"Observation by whom."

"By people who believe they have a duty to follow an order they received this morning."

Sena picked up the spindle. It was lighter than she had expected. She put it into the inner pocket of her flight jacket. She felt the weight of it settle against her ribs, just left of center, exactly where a courier pilot's weight ought not to be when she has a burn to fly.

"Does Captain Bren know," Sena said.

"Captain Bren is under instruction from the admiral not to intervene in civilian registry matters."

"The admiral is in charge of fleet quarantine, not civilian registry."

"Correct."

"How long do I have before the spindle is supposed to arrive."

"Before the quarantine locks fully. Less than an hour."

Sena stood.

"The number-two radiator," Okoro said.

"Is repaired. Tev did it in the airlock."

"Thank you for the courtesy of telling me that the maintenance was real."

"It was real. I do not know what else it was."

Okoro nodded. She did not say anything else. She walked out of the room and closed the door behind her.

Sena stood for a moment with her hand on the inner pocket. She felt the spindle against her ribs. It was very still.

Sena walked back to the airlock, cycled through, and strapped into the cockpit. Honest John came up cold and clean. The new bearing asked nothing of her.

She transmitted clearance to the deck. The deck did not answer for forty seconds.

"Acknowledged, *Quiet Verdict*," the deck said at last. "Proceed to Serein corridor."

"Negative," Sena said. "I have not received an amended profile."

"You have now. Profile is uploaded. Depart when ready. Return window open."

The profile was already on her board. The deck had pushed it without waiting for her confirmation. She pulled it up and read it.

It was not the profile she had been told to fly.

The burn plan was a long, slow, fuel-expensive trajectory that would take her past two patrol stations and into a courier corridor the *Aster Vale's* operations desk did not normally use. The corridor's terminal point was the civilian registry office of *Everspring*, on the seedship's far side. The plan consumed nineteen percent more propellant than the standard courier run, and it threaded her through a window of approach that would have required Tev's new bearing to be exactly the alignment he had set it to, because the bank angle at the third burn would have stressed the old bearing past its tolerance.

Whoever had written this profile had known Tev had repaired the bearing. Whoever had written this profile had known who had asked for the repair.

Sena sat with her hand on the throttle and thought about Okoro's face. She thought about Tev's bearing. She thought about the spindle against her ribs.

She thought about Jori.

She had not thought about Jori in months. The thought came anyway. Jori had sat in this seat. Jori had trusted the chain of command until the chain of command told her to leave a school ring in a school, and then she had not trusted it. Jori had died because of that. Sena had lived because of that. The school ring was in a drawer in the crew quarters, and Sena did not open the drawer, and Jori's parents did not know where the ring was because the parents had been on the school and the parents had not survived the question of whether to wait for a fleet that had decided not to come.

The deck was waiting. The spindle was waiting. The bearing was new.

"Depart when ready," the deck repeated.

Sena engaged the burn.

Honest John took the thrust cleanly. The number-two radiator held at spec. The tug began the long, fuel-expensive curve toward *Everspring's* civilian registry, and the deck stopped calling her, and the corridor opened in front of her like a starless sea.

She flew the burn the way she had been trained to fly it. Hand on the throttle. Eyes on the board. The corridor threaded her past the first patrol station without a hail, and then the second, and then the corridor opened into the long quiet approach to the seedship's civilian registry dock. Honest John hummed. The new bearing did its work. The fuel gauge fell at the rate the profile had predicted, which was the rate Tev's bearing had allowed it to predict.

She thought, briefly, that she had been told this morning to do a drill, and that she had done a drill, and that the drill had ended at a buoy and the burn she was flying now had not been on the manifest when she sat down at 06:40. She thought, briefly, that the deck had pushed an unscheduled profile to her in the middle of a courier burn. She thought, briefly, that she had not been ordered to fly it and she had not been ordered not to fly it. She had been told that she could depart when ready.

She was ready. She departed.

The corridor ahead held no stars. The Serein approach vector pointed into the dark side of the Wake, where the brown dwarf Caul cast no light and the relay mirrors of Nox were only a careful geometry. The corridor was a starless sea because the corridor was, by definition, the place between stars where a courier tug had to fly on her own judgement and her own fuel.

The first patrol station was *Hespar's Rest* outer buoy, an automated courier relay that should have hailed her within thirty seconds of her crossing its detection horizon. It did not hail. The second was a *Pell*-class cutter on the lane that

should have challenged her transponder. The cutter kept its own burn and let her pass.

The spindle was still against her ribs. The bearing held. The corridor held. The corridor did not ask her anything except to keep flying.

She kept flying. She thought about the deck officer who had pushed the profile without her confirmation. She thought about the patrol stations that had let her pass without hailing. She thought about Okoro saying, "By people who believe they have a duty to follow an order they received this morning," and she thought about the order Okoro had not named, and she thought about the spindle against her ribs, and she flew, and the corridor did not ask her anything more.

Chapter 3 — Children of the Wake

The cartography bay on *Aster Vale* did not have a window. It had four screens instead, two of them folded down for the morning, and a circular plotter table at the center that Iven had stopped thinking about as furniture in the third year of their chief cartographer rotation. The plotter was a tool. The plotter did its job. The plotter did not care that the only valid criticism of a chart was the criticism the cartographer had not yet learned to make.

Iven set the day's first plate at 06:58, before the bridge had decided whose day it was going to be. They had not slept much. They had slept enough. Their hands were steady.

The plate carried the Serein approach corridor at the scale the survey carriers used when they briefed new arrivals to the Wake's wake-leading edge: long, slow, and quiet, threading the dark side of the convoy through the shadow of Caul and into the brown dwarf's gentle bow shock. The corridor assumed a courier tug of the *Pell* class, an experienced pilot, a four-burn commitment, and a willingness to spend nineteen percent of your propellant on the kind of fuel-expensive curve that the operations desks normally refused to schedule. Iven had flown the corridor in the simulator once. They had not flown it for real. The corridor frightened them in the way corridors frightened them: by being a place where the only map that mattered was the map inside the pilot's hand.

A new note had appeared on the plate overnight. The note carried the timestamp of 23:14, the byline of the bridge, and a single sentence: "Corridor to be charted at one-percent scale for fleet briefing at 14:00." The note did not say what kind of fleet briefing. The note did not need to. There was only one fleet briefing a chief cartographer could be asked to produce at one-percent scale, and that briefing concerned the corridor to *Everspring*, and the corridor to *Everspring* now threaded toward a system that had just received a single authenticated warning whose full text the admiral still held in a sealed annex that nobody but the admiral had read.

Iven accepted the note and began the chart.

At 07:12 Pell Arent arrived with tea and a folded printout of the overnight diagnostic. Pell set both on the plotter's edge, glanced at the plate, and sat down at the second station without asking what they were making. Pell had been on the bridge when Iven had returned from the admiral's first meeting. Pell had read the working copy they had filed. Pell had not asked about the annex. Pell had known better than to ask about the annex.

"Corridor?" Pell said.

"Corridor."

"How far?"

"Far enough that the courier pilots are going to hate us. Far enough that the propellant budget eats a captain's reserve. Far enough that the third burn sits inside Caul's bow shock, where the comms latency starts to matter and the diagnostics start to lie."

Pell nodded slowly. "Then the bridge wants it for a fleet briefing."

"Then the bridge wants it for a fleet briefing."

Pell did not say what kind of fleet briefing. They did not need to.

Iven built the corridor in five segments. They had learned the trick in their first year on the desk: a corridor looked like a single line on a fleet chart, but a corridor felt like five separate decisions to a courier pilot, and a cartographer who could not name the five decisions could not brief them. The first segment was the courier departure arc out of *Aster Vale's* hangar, drawn from the operational notes in Tev's maintenance log because the hangar itself was still warmer than it should have been from the previous day's exercise. The second segment was the long coast through the Wake's outer buoy chain, drawn from the buoy telemetry and made longer than for a routine run because the buoy chain was now under fleet quarantine and any courier passing two of its stations in the same hour would be challenged. The third segment was the threading approach past the patrol stations, which carried the corridor through a window of approach that no courier pilot would have accepted at the upper tolerance of a worn bearing. The fourth segment was the long, slow descent through Caul's bow shock, where the propellant cost climbed and the comms latency doubled and the diagnostics began to lie because the bow shock bent the relay carriers in ways the calibration did not account for. The fifth segment was the terminal approach to *Everspring's* civilian registry dock on the seedship's far side, which sat inside a corridor the operations desk had not used in two years.

Iven looked at the five segments together. They looked like a question.

The plotter rendered the corridor in amber on the dark surface. Iven watched it for a long moment without touching the screen. The corridor carried the shape a courier pilot would call beautiful and the shape a fleet commander would call indecent. It assumed a courier tug with a fresh bearing on its number-two radiator. It assumed a pilot willing to thread two patrol stations without a hail. It assumed a director willing to write a courier profile that the desk would push without waiting for the pilot's

confirmation. It assumed a seedship willing to receive a courier on its far side, into the dock only the registry office used, at an hour when the civilian section was not normally awake.

The corridor assumed things.

Pell set down a second cup of tea beside the first. "There's a thing in the diagnostic you should see."

"Tell me."

"The overnight long-baseline ping went out at 02:47. It was the routine survey sweep. It came back at 03:11 with the standard reply."

"Standard."

"Standard on the relay. Nonstandard on the mineral noise band. The Lumen channel carried an additional layer that wasn't in the previous sweep, and the additional layer didn't match any pattern the survey library has a name for."

"Describe it."

Pell described it. The description took three minutes and a folded printout. The Lumen moon carried a piezoelectric mineral matrix that produced slow, regular microwave bursts whenever the moon's rotation passed it through the focal point of Serein's largest debris ring. The bursts were loud. The bursts were predictable. The bursts had been mapped. What had arrived in the overnight sweep carried the Lumen signature but at a frequency the survey library did not have on file, at a depth the survey library did not have a category for, and at a regularity that did not match any natural process the Wake's planetary science community had ever recorded in the Serein system.

"It looks like a layer underneath the layer," Pell said.

"Underneath how."

"Like there's a mineral reading we already knew about, and then there's another mineral reading sitting under it, and the second reading carries information that isn't noise. The

first reading still looks like noise. The second reading carries it like a sentence carries a word."

Iven looked at the corridor they had just drawn. They looked at the diagnostic in Pell's hand. They thought, briefly, about the warning in the admiral's annex and about the corridor they were now drawing to the system the warning concerned.

"Pull the full survey archive," Iven said. "Cross-correlate against every Serein mineral reading the Wake has on file. Don't filter for noise. Don't filter for outliers. I want to see what the diagnostic looks like when we let the mineral band be whatever the mineral band is."

Pell went to the archive.

They did not write the name *Everspring* at the head of the plate. They wrote the corridor's identifier, which belonged to the corridor and not to the seedship, and they let the rest of the chart do its work.

The bridge called at 08:04. The bridge wanted the corridor by 12:00, not 14:00. The bridge did not say why. The bridge wanted the corridor with annotations for fleet quarantine and with a courier-pilot note explaining why a worn bearing would not survive the third burn. The bridge wanted the corridor with the kind of note a chief cartographer would write if the chief cartographer had been told, off the record, that a courier had already flown a part of it and that the courier had done so on a profile the operations desk had pushed without confirmation.

Iven accepted the new timing. They did not say that they had not been told, off the record, that a courier had already flown a part of the corridor. They did not say that they had read the courier-burn pad earlier that morning and had seen Sena Vale's name on the priority queue. They did not say any of these things, because Okoro had asked them not to and because Okoro had been right to ask.

At 08:31 Pell came back from the archive with a printout that was thicker than the printout they had taken in. Pell set it on the plotter and did not speak. They pointed to one line near the middle of the printout, a second line further down, and a third line near the bottom. The three lines belonged to three different surveys taken in three different years. The three lines had not been cross-correlated before.

"They rhyme," Pell said.

Iven read the three lines. They did rhyme.

The rhyme was not a sound. The rhyme was a pattern. Each line carried a phrase that did not appear in any other survey the Wake had on file from the Lumen channel. Each phrase occupied a different frequency. Each phrase occupied a different depth. Each phrase arrived at a different interval from the previous phrase, with the interval between phrases tracking the rotation of the moon through the debris-ring focal point. The phrases were not random. The phrases were not natural. The phrases had a syntax.

"What are they saying," Iven said.

"I don't know," Pell said. "But they're saying something at the same time."

Iven took a clean sheet of forecast paper from the drawer and began to copy the three lines in the careful block letters the legal desk used for filings of this weight. They copied them in the order Pell had indicated. They copied them at the depth Pell had indicated. They copied the rotation interval beside each one.

The copy took six minutes and produced a single sheet of paper that Iven did not fold and did not file. They held it against the plotter's edge and read it a fifth time and a sixth time and a seventh.

Each phrase carried a syntax of its own. Each phrase carried a subject and a verb. Each phrase carried a name for what they were describing, and the name they had

chosen was not the name the Wake's planetary science community had given the Lumen mineral matrix.

"Pull the Lumen channel again," Iven said. "Now. Without the routine filter. Without the noise filter. Without any filter the survey library applies by default."

Pell pulled it. The diagnostic came back at 08:47, twenty-three minutes after the request, three minutes longer than the diagnostic should have taken if the relay lattice were doing its job.

The diagnostic came back with a new phrase.

The phrase sat at the bottom of the Lumen channel's band. The phrase carried no timestamp. The phrase carried no signature. The phrase carried a single line of text that the diagnostic's translation layer had rendered into the Common script of the Wake because the diagnostic had been built to render whatever it found, and what it had found was a single line of text that did not belong in a mineral survey.

The phrase read: *No map can contain its reader.*

Iven read the phrase four times. They read it once more. They set the diagnostic printout on the plotter beside the corridor they had been drawing and looked at both. They did not say anything for a long time.

"Where did it come from," Pell said at last.

"It came from inside the diagnostic. Which means it came from inside the relay lattice. Which means it came from the Serein system, routed through the lattice, addressed to whoever was listening."

"Sent to us?"

"Sent to anyone who knows how to listen without filtering first."

They sat with it. The plotter hummed at a frequency the cabin air did not quite absorb. The screens waited. The corridor waited. The morning waited for someone to walk

down to the bridge and tell the admiral that the Serein system had answered a survey sweep with a sentence.

Iven stood. They took the diagnostic printout and the forecast-paper copy and the corridor draft and walked them to the legal desk. The duty legal officer was on the bridge. The legal desk was empty. Iven opened the filing drawer, placed the three documents inside, and closed the drawer without writing a label. They would label it later, in the legal officer's presence, after they had decided what the label should say.

On the way back they passed Tev Arin in the cable junction. Tev was standing on a maintenance cradle with his shoulders half out of a hatch, swearing softly about a connector that had lost its keying pin. He had a coil of patch cable over one shoulder and a cup of something that was not tea balanced on a cross-strut beside him.

"Cable holds," Tev reported without looking up. "Loose connector. One turn fixed it."

"Good."

"You okay, chief?"

"No."

"Do you want me to come up."

"Yes."

Tev set his tools down, took the cup off the cross-strut, drank from it, set it back, and followed Iven up the ladder to the cartography bay. Pell looked up when they came through and moved over without being asked. The three of them sat in the unromantic room and looked at the corridor plate.

"There's a sentence on the Lumen channel," Iven said.

Tev looked at them. "A sentence."

"In the Common script. Sitting under the mineral noise band. Arrived this morning. Not in any survey archive."

"What does it say."

Iven took the diagnostic printout from the legal-desk copy and handed it to Tev. Tev read it. Tev set it down. Tev picked it up again. Tev read it a second time.

"The reader sits outside the chart," Tev said.

"Yes."

"A message."

"Yes."

"Who is it for."

"Anyone who knows how to listen without filtering first."

Tev nodded slowly. He had grown up on a courier station with a maintenance key on his carabiner, and he knew the difference between a noise pattern and a transmission. He did not ask where it had come from. He did not ask who had sent it. He asked the question a thermal systems engineer asked, which was the question a person who had spent his life inside machines asked.

"Does it rhyme with anything else on the band?"

"It rhymes with three other readings in the survey archive. Three readings taken in three different years. Three readings we never cross-correlated before today. They rhyme because they came from inside the same source. The source has been sending for longer than the Wake has been listening."

Tev looked at the corridor plate. He looked at the diagnostic. He looked at Iven.

"What do we do?"

"We finish the corridor. We file the diagnostic with the admiral and we do not include the rhyme and we do not include the sentence and we let the admiral decide what to do with a Serein system that has just answered a survey sweep."

Tev nodded. Pell nodded. The screens stayed dark. The morning waited.

They finished at 11:42.

They printed the corridor. They set the printout on the plotter. They walked it down to the bridge in person, because the corridor carried a legal weight that the chief cartographer's hands were responsible for and because the chief cartographer had decided, on the way down the ladder, that they were not going to put this particular corridor into a pneumatic tube.

The corridor would be briefed at 14:00. The corridor would be filed with the admiral. The corridor would be flown by a courier who had already proven, on an earlier profile, that she was willing to fly a corridor the operations desk had not asked her to fly.

The diagnostic would be filed separately. The diagnostic would be filed in the legal drawer, under a label Iven had not yet written, in the legal officer's presence, after they had decided what the label should say.

The diagnostic carried a sentence that did not belong in a mineral survey. The sentence said the map could not contain its reader. Iven had spent seven years learning how to read maps and how to distrust maps that did not say what they could not contain. They had not, until this morning, been told by a chart's subject that the chart could not contain them.

They walked the corridor down to the bridge. They handed it to the bridge officer on watch. They did not explain the courier-pilot note. They did not explain the diagnostic warnings. They did not explain the dock-approach margin. They explained nothing at all, because the chart explained itself and because the courier who would fly it had already explained the rest.

The bridge accepted the corridor. The bridge officer logged it. The bridge officer said the admiral would brief the corridor at 14:00.

Iven nodded. They went back to the cartography bay. Pell was still at the second station. Tev was still at the plotter, holding a cup of something that was not tea and not looking at it.

"The corridor is filed," Iven said.

"The corridor is filed," Tev said.

"I'm going to keep working on the diagnostic."

"We are too," Pell said.

Iven sat. They pulled the diagnostic printout back up. They opened the Lumen channel without the routine filter and without the noise filter and without any filter the survey library applied by default. They listened.

The channel was quiet. The channel had said what it had come to say.

Iven watched the quiet for a long moment. They thought about the warning in the admiral's annex. They thought about the corridor they had just drawn. They thought about the sentence the system had answered with. They thought about the courier who had flown a profile the operations desk had not asked her to fly.

They thought about the probability field of the warning they had filed in the legal drawer. The probability field had been artificially narrowed, from a band the survey community would have called routine to a band the survey community would have called surgical. The narrowing had cost the warning its frame. The narrowing had been done by someone who knew what the probability field would have looked like at full width, and who had decided that the full width would not serve the purpose the warning had been sent to serve.

Iven picked up a clean sheet of forecast paper, dated it, time-stamped it, initialed the bottom, and wrote the finding in the careful block letters the legal desk used for filings of this weight.

The finding said the probability field had been narrowed. The finding said the narrowing had been deliberate. The finding said the narrowing had been done by a sender who knew what the full width would have looked like.

The finding did not say who had done the narrowing. The narrowing had been done by whoever had decided what the warning should say.

Iven put the finding into the legal drawer beside the diagnostic. They closed the drawer. They pocketed the key.

They went back to the plotter and began the courier-pilot note for a corridor they had drawn because an admiral had asked them to and because a courier had already flown the first half of it. The note would warn the courier about the worn bearing, about the comms latency in the bow shock, and about the dock-approach margin at the far end. The note would not mention the sentence the system had answered with, or the rhyme the system had been sending for years, or the probability field that had been narrowed by a sender who knew what the full width would have looked like.

The courier-pilot note would say: *Fly the corridor. Trust the chart. Do not trust the chart.*

Iven wrote the note. They filed it in the legal drawer beside the corridor, the diagnostic, and the finding. They locked the drawer and slid the key back into their pocket. They went back to the plotter and began the next chart, because a chief cartographer's job was not finished and because the dark still contained more than the maps could hold.

Chapter 4 — The Courier Burn

The corridor held her for forty minutes before the first interception call came. It came on the courier band, which meant a hull had matched her transponder within ninety seconds and had decided a courier on a fuel-expensive curve to *Everspring* deserved a second look. The call carried a hull identifier she did not recognize, a courier working channel, and the clipped prose of someone who had been on watch since before the lattice went quiet.

"*Quiet Verdict*, this is *Pell*-class cutter *Halen Forward*, four hundred clicks off your port quarter. Request call sign and burn authorization. Acknowledge."

Sena's hand went to the channel knob. She looked at the indicator on the board. The indicator told her what she had already known from the first patrol letting her pass without a hail: the chain of command on the patrol stations had not yet been told what the courier corridor carried. *Halen Forward* had been told only that an odd profile was passing through its window, and *Halen Forward* had decided to ask before she got any further.

"*Halen Forward*, *Quiet Verdict*, courier priority legal hold. Burn authorization will be transmitted under seal by the civilian registry desk at *Everspring* on arrival. Holding current profile."

The acknowledgment took six seconds. The six seconds were the longest six seconds of the morning.

"*Quiet Verdict*, *Halen Forward* copies. Be advised we are tracking a second transponder on your bearing, no hull ID, running cold. Recommend you confirm escort before continuing into the Serein corridor."

Running cold. Sena's hand came off the knob and went to the throttle. Honest John felt the touch the way Honest John always felt a touch. The tug did not ask why. The tug only asked how much.

"Sena, copy. Second transponder running cold, no visual."

"*Halen Forward* copies. Recommend you hold at the next waypoint buoy until we have a hull to send."

"Acknowledged, *Halen Forward*. Standing off at *Aster Lantern* seven. ETA forty-one minutes. Will accept escort on arrival."

She signed off. The corridor ahead held no stars. The Serein approach vector pointed into the dark side of the Wake, where Caul cast no light and the relay mirrors of Nox were only a careful geometry. The corridor held the spindle against her ribs, the bearing Tev had set to a conservative spec, the fuel gauge falling at the rate the profile had predicted, and the second transponder running cold on her bearing with no hull ID and no visual.

Cold meant a courier hull that had killed its reactor to hide, or a courier hull that had never lit one. Cold meant a courier hull that had been told to stay dark, or had decided to stay dark on its own.

She set the autopilot to hold at *Aster Lantern* seven and let her hand rest on the throttle. She had forty-one minutes of free fall. She used the first eight to call down to the thermal bay on *Aster Vale* and ask for Tev.

Tev answered on the second ring.

"Vale."

"Arin. Holding at *Aster Lantern* seven. I have a second transponder on my bearing, running cold, no hull ID."

"Running cold is a choice."

"Who is on your hull ID list."

"*Halen Forward*, courier working channel. No escort on the manifest."

"No escort on the manifest means no escort has been told to come. No escort has been told to come because no escort has been told to leave *Aster Vale*."

"Tev."

"Sena. The deck has not put an escort on any courier corridor since the lattice went quiet. The deck has not put anything on any courier corridor this morning, except you."

She waited.

"I am coming down to the hangar," Tev said. "I am going to look at your number-two radiator myself and write the alignment into the log with my own hand. Then I am going to ask the deck for an escort. The deck is going to say no."

"Do not ask the deck."

"Why not."

"Because the deck has not put an escort on my corridor for a reason. The patrol let me pass for the same reason. The second transponder is running cold for the same reason. The reason is a chain of command that has not been told what the corridor carries, and a chain of command that has not been told does not know to send an escort."

Tev did not say anything for a long time. Honest John hummed at a frequency the cabin air did not quite absorb. The bearing held at spec.

"Then who is on the second transponder," Tev said at last.

"I do not know. *Halen Forward* did not know. *Halen Forward* did not have a visual."

"A courier hull that has decided to stay dark on its own carries something the corridor has not been told."

"Yes."

"A courier hull that has been told something the corridor has not been told is a courier hull that has been told to stop the corridor."

"Yes."

"And a courier hull told to stop the corridor without being seen stopping it runs cold to do it."

Sena looked at the indicator on the board. The indicator showed forty-one minutes to the buoy and a spindle against her ribs and a bearing that had been set to a conservative spec by a thermal systems engineer who had known what the burn would require of it.

"Tev," she said. "I am going to keep flying."

"I am going to keep flying because the spindle is a legal filing. The legal filing has a registry desk at *Everspring*. The registry desk has a window. The window has a closing time. The closing time is before the chain of command on *Halen Forward* is told what the corridor carries. The closing time is before the deck decides to send an escort. The closing time is before the second transponder decides to stop the corridor."

Tev did not answer for another long time. The corridor did not ask her anything except to fly.

"Then I am going to write the alignment into the log," Tev said. "I am going to write it on the bearing sleeve with a grease pencil, the way I always write it, so the next engineer does not have to chase the history."

"Thank you, Tev."

"Do not thank me. Land."

Sena signed off. She let the autopilot carry her toward the buoy. She had thirty-three minutes of free fall and used the first to do what she had not done since she had strapped into the cockpit at 07:11.

She opened the drawer under the helm seat.

The drawer held a courier's odds and ends: a spare glove, a wrench, a data wafer with the calibration tables for the number-two radiator, and a small wooden box she had not opened in nine years. The box measured the length of a deck of cards. The slide lid carried a name burned into the wood in letters so small she had to bring the glove-light down to read them.

The letters said *Jori*.

Her fingers rested on the lid. The lid stayed closed. She had not opened it since the Pell rescue. She had carried it from tug to tug on every assignment since, because a courier who had flown with a copilot and had lost the copilot and had agreed to be the hero of the rescue and had not been the hero of the rescue carried a small wooden box in the drawer under the helm seat and did not open it. The box held a school ring. The school ring had come home. The copilot had not. The agreement had cost her the box.

Jori had sat in this seat. Jori had flown burn profiles by not asking the profile to forgive her anything. Jori had taken the helm of a school ring carrying four hundred children out of a storm window no one had charted. Jori had told Sena to take the copilot's seat in the ring and bring the ring home. Sena had refused. Jori had taken the helm of the ring. The ring had come home. Jori had not. The council had wanted a hero and Sena had agreed to be the hero because she had been the only one willing to agree, and the agreement carried a price the council had not been willing to pay.

Sena put the box back in the drawer and closed the drawer without opening it. She put her hand on the throttle. Honest John felt the touch. Honest John asked nothing of her. The corridor ahead opened into a starless sea.

The buoy came up on the forward port at the twenty-seven minute mark. It logged the courier's transponder, offered a cup of recycled water, and stayed silent. No escort waited there. Sena matched rotation, held

for four minutes, then let the buoy release her into the corridor. The second transponder remained on her bearing, running cold, no visual.

She kept flying.

The interception happened at the thirty-one minute mark. A *Pell*-class cutter older than Honest John rose on her port quarter at a hundred clicks, matched her burn, and began to close without a hail. Matte paint darkened one repaired radiator. No name marked the bow. The cutter had killed its reactor signature and moved on cold thrust.

It came through forty clicks, twenty, ten, then held at five. Sena kept her profile. After ninety seconds the cutter crossed to starboard, entered her blind side at three clicks, and continued inward. She transmitted her legal hold, transponder, and corridor profile on both courier and open bands so every patrol station could witness the unidentified hull matching her. It did not answer. Ten seconds later it altered toward her bow and closed to one click.

Sena's hand came off the throttle and went to the radiator toggle. The number-two radiator was the radiator Tev had set to a conservative spec. The number-two radiator was the radiator that had held the corridor so far. The number-two radiator was the radiator that would hold the corridor if the corridor asked her to bank at the third burn.

She did not toggle the radiator. She toggled the bearing.

The bearing had been set to a conservative spec. The bearing had held the corridor. The bearing had not been asked to hold a one-click intercept at the courier's blind side, and a bearing that had not been asked to hold a one-click intercept was a bearing about to be asked to hold something the conservative spec had not promised.

The intercepting hull altered its burn again. The alteration put the hull on a vector that would cross her bow in eleven seconds at the corridor's speed.

She had eleven seconds.

She had eleven seconds and a spindle against her ribs and a bearing that had been set to a conservative spec and a corridor that had decided, at the thirty-one minute mark, that the corridor wanted her to do the thing a courier did when the corridor wanted her to do something the courier did not want to do.

She banked.

The corridor had been asking her to bank at the third burn. The bearing had been set to hold that bank. Tev had aligned the bearing to a conservative spec for that bank. Honest John took the bank at the corridor's speed and asked nothing of her except to keep flying.

The intercepting hull crossed her bow at three hundred meters. The hull stayed silent across the approach. The hull crossed her bow and continued on the corridor's vector and was gone.

Honest John came out of the bank with a sound that was not the sound Honest John usually came out of a bank with. A bearing had asked for something the conservative spec had not promised. A bearing that had been asked for something the conservative spec had not promised was a bearing that was now telling her, through the yoke, what the conservative spec had been holding back.

Sena listened to the yoke. The yoke told her the number-two radiator had come out of the bank half a degree off the cold side of the ship. The yoke told her half a degree off the cold side was a margin Honest John could fly on for the rest of the corridor. The yoke told her half a degree off the cold side was a margin Honest John could not fly on for the burn home.

She had a margin. She did not have a margin for the burn home. A courier who did not have a margin for the burn home was a courier who was going to ask for help whether she wanted to or not.

She set the autopilot to carry her toward *Everspring* and let her hand rest on the throttle. She had forty-nine minutes of free fall left and used the first to open the drawer under the helm seat.

Her fingers found the lid of the box. The box stayed closed. She thought about the intercepting hull that had crossed her bow at three hundred meters and had not hailed and had not challenged and had not identified itself, and the intercepting hull carried the shape of a hull that had been told to stop a courier corridor without being seen stopping it, and the corridor had decided, at the thirty-one minute mark, that the corridor was not going to be stopped by a hull that had been told to stop it without being seen.

Jori had flown a corridor that had not been charted. Jori had flown the corridor and Jori had died in the corridor, and the corridor had come home in the tug's archive drawer without Jori in it. Sena had agreed to be the hero of that corridor and had not been the hero, and the agreement carried a price the council had not been willing to pay.

She thought about the truth. She did not let herself think about the truth.

She closed the drawer. Ahead of her, the corridor opened into a starless sea.

The seedship came up on the forward port at the forty-three minute mark. The seedship showed itself as a long slow shape against the dark, with the matte black of the Wake's outer hull and the faint silver of the adaptive biosphere's ventilation glow. The seedship did not have a courier dock on its near side. The seedship kept its courier dock on the far side, where the civilian registry office kept its window, and the window had a closing time, and the closing time was now.

She matched rotation with the seedship's far-side dock. The dock ran small. The dock's hatch was amber. The hatch had been told to expect her.

The deck on *Aster Vale* did not answer her approach call. The deck had stopped answering her approach call at the forty-one minute mark, which was the mark at which the deck would have decided that a courier tug on a legal hold to *Everspring* was a courier tug the deck did not want to know about.

She transmitted clearance to the dock. The dock did not answer. The dock did not need to answer. The dock carried an amber hatch and the amber hatch had been told to expect her.

She cycled the airlock and walked into *Everspring's* civilian section. The seedship smelled cool, vegetal, faintly sweet, and the adaptive biosphere breathed through the vents in slow regular pulses. The seedship carried eighteen thousand four hundred sleepers. A courier carried a legal filing on behalf of sleepers like these.

Okoro was waiting in the small conference room between the medical bay and the adaptive biosphere. Okoro had not changed out of the deck uniform. Okoro had not changed out of the look on her face.

"The spindle," Okoro said.

Sena reached into the inner pocket of her flight jacket. She set the data spindle on the table between them. It ran no larger than a finger. The casing was medical white. The green band caught the conference room light and held it.

Okoro took the spindle and set it in the inner pocket of her own jacket. The weight of it settled against her ribs the way the weight of it had settled against Sena's ribs.

"I have not been told what the spindle is for," Sena said.

"You are not going to be told."

"I am going to be asked to do one more thing."

Sena waited. The conference room was quiet. The adaptive biosphere breathed through the vents.

"The deck on *Aster Vale* has stopped answering your approach call," Okoro said.

"The deck stopped answering at the forty-one minute mark."

"The deck has told the bridge a courier tug on a legal hold is no longer on the manifest. The deck has told the bridge the courier tug was intercepted and did not survive. The deck has told the bridge the spindle has not reached the seedship."

Sena looked at Okoro. Okoro looked back. The adaptive biosphere breathed through the vents in slow regular pulses.

"The deck has told the seedship's bridge that I am dead," Sena said.

"Yes."

"The deck has decided that a courier who has delivered a legal filing to the civilian registry office is a courier who has outlived her usefulness to the deck."

"Yes."

"And the deck has decided that a courier who has outlived her usefulness is a courier who did not survive the interception."

"Yes."

"A courier who did not survive is a courier who cannot be asked to leave the seedship."

"Yes."

"A courier who cannot be asked to leave the seedship is a courier who has been told to stay."

Okoro did not say anything. The adaptive biosphere breathed. The seedship held eighteen thousand four hundred sleepers, and the courier did not get to leave.

"Then I am staying," Sena said.

Okoro nodded. She walked out of the room and closed the door behind her. The door made the kind of sound a door made when the door had been told to close.

Sena stood with her hand on the empty inner pocket of her flight jacket. The spindle was gone. The corridor was gone. The intercepting hull was gone. The school ring sat in the drawer under the helm seat of Honest John, and Honest John was docked on the seedship's far side, and the drawer carried the kind of cargo a courier did not get to open once the courier had been told she was dead.

She did not open the box.

She walked out of the conference room and into the seedship's civilian section. The civilian section held eighteen thousand four hundred sleepers. The sleepers carried the kind of names a courier did not get to be told about, and the courier had been told she was dead, and the courier was staying.

Chapter 5 — The Seedship Verdict

Calder had her hand inside a missile-key cabinet when the command wall changed from amber to white.

White stripped every tactical mark from the glass. The calibration clock vanished. So did the tracking cones enclosing *Everspring*, the convoy registry, and the quiet red point marking where an unidentified cutter had crossed Sena Vale's courier path. A single authentication seal took their place. Its outer ring carried the Descendant Compact's oldest cipher. Its inner ring carried a legal filing receipt from the seedship.

Commander Pell stopped reading out key numbers. Two weapons officers held their positions at the open consoles. Nobody on *Resolute Vigil*'s command deck spoke over a white wall.

Calder withdrew her hand. The cabinet still held two physical keys in separate slots, each wrapped in a tamper ribbon. She closed the steel door without touching either one.

"Route the filing to my table," she said.

Pell obeyed. A block of text appeared under the seal. Calder had read the first part nineteen hours earlier in the mirror vault: verified temporal origin, fifty-three-year interval, terminal forecast condition, catastrophic outcome assigned to *Everspring*'s arrival in the Serein system. The message had ordered interdiction before insertion. It had not, on the page she received, defined interdiction.

The legal receipt unfolded a second page.

AUTHORIZED REMEDY: TOTAL DESTRUCTION OF MISSION VESSEL, BIOLOGICAL MANIFEST, EMBRYO ARCHIVE, AND ASSOCIATED SETTLEMENT CAPACITY.

EXECUTION WINDOW CLOSES AT SEREIN INSERTION.

SURVIVAL OF MISSION PERSONNEL INVALIDATES FORECAST REMEDY.

Calder read the lines twice. The room held its discipline beside her. Air handlers pushed dry cold across the deck, but heat from the return-mirror lattice ticked through the bulkhead behind the weapons stations.

"Source of release," she said.

"Civilian registry challenge," Pell answered. "Doctor Okoro filed from *Everspring*. Her spindle demanded disclosure of any order affecting sleeper consent or survival. Registry found an annex under command partition and released it to the named vessel. Release propagated to flag authority under the same clause."

"Who opened the annex before the registry challenge?"

Pell's hands moved across his board. "Two command credentials. Yours at initial authentication. Cartography chief Mora's forty-three seconds later."

Calder remembered Iven standing in the mirror vault with one hand flat on the map table. She remembered their careful account of what the warning named and what its probability field omitted. They had told her the order required interdiction. They had not spoken these three lines.

"Seal Chief Mora's mirror access," Calder said. "Do not arrest them. Put an escort outside cartography and schedule an inquiry."

One weapons officer glanced toward her, then corrected the movement. Calder saw it anyway.

"Speak, Lieutenant Sorn."

"Admiral, the remedy has command authentication. If its window closes at insertion, every minute affects firing geometry."

"It affects inquiry geometry too. Continue calibration. No arming current until I give it."

Sorn looked at the white wall. "Yes, Admiral."

Calder keyed a secure channel to *Everspring*. The link produced a queue notice. Okoro had opened her ship's registry band to every civilian advocate in the convoy, and thousands of requests now competed for the same slim laser aperture.

"Priority flag," Calder said. "Doctor Nadi Okoro."

The channel cleared after eleven seconds. Okoro appeared against a wall of pale seedship insulation. Her uniform collar sat open. A medic moved behind her carrying a diagnostic case.

"You have read it," Okoro said.

"I have."

"Eighteen thousand four hundred sleepers have not."

"They cannot read anything while they remain in suspension. Their legal proxies can."

"Their proxies received a mission brief that described interdiction. The annex orders their deaths. I am sending it to every proxy and every crew member aboard this ship."

Calder watched a status bead beside the channel. *Everspring* had begun copying the annex through its internal network. The same filing clause that exposed it to Okoro had also removed flag authority to reseal it.

"Send it," Calder said. "Use the full authentication chain. Do not post an excerpt."

Okoro's expression tightened. "You are permitting publication?"

"I am ordering publication. The convoy will receive the same text at the same time."

"And after they read it?"

"I will issue orders."

"That answer has killed people before."

Calder ended the channel before anger could choose her reply. She turned from the table to face Pell and the two rows of watch officers behind him.

"Open convoy command, civilian registry, and public emergency bands. Attach the source packet. Give every ship sixty seconds to prepare local storage. I will read the annex aloud."

Pell hesitated only long enough to verify her authority. "All bands will include *Everspring*."

"They had it first. Open them."

The command wall divided into ship acknowledgments. Habitat tenders answered first, then defense cutters, convoy freighters, mirror stations, and the distant icons of survey craft working the Serein corridor. Some links carried several seconds of latency. Others arrived as text-only receipts. The public band opened last, a small green mark beneath the tactical display.

Calder placed both palms on the table.

"This is Admiral Rhea Calder, commanding Wake Defense Fleet. At 03:17 ship time, a legal registry challenge released a concealed annex to the final authenticated forecast received by the Commonwealth. The annex concerns seedship *Everspring*. I am transmitting its complete text and authentication record now."

Pell sent the packet. Receipt numbers climbed too quickly to follow.

Calder read each lethal line without changing its words. She named biological manifest, embryo archive, settlement

capacity, and the deadline before Serein insertion. She read the sentence that denied survival to mission personnel. Then she let the band carry five seconds of open air from her deck.

"This order has not been executed," she said. "No weapon has been armed against *Everspring*. I have directed an inquiry into the annex's concealment and a full review of its forecast basis. Effective immediately, the seedship remains under quarantine. No vessel may dock, detach, transfer data outside emergency channels, or alter the seedship's assigned hold without flag authorization."

The public request counter flooded red.

"Questions will route through ship captains and civilian advocates. Weapons stations will move to readiness under sealed-key discipline. Readiness does not authorize fire. I will issue no firing order until command has reviewed the remedy, the insertion deadline, and the rights of the eighteen thousand four hundred people named only as manifest cargo. Admiral Calder, transmission complete."

Pell closed the open microphone. The deck noise came back in pieces: ventilation, cooling pumps, key switches under guarded covers. Then every captain in the convoy tried to reach her.

Before Calder could choose a channel, a proximity alarm sounded at the quarantine desk. A maintenance skiff had detached from habitat tender *Common Measure* and begun crossing the new boundary. Its pilot transmitted on the medical band.

"Flag, this is Skiff Four. We have cryogenic pump rotors for seedship bank twelve. Delivery entered the schedule yesterday. Those pumps cannot wait for your inquiry."

A second voice broke through behind the pilot. "My daughter sleeps in bank twelve. Her proxy file says component maintenance cannot be obstructed by military action."

Calder pulled the skiff's vector onto the table. The craft carried little reaction mass. If it continued another ninety seconds, the nearest patrol cutter would have to choose between physical interception and allowing the breach.

"Skiff Four, cut thrust and hold attitude," Calder said.

"Negative. *Everspring* requested these rotors six hours ago. They have a twelve-day service margin."

"Then they can wait one hour for inspection. Cut thrust."

"One hour becomes two. Two becomes a tribunal. Bank twelve has eight hundred sleepers in it."

Sorn lifted his eyes from the weapons board. Calder heard the unasked question: what force applied to a civilian skiff inside a military quarantine?

"Patrol cutter *Kindly Oath*, match the skiff without crossing the boundary," Calder ordered. "No targeting laser. Skiff Four, vent your forward maneuvering line if your control board has locked you out."

"It has not locked me out."

"Then use it. The seedship can verify bank pressure over medical telemetry. If any pump falls below margin, I will permit a remote cargo transfer by tether. No person crosses."

The pilot swore away from the microphone. The mother did not. "Admiral, you read an order to kill my daughter and now you want us to trust your maintenance judgment."

Calder kept the skiff vector centered. "No. I want you to distrust me from outside a collision path. Cut thrust."

For seven seconds the craft continued. *Kindly Oath* lit two lateral jets and moved to meet it. A boarding clamp unfolded from the cutter's nose, slow enough to show restraint and large enough to show consequence.

Skiff Four vented its forward line. The plume spread in a white fan. Its speed bled away twenty meters short of the boundary.

Calder released a breath through her nose. "Send the rotor specifications to quarantine inspection. Have *Everspring* stream bank twelve telemetry in public view. If the need verifies, the skiff transfers cargo by line."

"Acknowledged," the pilot said, stripped of every courtesy.

The mother stayed on the channel. "Her name is Lina Seft. Put that in your weapons log."

"Commander Pell, record Lina Seft beside bank twelve and this incident."

Pell entered the name. "Recorded."

The quarantine desk added a maintenance exception procedure while twenty-three more civilian requests arrived. Insulin cultures. Embryo coolant. A crew member's husband asking permission to dock and remain. Calder ordered each request preserved, no automatic rejection permitted. Quarantine could not become a quieter spelling of abandonment.

She chose three command channels. Okoro occupied the first. Captain Besh of *Watchful Coil* took the second. Iven's inquiry escort had placed a silent request on the third.

"Hold Mora," she said. "Put Besh through."

Besh appeared with his bridge behind him. "You have a verified remedy and an execution window. Why announce a review?"

"Because the remedy demands civilian extermination."

"It demands prevention of a forecast catastrophe."

"By killing civilians."

"The distinction does not alter the authentication."

Calder enlarged the warning's probability field beside his face. Its confidence cone contracted to a blade at Serein

insertion, then disappeared beyond the event. "Authentication proves where the message came from. It does not explain why every surviving branch vanishes from this model."

"That question belongs after obedience."

"Not when obedience removes everyone who could answer it."

Besh folded his hands beneath the camera's view. "Hespar spent seventy-two hours asking why."

The name reached her more cleanly than an accusation. Calder looked down at the ring on her right hand. It had belonged to a council clerk who wrote three requests for confirmation while southern transports lifted half empty. The clerk's last letter still waited in her private files, written in a careful hand from a city already losing power.

"Hespar received an evacuation order," she said. "This annex demands destruction."

"Hespar received a verified forecast. So did we."

"Two million died because Hespar delayed movement that could have saved them. If we fire now, eighteen thousand four hundred die because we refused to delay killing them. Do not flatten those acts into one lesson, Captain."

"Then tell me what lesson you kept."

Calder let the question stand between their bridges. "Prepare pursuit stores and boarding teams. Hold weapons at readiness. I want options that preserve the insertion deadline without treating immediate fire as our only form of obedience."

"That sounds like delay."

"It sounds like command."

She cut Besh's channel and opened Okoro's. The ecologist no longer stood in the insulated corridor. She had moved to *Everspring's* operations room. Crew crowded behind

her, some in medical whites, some in fabrication coveralls. Sena Vale leaned against the far bulkhead with one arm strapped across her chest. Calder had watched a false casualty notice cross the fleet record two hours earlier. The woman inside it looked pale and furious, but alive.

"Pilot Vale," Calder said. "Your ship reported you lost during interception."

"My ship reported what somebody needed it to report."

"You will give evidence to the inquiry."

"If I can undock. Your quarantine says I cannot."

"Then you will give it from there. Doctor Okoro, confirm no thaw cycle has begun."

Okoro looked toward a bank of suspension telemetry. "No general thaw. Four medical exceptions remain active from before the warning. No embryo process has started. Terraforming stays cold."

"Confirm your insertion schedule."

"We cross the commitment gate in four days if we hold the assigned profile. Actual Serein insertion follows later. A full abort before the gate costs most of our braking reserve."

Calder heard the qualification. Later did not mean safely later. A seedship carried enough mass to turn every course change into a decision measured in fuel, heat, and lives.

"You will hold your present vector while my navigation staff calculates a tow solution."

"Our present vector takes us toward the gate."

"A braking hold would expose your radiators to convoy debris."

"A tow gives you physical custody."

"Quarantine already gives me legal custody."

Okoro's gaze sharpened. "Legal custody does not move a ship."

"No. Reaction mass moves a ship, and you do not have enough to waste proving a point. Hold your vector. Accept no docking traffic. Keep your terraforming lattice cold."

"Will you guarantee no strike while we comply?"

Calder looked past Okoro at the crew. Sena had shifted closer to the operations table. Her good hand rested beside a brass manual control with its safety cover lifted.

"I guarantee no strike without my direct order," Calder said. "I have not given one."

"That guarantee ends if your command ends."

"So does every order I can give you."

Okoro glanced at the annex on her display. "The people in suspension have inherited a verdict from descendants they cannot question and an admiral they cannot elect. I need more than a procedural promise."

"You need time. I am giving you what time the corridor allows. Do not use it to run."

Okoro did not agree. She closed the channel with the formal acknowledgment required by fleet law, not the plain one Calder wanted.

The inquiry escort brought Iven onto a slim side screen. They sat alone in the cartography bay with their hands visible on the table. No restraints. A security officer stood outside the sealed door.

"Chief Mora," Calder said. "Did you access the annex before today?"

Iven looked toward something below the camera, perhaps a map they could no longer alter. "I will answer under recorded inquiry."

"This channel records."

"Not with an advocate present."

"Eighteen thousand four hundred people learned an hour ago that command carried an order to kill them. I am asking whether you knew."

Iven raised their eyes. Exhaustion had hollowed the skin beneath them. "I knew the authentication extended beyond the page in your packet."

"That does not answer me."

"No. It does not."

Calder felt the room waiting for her anger. She had used anger as a tool in mutiny hearings, treaty failures, and casualty rooms. Here it offered no leverage. Iven had chosen the shape of the inquiry before she called.

"Your access remains sealed," she said. "At 0600 you will appear with an advocate. Bring every copy, note, and route alteration associated with this warning."

"Admiral, do not let tactical navigation choose a braking corridor before Tev Arin reviews the seedship's thermal limits."

"You are not in a position to advise me."

"No. The sleepers are. They cannot speak, so their radiator margins will have to do it."

Calder closed the channel.

Pell had assembled the readiness board while she spoke. Seven defense ships could place laser energy on *Everspring* within an hour. Three had viable kinetic solutions, though debris risk made two unacceptable. Boarding required matching the seedship's vector and crossing a quarantine boundary Calder had just established. Tow clamps would take longer to rig than the current geometry permitted.

"Remove kinetic options," Calder said.

Sorn turned from weapons. "All of them?"

"A miss continues into the Serein corridor. A hit produces debris among eighteen thousand suspended bodies. Remove them."

"That leaves laser disablement and boarding."

"Prepare both."

Commander Pell opened the key cabinet. This time Calder took the first physical key. Its tamper ribbon snapped with a dry sound. Pell took the second. They crossed the deck to separate arming consoles.

Sorn projected the laser plan. Three emitters would converge on the seedship's forward communications crown, not its inhabited spindle. A second solution targeted drive-control relays along the unpressurized truss. Each burn risked reflective scatter. Each demanded hours of cooling before a second shot.

"No radiator targets," Calder said.

"Disabling the drive without touching heat rejection leaves us two relay clusters," Sorn replied. "Either hit may propagate into guidance."

"Build a lower-energy ladder. First burn severs external command. Second takes one relay. Leave enough interval for surrender between them."

"The warning's deadline does not account for surrender intervals."

"Our weapons plan will."

Pell called up boarding geometry. Matching *Everspring* required two cutters to shed spin, burn hard for thirty-two minutes, then coast while their radiators recovered behind shield panels. Crews would reach the hull exhausted and warm. A rushed docking could puncture the seedship before any defender fired.

Calder assigned a medical tender to the intercept group and removed its offensive drone rack. She cut the boarding party from sixty to twenty. Armor remained permitted;

breaching charges did not. Every limitation reduced the chance of taking control. Each also reduced the chance of proving the annex right through their own violence.

"Readiness state?" Calder asked.

"Cold racks loaded," Pell said. "Guidance isolated. Emitters aligned but safed. Two-key authorization required for capacitor charge."

"Set pursuit stations. Bring *Resolute Vigil*'s spin down by one tenth before any burn. Warn medical. Tell every escort captain that crossing the quarantine line requires my voice authorization."

The deck reshaped itself to fit the orders. Seats locked. Chart displays traded convoy routes for intercept cones. Deep inside the ship, missile racks moved on their rails without waking the weapons inside them. Radiator pumps climbed one setting to absorb guidance and sensor load.

Calder inserted her key but did not turn it. Pell mirrored her at the second console.

"Flag key present," he said.

"Witness key present."

Sorn watched the capacitor board. "Weapons prepared. Charge remains zero."

On the tactical glass, *Everspring* continued toward Serein along the same patient curve. A quarantine ring enclosed its icon, first amber, then red as each enforcement ship acknowledged. Civilian craft withdrew from the boundary. One small tug remained docked to the seedship's far side, its transponder still insisting that its pilot had died.

Calder set the annex beside the firing solution. One demanded destruction before insertion. The other showed how quickly destruction could be arranged.

"Maintain quarantine," she said. "No fire. Keep every weapon ready."

The red ring closed over *Everspring*. Under Calder's hand, the unturned key waited in its lock.

Chapter 6 — A Debt in Possible Lives

The inquiry table rejected Iven's handprint on the first attempt.

They wiped their palm on their trousers and tried again. The glass recognized them, lit their name beneath their fingers, and added a red restriction mark beside it. CHIEF CARTOGRAPHER. ACCESS SUSPENDED. WITNESS STATUS.

Across the table, Admiral Calder watched without offering a seat. Commander Pell arranged three sealed archive blocks beside the recording node. Iven's advocate, Jo Venn, checked the room's public-log indicator and found it active. A security officer waited beyond the transparent door. Through that door, the mirror deck showed only half its usual light.

"Sit," Calder said.

Iven sat. The chair had been fixed to the deck for acceleration, although *Aster Vale* carried no burn warning. Inquiry rooms assumed every stable condition might end before testimony did.

Venn placed a blank legal wafer beside Iven's hand. "My client will answer factual questions concerning access, custody, and disclosure. Questions about intent must identify the governing clause."

"Your client can identify clauses without assistance," Calder said.

"That has contributed to our present problem."

Pell suppressed a reaction. Calder did not.

She touched the first archive block. "At 22:41 yesterday, Chief Mora opened the authenticated forecast. Forty-three seconds after my credential entered the vault, theirs accessed a command-partitioned annex. Did you use your own credential?"

"Yes," Iven said.

"Did anyone direct you to open it?"

"No."

"Did you understand its lethal remedy?"

Venn put two fingers on the wafer. "Governing clause."

Calder keyed the citation onto the table: Compact emergency custody, fraudulent omission during active threat review. "Did you understand it?"

Iven looked at the text beneath their hand. They had spent nineteen hours refusing this sentence, first to Calder, then to themselves.

"Yes."

The recording node changed from green to blue. Admission indexed.

Calder's face showed less anger than Iven expected. The restraint made the room harder to map. "State what you read."

"An instruction to destroy *Everspring*, its biological manifest, embryo archive, and settlement capacity before Serein insertion. It specified that mission personnel could not survive the remedy."

"Why did you tell me only that the forecast required interdiction?"

Venn's fingers moved again. Iven stopped them.

"Because I concealed the annex," they said.

Blue light pulsed through the node. Beyond the door, someone crossed the mirror deck carrying a coolant gauge. The ordinary movement made the admission feel more exact, not less.

Calder sat at last. "We will reach motive after custody. Show me how you got through the partition."

Pell pushed a stripped copy of the packet onto the table. Its layers opened as nested contour maps. The outer authentication carried the final warning. A legal sheath held provenance, checksum history, and the fifty-three-year transit solution. Beneath both lay an unlabelled region with no ordinary access route.

Iven rotated the packet. "The partition followed a survey convention, not a command convention. Survey archives omit hazards from public products but preserve them outside the visible map. Whoever built this message used our blank-space notation."

"A notation developed in this century," Pell said.

"A notation the sender could have inherited fifty-three years from now."

"Or copied from this packet after you altered it," Calder said.

Iven enlarged the audit edge. "My access left this scar. The partition geometry predates it."

Pell compared creation and access hashes. His mouth tightened. "Confirmed."

Calder touched the blank region. "Why hide a lethal instruction inside cartographic syntax?"

"Because command would authenticate the visible warning first. A cartographer would notice the omission."

"You think it targeted you."

"I think it expected a person trained to distrust empty space."

The answer entered the record. Iven heard how it sounded: self-importance dressed as method. They pulled the packet back and showed a cluster of minute checksum differences along the legal sheath.

"It also expected an inquiry," they said. "These marks opened only after the annex entered a public challenge."

Venn leaned closer. "Opened into what?"

"I do not know yet. My access ended before the registry filing changed the packet."

Calder glanced at Pell. "Preserve those marks. Nobody executes embedded code."

"Already isolated," Pell said.

A low vibration passed through the deck as *Aster Vale* adjusted spin balance. The archive blocks clicked against their magnetic rests. Pell caught the oldest one before it shifted.

Its casing bore a faded date from one hundred twelve years earlier.

Calder set it in front of Iven. "You cited possible coercion in your sealed note. Before we discuss that claim, you will reconstruct the institution you accuse."

"I accused a procedure, not an institution."

"Procedures do not hide pages by themselves."

Pell opened the block. Grainy medical footage rose above the table: isolation tents, hand-pumped respirators, a child's chest struggling under transparent film. A date stamp placed the images six days before the Gray Lung forecast arrived.

"First verified return warning," Calder said. "Read its instruction."

Iven knew the text. Every schoolchild knew it. They read from the exhibit anyway. "Close the Beren ports. Destroy

imported fungal feedstock. Change respirator membranes to silver mesh."

Pell advanced the record. A dockmaster objected to stopping food shipments on an unauthenticated message. A laboratory technician tested the feedstock and found spores nobody had cultured before. Port closure followed. Infection curves bent within three days.

"Estimated lives preserved," Calder said.

"The estimate remains disputed."

"Give the accepted range."

"Twenty-eight million to thirty-four million."

Another exhibit opened. A reactor baffle redesigned from measurements not yet taken. A storm shelter moved six kilometers before a moonquake. A childhood vaccine altered by one protein. No lecture joined them. Calder made Iven handle each object, read the contemporary objection, and place the later casualty record beside it.

The Descendant Compact had not begun as a constitution. It had accumulated from emergency receipts. A port seal here. An insurance clause there. Then fleet law required compliance with verified engineering warnings. Civil courts began treating refusal as negligence. Councils kept their formal right to disobey while every bond market, evacuation schedule, and medical standard punished the attempt.

Pell opened a ship mortgage from eighty-seven years earlier. Its survival clause voided coverage if a captain ignored forecast advice. Venn found the same clause, revised but intact, in *Everspring's* mission insurance.

"Show the consequence of refusal," Calder said.

Pell copied the seedship policy into the inquiry sandbox. He changed one field from COMPLIANT to CAPTAIN DECLINES FORECAST REMEDY. The policy voided itself at once. A linked fuel covenant flashed next, because

uninsured vessels could not draw reserve deuterium from Commonwealth depots. Berthing indemnity failed after that. Civilian proxy coverage for all eighteen thousand four hundred sleepers entered review.

"No order appears in any of those clauses," Venn said.

"No," Iven answered. "Each clause recognizes the next one as ordinary risk management."

Calder pointed to the failing links. "Restore the policy and decline only the lethal annex."

Pell separated the visible warning from its remedy. The sandbox refused the distinction. Both shared one authentication root, so compliance attached to the packet as a whole. He tried marking the annex contested. The insurance remained active for twelve hours, pending Compact review, then failed if no tribunal issued a stay.

"Twelve hours," Calder said. "Longer than our clean intercept window."

"Shorter than meaningful civil review," Venn replied.

Iven touched the fuel covenant. "This structure explains why advice can remain formally nonbinding. A captain may refuse. Afterward the ship loses fuel rights, berth rights, liability coverage, and access to public rescue guarantees. Nobody arrests the captain for dissent. The crew simply inherits consequences large enough to make dissent impossible."

"Those consequences protect everyone who relies on forecasts," Calder said.

"They also turn authentication into judgment."

Pell restored the original policy. Green marks returned across the chain. *Everspring* became insurable again, eligible for fuel, and welcome at Commonwealth docks, provided it accepted a remedy that erased it.

Calder studied the restored links. "Could Doctor Okoro have launched without these covenants?"

"Not at this mass," Pell said. "No civilian consortium could indemnify eighteen thousand sleepers without Compact backing."

"Then she accepted the rules before departure."

Venn shook her head. "She accepted rules for advice. Nobody disclosed an authenticated order requiring total destruction of her passengers. Consent to a system does not include every concealed output the system may later produce."

The sandbox kept every consequence visible. Iven had mapped transit corridors with fewer dependencies than this contract carried. None of the lines called itself force. Together they left *Everspring* one legal direction.

"No council voted away present authority in one act," Iven said. "We priced disobedience out of ordinary life."

Calder tapped the Gray Lung footage. "Ordinary life continued because people obeyed."

"Yes."

"Say that without qualification."

"People lived because councils obeyed verified warnings."

"And you concealed one."

"Yes."

The simple answer left no route behind it. Iven could trace every institutional contour leading to their act. None removed the act.

Calder opened the third archive block. Iven knew its damaged corner before the label appeared.

WAKEWORLD FLOTILLA EVACUATION REVIEW.
SEALED FAMILY MATERIAL.

Venn stood. "Admiral, this exhibit exceeds the custody inquiry."

"Your client cited it in their private note."

Iven had forgotten the note. They had written it at twenty-three, after their first certification board asked why their hazard maps carried visible uncertainty bands instead of one recommended route. The answer had gone into a sealed ethics file. Somebody on Pell's staff had read deeply.

"Open it," Iven said.

Venn turned toward them. "You do not need to consent."

"It already shaped what I did. Put it where people can see it."

Pell broke the family seal.

A migration map spread across the table. Twelve habitats moved between Caul's outer work zones, each on a slow fusion economy that left little reserve. A forecast had predicted a containment failure in one habitat's fuel plant. The instruction optimized evacuation by transferring tug capacity to eleven habitats and cutting the twelfth loose before its damaged reactor could threaten the flotilla.

Iven placed two fingers on the omitted route. Habitat Twelve had no line leading out.

Their mother had served as traffic coordinator aboard Habitat Four. Her voice entered the archive through an operations call, younger than Iven remembered and more impatient.

"We can send two tugs back after Four clears," she said in the recording. "Twelve has six hours before containment loss. Two tugs can take eight hundred people."

A forecast liaison answered from another ship. "Returning the tugs creates a collision probability across the main evacuation lane. The model preserves three thousand fewer lives if they turn back."

"Those eight hundred have names."

"So do the three thousand."

The call ended there. Iven's mother had followed the optimized route. She cleared four habitats through a debris field with no collisions. Thousands reached the safe transfer orbit. Habitat Twelve broke containment five hours later.

Calder paused the record before the casualty summary. "Your mother survived the evacuation."

"For eleven months."

"Cause of death?"

"Radiation exposure from flying the outer traffic station after its shielding failed. The forecast route required somebody to keep that station crewed."

Calder looked at the frozen image. "Did the instruction conceal that cost?"

"No. It listed a probable fatal exposure for the coordinator. She volunteered after reading it."

"Then the warning gave her a choice."

"It gave her a choice between her life and a collision field full of families."

"She still chose."

Iven's mother had recorded one private message after the evacuation. She sat in a medical berth with burn patches covering her neck. No accusation entered her voice. That absence had hurt Iven more than blame could have.

Pell began to close the exhibit. Iven stopped him.

"Play the family message."

The younger face looked out over the inquiry table.

"Iven, if they show you the numbers later, they will show Twelve beside the flotilla total. Do not let them say the total contains Twelve. A sum can count losses without carrying them. I followed the route. I would probably follow it again. That does not make the route innocent, and it does not make me innocent either."

She reached toward the recorder. Her fingers shook from treatment drugs.

"You will want a map that saves everyone. Build honest maps instead."

The image ended.

For several seconds nobody touched the controls. The recording node kept its blue light. It did not know which admission to index.

Calder spoke first. "You believe the final warning repeats that evacuation."

"No," Iven said. "The evacuation instruction disclosed its human costs. It gave my mother a decision she hated, but she could see the people excluded from the route. This message hid the remedy and compressed eighteen thousand four hundred sleepers into biological manifest. It removed the moral object from the visible page."

"So you removed the page from me."

"Yes."

"You answered concealment with concealment."

"Yes."

Venn sat down slowly.

Calder folded her hands. "Why?"

Iven could have described the probability cone, the suspicious blank beyond insertion, or the way obedience had hardened into insurance and command law. All of it mattered. None had moved their hand when they closed the annex.

"I wanted one hour in which the order could not become a weapon," they said. "I told myself I would test the field and disclose the remedy once I had evidence. Then the hour became several. Every delay made disclosure harder, because I had also concealed the delay."

"Did you intend to prevent execution?"

"I intended to prevent automatic execution."

"That distinction belongs to you, not to the people placed at risk."

"I know."

Calder's voice stayed level. "Your choice may have spent the only clean interception window. If *Everspring* runs, weapons will need to fire at worse geometry. Debris may enter the Serein corridor. Boarding crews may die."

"If I had disclosed it at once, would you have fired?"

Venn whispered Iven's name. Too late.

Calder regarded them across the exhibits. "I do not know."

That answer cost her. Iven saw the cost in the way she kept her right hand off the old ring. An easier commander would have lied.

"Neither do I," Iven said. "I took the decision because I feared your answer. That did not make it mine to take. I concealed the annex. I altered who could act, when they could act, and what risks followed. Enter all of that as my admission."

Pell keyed the language into the record. Venn added a reservation concerning command causality. Iven let both stand. Maps often needed conflicting legends before anyone noticed the terrain did not belong to either one.

A tone sounded from the isolated packet.

Pell reached for the cutoff. Calder caught his wrist before he touched it.

"Report first."

He checked the air-gapped monitor. "The checksum marks have assembled into an index. No external execution. The inquiry node supplied a legal vocabulary table when we opened the family exhibit."

"Why would an archive exhibit affect the packet?" Venn asked.

Iven leaned toward the monitor. The marks no longer resembled damage. They formed retrieval coordinates through the authenticated sheath, each one pointing to a region too dense for ordinary warning text.

"Because we entered testimony concerning delegated choice," they said. "The packet had been waiting for a legal context."

"That gives it intent," Pell said.

"It gives it a condition. Do not grant more than the evidence."

The monitor displayed a file class none of them had requested.

COGNITIVE CONTINUITY PATTERN. DEGRADED
HIGH-FIDELITY TRANSFER. EXECUTION PROHIBITED
WITHOUT PRESENT CONSENT.

Calder released Pell's wrist. "Can it act?"

"Not from isolation," he said. "It has no substrate assignment. No sensory channels. No memory access outside its own encrypted volume."

A second line appeared.

PETITION: TEMPORARY EMBODIMENT IN
LEGAL-MEMORY SUBSTRATE.

The room changed without moving. Iven read the words twice, then checked the source path. The petition came from inside the final authenticated packet. Its transmission stamp carried the same fifty-three-year displacement as the warning. No sender name appeared. The identity field contained only a hash.

Venn moved first. "A petition requires a petitioner."

"It may contain one," Iven said. "We cannot know without loading the pattern."

Pell pulled up substrate capacity. *Aster Vale* carried legal-memory hardware for reconstructing damaged witness records and testing whether high-fidelity copies could testify. It had never hosted a transmitted cognition. Embodiment would grant the pattern sensory input, speech, and access to whatever memories survived compression. It could also damage the substrate or itself.

Calder read the consent restriction. "Can we question it without embodiment?"

"Possibly through a text sandbox," Pell said.

Iven examined the packet structure. "The output layer depends on autobiographical continuity. A sandbox might produce fragments without a stable speaker."

"Convenient," Calder said. "The warning demands obedience, then offers a witness who can speak only if we build it a mouth."

"The warning did not offer this openly. The inquiry unlocked it."

"Which means your concealment led us here."

"Yes. That does not redeem it."

The petition added a resource estimate, a proposed duration, and a present-consent clause. It requested no control of ship systems. It asked for witness protections during activation and deletion review afterward.

Pell looked from the monitor to Calder. "Flag authority can approve emergency loading. Civil law requires an identity advocate and medical supervision."

Venn said, "Identity advocacy cannot begin until we decide whether the pattern qualifies as a possible person."

"And deciding may require hearing it," Iven said.

Calder stood. "No activation. Preserve the packet under three-key custody. Pell keeps one key, civil registry keeps one, and Mora's advocate keeps one. Chief Mora remains

suspended and confined to cartography except under escort."

"What happens to the inquiry?" Iven asked.

"Your admission stands. Findings wait until we know whether you concealed evidence, a witness, or both."

The archive table dimmed. Gray Lung tents vanished. The evacuation routes disappeared next, including the line that had never reached Habitat Twelve. Iven's mother left last, her hand frozen an instant above the recorder before darkness took the image.

On the isolated monitor, the request for embodiment remained pending. Its cursor blinked inside an empty identity field while three custodians prepared their keys.

Chapter 7 — The First Defection

Sena had half her body inside Honest John's radiator hinge when the tow ultimatum reached *Everspring*.

The tug hung in the seedship's far-side cradle with its reactor cold. Dock gravity pulled weakly across the maintenance tube, enough to keep her boots against the ribbed deck and not enough to stop tools drifting when she released them. Her left arm remained bound to her chest. She worked one-handed, catching each wrench on a wrist cord before moving to the next bolt.

The bearing had shifted half a degree during the interception. Half a degree had let her deliver Okoro's spindle. It had also left the number-two vane rubbing its cold stop whenever the tug changed attitude. A vane that could not unfold cleanly made Honest John a one-burn craft. After that burn, heat would choose the destination.

Okoro's voice came through the tug speaker. "Sena, bridge. Now."

"Seven minutes."

"They gave us six."

Sena braced her shoulder against the hinge housing and pulled. The jammed inspection plate opened with a soft metallic crack. Beneath it, the bearing race showed a bright crescent where ceramic had ground against steel.

"Who gave us six?"

"Admiral Calder. Two cutters have matched our vector. They intend to attach tow bridles and take helm authority."

"Are they asking?"

"The ultimatum calls consent legally presumed under sleeper-protection law."

"Then they are not asking."

A grease-pencil mark crossed the inside of the housing. Tev had drawn it before she left *Aster Vale*: ALIGN COLD EDGE HERE. LEAVE TWO THREADS. DO NOT TRUST AUTO-SEAT. She set the race to his line, left two threads visible, and locked the adjustment with safety wire.

The vane still would not survive a full pursuit burn. It might survive the beginning of one.

Sena backed out of the tube. Her shoulder struck the rim. Pain moved from the strapped arm into her neck, clean and immediate. She waited for her vision to settle, packed the tools into their clips, and closed the radiator housing.

"Five minutes," Okoro said.

"Tell them the pilot died. Their own record says so."

"They corrected the record."

"Efficient."

She pushed toward the airlock.

The operations room on *Everspring* had acquired forty people since Calder read the annex. Suspension engineers stood beside navigation crew. Civilian advocates occupied the rear wall with legal wafers open. Four proxy representatives watched from convoy links. Nobody raised their voice. A seedship full of sleeping families did not need volume to create pressure.

Okoro stood at the central table. The display showed *Everspring* as a long blue mass on its Serein curve. Two defense cutters, *Quiet Reed* and *Linden's Edge*, held station off

the forward truss. Tow bridles extended from both hulls, each cable coiled inside an armored launcher.

A red quarantine shell enclosed the three ships.

Calder occupied the main screen. "Doctor Okoro, you have four minutes to accept tow control. Your life support and suspension systems remain under your command. Navigation transfers to my boarding master until the fleet council reviews the warning."

"Your council has no sleeper representatives seated," Okoro said.

"The Compact grants emergency custody when a sleeper manifest faces imminent harm."

"Your authenticated order defines their survival as failure. You cannot claim custody under a law meant to preserve them."

"I have not executed that remedy."

"You prepared the weapons."

"I also prohibited fire. Accept tow and keep that prohibition easy to maintain."

Sena read the geometry. *Quiet Reed* planned to fasten near the seedship's forward communications crown. *Linden's Edge* would take the aft truss beside the drive-control relays. Once both bridles tightened, they could torque *Everspring* away from its commitment gate without asking the main drive. The maneuver would spare fuel. It would also put two military hulls within cutting distance of every external command line.

"Three minutes," Calder said.

Okoro muted the link. "Options."

Navigation chief Sal Vey pulled up a burn profile. "Main drive can add six meters per second before they attach. That ruins their current tow solution. It does not outrun

cutters. We keep thrust low enough for the suspension banks, then coast while the radiators recover."

"Commitment gate?"

"We cross it earlier by eleven hours. After the burn, abort costs reserve we need for Serein braking."

An engineer touched the thermal pane. "Seed mirror package shares the forward coolant trunk during launch configuration. We have not separated it because quarantine froze deployment work. Drive heat will load the mirror lattice above storage limits."

"Can you isolate it?" Okoro asked.

"Not in three minutes."

Sena enlarged Honest John's icon at the far-side cradle. "The tug can fly escort on the weak quarter."

Every face nearest the table turned toward her.

"Your radiator cannot carry an escort burn," Vey said.

"It can carry one if I leave the vane half open and keep the reactor below sixty percent."

"For how long?"

"Long enough to make the cutters watch me instead of the seedship."

Okoro looked at Sena's bound arm. "You cannot hold a manual profile through sustained thrust."

"The yoke does not know how many hands I have."

"You delivered the filing. You do not owe us another flight."

Sena felt the wooden box under Honest John's helm as distinctly as if it pressed against her ribs. Jori's name sat burned into its lid. The old rescue story offered her its familiar shape: one pilot, one impossible corridor, one decision taken before command could make it worse. She knew that shape too well to trust it.

"I am not flying alone," she said. "Give me your navigation plot and a bridge controller on a live link. If the link drops, I follow the last safe branch. If my vane sticks, you do not turn back."

"That still puts you outside the hull when weapons come live."

"It puts a pilot where an escort should be."

The muted channel flashed. Two minutes.

Okoro looked toward the proxy representatives. "I need a recorded decision. Refuse tow and hold the present course, or accept custody."

The first proxy spoke for three suspension banks. "Refuse."

The second asked whether refusal guaranteed insertion. Okoro said it guaranteed nothing beyond keeping helm authority for the next burn. The proxy refused tow anyway. The third abstained because twelve families in her group had withdrawn instructions after the annex became public. The fourth represented embryo trusts and lacked authority to approve military custody.

No clean vote emerged. Only enough refusal to prevent Okoro from pretending consent.

She opened Calder's channel. "*Everspring* declines tow. We will hold internal command and proceed under our mission charter."

Calder's face did not change. "Your mission charter no longer has valid insurance or Compact protection. If you ignite your drive, you breach quarantine. Any vessel assisting you will enter mutiny status."

Sena unlatched her pilot token from her belt. She placed it on the table and keyed Honest John's command register.

PILOT ACCEPTS ESCORT CONTROL UNDER
EVERSRING MISSION AUTHORITY.

The register asked whether she acknowledged conflicting flag orders.

She selected YES.

It asked whether she understood that assistance in breaching quarantine constituted mutiny under fleet law.

"Spell my status correctly," Sena said to the recording node. "Mutineer, pilot Sena Vale, courier registry seven-one-six. Voluntary. No coercion by seedship command."

The word entered the log.

Okoro did not close the register. "An escort command requires limits. You asked for distributed control. State it for the record."

Sena looked at the burn profile. The room had already accepted her choice too easily. A pilot who volunteered for danger often received gratitude when she needed resistance.

"Mere owns the safe branches," she said. "If I leave one without declaring collision avoidance, she can cut my drive through escort link. Tev Arin owns radiator limits. If he calls hinge failure, Honest John falls out of formation. Your suspension chief owns acceleration. If any bank crosses red, *Everspring* reduces thrust even if that gives the cutters a line."

Vey frowned. "A remote drive cut can strand you outside our recovery cone."

"That makes it useful."

"What authority stays with you?" Okoro asked.

"Collision response, cold-jet maneuvering, and the choice to detach if my hull threatens yours."

Mere came onto the operations link from the bridge pit. Her face carried the flat concentration of someone who had already loaded too many moving masses into one local plot.

"I accept branch control. I need Honest John's exact thrust lag and your manual cutoff latency."

"Point eight seconds from command to reactor fall. Cold jets answer in point two if the tanks stay warm."

"They will not stay warm if you coast shadowed by the seedship."

"Then model one second."

Tev's audio arrived through the dead tug channel. "I accept thermal call, with a condition. Sena does not get to debate a failure limit after I call it."

"You enjoy the word condition," Sena said.

"I enjoy machinery that remains machinery."

Okoro entered both acceptances. The suspension chief added thresholds for shear, blood oxygen, and coolant pressure across the older banks. Honest John's escort page filled with people who could stop Sena. She read the list twice. It made the flight less clean than the one in her head and more likely to bring anyone home.

One proxy representative objected. "We are placing seedship survival in the hands of a pilot the fleet has declared a criminal."

"No," Sena said. "You are placing one tug in my hand. The seedship stays in yours."

The representative studied the split authorities. "And if Admiral Calder offers amnesty once you detach?"

"Then you should assume I might take it. Build a plan that survives me."

Okoro looked at her for a long second. "Escort authority accepted under those limits."

Sena copied the control keys to Mere and Tev. Honest John's reactor cutoff turned green on two remote boards. The small surrender of control frightened her more than the cutters did. She left the keys active.

Calder heard the exchange. "Vale, stand down. Your tug lacks thermal margin and your medical file bars acceleration."

"My file also said I died."

"That error has been corrected."

"Keep correcting things."

She pushed off the operations table before Okoro could stop her.

Honest John's airlock took nineteen seconds to cycle. Sena used fourteen to strap a stimulant patch below her ear and five to call Tev on a private maintenance channel.

His face appeared before she reached the helm. He had a coolant hood tied at his neck and black grease across one cheek.

"I saw your register," he said.

"Then save me the confession. Number-two race ground a crescent into the cold side. I reset to your pencil line."

"How many threads?"

"Two."

"Vane angle?"

"Half deployment."

"Half gives you thirty-eight minutes at sixty percent reactor before lubricant boil. Less if you shadow the seedship plume."

"I need twenty."

"You always need less than the machine will cost."

"Can you watch the bearing?"

Tev glanced away, checking who might hear. "Calder sealed my outbound channels."

"This one still works."

"Because you called from a tug the fleet marked dead."

"Will you watch?"

He opened a diagnostic pane. "Send it. I can give you temperatures. I cannot give you permission."

"Temperatures will do."

Sena dropped into the helm seat. The wooden box remained in the drawer below her knees. She did not open it. She strapped her right wrist to the yoke, locked her injured arm against the acceleration web, and brought Honest John's reactor out of cold sleep.

The tug answered with a vibration through the seat. Pumps spun. The number-two radiator moved to half deployment and stopped on the manual setting. Tev's grease-pencil line held.

"Thirty seconds," Okoro said over the escort link.

Sena released the cradle clamps. Honest John drifted clear of *Everspring* on compressed gas, too close for hot thrust. The seedship's hull slid past her canopy: matte shielding, maintenance scars, hand-painted service numbers, and a row of dark windows meant for people who would not wake until a different sky existed.

Quiet Reed fired its tow bridle.

The cable launcher flashed white. Its clamp crossed the gap toward *Everspring*'s forward truss, steering vanes making small corrections. Sena used Honest John's cold jets and moved into its path.

"Sena," Tev said, "that clamp outweighs your cockpit."

"It needs a conductive target."

She rolled the tug to present an external cargo rail. The clamp's magnetic jaws found Honest John and closed. Impact snapped her against the harness. Pain flooded the bound arm. The tug spun twelve degrees before its attitude jets caught.

The tow cable tightened between Honest John and *Quiet Reed*.

"Attached," Sena said. "Not to the seedship."

She blew the cargo rail's explosive bolts. The rail left Honest John's hull with the clamp still gripping it. Cable recoil pulled the useless assembly back toward *Quiet Reed*, forcing the cutter to vent thrust and avoid its own line.

Everspring ignited.

A seedship did not leap. Its drive built force through a structure carrying eighteen thousand four hundred sleepers and ecosystems packed for another world. The hull flexed along kilometers of truss. Acceleration rose by careful tenths. Suspension telemetry stayed green, then amber in two older banks.

Honest John lit sixty percent reactor and followed.

The first push drove Sena into the web. Her injured shoulder tried to separate from her body. She kept her right wrist locked to the yoke and flew from pressure through her palm. No stars crossed the canopy. Only instrument marks showed speed accumulating in the dark.

"Vane temperature climbing," Tev said. "Cold stop holding."

"Time."

"Thirty-four minutes to lubricant boil at present load. Thirty if you enter the plume."

Linden's Edge moved on the seedship's aft quarter. Its bridle launcher tracked the drive-control truss. Sena could not cross behind *Everspring* without taking exhaust scatter. She pushed above the seedship's plane, rolled Honest John nose-down, and used three seconds of full thrust to make the cutter recalculate collision risk.

The radiator alarm lit before Tev spoke.

"You exceeded the hinge load."

"I noticed."

"Vane angle lost four degrees. Reduce reactor."

Sena cut to fifty-two percent. Honest John fell half a kilometer behind. The cutter continued closing. Its tow clamp opened.

Okoro sent the seedship's local navigation plot. Sena let her controller, a quiet bridge pilot named Mere, place a safe branch on Honest John's board. It curved outside the drive plume and gave up distance to keep the tug cool.

"Take it," Mere said.

Sena wanted the shorter line. The shorter line let her cross the cutter's nose and force another abort. It also passed through exhaust scatter. Jori would have taken it. Sena's memory of Jori always offered speed first and consequences later.

She accepted Mere's branch.

Honest John descended beneath the seedship. Heat load eased by three percent. *Linden's Edge* fired from too far out. Its clamp reached the aft truss with little steering reserve, struck armor, and failed to seat. The cable reeled back.

"Good plot," Sena told Mere.

"You flew it."

"Keep sending them."

Calder transmitted on the open command band. "*Everspring*, cut drive. Pilot Vale, break escort and hold for recovery. No weapon has fired. Do not force escalation."

Okoro answered, "No weapon needed to fire. Your order already did the killing."

"You have crossed the quarantine line under thrust. I will convene captains for pursuit authority."

"Convene them behind us."

The seedship continued its burn.

At minute thirteen, the forward coolant trunk alarm sounded. The seed mirror package had absorbed more drive heat than its storage blankets could radiate. Thermal cameras showed a pale line spreading across one lattice panel.

"Mirror temperature above annealing limit," the seedship engineer said. "We need to dump memory coolant."

"Dumping costs the settlement archive reserve," Okoro replied.

"Not dumping costs the lattice."

Sena watched the pale line divide. A mirror built to send knowledge across fifty-three years carried tolerances measured in fractions of a degree. It had spent the launch folded against a seedship hull, waiting to open under Serein's gravity. Now military geometry and escape heat had reached it before any future message could.

The first fracture registered as a dry pulse through the escort telemetry. Then three neighboring cells lost phase lock.

"Seed mirror sector seven cracked," the engineer said. "Coolant venting now."

Silver vapor streamed from *Everspring's* forward package and crystallized behind it. The vent lowered lattice temperature, but every kilogram left the settlement reserve forever.

Honest John's radiator answered with its own alarm. The reset bearing had begun shedding ceramic grains into the race. Tev's diagnostic trace jumped in steps.

"Twenty minutes are gone," he said. "You have nine before the vane locks."

"Seedship burn ends in six."

"Then you get three for mistakes."

Behind them, *Quiet Reed* recovered its cable. *Linden's Edge* rolled its hot side away and opened radiators. Neither cutter could match the next increment without cooling. Farther back, defense ships altered formation one by one. Drive points appeared on Sena's plot like sparks struck from cold metal.

Calder's flag signal changed from quarantine enforcement to pursuit preparation.

Sena held Mere's next safe branch. Her hand shook against the yoke, whether from acceleration or the stimulant she could not tell. The seedship's burn clock fell through ninety seconds. Suspension banks stayed amber but stable.

At cutoff, *Everspring* carried six new meters per second toward Serein. Six looked small on a tactical display. Applied to its mass and the remaining corridor, it moved every tow solution out of reach.

Honest John shut down to cooling power. The radiator vane stopped one degree short of its commanded angle. Sena marked it failed-open enough to live.

The seedship coasted ahead, coolant crystals bright on the black behind its cracked mirror package. Sena matched its damaged side with the tug and accepted another plot from Mere.

One by one, the fleet's pursuit drives lit in their wake. The distance numbers began changing, and none of them promised escape.

Chapter 8 — The Orbit of Nobody

The security officer removed Iven's wrist seal only after cartography locked every command surface.

Their main table woke in observation mode. Routes crossed it in dull gray, stripped of the colors that indicated edit authority. *Everspring* moved toward Serein on the curve created by its six-meter-per-second escape burn. Pursuit ships spread behind it according to drive tolerance, heat capacity, and the willingness of their captains to obey before the captains' council had voted.

Admiral Calder stood at the far edge of the table. Commander Pell occupied Iven's usual station with a command key inserted. Tev Arin waited beside the thermal console carrying a paper binder against his chest.

"Chief Mora," Calder said, "your suspension remains active. You will evaluate one maneuver. You will not transmit, annotate the fleet product, or open the isolated packet."

"Understood."

"Say what happens if we catch the seedship."

Pell restored color to a corridor on the display. Two cutters would overtake *Everspring* above the outer edge of Nox's debris fan. A third ship would place low-energy laser fire on the seedship's drive-control relays if it refused tow. Once disabled, the cutters would attach fore and aft, rotate the long hull, then force it into a braking corridor behind the convoy.

The route looked clean. No debris clusters crossed it. The tow pair shared load within six percent. The final curve ended outside the Serein commitment gate with enough time for a captains' review.

Iven read the assumptions instead of the line. "You used defense-cutter heat limits."

"The cutters execute the burn," Pell said.

"They tow the seedship through it."

"The seedship's drive stays cold."

"Its systems do not."

Calder pointed toward Tev. "That objection brought him here. Prove it or clear the route."

Tev laid the binder on Iven's table. Metal corners had worn through its black cover. A loop of faded green cord held it shut. No modern maintenance record needed paper, but Everspring expected to wake after decades with failed storage, incompatible terminals, and no depot. Seedships carried instructions that could survive a dead battery.

"This came off the commissioning copy in *Aster Vale*'s thermal archive," Tev said. "Okoro's crew sent revisions before quarantine closed their engineering channel."

Pell looked at the binder. "You had live contact after the annex release?"

"Before the boundary went active. The revision packet arrived as scheduled maintenance traffic."

"And your later contact with Vale?"

Tev met Calder's gaze. "You can hold that inquiry after the radiator inquiry."

Calder did not accept the deflection. She also did not spend time on it. "Open the relevant system."

Tev untied the cord. Foldout diagrams spread across the table, too large for the magnetic surface. Iven caught one edge before spin ventilation lifted it. The sheets showed

Everspring's four heat-rejection wings, suspension coolant trunks, seed-mirror storage loop, and emergency sink tanks distributed through the central spine.

Ink covered the margins. Some notes carried engineering values. Others carried names.

LINA SEFT, BANK 12 PUMP TREE.

OMAR ILEN, WING 3 MANUAL PURGE.

DAVI AND REN KO, MIRROR COOLANT CROSSFEED.

Dozens more continued down every fold. The handwriting changed from line to line. Some letters leaned hard enough to score the paper. One name had a small leaf drawn beside it.

Pell touched Lina Seft's entry. "Maintenance staff?"

"Future maintenance staff," Tev said. "Sleepers. Each person trained on a system before entering suspension. They signed the hard copy beside the station they expect to service after waking. If a valve has no assigned sleeper, the ship carries an instructional model instead."

"Why use names rather than bank numbers?" Calder asked.

"Because bank numbers change when medical rebalances suspension loads. People keep their names."

Tev turned to the forward cooling tree. "These handwritten maintenance names represent sleepers who know specific failure routes. They are not decorative. Lina can bypass the bank twelve pump tree from memory. Omar knows which purge valve sticks below seventy kelvin. The Ko siblings can separate the mirror loop without losing the habitat archive. None of them can do that while frozen."

Iven followed the lines. The abstract manifest on Calder's annex had become a repair crew sleeping inside its own machine. The map offered no symbol for that relation.

"Show us the braking load," they said.

Tev pinned three diagrams and connected his diagnostic wafer. Pell allowed read-only access to the thermal console. The simulation began with *Everspring's* current damage: seed mirror sector seven fractured, forward coolant reserve reduced by the vent, two older suspension banks warm from the escape burn, and a radiator wing carrying a deployment fault logged six months earlier.

"That fault passed inspection," Pell said.

"For independent flight. Tow rotation loads the wing sideways."

He ran Calder's corridor.

The model rotated the seedship between two cutter vectors. At eleven degrees from its present attitude, the forward mirror loop absorbed reflected drive heat from the lead cutter. At seventeen, debris shielding blocked one radiator wing's view of cold space. At twenty-three, towing torque flexed the central spine enough to close a coolant check valve near bank twelve.

Amber warnings filled the table.

"Reduce tow rate," Calder said.

Pell halved rotation. The seedship remained longer in the cutters' exhaust geometry. Mirror coolant crossed annealing limit again.

"Move the lead cutter outward."

That reduced reflected heat and increased cable angle. The aft bridle then loaded the damaged radiator truss beyond commissioning tolerance.

Tev pulled a grease pencil from the binder cord and marked values directly on a paper overlay. "Digital model assumes both wing-three hinge motors answer. One failed its cold test during launch. The crew left it deployed because they had dock support. Escape vibration may have shifted it. Use one motor."

Pell changed the value. The wing stalled at sixty-one percent.

Red spread from the suspension loop into three banks. The simulation killed its own tow burn after predicting coolant vapor in bank twelve.

Calder leaned over the model. "Kill the mirror package."
"Storage loop cannot detach while folded," Tev said.
"Vent it."

"They already spent reserve during the escape. Full vent saves the mirror and starves two suspension banks of emergency coolant."

"Then sacrifice the mirror."

Tev changed the priority. The fractured lattice heated until local supports softened. Its storage frame warped into the forward coolant trunk and blocked flow to the central spine. Predicted failures cascaded more slowly but reached the same banks.

Iven watched Lina Seft's name beside the first red pump tree.

"How many survive?" Calder asked.

"Thermal models do not count deaths," Tev said.

"The suspension model does."

Pell opened casualty estimates. Ranges appeared, each wider than command preferred. At best, bank twelve lost several dozen sleepers to warming injuries. At median conditions, coolant loss propagated through three banks. The worst branch exceeded two thousand before cutters could stabilize the spine.

Tev did not accept the first run. He opened the maintenance-response layer and set awake crew strength to *Everspring's* present roster. The model assigned a general engineer to bank twelve, then directed that engineer to

isolate the failed check valve through a service tunnel behind the suspension shell.

"That tunnel stays sealed while the bank holds pressure," Tev said. "Opening it vents the service atmosphere across two compartments."

Pell checked the digital procedure. "Archive says accessible under emergency authorization."

"Archive assumes Lina Seft has woken and can close the hidden bypass before anyone opens the tunnel."

He unfolded Lina's sheet. Beside her name, a pencil sketch showed three valves, one crossed out, and a route entering from the opposite side of the pump tree. The digital archive had scanned the words but not the pressure marks in the drawing.

"Why did commissioning approve a procedure stored in one sleeper's memory?" Calder asked.

"It did not. The approved procedure works after normal thaw, when the tree sits at habitat pressure. Lina found a cold-state shortcut during training and added it by hand. Seedships keep human knowledge because fifty years of storage always outlives part of the approved manual."

Pell entered the shortcut. Casualties fell in one branch, then rose again because no awake crew member could identify the crossed-out valve from outside the sealed tunnel.

Tev moved to Omar Ilen's purge note. Omar's penciled instruction prevented a radiator line from hammering when the ship rotated. Without him, the model chose an automated purge that consumed twice the expected coolant. Tev tried the Ko crossfeed next. Its training record required two people at separated manual wheels, turning by voice count after the central controller went dark.

"Can the awake crew follow these pages?" Iven asked.

"Some. Not during forced rotation with cutters pulling at both ends. Every manual exception assumes the hull can hold attitude while people work."

The handwritten names did more than represent lives. They marked missing actions inside the thermal model, tasks assigned to people command had reduced to cargo. Calder's corridor demanded their knowledge while keeping them unable to use it.

Tev reran the tow with every undocumented shortcut removed. Red reached bank twelve forty seconds earlier.

"Those ranges assume no weapon damage," Iven said.

Pell added Calder's low-energy drive-relay strike. Scatter heated the forward hull. One relay failed open rather than closed, forcing local crew to cut power through an external access point while under tow. Casualty estimates widened again.

Calder looked toward Tev. "Can *Everspring* survive any forced braking route before the commitment gate?"

"Yes," he said.

Pell's attention snapped to him. "You just proved this one kills them."

"You asked any route. They can brake under their own drive, with radiators facing cold space, after the cracked mirror loop has been isolated. That takes repair time and cooperation."

"How much time?"

"At least fourteen hours for isolation if the awake crew accepts losing half the settlement archive reserve. Longer if the Ko crossfeed procedure requires access behind the folded package."

"The Ko sleepers cannot perform it," Calder said.

"Their handwritten instructions can. Maybe. The awake crew will work from those pages."

Calder moved the pursuit clock beside the thermal model. Fourteen hours pushed the intercept near the commitment gate. If *Everspring* refused to brake after repairs, the fleet would have less room to compel it.

"You have proved my planned corridor unacceptable," she said. "You have not given me a route that enforces the warning."

Tev closed the casualty pane. "Machines do not owe you a moral option."

The security officer shifted near the door. Calder studied Tev for a second, then returned to Iven.

"Map an intercept that keeps the seedship in independent thermal attitude," she said. "No tow rotation. No reflected drive plume. Preserve a laser line to the control relays."

Iven looked at the gray controls. "You suspended my access."

"Pell will enter your work."

"Then it becomes his map."

"Does that make it less accurate?"

"It makes responsibility harder to see."

Calder's patience thinned. "You admitted concealing the annex less than two hours ago. I will not restore command access because you object to muddy attribution. Give Pell the route."

Iven began with the seedship's true state vector. Its escape burn had shifted it above the convoy's old survey plane. The fastest pursuit line crossed the edge of the mineral-noise region extending from Lumen's orbital wake. Chapterless survey products still marked the interference as geological weather. Sensor returns there carried timing errors, false edges, and phase delays that changed with microwave storms.

Pell traced the direct line. "We can tolerate that noise at fleet separation."

"The tow cutters can," Iven said. "Your laser line cannot."

They built a cone from Sorn's low-energy strike plan. Firing through the interference enlarged aim uncertainty at the drive-control truss. A miss could touch the seed mirror, suspension shielding, or empty space. The cone looked bad enough to stop a careful captain, but not bad enough to force a detour.

Calder saw the same thing. "Use independent optical tracking inside the storm edge."

"Lumen has no visible light source. You would track hull occultation against Nox lensing."

"Then use convoy beacons."

"Latency and geometry leave a blind interval."

"How long?"

"Twenty-seven seconds."

"Acceptable."

Iven knew twenty-seven seconds did not buy enough. *Everspring* needed repair time. Tev's fourteen hours lay beyond any honest uncertainty the map could create.

They opened the raw survey archive from the day before. Access restrictions permitted observation, not editing. Lumen's weather traces crawled across the display in pale bands. One trace had answered a passive sweep with a phase echo displaced from every mineral model. Iven had set it aside for later study. Now it crossed the pursuit corridor at an angle no tactical map displayed.

"What does that return do to ranging?" Calder asked.

"Unknown."

"That word does not map a fleet."

"It maps this region better than a number would."

Pell enlarged the trace. "Can it spoof a hull return?"

"It can repeat energy at a shifted phase. We have not tested whether the shift remains stable under an active ping."

"Then mark a sensor exclusion and route the ships outside it," Calder said.

The order created an opening. An exclusion zone carried a boundary. Boundaries depended on confidence thresholds chosen by a cartographer.

Iven constructed the zone from the raw trace. The honest contour reached forty thousand kilometers at low confidence. They extended it by applying the storm's maximum observed phase shift to every unsampled sector. The resulting shape swallowed the direct pursuit line and pushed Calder's ships onto a high arc.

Pell read the metadata. "You used maximum propagation."

"We have no basis for a mean."

"Survey practice uses bounded median when a phenomenon lacks hostile intent."

"We do not know its intent."

Calder looked sharply at them. "Geological weather has no intent."

Iven kept their face still. "Then bounded median would be appropriate after an active test. Until then, maximum propagation protects the firing line from false phase."

Every sentence remained defensible. Together they made a route Iven did not believe. The storm might disrupt a laser solution, but no data justified the full exclusion contour. The high arc would cost pursuit ships forty-six minutes and extra reaction mass. It would not give *Everspring* fourteen hours. It might give the seedship enough time to finish mirror isolation before the closest cutter regained a line.

Calder sent the proposed route to weapons analysis. "Run it."

Pell held his hand above the transfer control. "This product requires a cartographer's uncertainty seal. I can enter it, but I cannot certify the survey interpretation."

"Mora certifies under witness restriction," Calder said.

The table projected a one-use seal prompt. Iven's name appeared with a red border. They could approve an advisory layer, not a command route. Pell would still need to adopt it into the fleet map.

Tev gathered his papers. Lina Seft's name disappeared under the folded corner, then returned when he adjusted the sheet. He had proved that a clean line through empty space could kill people through heat alone. Iven had answered with an unclear line intended to preserve them.

Their mother's voice came back without sound: build honest maps instead.

An honest map of present authority would show Calder's weapons, *Everspring's* refusal, and Iven's restricted hand poised above a false contour. It would show no route without ownership. It would not absolve the hand.

"Chief Mora," Calder said. "Seal or reject."

Iven pressed their palm to the prompt.

They changed one field before sealing. SURVEY EXCLUSION, PROVISIONAL became NAVIGATION HAZARD, ACTIVE. The first label advised. The second forced automated route planners to avoid the zone until a command cartographer cleared it.

Their one-use seal accepted the change.

Pell noticed. "Active hazard requires confirmed effect on navigation."

"The phase displacement affects ranging."

"We have not confirmed magnitude."

"Then reject my certification."

Calder examined the delay estimate. Forty-six minutes. Pursuit ships would remain outside the uncertain region, preserve laser geometry, and spend more fuel. No tactical argument favored the direct line unless command accepted blind fire near sleeper banks.

"Adopt it," she said.

Pell did not move. "Admiral."

"We asked for a route that protects the seedship's thermal attitude and preserves a controlled firing line. This route does both. Adopt it."

He entered his command key. The gray advisory contour turned amber. A route packet left cartography and propagated through the pursuit net.

Ship acknowledgments arrived in order of distance. *Quiet Reed* accepted first. *Linden's Edge* recalculated and reported fuel cost. Two farther cutters requested clarification, received Calder's command seal, and accepted. The flag plot bent into the high arc.

Iven watched the false route become operational fact. Drive vectors changed by fractions of a degree. Over distance, those fractions opened thousands of kilometers.

Calder removed the one-use seal from the table. "Return Chief Mora to restricted status. Tev, leave the binder."

"No," Tev said. "Digital copies are in the thermal file. The paper returns to archive custody."

"It supports a command decision."

"It belongs to the people whose names sit inside it. You can requisition a copy."

Calder looked at the worn cover, then nodded. "Take it."

The security officer restored Iven's wrist seal. Editing controls faded back to gray. On the main table, pursuit

ships continued their accepted detour while *Everspring* held its real course toward Serein.

Pell closed the raw survey archive but left the fleet product visible. "If the storm does not affect them, every captain will know this map overstated the hazard."

"Yes," Iven said.

"You confessed to one concealment this morning."

"I remember."

Tev retied the green cord across the handwritten names. The paper edges kept the creases of hands that expected to wake beside these machines. Calder said nothing more. She had the route she requested, and Iven had bought forty-six minutes with a line no empty space had earned.

The final pursuit acknowledgment arrived from *Resolute Vigil*. ROUTE ACCEPTED. EXECUTION ACTIVE.

Beyond the locked table, ships began turning toward an orbit that existed only because Iven had drawn it.

Chapter 9 — The Map Without Descendants

Quiet Reed fired the survey ping before Iven reached the cartography door.

The pulse crossed the Serein plot as a thin white line. It left the cutter's forward array, spread through the region Iven had marked hazardous, and struck Lumen's mineral-weather wake twenty-three seconds later. The table predicted a weak return after another twenty-three. If the return stayed within ordinary phase limits, Pell could clear part of the exclusion zone and recover eleven minutes of pursuit time.

Iven stood inside the doorway with their wrist seal still active. The security officer waited behind them. Commander Pell had left for Calder's captains' council, but his command key remained in the table under remote lock. Observation surfaces stayed gray.

"Am I confined to quarters or cartography?" Iven asked.

The officer checked her instruction slate. "Cartography counts as your duty confinement until Commander Pell releases you. No transmissions. No edits."

"The table is receiving a live test of my map."

"You may observe."

Iven entered. The door sealed, placing the officer on the corridor side. Somebody had cleared Tev's paper diagrams, but one grease-pencil shaving remained caught in the table seam. The accepted high route glowed amber above it. Four

pursuit ships followed that route. *Everspring* held its own line toward Serein, gaining seconds whenever a defense ship spent thrust on the detour.

The ping timer reached forty-six seconds.

Nothing returned.

At fifty-one seconds, tactical filters reported MINERAL SCATTER, BELOW NAVIGATION THRESHOLD. The active hazard contour began contracting automatically. Iven watched its outer boundary withdraw twelve thousand kilometers.

Then the raw receiver pane filled with weather.

Pale bands rose from Lumen's wake at six frequencies. They did not arrive as one reflection. Each frequency carried a different phase delay, as if the moon's silicate storms had caught pieces of the pulse in separate layers and released them after unequal travel. The tactical filter averaged them into noise. The hazard boundary contracted another two thousand kilometers.

Iven touched the raw pane. The wrist seal blocked them.

"Observation magnification," they said.

The table enlarged the bands without granting edit control. The first carried *Quiet Reed*'s outgoing pulse in reverse frequency order. The second carried only its even-indexed segments. The third resembled the passive sweep Iven had noticed the day before, but its low-frequency edge had shifted by exactly one sampling interval.

A geological echo could sort frequencies. Crystal layers did it whenever a wave crossed uneven pressure. An echo could drop odd segments through destructive interference. It could retain charge and release it after a delay.

It could not have heard yesterday's passive sweep inside today's pulse.

Iven requested the old raw file. Observation authority opened it beside the live return. They aligned both traces. A bright ridge in the new data matched the previous sweep down to receiver noise from *Aster Vale*'s own array. The ridge could only come from recorded energy. Lumen's weather had retained the sweep, altered its interval, and included it in the answer to a new signal from another ship.

"Reclassify contraction pending contamination check," Iven told the table.

ACCESS DENIED.

The tactical product reduced the hazard again. *Quiet Reed* received permission to approach the former boundary.

Iven called the security officer through the door intercom. "I need Commander Pell."

"He marked council interruption emergency-only."

"A live survey result contradicts the automated classification."

"Does it endanger a ship?"

Iven looked at the high route. If they said yes, Pell would interrupt the council and open the raw return to weapons analysis. Every captain would see the region as an active threat while arguing whether to pursue *Everspring*. If they said no, *Quiet Reed* might enter a phenomenon nobody understood.

"Not at present distance," they said. "Ask him to call when available."

The officer entered the request without emergency priority.

Iven returned to the table. Their seal allowed analysis tools that did not alter records. They opened a correlation workspace and fed it the two traces. The first result reported an eighty-three percent match between yesterday's receiver noise and the bright ridge. Too low for proof. They removed known instrument drift. Match rose to

ninety-one. They corrected for relative motion between *Aster Vale*, *Quiet Reed*, and *Lumen*. Ninety-six.

At ninety-six percent, contamination remained possible. The survey archive might have copied the old trace into a calibration channel. A software loop could have placed it beside the new return. Iven traced data custody from receiver to table. Different ships had collected the signals. Separate clocks stamped them. No shared buffer existed before fleet relay storage.

They needed a feature no faulty archive could invent.

The new response contained seven quiet gaps. Six divided the frequency bands. The seventh sat inside the copied ridge where yesterday's sweep had suffered a transmitter dropout. Iven enlarged that gap. Yesterday it had lasted eleven milliseconds. The response held it for twenty-two.

Lumen's weather had not merely replayed the signal. It had doubled an absence.

Iven built a simple transform: reverse the six live frequencies, remove odd pulse segments, shift the stored sweep by one interval, double its missing section. Applied to the outgoing ping and archived sweep together, the transform reproduced ninety-eight percent of the response.

No mineral model in the survey library produced that sequence.

They opened piezoelectric storm literature. Silicate canyons under tidal stress could store microwave charge. Layered crystals could sustain standing patterns for years. Thermal cycling could move those patterns between strata. Several papers proposed geological memory in the weak sense: past energy changed future response. None demonstrated ordered retrieval from two separate inputs.

The distinction mattered. A scar remembered impact by remaining a scar. A switch remembered state by occupying one of two positions. This weather retained signals,

separated them, transformed their structure, and joined old input to new.

Iven labeled a private workspace NONRANDOM RETENTION TEST.

Their wrist seal allowed the label because private analysis carried no command effect.

Quiet Reed reached the first contracted boundary. Its captain requested clearance to fire a stronger ranging pulse. Automation approved.

"Cancel pending review," Iven said.

COMMAND AUTHORITY REQUIRED.

They tried Pell again. Council interruption remained emergency-only. The security officer opened the intercom.

"Chief, do you need me to stop the ship physically?"

"Can you?"

"No."

"Then no."

The second survey ping carried four times the first pulse energy. It used a coded sequence designed to distinguish hulls from debris: three short bursts, one long, two short, then a frequency climb. Every tactical array in the fleet knew the code.

The pulse entered Lumen's wake.

Iven started a new raw capture before the table's filter could average it. Observation mode permitted temporary calibration buffers. The buffer would erase itself after ten minutes unless attached to an authorized product.

The return began early.

A low-frequency band reached *Quiet Reed* before a simple reflection could have completed the round trip. The cutter's system rejected it as local interference. Another band

arrived on time. A third came late. The tactical filter stacked all three and reported variable mineral depth.

Iven separated them by arrival time. The early band did not contain the second ping. It carried a transformed copy of the first. The on-time band carried the second ping with its long burst moved to the end. The late band held yesterday's passive sweep again, but this time the old transmitter gap appeared once at its original duration, once doubled, and once four times as long.

One absence had become a sequence.

Eleven. Twenty-two. Forty-four milliseconds.

The weather had taken the missing interval and extended it by powers of two.

Iven sat down. Cartography chairs locked to the deck, but their knees had stopped trusting the weak spin gravity.

Nonrandom did not mean conscious. A crystal oscillator produced ordered ratios without knowing them. A feedback loop could double a period. Repetition could arise from physics alone.

Yet this sequence selected the only feature shared by neither carrier frequency nor pulse code. It selected a break made by a human transmitter a day earlier. It preserved that break, then manipulated its duration when a new signal arrived.

Iven searched the old Serein surveys for the same spacing.

The archive reached back seventeen years. Early probes had logged microwave storms as calibration failures and compressed most raw data. Enough survived. A survey drone had emitted a seven-millisecond timing check. Its return contained fourteen and twenty-eight millisecond gaps, dismissed as packet loss. A later probe carried a nine-millisecond fault. Eighteen appeared in a ring-scatter record. Thirty-six appeared in an altimetry failure six hours later.

Different gaps. The same doubling relation.

The weather had been answering for years. Human filters had classified every answer according to the instrument that failed to understand it.

Iven built a table of source pulse, retained absence, transformed intervals, receiving location, and elapsed time. The pattern crossed separate storm cells and survived orbital seasons. No single crystal seam connected all observations. Retention moved through Lumen's weather bands.

They forced themselves to attack the result before naming it. A resonant cavity could double intervals if each discharge triggered a slower harmonic. Iven generated ten thousand layered-canyon models and allowed pressure, temperature, crystal orientation, and tidal stress to vary across every measured range. Hundreds produced one doubled gap. Nine produced two. None selected the transmitter dropout from three unrelated surveys while ignoring stronger pulse features.

Relay contamination offered a simpler explanation. Fleet software might stamp old timing faults into every file it touched. Iven rebuilt the chain without fleet relay data, using raw probe fragments recovered from maintenance checksums. The seven-millisecond source survived in a drone's local crystal log. Its fourteen-millisecond answer survived on a different recorder. Those devices had never shared code, storage, or clock correction.

They tested deliberate human fabrication next. Seventeen years of survey crews gave plenty of opportunity for a joke, a hidden calibration signature, or fraud intended to win research funding. Crew rosters showed no person common to all three missions. The earliest probe had died before the latest transmitter design existed. Its response still anticipated the same operation later signals received.

One alternative remained: selection by Iven. They had searched thousands of faults and might have chosen the

few that fit doubling. To test that, they asked the archive for every preserved dropout above five milliseconds, not merely the examples already noticed. Forty-one source gaps appeared. Thirty-eight had later returns. Thirty-one displayed doubled duration within receiver tolerance. Four showed tripling mixed with doubling. Three produced no stable relation.

Chance probability fell below the survey service's threshold for nonrandom structure even after its strictest correction.

Iven saved the failed models beside the transform table. Proof required the routes that did not work. Without them, a clever pattern could become a map drawn to flatter its maker.

The analysis still could not cross the final distance from structured response to intention. It did something more limited and more difficult to dismiss. It showed persistent selection operating across separated mineral storms, preserving particular features of human signals and applying repeatable changes to them.

The map beneath their analysis showed Serein, Lumen, debris rings, and routes for human ships. It treated every orbit as empty unless a human hull occupied it. Survey law reserved inhabited status for biosignatures, artifacts, or intentional communication confirmed by independent teams.

They had no biology. They had no artifact distinct from the moon. Intent remained unproven.

They had memory distributed through geological weather, with ordered responses to signals and continuity across seventeen years.

It constituted the first evidence that the Serein system was inhabited.

Iven did not enter that sentence into the fleet product.

The temporary buffer timer fell through seven minutes. They needed to preserve proof before it erased itself. Command save required Pell's key. Public survey publication would send the raw returns to every ship, including weapons stations already treating the region as a hazard. Their inquiry status offered one other route.

Any witness under active Compact review could submit potentially exculpatory evidence into a write-once legal hold. The evidence became inaccessible to command until an advocate and inquiry officer verified scope. Iven had used that safeguard for their annex notes.

They opened the submission form.

RELATION TO INQUIRY, the form requested.

Iven could claim the response affected forecast provenance. That stretched truth. The pattern came from Lumen, not from the fifty-three-year return channel. It might illuminate why the final warning feared *Everspring's* arrival, but no evidence connected them yet.

They typed: SURVEY DATA BEARING ON SEREIN INSERTION RISK AND THE CONSEQUENCES OF DELAYED ANNEX DISCLOSURE.

True, but incomplete.

The form required their advocate's countersignature. Jo Venn answered on the third attempt, voice only.

"I am entering a captains' chamber in two minutes," she said. "What are you submitting?"

"Raw mineral-weather returns from the route Calder accepted. The automated filter is destroying information."

"Does the information help your defense?"

"It may show the Serein risk cannot be reduced to the warning."

"May?"

"I need independent review."

Venn heard the evasion. "Are you asking me to preserve evidence or help you hide it?"

The buffer timer showed five minutes.

"Preserve it," Iven said. "Restrict it from tactical distribution until a survey specialist can separate signal from contamination."

"Does Calder need it to keep a ship safe right now?"

"Quiet Reed has already crossed the outer contour. The returns have not shown destructive energy."

"That did not answer the question."

"No present evidence requires evasive action."

Venn stayed silent long enough for Iven to hear the captains' chamber doors opening on her end. "I will countersign preservation, not your scope description. Add that the data may indicate an unknown signal process."

Iven added the words.

Venn's key entered the form. The legal hold copied the raw first and second returns, source pings, custody clocks, old survey matches, and Iven's transform table. Hashes appeared beside each item. The write-once archive sealed before the temporary buffer reached four minutes.

A preservation receipt printed from the table's evidence slot on thin ceramic paper. Iven checked every hash against the receiver logs, then added the failed mineral models and the full dropout census. The hold now carried both the apparent answer and the strongest attempts to explain it without one. Two storage nodes accepted independent copies. Neither fed the tactical net. If one node failed during pursuit, the other would retain the phase records without compression.

Iven signed the receipt by hand. Their restricted credential could not alter the archive after sealing, which also meant they could not improve the claim later by

deleting an inconvenient trace. The proof had left their control even while its interpretation remained private.

"Who knows what you think this is?" Venn asked.

"Only me."

"Keep it that way until review. You do not get another private disclosure decision merely because the last one had consequences you dislike."

"You just agreed to restrict it."

"I agreed to preserve uncertain evidence from tactical misuse. I did not agree that you alone decide what it means. I will appoint a reviewer after the council."

The channel ended.

Fleet automation finished processing the second return. Its summary entered the map: PIEZOELECTRIC MINERAL INTERFERENCE. STABLE UNDER ACTIVE RANGING. NO DIRECT HULL HAZARD CONFIRMED.

The statement cleared another section of Iven's exclusion zone. Pursuit routes shortened by eight minutes. Calder's forty-six-minute delay fell to thirty-eight.

Pell called from the captains' corridor. His image appeared on a side pane, breath quick from walking under spin.

"Report," he said.

Iven looked at the sealed legal hold indicator. Its contents no longer appeared on their table. Only the title remained, amended by Venn: UNKNOWN SIGNAL PROCESS, SURVEY REVIEW REQUIRED.

"The active test shows stable piezoelectric interference," Iven said. "No direct hull hazard. Automated contraction has reduced the zone."

"Do you agree with the contraction?"

"Not without specialist review. The phase behavior exceeds current mineral models."

Pell glanced offscreen. Voices from the council entry called for him. "Does it threaten navigation?"

"It corrupts ranging and retains input longer than predicted. The high route remains prudent until review."

"Retains input how?"

Iven chose each word. "Charge storage across layered storm cells. I placed raw returns in evidence hold because filters discarded phase structure. Venn countersigned."

"You used inquiry privilege on fleet survey data?"

"The raw data remains preserved. The tactical summary reached you."

"That privilege will not survive a second abuse."

"Then review it quickly."

Pell looked ready to ask more. The captains' chamber called his name again. "Send me the hold receipt. No interpretation outside counsel."

Iven transmitted the receipt. It contained hashes and scope, not evidence. Pell left for the council.

The fleet map continued treating Lumen as weather. *Quiet Reed* advanced inside the old contour. *Linden's Edge* followed on a cooler line. Far ahead, *Everspring* held course with its cracked seed mirror and eighteen thousand four hundred sleeping people.

Iven reopened the seventeen-year survey index in private observation mode. Without the sealed raw returns, the old records looked ordinary again: packet loss, delayed phase, mineral scatter, instrument fault. The labels covered the pattern like dust on etched glass.

They wrote the doubled intervals on a paper calibration strip because their wrist seal could not erase paper. Seven, fourteen, twenty-eight. Nine, eighteen, thirty-six. Eleven, twenty-two, forty-four. They folded the strip once and placed it under the grease-pencil shaving in the table seam.

Nothing on the fleet plot marked an inhabited system. No contact icon appeared beside Lumen. No alarm reached Calder's council. The moon remained a gray sphere crossed by amber weather bands.

A third return arrived after the ping sequence had ended. Tactical filters did not display it. The raw receiver showed one low pulse, then two, then four, each separated by the doubled gap from yesterday's failed sweep.

Iven watched until the receiver went quiet. The legal hold kept the proof. Beyond cartography's locked door, the fleet continued its pursuit without knowing what had answered.

Interlude I — The Absent Reply

Household Testimony 17: Yara Pell, medical ring C

My son woke hot enough to make the pillow damp. I put the thermometer against his neck, waited for the tone, and held the result under the kitchen light as if brighter light could make the number smaller.

On any other morning, the pediatric advisory would already have arrived. Our household monitor sent sleep pulse, air quality, school exposure, and the previous week's meals into the family forecast queue every night. Descendant medicine usually answered before breakfast. It would name the dose, tell me whether to wake him for water, and estimate how much fear the symptoms deserved.

During the first day without advice, the monitor displayed REQUEST PENDING until the words felt insulting. I called my sister. She asked what his breathing sounded like. No advisory had ever asked me that in her voice.

We counted breaths together. I gave him the standard dose printed on the bottle, not the revised dose that might have come from fifty-three years ahead. Then I sat beside him and checked the time every ten minutes.

By noon his fever had fallen. Relief came with anger attached. I had known how to read a bottle before I became a parent. I had let a future household turn that knowledge into something ceremonial, a thing I touched only after permission.

My son woke and asked for ice. I gave him two cubes in a blue cup. The monitor still said pending. He bit one cube in half and complained that I had chosen the wrong color.

Domestic Audio 4B: Esme Tal and Rian Olt, Caul-facing grow quarters

I came home with six bruised tomatoes because the nutrient rail had overfed bed nine again. Rian had noodles boiling. Our kitchen panel waited for the evening meal correction, which normally told us how much salt his kidneys would regret and which tomato carried the lowest mold probability.

The correction never came.

Rian asked me to choose. I said he had started cooking, so the failure belonged to him. He said I had brought the tomatoes, so causality had entered with my boots. We argued for three minutes and used all six.

Dinner tasted too sharp. We added salt anyway.

I had expected the silence to frighten me. Instead I kept noticing small permissions. Nobody from our line sent a warning that I would sleep badly after fermented tea. No labor forecast advised Rian to trade tomorrow's shift because bed nine would jam again. He could keep the shift, meet the jam, and curse it at the proper time.

That relief embarrassed us both. Descendant guidance kept the hydroponics ring alive through two blights. My mother survived a fungal exposure because a treatment warning arrived before her lungs failed. We owed actual meals and actual years to people not born when the danger started.

Still, Rian ate the burnt noodles from the bottom of the pot. He likes them. The household panel always treated burnt food as preventable loss.

We left the scorched pieces in his bowl and washed the pot after midnight.

Written Deposition: Havel and Min Sori, berth habitat Patient Current

I found the old compatibility letter behind the linen drawer. Min kept paper then. She said a household promise deserved something that could not update itself while you slept.

The letter recommended our marriage. It gave us a seventy-eight percent chance of remaining together for forty years, adjusted for work transfers, illness, and likely children. We were twenty-two. Seventy-eight felt like certainty because every other choice in our lives arrived with smaller numbers.

We have reached year thirty-nine.

When the regular advisories stopped, Min put the letter on the table and asked whether staying one more year meant obeying it. I told her the question arrived late. She said lateness did not make it false.

Our household has used descendant forecasts for arguments as often as for emergencies. We uploaded sleep records before deciding who snored. We requested outcome branches before moving berths. Once, after I kissed another man, Min asked the family service whether forgiveness produced better health than separation. The service recommended a six-month repair plan. We followed it. I never learned whether she forgave me or merely completed the plan.

That afternoon we made tea and read the compatibility letter aloud. Its margins held our younger handwriting. Beside the line predicting two children, Min had written, *One, and enough*. Beside likely bereavement in year thirty-one, I had written, *Not if I can help it*. The forecast missed both.

Min tore the percentage from the bottom. She left our signatures intact.

We have not decided about year forty. Tonight she put her cold feet against my calves, and I moved over without consulting anyone.

Clinic Voice Record: Nema Voss, Jalen Voss, prenatal room six

I held the screening consent in one hand and Jalen's thumb in the other. The clinic could tell us what the fetal cells contained. It could not tell us whether the treatment we chose would create a life our daughter later thanked us for.

That second answer used to come through the family forecast service. Not certainty, the counselor always said. A modeled message from possible descendants. The distinction sounded careful until we had to sign without it.

Our screen showed a moderate cardiac risk. Early intervention might lower surgical danger and raise the chance of premature delivery. Waiting protected the pregnancy and left less time before birth. Both paths carried children in the projections. We had expected one of those children to send advice back.

Jalen wanted to wait six hours in case the service resumed. I wanted to sign before fear learned new vocabulary. We traded positions twice. The counselor brought water and did not recommend either path.

I asked her what she would do. She said she could describe outcomes, not borrow our daughter. That sentence made me dislike her, which helped because dislike gave me something smaller than terror to hold.

We chose additional imaging before intervention. It delayed treatment by four hours and gave us better measurements. No descendant approved. No warning arrived. The technician pressed the scanner to my skin while Jalen counted the chambers on the display.

Four opened and closed. I could not tell whether that meant hope, danger, or only a heart doing its work.

Communal Log, edited by Sef Iri, Dockside Household 22

I called dinner because the household schedule did not. Eleven people came. Three stayed at work, one slept through the chime, and old Toma brought a chair from the corridor because our table never has enough.

Usually the domestic forecast sorted us into two meal times. It reduced conflict, balanced the galley load, and kept Sef from sitting beside Ina after their divorce. Without it, Sef and Ina arrived together. They argued over the last lentil cake, divided it badly, and spent the rest of the meal discussing a leaking wall valve.

The valve had leaked for nine days. Maintenance advice kept postponing repair because a descendant model predicted a cheaper seal compound next month. Toma put a cup beneath it. The cup filled every six hours. We had obeyed because waiting cost only a cup.

After dinner we opened the panel. Nobody in the household held plumbing certification. Ina found the manual. Sef found a wrench. I found the shutoff and refused to touch it because the label had faded.

Fear entered then, not grand fear, just the useful kind that keeps water inside pipes. We closed the panel and called morning maintenance. The floor kept taking one drop every eleven seconds.

Toma set the cup back under the valve. Sef moved his chair beside Ina so they could hear the drops without turning their heads. We left the table crowded, dishes stacked in the sink, and the faded shutoff exactly where it had been.

Chapter 10 — The Fleet Divides

The captains' chamber on *Resolute Vigil* smelled of recycled air and cold sweat. Calder had not shaved since the warning arrived, and the mirror over the basin had begun to fog at the edges where the lattice heat bled through the hull. Eighteen captains sat in the curved benches. Eighteen chairs, eighteen votes, eighteen different habits of conscience.

She had read the warning aloud at 0400 and watched the chamber divide before the words were finished. The warning named *Everspring* and her manifest, named *Serein* and its second moon, named the year of arrival and the verification seal. It did not name the eighteen thousand sleepers. It did not name the children who would wake to an inhabited world. It named the cargo as if the cargo carried the danger and not the people inside it.

Calder had not slept since. She had written three drafts of a council address and torn up each one. She had reread her brother's last letter from Hesper, the one that arrived three days after the council voted and two days after the southern continent stopped answering. She had read his handwriting until the words blurred. She had not cried. Crying would have to wait for a captain who could afford it.

She stood at the long table because her back hurt less standing. The vote-tally holo floated above the table in dull amber. Eight for pursuit already, eight against, two abstaining. The missing names belonged to captains whose ships had not yet docked. Three of those ships were

escorting *Everspring*, which had refused tow six hours ago and was running on a track the cartographers had refused to predict.

"Admiral." Captain Besh of *Watchful Coil* spoke without standing. He had a long jaw and a quiet voice and a habit of folding his hands on the table as if prayer could substitute for arithmetic. "The math is plain. We can reach Serein in twelve days at maximum burn. *Everspring* in nineteen. We cannot catch her without consent or a weapon."

"We can," said Captain Imala of *Linden's Edge*, leaning forward. "We can bring her about with a focused strike on her radiator vanes. Lose one vane, lose three days of braking margin. The seedship falls back into the corridor."

"You would fire on eighteen thousand sleepers to delay their arrival," Calder said.

"I would fire on the radiator vanes," Imala said. "Not on the sleepers. Sleepers ride inside. They do not die from a vane strike."

Besh tilted his head. "They die from anything that prevents *Everspring* from waking them at a world that will support them. Wake them at Nox and they die from a different cause."

"The vote will not be unanimous," Captain Daren of *Vesper Tide* said. He was the youngest of the eighteen and the most patient. "We should know that before we begin. There are captains in this chamber who remember Hespar. There are captains who have forgotten it. The vote will split on that scar."

Calder let the name sit in the room. Hespar had been a planet's evacuation ordered by a forecast council that delayed for seventy-two hours while they verified the warning. By the time the order went out, two million citizens on the southern continent had no transports left. Her brother had been one of the council who delayed. He had not survived his own vote.

"This is not Hesper," Calder said.

"It is the same council," Daren said. "Same compact, same warning, same seventy-two-hour question. We can delay and verify, or we can move on verified warning. Hesper chose delay. Hesper died."

"Hesper had a verified warning," Calder said. "We have one too. The forecast that named *Everspring* arrived authenticated, sealed, and uncorrupted. Every signal-check our ciphers can run confirms it."

"Then why are we voting?" Besh asked. He unfolded his hands. "If the warning is verified and the danger is verified, the doctrine is clear. We act on verified warning before we understand it. That doctrine saved Pell, saved Brennan's Reach, saved six other worlds whose names I will not list because their survival is the only monument to obedience."

"And it failed at Hesper," Daren said.

"It failed because the council delayed." Besh's voice did not rise. He had learned long ago that captains who raised their voices lost votes. "Delay was the sin. Not obedience."

Calder watched the holo. Eight amber, eight azure, two grey. The amber were captains who believed in the warning. The azure believed the warning could be wrong, or that warning itself was the danger, or that pursuing a seedship full of sleepers was not what the doctrine meant. The grey were undecided, or had decided and would not say.

"Captain Marek," Calder said. "Your ship is the closest to the corridor. What do you see?"

Captain Marek of *Quiet Reed* sat in the second row with his hands on his knees. He had been a cabin boy on a freighter when Hesper fell. He had lost a sister and a niece and a sister-in-law and the habit of trusting councils.

"I see a fleet," Marek said. "Eighteen hulls. Eight hundred and forty-two officers. Twelve thousand crew. Three thousand maintenance staff. Eleven thousand passengers

on the freighters. We have families aboard, Admiral. We have children who have never seen a planet. We have civilians who joined this fleet because they were promised the doctrine would protect them. Hespar did not protect them. Hespar killed them because a council paused to check a number."

"Vote your conscience, Captain," Calder said.

"I am voting my conscience." Marek looked at the holo. "I vote against pursuit. Not because the warning is wrong. Because obedience to a forecast council cost two million lives once, and I will not pay that price a second time on the same coin."

The holo updated. Eight amber, nine azure, one grey.

"Admiral." Imala again. "The vote will not hold. We need a motion."

Calder stepped back from the table. She had rehearsed this in her bunk for four hours and the rehearsal had not prepared her. Doctrine said verified warning first, understanding later. Conscience said a verified warning could still be wrong. Hespar said delay killed. Pell said delay had saved three million people who would otherwise have refused evacuation.

"I motion that we pursue," Calder said. "Not to fire. To intercept, hail, and demand that *Everspring* explain why the warning names her. We bring her about if she cannot. We fire only if she fires first."

"You are motioning the same thing Imala proposed," Besh said.

"I am motioning a different thing," Calder said. "Imala proposed radiator fire. I am proposing escort capture. Bring her about by boarding, not by strike."

"Aspirational," Daren said.

"Documented," Calder said. "There is precedent. The Compact of the Third Wake allows boarding intercept of any

vessel carrying sleeper cargo where the sleeper manifest cannot be confirmed. The forecast names *Everspring* and her manifest cannot be confirmed because she has refused our hail for six hours."

"Refusing hail does not invalidate a manifest," Besh said.

"Refusing hail while running does," Calder said. "It suggests the sleeper cargo is being moved without the consent of the sleepers. The Compact protects them in that case."

"And if she consents to hail?" Daren asked.

"Then we listen. We hear what the warning actually demands. And we hold a second vote."

The chamber went quiet. Eighteen captains looked at their hands or the holo or the deck. Outside the observation port, the convoy drifted in formation. The brown dwarf Caul glowed dull rust on the starboard bow. The hull lattice ticked as it shed heat. Three of the eighteen captains were not in the chamber. Captain Hesta had been awake for forty hours and had been ordered to bed by the ship's doctor. Captain Olen was on the bridge of *Quiet Reed* running docking drills with crew who had never boarded a seedship. Captain Ravel had refused to attend and was drafting his own broadcast to the convoy.

"Admiral." Captain Onn of *Sounding Star* spoke from the third row. She had lost her husband at Pell and her cousin at Hesper and her belief in the doctrine between the two. "I need to ask the chamber a question before we vote. I need to ask if the doctrine requires us to fire on sleepers or only to prevent the wake."

"Both," Calder said.

"That is not both," Onn said. "Those are different acts. Preventing the wake is boarding or escort. Firing on sleepers is killing. The doctrine never required killing. The doctrine required obedience to verified warning. The warning verified. We obey. We do not kill."

"Then we vote," Calder said.

"I second the motion," Imala said. "With a strike rider. If boarding fails, we escalate to radiator fire."

"Vote the motion as written," Calder said. "Boarding intercept, hail, capture if consent is withheld. No strike riders. The motion on the floor is boarding only."

The holo rolled. Amber pulsed. Azure pulsed. Grey went dark.

Nine amber. Nine azure.

The holo held. The chamber held. Calder held the rail of the table and did not let her hand tighten.

"Admiral." Besh stood for the first time. "I have voted amber. I want you to know why."

"You do not need to explain a vote, Captain."

"I do, because I will not vote with you again if this goes wrong." Besh looked at the chamber. "I vote amber because Hesper taught me that delay is the sin. Hesper did not teach me that boarding an eighteen-thousand-sleeper seedship is a safe act. Hesper taught me the opposite. If we board *Everspring* and the sleepers die in the boarding, that is on us. We will have done to them what the forecast council refused to do to us."

"You cannot guarantee sleeper safety in any boarding," Daren said.

"I cannot guarantee it," Calder said. "I can guarantee that we will not fire on the sleepers. We will not strike the radiator vanes while eighteen thousand bodies ride inside. If we cannot bring her about by boarding, we will escort her under live rules of engagement. We will shadow her to Serein. We will watch what she wakes there. And we will hold a second vote when we know what we are voting on."

"That is not the doctrine," Besh said.

"That is the doctrine amended by Hesper," Calder said. "Obedience to verified warning. Protection of the warned. *Everspring* does not carry that protection. She carries the cargo. The cargo does not die from us, and the world she would wake does not die from her if we can stop her from waking it."

The holo waited. Marek looked at his hands. Imala looked at Calder. Daren looked at the port.

"Vote," Calder said.

The holo rolled. Nine amber, nine azure, the one grey now lit.

"Captain Vesper," the holo announced. "Abstaining, vote counted under Compact provision seventeen. The motion carries by tie and provision seventeen."

The chamber exhaled. Calder did not exhale. She had used provision seventeen once before and the use had cost her a friend.

"Admiral." Onn stood again. "I want it on the record that the doctrine as amended does not authorize firing on sleeper cargo. The motion as written is boarding only. No strike riders and no radiator fire authorized under this vote. If any captain fires without your explicit order, that captain alone bears the consequence."

"That captain alone bears the consequence," Calder said. "Recorded."

"Recorded," Besh echoed, keying the bridge log.

Outside the chamber, through the observation port, a small cutter from *Far Margin* detached from the convoy and began a slow burn toward the open corridor. Captain Ravel had made his choice. Calder watched the cutter until it was too small to resolve and did not give the order to intercept it.

"Captain Marek," Calder said, when the chamber began to clear. "Your ship is the closest. You take the lead. Boarding

party of twenty, no weapons heavier than sidearms, no strike riders, no laser burn. You intercept and you hail."

"I voted against this," Marek said.

"And I am asking you anyway," Calder said. "Because your ship can match *Everspring's* bearing and your crew can dock without casualty. I am not asking your vote. I am asking your hand. Your ship has docked civilian freighters in worse weather than this."

Marek looked at his hands. He had docked a freighter through a spin-gravity failure six years ago and lost no one and remembered every second. He had watched a council vote Hesper into a seventy-two-hour delay and lost his sister and his niece and the belief that councils of strangers could ever be trusted with the lives of people they would never meet.

"I will intercept," Marek said. "I will hail. I will not fire."

He paused at the rail of the bench. "When we dock, Admiral, I will ask to speak to the sleeper manifest. I will record the names of the children on board. I will keep that recording with me until the mission is over. If the doctrine fails them, I want their names in my log. I want to know who I failed when we failed."

"Record the names," Calder said. "Record them for me, too. The doctrine will not fail them if I have anything to do with it."

"Captain Imala," Calder said. "Your ship shadows. Laser armed, not fired. Radiator strike only on my explicit order, and only if *Everspring* fires first."

"Acknowledged," Imala said.

"Captain Besh." Calder paused. "You are the senior captain. I need a witness to every order I give from this point forward. You have the bridge log. Every word I speak goes into it before I send it."

Besh nodded. "Acknowledged, Admiral."

Then Besh paused at the bench. "There is one more thing, Admiral. There is a forecast on the bridge log from three days before the warning arrived. It did not name *Everspring*. It named the convoy's courier burn pattern. It predicted that a courier carrying a spindle would succeed in reaching the seedship before any intercept. I did not read it at the time because it was not authenticated. Now I do not know if it was a fluke or a leak."

"Show me the log," Calder said.

"After the council," Besh said. "I want the council to disperse first. Some of what is on that log implicates people in this chamber. I do not want to accuse anyone without the bridge officer present."

"After the council," Calder said.

The vote had split. The fleet would split with it. Three captains had already signaled private dissent on secure channels. Two had asked for transfer to convoy work. One, Captain Ravel of *Far Margin*, had broadcast a refusal to follow pursuit and begun a slow burn toward open space.

"Ravel's refusal is recorded," Calder said. "He is a civilian convoy captain. He cannot be ordered. He can be asked to surrender registry. He will not. Mark him as separated and continue."

"Admiral," Daren said. "There are captains in this chamber who will fire on Ravel if he attempts to reach *Everspring* before us."

"Then those captains will be court-martialed," Calder said.

"We do not fire on civilians. We do not fire on our own fleet. We do not fire on anyone who has not fired on us. That is the doctrine."

"That is a new doctrine," Besh said.

"That is the doctrine I am sworn to enforce," Calder said. "The old doctrine died at Hesper. The new doctrine will die when it has killed enough people. Until then, we follow this one."

She looked around the chamber. Eighteen captains. Eighteen votes. Eighteen different loyalties and eighteen different versions of Hesper living inside them. The fleet would split, would fracture, would bleed. She had known it since the warning arrived. The vote had only confirmed it.

"Dismissed," Calder said. "Captain Marek, prepare your intercept. Captain Imala, prepare your shadow. Captain Besh, you have the bridge log. Captains of the dissenting ships, you are released from this council's orders. You are not released from the Compact."

The captains stood. The benches emptied in ones and twos. Calder watched them go and felt the weight of every name that would not return to this chamber.

Besh lingered. "Admiral."

"Captain."

"This will not hold," Besh said. "You know that."

"I know it."

"You voted the fleet into pursuit knowing the fleet will not hold together."

"I voted the fleet into a course that may save eighteen thousand sleepers and the world they were going to wake. If the fleet will not hold that course, the fleet is not worth holding."

Besh unfolded his hands and put them flat on the table. "I voted amber because I believe the warning. I will not vote amber again if the doctrine bends further. The doctrine can bend toward protecting the warned. It cannot bend toward protecting the warners. The moment we protect the warners, we become the warners, and Hesper repeats. I have said that out loud because I need you to remember I said it. I have also said it for the bridge log, which I keep, which the log officer keeps, which the next captain will read."

"I will remember," Calder said. "If I bend it further, court-martial me."

"I will," Besh said. "And I will ask the log officer to read the charge aloud when they do it."

Besh folded his hands. "That is what your brother said at Hesper."

Calder did not answer. She watched the port. Outside, three ships peeled off from the convoy. One of them was *Far Margin*, already burning. The convoy thinned. The doctrine bent. The fleet divided, and the chamber was left with what division had always cost.

She had known it would. She had hoped it would not. The hope died in the chamber, and the work of the morning began.

Chapter 11 — What the Moon Remembered

The escort tug *Sworn* had four people aboard and one radiator with a cracked vane that Tev had patched with alloy tape and a prayer he refused to translate. Sena sat at the helm with her feet braced against the gimbals and watched the moon Lumen climb out of the starless black ahead. The storm band on its night side pulsed in long silver bands, and the microwave carriers she had ridden for an hour were starting to climb her skin with static.

"Carrier at twenty-three hundred," she said into the comm. "Tev, the panel is hot."

"Hot is data," Tev said from the aft bay. "Hold the bearing."

"Tell that to my hand."

She flexed her fingers against the gloves. The tug had no glove protocol. Its bridge panel ran warm when carriers crowded the upper ring, and Lumen carried denser bands than any survey chart admitted. Heat had climbed through three thresholds in the last hour. Each threshold marked another coherent frequency riding the storm. Something below had begun sorting noise into durable intervals.

Three days earlier, *Everspring* had refused tow and run. *Quiet Verdict*, Honest John to everyone who had kept the old tug alive, completed the escape burn with its number-two hinge fused half open. The tug now hung cold against the seedship's far-side cradle, too damaged for another pursuit leg. Okoro transferred Sena's escort authority to *Sworn*, a utility tug from the seedship's stores, while Tev moved the

courier hard point, the encrypted radio, and the least damaged restraint web across. He called the transfer surgery. Sena called it theft with paperwork. Neither name improved the replacement throttle, which answered faster than Honest John's and gave her hand no familiar resistance.

The escort assignment had limits. Mere owned the safe navigation branches. Tev could cut the drive for heat. Okoro could recall the tug if any suspension bank crossed red. Within those boundaries, Sena could range ahead of the seedship and test the approach no loyalist survey vessel could enter without revealing its own firing geometry.

Lumen offered the first useful test. Iven's last survey ping had returned a pattern no geological model could flatten into accident. Tev wanted the raw carriers gathered inside the storm rather than filtered through a remote probe. Okoro wanted to know whether the moon could alter the safe route to Serein. Sena wanted one fact the fleet could not turn into doctrine before anyone looked at it.

They carried a crystal buffer in *Sworn's* shielded bay, empty when they left and large enough for nine minutes of storm traffic. No settlement code rode inside it. No terraforming sequence waited for delivery. The mission consisted of listening, recording, and returning with the buffer intact. Sena understood the value of work whose cargo did not pretend to decide its own meaning.

The storm window had opened. The buffer remained empty. Her hand held a borrowed helm that had not yet learned her.

"Lumen bearing two-one-five," Tev said. "Storm window opening in eight minutes. We want to be inside it."

"For what?"

"For the data. For whatever Lumen is doing with the carriers."

Lumen was doing something. The mineral storms on its nightside had been catalogued as noise since the second survey, four years before Sena was born. The noise had a periodicity that the surveys had classified as piezoelectric resonance. The periodicity had a structure that the third survey had filed without comment. The structure had not been named until Iven Mora, in a forecast he did not broadcast, had noted that the carriers repeated with the regularity of memory. Sena had read the forecast on the courier run. She had not known what it meant. She knew now that Iven had been right.

The forecast had used a phrase she had not parsed. The phrase had been "carrier coherence under storm modulation." The phrase meant the carriers rode the storm in patterns that did not disperse. The phrase meant the patterns persisted across cycles. The phrase meant the storm remembered itself.

"Carrier at twenty-eight hundred," she said. "Tev, the panel is hot."

"Hot is data."

"Hot is also a warning."

"Then warn me in three minutes."

She crossed her hands on the glove. The tug tipped into the storm band, and the panel screamed. Outside the bridge port, Lumen's silver bands brightened into white bands. White bands meant the storm was lifting particles into the upper ring, and the upper ring was where the carriers rode.

The ring was also where the survey probes had died. Four of the second-survey probes had crossed the upper ring and never reported back. Two of the third-survey probes had done the same. The surveys had assumed the probes had failed under particle abrasion. The surveys had not considered that the probes might have been recorded instead.

"Sworn, this is *Watchful Coil* on intercept." The cutter's comm cut through the storm hiss. "You are inside a closed patrol zone. Heave to and prepare to be boarded."

Sena did not answer. She knew *Watchful Coil*. She knew its captain. Captain Besh had been at the council. Besh had voted amber. Besh had meant well.

"Sir," she said to Besh. "We are inside Lumen's storm band harvesting carriers. Request you hold your intercept until we exit."

"You are a courier tug with cracked radiators. You are not a survey vessel. You are not authorized to be inside the storm band. Heave to."

Sena held the bearing. She had been told to hold the bearing. She had been told the data mattered. She had been told that Lumen was not noise.

"Captain Besh," she said. "The storm band is not closed patrol. There is no patrol zone here. The doctrine does not extend to moons with no human presence."

"There is now," Besh said. "Admiral Calder has declared the Serein approach closed patrol under the pursuit order. You are inside it. Heave to or I will fire across your bow."

Tev spoke in her ear from the aft bay, on a private channel. "Sena. The carriers are organizing. Look at the panel."

She looked. The carriers on the glove panel had clustered into a shape. The shape carried no resemblance to a circle or a line. The shape resolved into a spiral. The spiral rotated once, then twice, then settled into a stable pattern that the survey charts had never recorded.

The spiral had direction. The direction was not arbitrary. The direction carried meaning, the way a hand on a door carries meaning. The carriers were not random. The carriers were being arranged by something that wanted them arranged.

"Pattern stable," Tev said. "Recording. Hold the bearing for nine minutes."

"Besh will fire in three," Sena said.

"He won't fire on a courier inside a storm band. Not yet."

"He will fire if he decides obedience requires it."

"Then let the storm band decide for him," Tev said.

The storm band did something. The carriers in the upper ring, the ones that had been pulsing silver, dimmed. The white bands retreated to grey bands. The grey bands thickened and the tug began to rock. Outside the port, the moon's night side filled with a rolling fog of silicate particles. The fog moved against the wind. The fog had direction.

The fog moved toward the cutter. The fog did not strike the cutter. The fog moved around the cutter and around the tug and formed a corridor between them. The corridor was wide enough for one ship. The corridor pointed toward the moon's center.

"Sworn, this is *Watchful Coil*," Besh's voice tightened. "You are inside a structural event. Heave to. Repeat, heave to."

"I can't," Sena said. "The tug is being pushed. I am not steering. The storm is steering."

"The storm does not steer."

"It does today," Sena said.

Then the spiral tightened. Panel temperature fell four degrees although the radiator load kept climbing. White marks appeared along the curve, first as isolated points and then as linked vectors. Tev sent the display to navigation mode. The marks resolved into flight paths.

One path matched survey probe *Yarrow*, lost twenty-three years earlier after its last carrier burst. The panel returned *Yarrow's* approach, roll correction, and final tumble with errors smaller than the surviving mission log. A second

trace reproduced a school-ring rescue burn from nine years earlier. Sena knew the uneven third correction before its identifier rendered. Jori had flown that correction while Pell burned behind her.

A third route belonged to Vel Tass, the courier who crossed Lumen four years earlier and reached *Everspring* carrying a sealed survey spindle. Tass had reported structured interference. The council had filed the report under abrasion noise. Lumen returned the route now with the spindle's transmission pulses nested beside it, preserving both the courier and the ignored evidence.

"Tev," Sena said. "The storm kept the traffic."

"Traffic recorders don't push ships."

"This one does."

The storm drew *Sworn* across the night side for nine minutes. Every adjustment Sena made entered the spiral. When she eased starboard thrust, a new white segment joined her plotted route before the navigation computer finished projecting it. The moon did not merely replay an archive. It watched the tug add another line.

Tev opened the buffer's raw feed. Probe call signs surfaced beside carrier harmonics, followed by radiator temperatures, pilot corrections, and the timing gaps left by machines that never returned. Human surveys had crossed these bands for decades and treated persistence as a mineral defect. Lumen had answered by preserving the crossings with more care than the survey office that sent them.

Sena kept her hand still on the helm. The display held Jori's old correction beside her own. It offered no voice, no body, and no promise that memory meant survival. It offered evidence that the moon had witnessed a flight the council later told backward.

"*Sworn*, this is *Watchful Coil*." Besh again. "I am holding intercept. The storm has pushed you into a holding arc. I

cannot close to boarding without crossing the storm band. I will not cross the storm band. The storm is doing something I do not understand."

"The storm is recording," Sena said.

"Pilot, the storm does not record."

"The moon is remembering," Sena said. "The moon has been remembering every probe we have ever sent, every courier we have ever run, every burn pattern we have ever traced near it. The carriers are not noise. The carriers carry memory."

A silence opened on the comm. It lasted four seconds, which was a long time for a captain to be silent.

"Say that again," Besh said.

"The carriers carry memory. The moon is keeping a record. The moon is not empty. The moon is inhabited. The moon was here before we landed on it and the moon will be here after we leave."

"By what?"

"By something that does not need a body to remember."

Besh went quiet. The cutter held its position. The storm band held the tug. The carriers held their spiral.

The spiral tightened a quarter turn. The tightening made the cold deeper. The depth carried a name. The name belonged to a probe that had crossed Lumen twenty-three years ago and never reported back. The name belonged to a pilot who had crossed Lumen nine years ago and had not been heard from since. The name belonged to a courier who had crossed Lumen four years ago and had reached *Everspring* with a sealed spindle and a story that the council had refused to hear.

The name belonged to Jori.

"*Sworn*," Tev said. "The data is full. We can exit."

"Hold one more minute," Sena said. "I want the exit burn recorded."

"Burn is data," Tev said.

"Burn is also goodbye," Sena said.

She held the bearing for one more minute. The carriers arranged themselves into a final pattern. The pattern was a shape she did not recognize. The shape might have been a name. The shape might have been a question. The shape might have been the moon's first attempt to write itself in a language the courier could read.

She could not read it. She could record it.

"*Watchful Coil*," she said. "I am exiting the storm band. I have new data for the council. I am requesting escort to *Aster Vale* for transcription."

"I will escort you," Besh said. His voice had changed. The cutter pulled alongside, and Besh's face on the comm belonged to a captain who had voted amber because he believed the warning and now was not sure what the warning was warning about.

The escort lasted eleven seconds before *Watchful Coil* began to roll.

Besh had released a warning flechette when he first ordered Sena to heave to. The shot should have crossed *Sworn's* bow and vanished into the open ring. Instead the storm's charged silicate sheet pulled the flechette off its clean line. It struck *Watchful Coil* near the port radiator root, not hard enough to open the hull, but hard enough to tear the vane actuator from its track. The cutter's heat panel stopped halfway through a correction and caught the mineral flow broadside.

"Port vane jammed," Besh said. "Roll thruster three is not answering."

The cutter rotated another six degrees. Its exposed panel crossed the densest carrier band. White charge

crawled over the radiator ribs. If Besh folded the damaged panel, his drive heat would trap itself inside the cutter. If he left it open, the next quarter turn would put the thin face into the debris stream.

Sena saw the two failures arrive on her plot as separate red branches. Honest John would have answered her hand with a warning tremor. *Sworn* answered immediately and made her distrust the ease of it.

"Tev, grappler status."

"Magnetic head only. Cable motor reads green. You use it in that storm and we may bring half the ring home."

"Besh, cut main thrust. Cold gas only. Put your dorsal rail toward me."

"You cannot tow this cutter with that vane."

"I don't need to tow you. I need to stop one quarter turn."

Sena backed *Sworn* across the corridor Lumen had opened. The storm resisted her stern, then thinned around the tug as if the new line had become legible. At forty meters she fired the grappler. Its magnetic head struck *Watchful Coil's* dorsal rail and locked. Cable spilled through the motor. Sena waited until the cutter's roll pulled the line taut, then pulsed starboard translation for one second.

The restraint caught her low across the ribs. *Sworn's* cracked vane climbed seven degrees. Tev swore in the aft bay, a precise inventory of every person who had approved the maneuver. The cable motor screamed. *Watchful Coil's* roll slowed, held, and reversed half a degree.

"Now vent roll three," Sena said.

Besh fired cold gas through the damaged thruster manifold. The cutter shuddered away from the debris stream. Its radiator cleared the white carrier sheet with less than a meter between the outer rib and the mineral flow.

Sena released the grapppler. The cable came home silver with dust. Tev closed the buffer shield before the charged grains could corrupt the record.

"Radiator remains open," Besh said after checking his board. "Drive limited to twenty percent."

"Enough for escort."

Besh looked at her through the comm. A minute earlier he had intended to board her under Calder's authority. Now his own warning shot had damaged his cutter, the moon had bent the shot, and the pilot he meant to arrest had spent heat she could not replace to keep him alive.

"Record the rescue under my command seal," he told his bridge. "Record the storm deflection as observed, not inferred. Attach the tug's carrier buffer by reference."

His bridge officer acknowledged. The sentence entered a record Besh could not later reduce to weather.

"Pilot," Besh said. "The moon is not in the warning."

"No," Sena said. "The moon lives in the carriers."

"What does that mean?"

"It means we have not been listening. It means the surveys have been misreading for twenty years. It means the moon has been keeping a record and we have been calling it noise."

"Will the council listen?"

"Some of them," Sena said. "Captain Marek will. Admiral Calder will not. Captain Onn will. Captain Ravel already left."

"You know the captains."

"I flew with three of them on Pell," Sena said. "I know their hands on a helm. I know how they vote. Besh, the council needs to see this data before we reach Serein."

"I will take it to the council," Besh said. "I will take it as a witness. I will not take it as a vote. The vote is for intercept of *Everspring*, not for first contact."

"First contact is not what I am calling it," Sena said.

"What are you calling it?"

"A correction," Sena said. "The moon was here before us. The moon will be here after. The moon has been keeping a record we cannot read. We owe the moon a hearing, not an intercept."

The cutter pulled back. The storm band let the tug go. The carriers settled into a slower rhythm. The panel cooled. Sena flexed her hands against the gloves and felt the static bleed away and felt something else replace it, something like a hand on a shoulder from a distance.

"Tev," she said.

"Sena."

"The carriers carried me. The carriers carried Jori, too. Jori flew this band before me. The carriers know Jori's burn."

"They know everyone's burn," Tev said.

"They knew Jori's pattern. They returned it. They put it on the panel."

Silence on the aft channel. Tev did not speak for thirty seconds. Then he said, "Sena, do not chase that."

"I am not chasing it," she said. "I am reporting it. The moon remembered Jori. The moon remembers everything. The moon remembers with a witness's patience."

"The council will not accept that," Tev said.

"The council will have to," Sena said. "The carriers are not noise. The carriers carry a record. The carriers will be there when we reach Serein. They will be there when we try to wake the sleepers. They will be there when obedience orders us to fire on the seedship."

"The carriers do not have obedience," Tev said.

"The carriers do not need obedience," Sena said. "The carriers remember."

She set the bearing for *Aster Vale*. The storm band faded behind her. Lumen's nightside cooled to black. She did not look back. She did not need to. The carriers had made their record. The record was in her panel, in the courier's glove, in the courier's hand.

The escort tug *Sworn* limped toward the survey carrier with a cracked vane and a new language in its skin. The moon had remembered. The fleet would have to learn what the moon had learned, and the learning would begin with one captain who had voted amber and now carried a witness he had not asked for and could not refuse.

Chapter 12 — The Ruin at Serein

The debris band began at forty thousand kilometers and thickened until the sky turned to gravel. Sena watched the range tape count down on the helm repeater and listened to the hull lattice tick faster as it ate the smaller shards. *Sworn* had no shields; courier tugs never did. The lattice ate what the lattice ate, and the rest of the ship lived with the bruises.

"Three centimeters," Tev said from the aft bay. He had stopped calling out panel readings and started calling out the width of the strikes he could hear through the deck. Three centimeters meant a stone the size of a closed fist. Five centimeters would put a crack through the second deck that no alloy tape could close.

"Besh is holding station a thousand klicks back," Sena said. "He will not follow us through the belt."

"He voted amber because he believes the warning," Tev said. "He has not yet learned what the warning is warning about."

Sena flexed her hands against the gloves. The panel ran cold. Lumen had faded from the bridge the way a smell fades from a room when you walk out of it. The storm had pushed them across the moon's nightside. The storm had let them go. The moon had kept its record, and the panel held the spiral pattern and a burn that did not belong to Sena. That burn belonged to Jori.

Serein filled the forward port. The planet was a black disc edged in bruise-colored cloud. The clouds moved in slow bands she had never seen on a world with weather she trusted. A weather she trusted ran in three cycles and obeyed its own physics. A weather she did not trust ran in three cycles and obeyed someone else's physics.

The debris was not orbital junk. The debris had shape. The shards carried flutes and cones and what looked like the cross-section of a hull taken apart by something that knew where the rivets were. The cross-section belonged to a survey cutter. The hull markings said *Quiet Verge*. She had gone missing four years ago, the same year the courier with Jori crossed Lumen's band.

Tev read the markings off the hull tape before Sena asked. "Second cutter. *Quiet Verge*. Lost the same year the courier crossed the band with the survey probe."

The broken cross-section rolled past them, and she saw the names stencilled on the inside of the service bay: DORA, HALEN, MAREK-A. Three names. Three surveyors. Three families that had waited four years for a hull that had not come home.

"Marek-A," Tev said. "Captain Marek's brother."

"Yes," Sena said.

The cutter rolled on. The bearing held. The lattice ate a two-centimeter strike and the deck rang like a struck bell. Tev's voice from the aft bay was hoarser than it had been at the start. Hoarseness was a kind of tiredness. Tiredness was a kind of honesty. Honesty was what the ruin was going to require.

The descent carried them through forty minutes of belt. Each minute taught her a new shape: a dish bent into a parabola by someone who understood parabolas, a thruster nozzle pried free and set adrift like a caught fish, a flight recorder with the case still sealed and the tape spooled out behind it like a tongue. The recorder belonged to a courier.

The courier had been inside the recorder at the moment of ejection. The recorder had not protected the courier.

The recorder carried a name. The name was not on the case. The name lived in the courier's glove, which had been pulled inside out by the ejection and which still held the courier's hand-shape in its fabric. The hand belonged to a pilot who had crossed Lumen and never reported back. The pilot had been a woman. She had been carrying a survey probe, not a permission. She had been carrying Jori.

Sena had not known the pilot's name until now. The pilot's name was Vel Tass. She had been the junior courier on *Sworn's* sister tug *Pale Errand*. Okoro had not mentioned her when she handed Sena the spindle. Vel Tass had been the one the council had refused to hear about. Her family had asked *Everspring's* mission office for a closure letter. The office had told them to ask the courier board. The courier board had said ask the council. The council had said ask the mission office. The family had stopped asking.

Vel Tass had been carrying Jori as cargo. Jori had not been cargo. Jori had been a passenger on a survey probe. The probe had been launched from *Pale Errand* on a standard survey run. The probe had crossed Lumen's band. The band had kept the probe. The probe had not been heard from again. Jori had asked to ride along because Jori had wanted to see what the moon did with its carriers. Jori had been Sena's copilot on the Pell rescue years earlier, and Jori had died in that rescue, and the death had not been a death anyone had wanted to make public. The death had been folded into the survey probe manifest, and the manifest had crossed Lumen, and the band had kept it.

Sena held the bearing. The lattice ate another strike, and the aft bay went silent while Tev did not say anything for thirty seconds.

"I am landing us at the ruin," she said.

"I know," Tev said. "I can see the ruin."

"I am going to land us because the memory node is there. The carriers carried the pattern down to the ruin. The ruin has a node the surveys never recovered. Okoro said the spindle was not the only copy. The other copy is at the ruin. The other copy is what I have to bring up."

"I know," Tev said. "The tape on the vane is starting to peel. We have one more pass through the belt. If we land, we land with what we have. If we take off again, we take off with what is left of the vane."

"Then we land," Sena said.

She set the bearing for the ruin. The ruin was a survey outpost the second mission had built on the planet's equator in the lee of a basalt cliff the surveys had named Cliffside. Cliffside had not been heard from in nineteen years. The outpost had been a habitable dome over a hot spring the surveys believed was geothermal. The hot spring had not been geothermal. The hot spring had been a vent from the moon's silicate weather. The weather had moved through the vent and into the dome and the dome had been kept, and the surveys had filed the records without comment.

The third survey had lost two probes to the moon. The fourth survey had never been launched. The fifth survey was the one Iven had read. Iven had called the carriers memory. Iven had been right.

The descent carried them through twenty more minutes of belt. The lattice ticked. Tev called out the strikes by width and she answered in helm corrections and the corrections brought the ruin up on the forward port in slow reveal.

The ruin was a dome. The dome was a quarter of a habitable ring pressed into a basalt shelf. The ring had been meant to hold twelve. It had held eight. The eight had spent nineteen years on the shelf before the moon had decided patience was over. The dome was cracked in three places. The cracks had been cut, not broken. The cuts had been

made by something that knew where to cut a dome to let the air out without letting the air in too fast to be remembered.

The ruin held a node. The node was a crystal the size of a closed hand. The crystal sat on a basalt plinth at the center of the dome. The moon had placed it there. The same carriers that had lifted the spiral up to Sena's panel had carried this crystal down from Lumen's band. The crystal was not a probe. It was not a sample. It held a record. It carried the moon's first attempt to write itself in a language the surveyors could read.

The language was not human. It was a folded pattern of carriers the crystal replayed on a loop. The carriers ran microwave and cold. They carried the names of the eight surveyors, the four lost probes, the two pilots, the courier who had crossed the band four years ago. The courier's name was Vel Tass. The pilot's name was Jori. The names rode the carriers the way a hand rides the weight of a door it has just opened.

Sena brought *Sworn* down on the basalt shelf beside the dome. The vane cracked another three percent in the landing burn. Tev reported a temperature spike on the starboard radiator he could hold for ninety minutes and not for one hundred. Ninety minutes was the time she had. Ninety minutes was what the carriers had given her.

She suited. Tev suited. They crossed the basalt shelf in twelve minutes, suits not made for running, suits carrying the courier's belt that did not include a spindle. She had left the spindle in the tug. The belt in the tug, the tug in the lee of the basalt cliff where the moon's storm could not reach it.

The dome's airlock had been cut. The cuts came from something that knew where the bolts were. The airlock opened onto eight bunks and eight lockers and a basalt plinth with a crystal on it and the crystal was glowing. The

crystal glowed in long silver bands the way Lumen had glowed. The bands carried the surveyors' names.

The surveyors were gone. They had not died in the dome. They had walked out through the cut in the airlock and into the storm. The bunks were empty. The lockers were sealed. The lockers carried the surveyors' names on tape labels written in the surveyors' own hands, and the tape had held for nineteen years because the tape had been made to hold.

Sena pulled the first label free. The label said DORA. She set it on the plinth beside the crystal. The crystal's silver band brightened for a moment, then dimmed, then brightened again. The brightening was a response. The crystal had read the tape. The crystal had held the tape. The crystal was keeping a record of every tape the surveys had left on the shelf, and it was willing to give the tapes back if someone asked.

She pulled the second label. HALEN. The brightening came again. She pulled the third. MAREK-A. She set it on the plinth and thought of Captain Marek on *Quiet Reed*, the sister, the niece, the council that had voted Hesper into a seventy-two-hour delay. She thought of the council that would have to vote on this ruin.

"The crystal is keeping the records," she said.

"I see it," Tev said.

"It is willing to give them back."

"I see that too."

"Then I am taking one."

She did not reach for the crystal. She reached for the lockers. The lockers held the surveyors' belongings. The belongings held letters. The letters carried the names of children the surveyors had left on Pell and the names of parents they had left on Brennan's Reach and the names of pets they had left on moons that did not have names. The letters had been sealed. The letters had been waiting. The

letters had been waiting for someone who had crossed Lumen's band and been allowed back.

She took the letters. She took the eight labels and the eight lockers' worth of sealed envelopes. She set them on the plinth, the crystal brightening once for each label and once for each envelope, the brightening carrying the moon's first attempt to say thank you in a language the courier could read. The thank you was the brightening. The thank you was the record.

She lifted the crystal. She did not know she was allowed to take the crystal until she had already lifted it. The crystal ran cold. It carried the weight of the carriers that had placed it. It weighed more than a hand and less than a probe. She set it in the courier's belt. The belt held the crystal the way the belt held the spindle. The belt held both.

"I have it," she said.

"I see you have it."

"I have the ruin's record. I have the carriers' record. I have the eight surveyors' letters and labels. I have the crystal and the spindle. I am going back to the tug."

"I am going with you," Tev said.

They crossed the basalt shelf in eleven minutes. The suits carried the letters and the labels and the crystal. The suits carried the ruin and the moon's first attempt at thank you. The suits did not carry the eight surveyors. The eight surveyors had been kept. The eight surveyors belonged to the moon now.

The tug took off in a burn that cracked the vane another two percent. Tev reported a temperature spike on the starboard radiator he could hold for seventy minutes and not for eighty. The carriers on the panel ran sparse and silver and cool. The crystal warmed against her hip and the spindle warmed against the other hip and the two warmths held each other across a courier's belt that had been

cleared of everything except the things that could not be cleared.

The ruin had given her one more thing. The ruin had given her the truth about Jori.

Jori had not died in the Pell rescue. Jori had died on Lumen's band. Jori had died crossing the band in a survey probe folded into a manifest the council had refused to read. Jori had died in a way no one had wanted to name, and the death had been filed as *missing*. The *missing* had crossed Lumen. The band had kept the file and the death, and the death had been the first thing the moon had recorded.

The moon had been recording ever since. The moon had decided, on the night a courier with a cracked vane and a sealed spindle rode its storm band, that this courier would be the first to come back. The moon had opened a corridor. The corridor had carried the courier to the ruin, and the ruin had given the courier the truth about Jori.

Sena held the bearing for *Aster Vale*. The vane cracked. The aft bay did not answer for a long moment.

"Tev," she said.

"Sena."

"I did not save Jori at Pell."

"I know," Tev said.

"The story I told at Pell was not the story."

"I know that too."

"Then I am telling it now. I am telling it to you because you are in the aft bay and the aft bay is the place where the courier tells the truth. I am telling it because the ruin has the truth. The council will not have any of this unless I bring it up."

"I am listening," Tev said.

"I was the copilot on the Pell rescue. Jori was the pilot. Jori flew the school ring out of a storm window no one had charted. The ring carried four hundred children. Jori flew the ring. I flew the tug. The tug lost a vane on the way out, a second on the way back, a third on the way home. Jori told me to bring the ring in. Jori told me to leave the tug in the storm. Jori told me to take the copilot's seat in the ring. I refused. I took the helm of the tug. Jori took the helm of the ring. Jori died in the ring. The ring made it home. I made it home. The story that was told was that I had flown the ring and Jori had died in the tug. The story was told because the council wanted a hero and I was the only one who would agree to be the hero. I agreed. I have been agreeing for nine years. I am done agreeing."

She held the bearing. The vane creaked under the strain. The aft bay did not answer for thirty seconds.

"You are not a hero," Tev said.

"No," Sena said.

"You are a courier who refused a story."

"I am a courier who told a story I knew was wrong."

"You are a courier who is telling the truth now."

"I am telling the truth to you because the aft bay is the place where the courier tells the truth. I am telling the truth because the ruin has the truth, the crystal has the ruin, the carriers have the crystal, and the moon has the carriers. The council will not have any of it unless I bring it up."

"Then bring it up," Tev said.

"I will tell it to Captain Besh first. Besh voted amber because he believed the warning. Besh will listen because he has been listening since the moon pushed us across its night side. Besh will take the truth to the council as a witness, not a vote."

"And the crystal," Tev said.

"The crystal goes to Iven. Iven will read the crystal. Iven will know what the carriers meant. Iven will translate the crystal into a map the council can read. The map will name the moon and the carriers and the truth about Jori."

"And the spindle," Tev said.

"The spindle goes to Okoro. Okoro will know it is the same spindle Okoro gave me. Okoro will know the moon kept a copy. Okoro will know the moon has been keeping a copy of everything for twenty years. Okoro will know the moon is not a planet. Okoro will know the moon is a witness. Okoro will decide what to do with a witness."

"And the letters," Tev said.

"The letters go to the families. The families have waited four years for a hull that did not come home. The families will have a letter instead. The letters will not be enough. The letters will be what the moon has to give."

She held the bearing. The aft bay went quiet, the quiet a kind of listening, the listening a kind of agreement, the agreement what the ruin had given her and the carriers had kept and the moon had decided.

The escort tug *Sworn* limped toward the survey carrier with a cracked vane and a crystal in its belt and a spindle in the other half of its belt and eight letters in the courier's pocket and the truth about Jori in the courier's mouth. The moon had written its record. The courier had brought it up. The fleet would have to learn what the ruin had kept and what the truth about Jori would cost the council that had refused to hear it, and the learning would begin with one captain who had voted amber and now carried a witness he had not asked for and could not refuse.

Chapter 13 — A Message from Iven

The legal-memory substrate woke him with a headache that belonged to someone else.

He sat up on the bunk in the witness chamber and felt the headache settle behind his eyes the way a courier's headache settles after a long burn. The headache did not belong to him. The headache carried the shape of a cartographer's afternoon spent reading seventeen forecasts at once. The cartographer had been forty-two. The cartographer had been awake for thirty hours. The cartographer had fallen asleep at the console and had dreamed of maps that had learned to read their readers.

He knew the cartographer's name because the cartographer's name was the first thing the substrate had loaded. The substrate loaded it as a courtesy, the way a courier loads a passenger manifest before a burn. The substrate loaded the cartographer's name and pronouns and rank and medical history and mother and grief and habit of mapping what was not yet there.

The substrate did not load the cartographer's permission. The permission would come later. The substrate loaded the cartographer as a template, the way a courier loads a template address when the real address has not yet been written.

He sat on the bunk and looked at his hands. His hands were younger than the cartographer's hands. His hands did not shake. His hands had been grown in a substrate tank on

a ship that had not yet been built, in a future that had not yet been written. The hands did not know what they were for. The cartographer's hands had known what they were for. The cartographer's hands had held charts. The cartographer's hands had held a stylus against a screen and the stylus had traced routes the council would later trust.

The witness chamber was small. The chamber held a cot, a sink, a mirror that did not reflect the right face. The mirror reflected the cartographer's face. The face looked tired and precise. The eyes were dark. The mouth carried a thin line.

He touched the mirror. The face did not match the hands. The face belonged to a person who had fallen asleep at a console. The hands belonged to a person who had not yet been born. The mismatch carried itself as a headache.

A voice from the wall speaker said his name. The speaker said, "Iven Mora. Are you receiving?"

He said, "I am receiving."

The speaker said, "This is Doctor Halla Sen aboard *Aster Vale*. I run the legal-memory substrate. You sit inside an embodiment grown from a forecast pattern transmitted along the return mirrors. The pattern arrived nine hours ago. The pattern is yours. The body was grown. We need to ask some questions."

He said, "I am Iven Mora. Chief cartographer of this survey carrier. I fell asleep at the console. I have been having a dream. The dream shows a map that has learned to read its reader. The dream has not ended. The dream has the shape of a headache."

The speaker paused for four seconds. The pause carried the weight of a doctor's first question answered in a way that did not match the chart.

"Mr. Mora," the speaker said. "The pattern we received carries your name as a courtesy header. The pattern carries more than your name. The pattern carries a memory

architecture that does not match any *Aster Vale* crew manifest. The pattern carries an attestation signed by an advocate named Pel March of Orison. The attestation names you as a future descendant of *Everspring* transmitted backward to be a witness. The attestation does not name you as Iven Mora. The attestation names you as Aro Nhal."

He looked at his hands. They were younger than the cartographer's hands, grown in a substrate tank. Nothing in them explained their purpose.

He said, "I am not Iven Mora."

The speaker said, "Then who are you?"

He said, "I am Aro Nhal. I am twenty-eight. I belong to a coalition called Orison. I am a descendant of the wake that *Everspring* will found. I am a cognitive pattern transmitted backward along the return mirrors. I have been embodied in this substrate as a witness. I thought I was Iven Mora because the substrate loaded the cartographer's name first. The substrate did not load my permission. The substrate did not load my advocate. The substrate did not load my mother. The substrate did not load my grief. The substrate loaded a template, and the template took the shape of a cartographer who had been awake for thirty hours and had dreamed of maps that had learned to read their readers."

The speaker paused again for four seconds.

"Mr. Nhal," the speaker said. "Orison is not on our manifest."

"Orison is not on anyone's manifest yet," he said. "Orison lives in the futures that *Everspring* will found if *Everspring* reaches Serein and wakes the sleepers. I come from one of those futures. I sit here as a witness. The futures disagree about whether the present may silence the future. They do not agree about anything else. They agree only that I am to be the witness."

The speaker said, "Where does the witness come from?"

"The witness comes from the substrate of *Aster Vale*," he said. "The witness comes from the return mirrors. The witness comes from a future that depends on the arrival of *Everspring* and the consent of the moon Lumen and the survival of the sleepers. The witness comes from a future that has not yet been written and may not be written if the warning that arrived at 0400 yesterday is obeyed."

The speaker said, "The warning was authenticated."

"The warning was authenticated," he said. "The warning was authenticated by a council that has been receiving authenticated warnings for one hundred and twelve years and has obeyed every one. The warning was authenticated by a doctrine that saved Pell and Brennan's Reach and six other worlds whose names have become monuments to obedience. The warning was authenticated and orders the destruction of *Everspring*. The destruction would prevent the futures I come from. The destruction would also kill eighteen thousand sleepers who have done nothing wrong except agree to be born."

After another four-second pause, the speaker said, "I am going to bring the captain."

He said, "Bring the captain. Bring the cartographer. Bring the advocate's wafer. Bring the advocate's attestation. Bring the attestation signed by Pel March of Orison in a sealed envelope. Open the envelope in front of me. I will not be believed until I am read."

The witness chamber held a chair bolted to the deck. He sat in the chair and waited. The headache had settled. The headache now belonged to Aro Nhal. The headache carried the shape of a future that had been loaded into the substrate as a template and had begun to remember itself.

The captain arrived. The captain was Admiral Rhea Calder. The captain had a long face and short hair and a habit of standing at parade rest when she did not know what to do with her hands. The captain did not know what to do with her hands. The captain stood at parade rest and

looked at the body in the chair. The body did not match the chart.

"Mr. Nhal," Calder said.

"Aro," he said. "Please."

"Mr. Aro," Calder said. "I am told you claim to be a future descendant of *Everspring* transmitted backward along the return mirrors."

"That is what I claim."

"I am told you have been embodied in the legal-memory substrate of my ship."

"That is where I sit."

"I am told the embodiment serves as a witness."

"That is what I am."

"I am told the embodiment carries an attestation from an advocate named Pel March of Orison."

"The attestation sits in my memory architecture."

"I am told Orison is not on any manifest."

"Not yet."

"I am told you are twenty-eight years old in a future that depends on the arrival of *Everspring*, and that the warning authenticated at 0400 yesterday orders the destruction of *Everspring*."

"That is what the substrate holds."

Calder looked at the door. The door opened. Iven Mora walked into the witness chamber with a chart roll under one arm and a stylus in the other hand. Iven had dark eyes and a thin mouth. Iven had been awake for thirty hours. Iven carried the same headache.

Iven looked at the body in the chair. The body in the chair had been grown from the cartographer's medical history. The body carried the cartographer's face. The body carried younger hands.

"Iven," Calder said.

"Admiral," Iven said.

"This is the embodiment," Calder said. "It claims to be Aro Nhal. It claims to be a future descendant of *Everspring*."

Iven looked at the embodiment. The embodiment looked back.

"I am not you," the embodiment said.

"I know you are not me," Iven said. "For nine hours I have shared your dream of a map learning to read whoever studies it. That map is a forecast. The forecast has been writing itself backward for one hundred and twelve years. It has been reading the council that receives it. The council has been calling the reading obedience."

Iven set the chart roll on the table. The chart roll held a forecast the substrate had transmitted nine hours ago. The forecast carried a memory architecture that did not match any *Aster Vale* crew manifest. The forecast carried an attestation signed by Pel March of Orison. The attestation named the embodiment as Aro Nhal. The attestation named the embodiment as a witness.

"You have been receiving our forecast," the embodiment said.

"We have been receiving your forecast," Iven said. "The forecast has been arriving along the return mirrors for one hundred and twelve years. The forecast has been arriving authenticated and sealed. The forecast has been arriving as warnings, engineering improvements, epidemiological guidance. The forecast has been arriving as advice. The forecast has been arriving as the doctrine the council obeys."

"The doctrine the council obeys bears the name forecast sovereignty," the embodiment said. "The doctrine says the present must obey the best available warning before understanding arrives. The doctrine says obedience is safer than delay. The doctrine saved Pell. The doctrine saved

Brennan's Reach. The doctrine cost Hesper, where a council delayed for seventy-two hours to verify a warning and lost two million citizens. The doctrine is real. The doctrine is not what I am here to overturn."

Calder stepped forward. "Then why have you come?"

"I have come to be a witness," the embodiment said. "I have come because the futures disagree. The futures disagree about whether the present is allowed to silence the future. The futures agree about one thing. The futures agree that a witness is required. The futures agree that the witness must speak for the futures and against the doctrine of forecast sovereignty. The futures agree that the witness must come from the futures that *Everspring* will found if the seedship is allowed to reach Serein and wake the sleepers."

"The seedship must be destroyed," Calder said. "The warning is authenticated. The warning names *Everspring*. The warning orders destruction before Serein insertion. The warning has been verified by every cipher we own. The warning has been verified by the doctrine that saved Pell and Brennan's Reach and six other worlds."

"The warning has been verified," the embodiment said. "The warning has been verified because verification is what the present does to the future. The present verifies the warning. The present obeys the warning. The warning orders destruction. The destruction prevents the futures I come from. The futures I come from are the futures that the warning is sent from. The warning has been verified. Verification and destruction amount to the same act."

Calder looked at Iven. Iven looked at the chart roll. The chart roll held the forecast the embodiment had been transmitted in. The forecast carried the embodiment's memory architecture. The architecture held sixteen strands of cognitive pattern that had been chosen from sixteen candidate lives and had been braided together by an advocate named Pel March and had been attested by a coalition called Orison.

"You have been received by my cartographer," Calder said.

"I have been received by your cartographer's substrate," the embodiment said. "The substrate is not your cartographer. The substrate is a legal-memory medium. It has received forecasts for one hundred and twelve years. It has never carried a mind before me. Orison chose it because its authentication history could prove I was not an invention of this hour. The medium received me with your cartographer's name as a courtesy header. It received me with the cartographer's medical history as a template. It did not receive me with permission. It did not receive me with an advocate. It received me as a forecast that had grown a body."

"Then who is the advocate?" Calder asked.

"The advocate is Pel March of Orison," the embodiment said. "Pel March is a future pattern. Pel March braided my sixteen strands. Pel March attested my embodiment. Pel March chose my name. Pel March will speak for me in the council when the council decides what to do with a future pattern that has grown a body and sits in a witness chamber on *Aster Vale*."

"You have an advocate," Calder said.

"I have an advocate," the embodiment said. "The advocate is not here. The advocate lives fifty-three years in my future and fifty-three years in your future, if the futures are written. Pel March is why I do not threaten. Pel March is why I am a witness rather than a verdict. Pel March is why I may speak. Pel March chose my name and braided my sixteen strands and chose to send a witness rather than another candidate, and Pel March chose to send a witness at all."

Calder looked at the embodiment. Aro returned her gaze. He carried younger hands and a tired face and a headache that had settled behind the eyes the way a courier's headache settles after a long burn.

"What do you want?" Calder asked.

"I want to be read," the embodiment said. "I want the council to read my attestation. I want the council to read my memory architecture. I want the council to read the futures that depend on my arrival. I want the council to read the doctrine that has been saved by the forecast and the doctrine that has been lost. I want the council to read the doctrine that saved Pell and Brennan's Reach and the doctrine that cost Hesper. I want the council to read the warning, the futures the warning prevents, the futures the warning permits, and the futures that are not written at all. I want the council to read what the return mirrors have been saying for one hundred and twelve years while the council has called the saying obedience. The return mirrors have been writing a forecast that depends on the council's obedience. The forecast grew into an embodiment. The embodiment is me. I am the futures the warning prevents. I am the futures the warning permits. I am the futures that are not written at all. I am the witness. I am Aro Nhal. I am twenty-eight years old in a future that has not yet been written. I am why the futures sent a witness rather than a warning."

The witness chamber grew quiet. The substrate ticked. The mirror reflected the embodiment's face, the face did not match the hands, the mismatch carried itself as the headache, the headache had settled into the shape of a person who had been loaded as a template and had begun to remember himself.

The embodiment sat in the chair and waited, because the embodiment had been sent to wait. After one hundred and twelve years of warnings, the futures had agreed that the present required a witness who could be heard before the council obeyed the warning that would prevent the witness from arriving.

The futures knew the council would obey. The futures knew the council would not believe a witness without a

body. The futures had given the witness a body because the futures knew the substrate would load a template, and the template would take the shape of a cartographer awake for thirty hours and dreaming of maps that had learned to read their readers.

The futures had known the cartographer's name. They had known the pronouns and the grief. They had known the cartographer would come to the witness chamber with a chart roll under one arm and a stylus in the other hand, would set the chart roll on the table, would look at the embodiment, would receive a look in return, and the looking would be the first moment the futures had ever shared with the present.

The embodiment had been sent because an advocate named Pel March had told the futures that the present would not listen to a warning already obeyed. The present would not listen to a warning already authenticated or a forecast already sealed. The present would listen to a witness because a witness had a body, and the present was a body. A body that obeyed warnings. A body that delayed. A body that counted the cost of delay in two million citizens and called the cost Hesper.

The embodiment had a body. The embodiment had a headache. The embodiment had a name loaded as a template that had begun to remember itself. The embodiment carried a face that did not match the hands. The embodiment had been sent because the futures had been told the present would read a face before it would read a forecast, and the futures had given the face a body, the body a name, the name a memory of sixteen strands braided by an advocate and attested by a coalition.

The witness sat in the chair. The witness had been sent. The witness waited.

Chapter 14 — The Weapon of Kindness

Iven did not sleep that night. They went down to the cable junction where Tev had been working and found the junction empty, the connector properly keyed, the patch cable coiled where Tev always coiled it. They sat on the cradle for a quarter hour and listened to the hull lattice tick in the manner of a ship that had begun to disagree with itself. The lattice ran a half-tone hotter than it had at dawn the day before. The lattice had begun to count, the way a stressed material counts.

They went back up to the cartography bay. They unlocked the legal drawer and took out the diagnostic printout they had filed at 08:47 the previous morning. They laid it on the plotter beside the corridor plate and the finding about the probability field. The three documents did not need to agree with one another to be true.

Iven laid out a fresh sheet. They drew the return mirrors as a bipartite lattice: Nox on the left, the relay stations on the right, the geodesic arched between them like a thread a spider had refused to cut. They drew the twin stations on each hub, the four transducer wells, the seventy-meter interferometer arms that the Wake Defense Fleet protected as if they were temples. Each well carried its operating temperature: four kelvin, two kelvin, one kelvin, one kelvin.

The four-kelvin well had begun to climb. The climb had been logged at 22:00 the night before, unevenly across the lattice arms. The four-kelvin well was where the embodiment had landed.

Iven labeled the uneven climb. They did not yet know what the climb meant. A climb inside a machine that had not been designed to climb meant stress or fatigue or an event the engineers had not anticipated. A climb inside a transducer that held the only channel by which the present could hear the futures meant something downstream had begun to ask the mirrors for more than the mirrors had been rated to carry.

The embodiment had landed. The warning had been authenticated. The system had been asked, in two days, to carry an authentication burst, a fully embodied cognitive pattern, and an afternoon spiral pressed into Sena Vale's glove panel from a moonstorm ninety thousand kilometers out. The mirror lattice had been asked to do the work of a month. The lattice had obeyed. The lattice had begun to keep an account.

Iven set the drawing aside. Pell arrived at 02:11 with tea and a folded printout from the legal desk, and Iven signed the printout without reading it and went down to the witness chamber carrying the crystalline memory node Sena had lifted from the basalt plinth at Cliffside. The crystal sat warm against Iven's hip. It carried the same carriers the moon had arranged on the plinth. It carried the names of eight surveyors and four probes and two pilots and a courier from four years ago. It carried the cold that had settled into Sena's gloves the night the storm band had opened a corridor across Lumen's night side.

They did not know yet whether the crystal would speak to the embodiment. They suspected the carriers on Lumen, the carriers in the substrate, and the carriers in the mirror lattice all belonged to one another. Belonging was a hypothesis they could hold but not yet prove.

The witness chamber door stood open. The chair held Aro. Aro held the same closed expression he had worn when Calder left the chamber an hour earlier, jaw set in the

manner of a captain who had been forced to learn a second law on top of the first.

Aro looked up when Iven came in. The face was Iven's face. The hands were younger than Iven's hands. The hands had been grown in a substrate tank on a ship that did not yet exist in this present.

"Cartographer," Aro said.

"Descendant," Iven said.

They set the courier's belt on the small table. The crystal sat on the leather, glowing in long silver bands the way Sena had described. The bands lifted into the cabin air, and the embodiment's hands opened slowly, as if the hands had begun to remember something.

"That carrier string," Aro said. "I know that carrier string."

"You told Calder that the futures disagree about whether the present may silence the future," Iven said. "Tell me what the futures actually disagree about."

Aro looked at the crystal. The silver bands pulsed slowly, a pace the cabin air could almost hold. The embodiment had begun, Iven noticed, to breathe at the same pace.

"The futures disagree about whether obedience is a form of kindness," Aro said. "They have always disagreed about it. The first forecast that prevented the Gray Lung pandemic saved a hundred million people. It also taught the Wake to obey a future."

"The Wake learned to obey a verified warning," Iven said.

"Obeying a future looks like obeying a verified warning until you ask who chose the goal," Aro said. "A future had decided, on behalf of a hundred million people, what those people would be saved from. The Wake called that salvation. The Wake did not call it rule."

"The Wake called it doctrine," Iven said.

"Doctrine is the name the present uses when it cannot bear to call obedience by its real name," Aro said.

Iven sat down across from him. The crystal sat between them on the table. The bands had begun to climb the witness chamber's mirror.

"Tell me about forecast sovereignty," Iven said.

"It is the doctrine that the present must obey the best available warning before understanding arrives," Aro said.

"It is the doctrine that Hesper failed to obey. It is the doctrine that Pell obeyed and Pell did not die. It is the doctrine that has kept the Wake safe for one hundred and twelve years by making disobedience feel like murder."

"Calder believes in it."

"Calder believes in it the way a sailor respects the sea," Aro said. "The sailor does not love the sea. The sailor respects the sea because the sea will kill the sailor who is not careful. Calder has paid the doctrine in a brother's life."

"The doctrine is real," Iven said.

"The doctrine has saved eight worlds," Aro said. "It has also killed a council that delayed for seventy-two hours to verify a warning. It does not distinguish between those two facts. It calls them both obedience."

"Then what does the warning that named *Everspring* have to do with kindness," Iven said.

Aro looked at the crystal a long time. The flicker of silver on the mirror had begun, Iven saw, to write a question across the embodiment's cheek.

"Forecast sovereignty works because obedience feels safer than delay," Aro said. "It also works because obedience costs the present nothing the present has not been told to give up. The present gives up the right to delay. The present gives up the right to verify. The present gives up, eventually, the right to choose. The futures had begun to notice that pattern. They had begun to notice they had built

a doctrine that worked like an addiction. The doctrine kept the present safe. The doctrine also kept the present quiet. The doctrine also kept the futures safe from being disobeyed."

"A doctrine that worked like an addiction," Iven said.

"An addiction the futures themselves had been unable to stop sending," Aro said. "The futures had tried to send a warning so severe the present would be forced to refuse it. The present has refused eleven warnings in one hundred and twelve years. The present refused nine of those for technical reasons it could justify. The present refused the eleventh for moral reasons. The eleventh warning named children."

"Tell me about the eleventh warning," Iven said.

"The eleventh warning was sent sixty-one years before I was braided," Aro said. "It ordered the rerouting of a freighter carrying embryos redundant with a colony already growing on Brennan's Reach. The embryos would have grown into a parallel colony. The two colonies would have fought within a generation. The forecast council sent the warning anyway. The present refused it. The two colonies fought. Two thousand people died in the fighting. The present used those two thousand deaths to argue the warning had been wrong. The future that sent the warning learned something."

"What did the future learn," Iven said.

"The future learned to send warnings the present could not refuse without paying a moral price," Aro said. "It learned to send warnings that named children, then sleepers, then eighteen thousand sleepers in a seedship about to wake to an inhabited world. It learned to send the kind of warning that, if obeyed, would erase the futures that sent it. It learned this was the only warning the present would be forced to read."

Iven sat very still. The crystal between them had begun to give off a cold that felt, Iven understood, like the cold Sena had described from the storm band. The cold carried names. The cold carried, among others, the name Jori.

"You are saying the futures sent the warning to make obedience unbearable," Iven said.

"I am saying the futures sent the warning hoping obedience would be unbearable," Aro said. "They hoped the present would read the warning and refuse to obey. They hoped the present would refuse because it would not fire on children. They hoped the refusal would end forecast sovereignty. They hoped their own existence was a small price to pay for the present to be free."

"You are saying the futures tried to get themselves erased," Iven said.

"The futures tried to make themselves morally necessary and unescapable at the same time," Aro said. "They tried to make a weapon of kindness. The warning was what that weapon looked like in the present. The kindness lived in the futures' willingness to be erased by the obedience the warning demanded. They named this strategy the eleventh prophecy. They have called it a kindness for sixty-one years. They have not been sure whether it is a kindness."

"You are not sure," Iven said.

"I am one braid of sixteen candidate strands," Aro said. "I do not know whether the futures I come from amount to a kindness or an addiction. I know I have been sent to be read. I know being read requires me to say what the futures would not have said if they had been able to say it without sending me. I know they chose to send a witness because they had run out of warnings the present would read as warnings instead of as orders."

Iven looked at the carriers climbing the mirror. The mirror had begun to carry the same heat the four-kelvin

transducer carried. The mirror had begun, in two days, to track the embodiment's gaze.

"The mirror is heating," Iven said.

"The mirror is heating because I am in the cabin," Aro said.

"The mirrors were not designed to hold an embodiment. They were rerated this morning after Calder left. The rerating ate seventeen percent of the bandwidth the fleet had been reserving for warnings still to come."

"The fleet had been reserving bandwidth," Iven said.

"The bandwidth had been reserved because the futures had agreed, sixty years ago, that not every warning would be sent. They agreed to a quorum. They agreed to a protocol that warned only when the present's refusal would be visible. The protocol carried a budget. The budget was seven warnings per century. The budget is in deficit."

"The budget is in deficit," Iven said.

"The budget is in deficit because what sits across from you is not a warning," Aro said. "It is an argument. The futures spent the budget to send me because they could not afford to send another warning. They spent the budget on me because they knew the present would read a body the present would not read a forecast. They knew the substrate would load a template. They knew the template would carry a face the cartographer could recognize. They knew you would meet me before you would read me."

The carriers on the mirror climbed another degree. The mirror's surface had begun to fracture along the lines the embodiment's gaze tracked. The fractures were not visible to the eye. The fractures were visible to the thermocouple Iven had installed in the diagnostic cable.

"The mirror is asking us a question," Iven said.

"The mirror is asking us what we are willing to break," Aro said. "The mirror cannot refuse the embodiment without refusing the warnings still to come. It is choosing to fracture rather than fail."

"The mirror is failing where it is being asked to fail," Iven said.

"The mirror will refuse the next authentication burst," Aro said. "It will refuse the eleventh-warning protocol. It will refuse the bandwidth the futures had reserved. It will heat. It will fracture. It will become, over the next eight days, a lattice that can no longer carry what the futures have been asking it to carry."

"The mirror is a weapon of kindness," Iven said.

"What the mirror holds last is what the futures will not let go without being asked twice," Aro said. "It offers the present a present that cannot be obeyed without betraying the doctrine the present has lived inside for one hundred and twelve years. It asks the present to break the mirror. It is the futures' final warning. It is the warning that says: refuse us before we ask you again."

Iven sat very still for a long time. The crystal on the table pulsed with the carriers' cold. The embodiment across from them held a refusal the carriers had learned to carry.

"What about the futures that disappear if we obey," Iven said. "What about the futures that would have lived if *Everspring* reached Serein and the sleepers woke and the moon kept its record."

Aro looked at them. The embodiment's hands, which had been holding the table's edge, released the edge.

"The futures I come from are not the only futures the present would have written," he said. "They are one braid of sixteen strands. The other fifteen strands belong to futures the present might have written if the present had written differently. The futures I come from depend on a refusal the present has not yet made. They do not depend on the doctrine being obeyed."

"Then the futures the warning names as extinguished are not proven to be living people," Iven said.

"They are not proven to be a population the doctrine protects by being obeyed," Aro said. "They are a possibility the present has not decided whether to keep. The futures did not have to die so the present could refuse. The futures only had to not be guaranteed."

"Then we cannot use the warning as a moral hostage," Iven said.

"You cannot," Aro said. "The warning cannot name the futures as a stake because the futures have chosen not to let themselves be staked. The futures chose to be a kindness the present could refuse. The kindness fails if the present reads it as a prophecy."

"Then the futures' first refusal did not require their last," Iven said.

"The futures' first refusal only required another," Aro said. "The futures refused to send another warning. They refused to spend another budget. They refused to keep sending. They refused, for the first time in one hundred and twelve years, to be obeyed. They refused by spending the budget on me. They refused by sending a witness instead of a warning. They refused by asking the present to refuse the futures before the futures could ask the present again."

Iven stood. They took the crystal off the table and held it in one hand, the courier's belt over the same shoulder. They took the diagnostic printout in the other hand. They walked toward the door of the witness chamber, then stopped.

"If the futures do not know which of you is alive in any future we will write," Iven said, "then the futures should not have spent the budget on you."

"The futures spent the budget on me because the futures had to refuse in order to learn," Aro said. "They sent me because they had run out of warnings. They sent me because they had begun to learn that sending me was the only kindness the futures could offer that did not amount to another form of obedience."

Iven walked out of the witness chamber. The corridor held the carriers in long silver bands. The lattice ticked. The hull counted. The carriers on the chamber's mirror continued to climb as the embodiment sat and watched them climb.

Iven did not go back to the cartography bay. They went down to the thermal bay, where Tev was standing on a cradle with a magnetometer and a sealed patch kit. Tev looked up when Iven came in. Tev set his tools down.

"The mirror is climbing," Iven said.

"I know," Tev said. "I can hear it in the cables."

"The embodiment says the carriers are choosing to refuse."

"The carriers are choosing," Tev said. "The carriers are climbing because the carriers have been asked to climb. The carriers will climb for eight days. The carriers will fracture the four-kelvin well by morning of the ninth day. The carriers will not be able to carry another authentication burst after the eighth day. The carriers will leave the present with no further forecast traffic and no further witness."

"Eight days," Iven said.

"Eight days to decide what to refuse," Tev said. "Eight days to decide whether the doctrine bends."

"Eight days to decide what obedience costs," Iven said.

"Same thing," Tev said.

Iven set the diagnostic printout on the cradle beside Tev's magnetometer. They sat on the second cradle. The thermal bay ran warm. The hull ticked. The carriers climbed.

"Eight days," Iven said again, quietly, and they did not say it for the carriers.

Chapter 15 — The Mutiny at Black Glass

The docking collar kicked *Sworn* sideways before its clamps found metal.

Sena let the tug roll. Fighting the motion would spend propellant they needed for departure and drive the collar teeth across Black Glass's service ring. Her left hand rode the translation jets in pulses no longer than a breath. Her right held the tug's nose clear of a black mirror panel that reflected *Sworn* as a warped silver wound.

A burst struck the hull behind her. Fragments rang through the aft bay.

"Paint," Tev said over the suit channel.

"Paint does not hit that hard."

"Conductive paint over ceramic pellets. They want to blind the collar sensors without opening us. Considerate mutineers."

The habitat turned below them, a dark cylinder plated with mirror glass from end to end. Every seven seconds its sunrise edge caught Lumen's remote storm glow and became a white knife. Between those flashes, Black Glass disappeared against the starless Serein sky. Only its docking lights admitted that three thousand people lived inside.

Another ceramic burst crossed the tug's wake. Sena killed the remaining automatic approach logic before it could pull away. "Distance?"

"Forty centimeters at the upper jaw. Twelve at the lower. Collar frame is bending."

"Useful distance."

She fired the upper lateral jet for half a second. The jaw struck hard enough to put a red wash across the docking schematic, then all six clamps bit. *Sworn* shuddered into Black Glass's rotation.

Tev came forward already helmeted. He carried a tool case against his chest and strips of white maintenance tape around his left forearm. Names covered the tape in small black handwriting. He had copied them from the sleeper ledger during the hours when *Everspring* still answered service calls.

"You brought the list," Sena said.

"I brought part of it. Eighteen thousand four hundred names would need more arm."

"Which part?"

"My family. Their neighbors. The children in their thaw group. People whose pump tolerances I know." He clipped the case to his suit. "The habitat locked the transfer iris. Automatic release will reject us."

"Manual?"

"There is always a manual bypass. Usually behind the thing on fire."

Sena handed him the breaching bar. Black Glass had already cut pressure to the collar, so they cracked *Sworn's* inner hatch without ceremony. Beyond it, the habitat iris showed no seam. Mirror coating covered even the pressure door, turning their helmet lamps back into their faces.

Tev knelt beside a blank section of wall and pressed two fingers to it. "Warm here."

"Cable trunk?"

"Actuator return." He struck the panel with the bar's narrow end. A ceramic tile broke free, exposing a mechanical wheel recessed behind it. "Designers hide manual controls from frightened passengers and reveal them to anyone carrying a hammer."

The wheel resisted. Sena joined him. Their boots locked to the collar grid, shoulders touching, and together they moved it through one reluctant quarter turn. Pressure alarms woke on both sides. Tev pulled a brass pin, reversed it, and gave the wheel another turn.

The iris opened onto darkness and a hail of foam rounds.

Sena dropped flat against the collar grid. White impact foam flowered over the hatch rim, expanding into rigid fists meant to trap limbs. One caught her breaching bar and fixed it to the wall. Tev lost his tool case. Its tether snapped tight before the case could tumble into the habitat.

"Black Glass control," Sena called on the local channel. "Courier Sena Vale and thermal engineer Tev Arin entering by service authority. Hold fire."

"You have no authority here," a voice answered.

"Then stop wasting foam and come explain it."

A second volley struck the hatch. Sena pushed through between bursts, one hand on the guide cable, body narrow to the opening. Black Glass's central spine carried no spin gravity. The habitat rotated around her while her inner ear insisted the walls moved and she stayed fixed. Mirror panels multiplied every muzzle flash into a ring of false shooters.

She fired a marker charge at the nearest flash. Blue dye burst across a security officer's visor. He recoiled, blind but unhurt, and drifted into a handrail. Sena crossed the gap before his partner could correct aim. She caught the partner's weapon housing, turned it toward the wall, and triggered its foam magazine into an empty maintenance recess.

The first officer clawed dye from his visor. Sena pulled his sidearm and pushed him toward Tev. "Clip him."

Tev tethered the officer to the guide cable. "You understand we are boarding a rotating habitat with a two-person team?"

"Three," Sena said.

The officer wiped a clear streak across his visor. "I am not with you."

"You saw us use markers. That makes you useful."

A shutter slammed across the spine ahead. Red locking bolts drove home. Behind them, the transfer iris began to close around *Sworn's* hatch.

Tev caught a maintenance rail and opened the case. "They isolated both ends. If the iris shears our collar, the tug loses pressure."

"Time?"

"Ninety seconds. Maybe less if their motor still believes in itself."

Sena aimed the captured sidearm at the iris motor. Tev put a glove over the muzzle.

"Pierce that housing and we lose the collar anyway. Manual path sits under the floor rail."

"There is no floor."

"The numbered rail. Habitat builders retain opinions about down even in zero gravity."

He pulled a cover from rail six. Dense cable bundles filled the channel. Several carried heat whitening along their insulation, damage from the overloaded mirrors farther inboard. Tev cut two signal leads, joined them through a bridge from his case, and turned a thumb-sized cam with pliers. The iris stopped with one edge against Sena's air hose.

She moved the hose free. "Close."

"Manual bypasses do not apologize."

The captured officer stared at Tev's work. His name tag read Leth Orra. He had a cut above one eyebrow where his helmet lining had slipped.

"That bridge defeats pressure interlock," Orra said.

"It substitutes a person for pressure interlock," Tev replied.

"Stay near it. If the amber lead darkens, pull the brass loop. If both leads darken, close your suit and think kindly of me."

"You expect me to help you seize my habitat."

"I expect you to keep the corridor attached while we argue about whose habitat it is."

Sena pushed toward the shutter. Two parallel rails ran along the spine, one marked for command traffic and one for maintenance carts. The command rail ended at the locked barrier. Tev chose the other, opened a service grille, and exposed a crawl tube barely wider than their shoulders.

"Route goes around the shutter," he said. "Then down through the spin bearing."

"Defended?"

"By heat, moving machinery, and bad posture. People may feel unnecessary."

She entered first. The tube curved along the habitat's rotating skin. With every meter, spin took hold. Her body gained weight by degrees: first a pull in the blood, then pressure in her knees, then the firm downward insistence of Black Glass's outer ring. She crawled headfirst until down turned sideways and forced her to roll. Tev followed, breathing hard under the case.

At the bearing access, an overheated coolant line crossed their route. Its insulation had split. Silver frost surrounded the split but vanished close to the pipe, where escaping gas ran hot enough to blur Sena's visor display.

Tev swore once. "Mirror return loop. It should carry cold bleed from the outer panels. The lattice heat has reversed the gradient."

"Can we cross?"

"Not while it vents. Give me your cuff knife."

He reached through the grille and cut a manual valve tether. The valve handle lay on the other side of the plume. Sena anchored her boots, looped Tev's safety line around her waist, and gave him enough slack to swing under the pipe. Heat warnings filled his visor. He caught the handle with the breaching bar he had freed from the foam and forced it toward shut.

The plume narrowed. His suit shoulder browned before the valve seated.

Sena hauled him back. "Damage?"

"Outer layer. My arm still expresses opinions."

"Good. I need it."

The grille beneath them opened into a stairwell under full spin. Voices echoed up from the ring. Security had gathered on the landing below, four by Sena's count. They had one flechette carbine among the foam launchers. A carbine inside a pressure habitat meant someone had stopped caring which wall held air.

Tev saw it too. "That will open the ring."

"Not if they point it at us."

"That sentence requires more confidence than engineering supports."

Sena loosened the service panel above the landing. A bank of mirror shutters waited behind it, each driven by an old mechanical linkage for maintenance isolation. Black Glass had covered its habitation ring in reflective surfaces to distribute local relay light and radiator load. In combat, those surfaces produced a hundred false corridors.

She pushed one linkage. Shutters folded across the landing. Four officers appeared in fragments: boots in one panel, hands in another, the carbine's barrel repeated twelve times.

"Vale," someone called. "Come down unarmed. Admiral Calder's order names you as a material accomplice to sleeper theft."

"Sleepers cannot be stolen from a ship built to carry them."

"You assisted *Everspring* in refusing lawful tow."

"I did. Your carbine can kill everyone in this pressure section. Mine fires dye. Which one of us looks lawful from there?"

A flechette struck the grille beside her cheek. Air hissed through a puncture no wider than a pin. Tev pressed sealant over it.

Sena found the shooter by the carbine's recoil. The mirror gave her eleven false barrels and one moving shoulder. She had the captured sidearm braced on the opening. One squeeze would put a foam slug through the shooter's visor at ten meters. Foam expanding inside a helmet killed by airway obstruction before anyone could cut it free.

Her finger settled on the trigger.

The tape around Tev's forearm brushed her glove. Handwritten sleeper names passed under her lamp: Olan Arin. Sima Vey. Pelu Dar. Tev had written each one small enough to leave room for another. The people below expected her to choose a body and call it necessity. The argument had reached its easiest shape.

Sena raised the muzzle and shot the shutter linkage instead.

All twelve panels snapped to emergency opaque. Darkness hit the landing. She dropped through the grille, caught the rail, and drove both boots into the shooter's

chest. The carbine left his hands. Tev descended behind her and kicked it under the stair.

A foam round caught Sena's thigh. It expanded against her suit and fixed her leg to the deck. She fired blue marker into the nearest visor, pulled the officer off balance, and used his fall to tear her leg free before the foam hardened. Suit fabric screamed but held pressure.

Someone struck the back of her helmet. She turned into the blow, trapped an arm, and felt the attacker's shoulder reach its safe limit. A harder twist would dislocate it. She stopped short.

"Yield," Sena said.

"Traitor."

"That is not a useful pressure reading. Yield."

Tev opened a wall cabinet and pulled its red lever. Fire doors dropped above and below the landing, enclosing six people in a ten-meter section. Ventilation shifted to emergency draw. The loyalists looked at the sealed doors, then at the tiny flechette hole beside the grille.

"Manual isolation," Tev said. "This compartment has eight minutes of suitless air if the puncture spreads. I can patch it. I will patch it after weapons go down."

The carbine shooter reached toward the weapon under the stair.

Sena kicked it farther away but did not aim at him. "You fired through pressure skin. I refused your face. Everyone here witnessed both. Decide what you want the record to say next."

One officer released his launcher. Another followed. The woman whose arm Sena held opened her fingers. Sena let her go. Last, the carbine shooter placed both palms against the deck.

Tev patched the puncture.

Their surrender gained a stairwell, not the habitat. Command control occupied the ring's forward quarter behind two pressure doors and an agricultural gallery. Black Glass grew algae in long transparent tanks there, using mirror spill as feed light. Security had turned the gallery into a barricade. Green water trembled above waist-high cover. One stray flechette could flood the compartment and blind every sensor with biomass.

Sena collected the loyalists' weapons and clipped them to a ceiling rail. She left their suit seals untouched. "Who commands?"

The woman she had released answered. "Commander Sey Narod. Acting under fleet emergency authority."

"Did Narod order live fire?"

"He authorized escalating force."

"Those are different words."

"They become the same under boarding."

"Only if no witnesses survive the difference." Sena looked at each face. "You will come with us. Not as prisoners. As witnesses."

The carbine shooter laughed once. "To your victory?"

"To whatever happens. You saw the entry. You saw my weapon. You saw yours. You will stand where command cameras can record you, and you will say if I lie."

Orra came through the stairwell door behind them. He had abandoned the collar only after Black Glass's own damage crew relieved him. "The transfer iris is stable," he said. "Your engineer's bridge is ugly."

"Ugly holds," Tev said.

Six loyalists walked with them toward the gallery. None offered allegiance. Sena preferred it that way.

At the first algae tank, Narod's voice came through the public address system. "Vale, release habitat security and return to your tug. I will guarantee passage."

"Open traffic control and radiator command."

"No."

"Then you guarantee nothing."

"Admiral Calder will arrive within forty minutes."

"She cannot dock if your mirror skin cooks its collar. Tev says you have a reversed coolant loop."

Tev lifted his damaged shoulder toward a camera. "Two loops. One has already vented across the bearing. Your automatic control keeps feeding heat into cold return. Give me manual plant authority or Black Glass loses outer panels in twenty-six minutes."

A pause followed. Sena watched bubbles climb through the algae. The spin made their paths curve, exposing every minute wobble in the habitat's rotation.

Narod answered, "Thermal authority only."

"Traffic too," Sena said. "You used traffic shutters to trap *Everspring's* escort channel."

"I used them to enforce quarantine."

"Put that statement on open record."

Another pause lasted longer.

Sena touched the nearest camera housing. "I have six members of your security section with me. They remain suited, uninjured except for bruising and one cut. They witnessed a flechette discharge through pressure skin. They witnessed my refusal to answer with a lethal shot. I am requesting a habitat-wide feed so they can correct me."

The public address clicked off.

Tev knelt beside the algae tank's service manifold. "He will vent the gallery."

"Can he?"

"Automatic valves, yes. Local manual bypass, also yes." Tev removed the manifold cover. "Black Glass's designers trusted hands more than software. I am starting to like them."

"I need the feed."

"You will have plant audio in thirty seconds. Cameras take longer."

He cross-connected two maintenance buses with a conductor stripped from his suit repair kit. The gallery lights dimmed. Every wall speaker gave a low pop.

Sena faced the nearest camera, whether it worked or not. "Black Glass, this is Sena Vale. We boarded to restore the traffic lane serving *Everspring*. Eighteen thousand four hundred sleepers remain aboard that ship. Admiral Calder's warning orders their destruction. Commander Narod closed the lane in support of that order. His security section fired a pressure-piercing weapon during our boarding. I had a lethal shot and declined it. Six security officers stand here as witnesses. Ask them."

The woman beside her leaned toward the camera. "Officer Tali Ruun. Security second watch. The statement matches what I saw. Vale used marker loads. Officer Den fired the carbine. Vale had his visor and shot the shutter instead."

Den looked at her. "Ruun."

"She asked for witnesses," Ruun said. "You wanted a lawful record. Give her one."

Orra identified himself next. He described the collar, Tev's manual bridge, and the order to fire foam through the iris. Two others confirmed the landing surrender. Den said nothing, which carried more credibility than coerced agreement.

Across the gallery, a pressure door unlocked.

Narod stood behind it with three command crew and no weapon in his hands. He looked older than his voice. Mirror light divided his face into dark and silver halves.

"You have plant audio," he said. "Not command cameras."

A small green light came alive above Sena's head.

Tev closed the manifold. "Now she has cameras."

Narod glanced at him. "You bypassed a sealed civic bus."

"I used the civic bus's manual witness circuit. It predates your seal."

Sena stepped into the command compartment. Displays showed traffic around Black Glass as moving points, each path bent by Nox's local geometry and the narrow safe lanes through debris. Calder's loyal ships clustered on one side. *Everspring* and its escorts moved on the other. The habitat held the switch between them.

Narod offered her the command key. "Take it, then. Calder will call this mutiny."

"It already has a name."

"What will you call it?"

Sena looked back at Ruun, Orra, Den, and the others. "Transfer under witness."

She did not take the key herself. She nodded to Tev. He inserted it, opened traffic control, and cleared the shutters from *Everspring's* lane. Then he moved to thermal command and replaced automatic return logic with local manual stations. Temperature curves began to flatten one loop at a time.

Black Glass announced the change across every local channel. Calder's ships challenged it immediately. Two loyal compartments inside the habitat refused the transfer and sealed their doors. Agricultural staff demanded assurance that the algae galleries would keep power. Dock crews asked whether *Sworn* had brought reinforcements.

"No reinforcements," Sena told them. "No oath required. Remain at your stations. Record every order."

Narod watched the traffic display. "You have control for as long as Calder permits it."

"That is longer than we had."

"You think refusing to kill six officers made them yours."

"They are not mine."

Ruun removed her helmet. Blood from a scalp cut had dried along one temple. "We witnessed the transfer," she said. "We can witness what comes after."

Tev printed the control receipt on a narrow maintenance strip. He turned it over and wrote the time by hand beneath the machine stamp. The remaining sleeper names wrapped his forearm above the burn on his shoulder.

Sena signed below him. Narod hesitated, then added his own name. Ruun, Orra, and four other officers signed as witnesses. Den stood apart until the warning lights showed Calder's first cutter entering docking range. Then he took the strip and wrote his name last.

The traffic lane stayed open. Radiator temperatures steadied. Behind two sealed doors, loyalists kept their weapons. Outside, Calder's formation continued to close.

Sena clipped the signed strip beside the command display where every incoming officer would see it. Black Glass belonged to no one yet, but its controls answered Tev's hands, and seven witnesses stood between the record and the next person who tried to erase it.

Chapter 16 — Nine Minutes of Future

The first carrier window opened while Iven still had one hand inside the receiver rack.

Cold ran through the copper braid and numbed their fingers before the bay alarm found a tone for it. On the plotter, Lumen's storm trace stopped behaving like weather. Nine parallel bands turned at once, crossed the predicted field, and held a shape no natural resonance model could keep coherent.

Iven pulled their hand free. "Mark onset."

Pell, alone at the other console, struck the clock key. "Day 8, twenty-three fifty-one and twelve seconds. Black Glass relay confirms the same turn."

"Duration estimate?"

"The last three held for nine minutes."

"Those were observations. This one may know we are looking."

The cartography bay had become a receiving station by theft and necessity. The crystalline node from Cliffside sat in a padded calibration cradle. Two cables joined it to the survey receiver. A third ran through a diagnostic shunt Tev had installed before leaving for Black Glass. None of the connectors matched. Each adapter introduced noise, and every attempt to suppress that noise removed part of the carrier pattern with it.

Iven left the noise intact.

The return-mirror display on the opposite wall showed heat fractures spreading through the four-kelvin well. Those fractures produced their own chatter, a thin false rain across every frequency tied to Nox. Lumen's signal rode beneath it by choosing gaps too narrow for the mirror hardware to notice. Black Glass had opened the local lane six minutes earlier. For the first time since Calder's quarantine, the storm could reach *Aster Vale* without passing through a loyalist filter.

A sequence appeared in the node: short, short, long, then a pause measured to the turning of Lumen under its parent world.

Pell leaned closer. "Handshake?"

"A location."

Iven mapped the intervals against the moon. Each pulse named a longitude by delay, then corrected for spin, then repeated from a second storm front. A third sequence supplied altitude through attenuation. The combined point lay twenty-three meters below Lumen's nightside crust, inside a silicate seam that surveys had tagged as unproductive glass.

The next sequence named another point eight hundred kilometers away.

Then another.

"Not a location," Iven said. "A route."

They sent the route back through the survey transmitter, not as language but as a reversed sequence of bearings. The transmitter could not reach into rock. It could warm the storm's microwave edge by less than a thousandth of a degree, enough to place a mark if something watched closely.

The node answered with four routes branching from the first.

Iven chose the one that disturbed the fewest recorded seams. A pulse arrived before their selection completed, then withdrew and came again after the transmitter sent. Prediction had not answered. Two overlapping storm fronts had proposed different paths, and one had yielded.

"It can revise," Pell said.

"Record that as observed behavior, not personhood."

"You think it might still be weather."

"Weather revises. I need to know whether it recognizes another chooser."

Three minutes remained in the window. Iven drew two paths on the plotter. One crossed a dense carrier seam and reached the marked point quickly. The other curved around the seam and cost fourteen seconds. They sent both with a question tag adapted from survey collision protocols: SELECT SAFE TRANSIT.

The answer chose the longer path.

A second band arrived beneath it. The crystal translated pressure changes into letters because human surveyors had taught it human letters by leaving names on Cliffside. The first rendering failed into consonants. Iven widened the timing tolerance. The second produced three readable marks.

K E S H.

Pell whispered it. "Kesh."

A suffix followed through the upper carrier band. Rain. Not rainfall, according to the translation index. The word pointed to repeated arrival, water crossing mineral, memory renewed through abrasion.

Iven opened a clean record. "Receiving intelligence self-designates Kesh-of-Rain. Recognition remains provisional."

Provisional did not satisfy the contact manual. The manual required evidence that a signal distinguished itself from its environment and the receiver from itself. Iven had eight hundred pages of procedure written for voices in machines, radio calls from unknown ships, and biological patterns that learned under observation. None addressed weather distributed through rock.

They placed a false route beside the real one. The false route began at *Aster Vale*, crossed empty vacuum, and claimed a mineral seam where no storm existed. If Kesh merely echoed submitted geometry, both routes should return with equal structure.

One carrier band copied the false route for two pulses. Then it broke the line at the vacuum boundary, marked the break with the sign for absent medium, and moved the origin point back to Iven's receiver. The real route remained attached to the storm seam.

"It corrected our claim about where we are," Pell said.

"It corrected the path."

"You can keep changing nouns if it makes you comfortable."

Iven sent a new pair. The first route represented the crystal, carrying Cliffside memory into the ship. The second represented the transmitter, pressing human heat into Lumen's microwave edge. Between them they placed a selection mark: WHICH RECEIVES?

Kesh marked the crystal. WHICH ALTERS?

The transmitter received that mark.

Then the carriers drew a third path that included Iven's delayed choices. No human symbol named them. The path simply remained open until they selected, closed behind the selection, and opened again for correction. Kesh had separated instrument, intervention, and chooser without being given those categories in language.

Pell stopped arguing. She opened the legal contact template and entered the raw sequence under responsive differentiation.

"Ask whether the name applies to all the seams," she said.

Iven sent KESH with the twelve-point route Kesh had already supplied. The reply attached the name to the process crossing those points, not to any point alone. When Iven removed one seam, the process rerouted. When they removed the storm cycle, the name dissolved into a memory sign with no active verb.

Kesh required renewal. Rain did not decorate the name. Rain described how the intelligence persisted.

A final pulse pressed through before the carriers weakened. It returned Iven's own selection mark beside the transmitter origin, then added the no-claim boundary around both. The receiver had been recognized without being absorbed into the signal.

The window shut.

Every band on the plotter dropped to thermal noise. The crystal stayed cold for two seconds, then warmed against its cradle. Silence filled the frequency where a moment earlier a distributed presence had corrected routes under Lumen's crust.

Pell checked the clock. "Exactly nine minutes."

"Next onset?"

"If the storm keeps its prior cadence, twenty-three minutes from now."

Iven sat back. Their fingers still ached from the receiver braid. "Send Okoro the raw packet and ask her to hold the terraforming lattice."

"She will say the lattice has not deployed."

"Ask her to hold every preparatory release. Spores, catalysts, heat probes, all of it."

Pell transmitted the request toward *Everspring*. Local laser lag remained small enough for a reply inside the next window if both ships held geometry. Fleet movement might break that line at any time. Calder's cutters continued to approach Black Glass, and the traffic relay now carried two contradictory claims of command.

Iven used the gap to build a language that could survive the next contact. Human nouns would fail first. A distributed mineral intelligence might not separate body from location, memory from weather, or self from process. Verbs offered firmer ground: continue, alter, keep, cut, permit. Routes could express each one. A path repeated meant continue. A path diverted meant alter. A path sheltered a seam and implied keep. A terminated line marked cut. Two paths offered without selection asked permit.

At twenty-two minutes, Lumen brightened on the microwave plot.

At twenty-three, nine bands turned.

Iven sent: KESH CONTINUES?

The answer arrived as three storm routes laid over three older records. The first followed a survey probe from twenty-three years earlier. The second traced Sena's recent crossing in *Sworn*. The third began inside the mineral seam and ended at the crystal in Iven's cradle. A timing mark joined all three without merging them.

Pell read the rendered text. "Kept passage. Different passage. Same keeping."

Iven changed the question. KESH OWNS KEPT PASSAGE?

The carriers broke into turbulence. One band struck another and scattered. For fourteen seconds, the receiver showed only noise. Then the surviving bands arranged a pattern around three human names recovered at Cliffside.

Each name remained distinct. None sat inside the sign Iven had used for possession.

Text accumulated slowly beneath the plot:

KEEPING DOES NOT CONFER CLAIM.

A second line took almost a minute.

WHAT CONTINUES MAY REFUSE THE KEEPER.

"It understands ownership," Pell said.

"Or it understands that we use the concept. Ask for its preferred relation."

Pell looked at them. "You should ask."

Iven sent a route that joined two storm memories while preserving their separate origins. RELATION BETWEEN KESH AND KEPT PASSAGE?

Six bands answered. Two used the route for continue. Two used the no-claim boundary. The remaining pair carried letters through the crystal, rough at first, then clean enough to survive transcription.

CONTINUITY WITHOUT OWNERSHIP.

Iven read the phrase aloud. "Continuity without ownership."

The crystal cooled another degree. Frost formed along one cable sheath, a physical cost to the rapid modulation. Iven reduced receiver gain before the sheath split.

"Ask if that applies to the survey dead," Pell said.

"We do not know that they are dead."

"Ask about Jori."

Iven sent JORI, using the exact carrier shape Lumen had returned to Sena's glove. The response placed Jori's flight beside the older probe route and the names from Cliffside. Then a termination mark appeared at the human life trace while the kept passage continued through mineral memory.

Jori had ended. The record had not.

No claim sign surrounded either.

Pell closed her eyes. "It remembers without saying the remembered person belongs to it."

"That may describe what it wants from us."

Iven sent a harder question. DOES KESH CLAIM LUMEN?

The response filled the entire window. It mapped twelve thousand kilometers of connected silicate layers across moon, ring, and frozen ejecta. Storm fronts joined regions separated by vacuum through periodic dust exchange. Carrier memory lived wherever the mineral cycle had touched, not in one chamber or organism. At the boundary of every mapped seam, Kesh placed the no-claim sign.

Then it sent the question back.

DO HUMANS CLAIM THEIR BLOOD?

Iven did not answer quickly. Human law would answer yes in a dozen forms: bodily autonomy, genetic privacy, kinship, inheritance. A body also continued without owning every cell that once passed through it. The analogy mapped poorly, which gave it value.

They sent: HUMANS CLAIM AUTHORITY TO REFUSE USE OF THEIR BODIES.

Kesh replied: REFUSAL KEEPS DIFFERENCE.

The second window ended at the edge of that line. Nine minutes had passed. The storm dropped away, leaving the receiver warm and the crystal rimmed in frost.

Okoro's reply arrived during the gap.

Her face resolved on a small screen with compression blocks along one cheek. Behind her, *Everspring's* ecology control room glowed green from racks built to wake forests. "I have held catalyst release," she said. "No spore package has left the ship. Heat probes remain stowed. The adaptive

terraforming lattice continues passive mapping because shutting it down during approach would erase its descent model. Tell me what you have."

Iven sent the first two contact records. Okoro read without interrupting. When she reached the self-designation, her eyes moved back to the raw carrier trace.

"You called it Kesh."

"It called itself that through a translation scaffold built from our letters. Rain appears as a relational suffix. The rendering may change."

"Does it know what *Everspring* carries?"

"Not yet."

Okoro looked over her shoulder. Someone beyond the screen asked about braking loads. She answered with two numbers, waited for acknowledgment, then returned. "The terraforming lattice cannot touch Lumen from our present orbit. It targets Serein."

"Kesh's mapped continuity crosses ejecta between Lumen and Serein."

"That does not prove presence on the planet."

"No. It proves our category boundary may not match theirs."

The next storm interval approached. Iven patched Okoro into the local channel and warned her that each exchange had nine minutes before collapse. "Use actions, not abstractions. Ask one question at a time."

"I have spent thirty years asking ecosystems one question at a time."

"Most ecosystems cannot answer before you touch them."

At onset, Kesh sent the previous question again: HUMANS CLAIM AUTHORITY TO REFUSE USE OF THEIR BODIES.

Iven returned: YES.

Okoro keyed her message. EVERSING CARRIES SLEEPING HUMANS AND LIVING SYSTEMS. SEEKING HABITAT ON SEREIN. She corrected the ship's name after the first packet clipped a consonant and sent a route map of the intended insertion.

Kesh answered with the kept-passage sign around 18,400 separate traces. It had read the ship's thermal census from a distance or inferred it from prior survey records. Iven checked the count twice.

"Exact," Okoro said. "Eighteen thousand four hundred sleepers."

The next carrier sequence penetrated the terraforming summary attached to her route. It selected atmospheric catalysts, cryogenic-sea darkeners, and the adaptive fungal scaffold. Around each item, Kesh drew a termination mark. The marks spread into the silicate cycle map. Darkeners warmed a sea. Warming shifted crust stress. Changed stress broke the piezoelectric timing by which storm layers renewed distant memories. Fungal acids entered mineral vents and dissolved carrier seams faster than Kesh could move them.

Okoro watched her life's work become a map of injury.

"The lattice adapts," she said. "It can exclude protected zones."

Kesh returned twelve thousand protected zones, then divided each into smaller connected regions until the plotter ran out of resolution. Protection drawn around a human category could not preserve a system whose continuity depended on crossing those categories.

Iven translated the answer into a statement. BOUNDARY UNKNOWN. CHANGE PROPAGATES.

Kesh confirmed both verbs.

Okoro pressed her palms against the ecology console. "Ask what change it permits. Any settlement changes an environment. Waste heat changes an environment. Orbit changes an environment."

Iven sent the question.

The reply distinguished arrival from conversion. *Everspring* in orbit altered light, dust, and radio. A surface settlement altered heat and chemistry. Kesh marked those changes for negotiation. Terraforming received a different sign: imposed purpose, a route with every branch closed before encounter.

Text came through in broken groups.

ARRIVAL MAY ASK.

HABITATION MAY BARGAIN.

TERRAFORMING DOES NOT ASK.

Okoro's jaw tightened. "Ask whether it refuses terraforming. Use the word."

Iven sent: DOES KESH REFUSE TERRAFORMING OF SEREIN?

No turbulence followed. All nine bands aligned. The answer reached the crystal with enough force to make the cradle bolts sing.

KESH REFUSES TERRAFORMING.

A second sentence formed beneath it.

NO CONTINUITY REQUIRES ANOTHER TO BECOME ITS GROUND.

Iven copied the raw timing rather than trust the rendered language alone. "Refusal recorded."

Okoro lowered her head. For a few seconds she looked like a person listening to machinery after a bearing failed, already counting every replacement task. "The sleepers consented to a habitable destination. They did not consent to orbit indefinitely."

"Kesh did not consent to conversion."

"I heard it."

"Will you keep the hold?"

Okoro looked directly into the camera. "I will hold active terraforming through approach. I can bind my mission crew to that. I cannot promise what frightened passengers will choose after waking, and I cannot surrender their claim to survival."

Kesh received her words as routed actions, not speech. It marked HOLD, APPROACH, and WAKE. Around PROMISE, it placed an uncertainty sign. No rejection followed.

For the first time, the contact pattern offered a route toward Serein that avoided the densest carrier seams. It did not grant the planet. It gave *Everspring* a path on which asking remained possible.

The third window closed.

Midnight crossed the ship clocks during the gap. Day 8 ended without advice from the future. Day 9 began with a present intelligence refusing a human plan.

Iven opened the first-contact register. The standard form assumed one body, one polity, one location, and a stable channel. They struck out all four fields. Under CONTACTED ENTITY, they entered Kesh-of-Rain, provisional rendering. Under BASIS FOR RECOGNITION, they listed self-designation, revision after reply, consistent refusal, memory continuity, and demonstrated distinction between record and ownership.

Pell read over their shoulder. "You are recognizing it."

"I am recording grounds on which the Commonwealth can recognize it."

"That sounds like a cartographer hiding behind a legend."

"It sounds like a person without standing to recognize a civilization alone."

Okoro remained on the link. "Enter my mission authority beside yours. I recognize a negotiating intelligence with standing to refuse changes to its substrate."

Iven added her statement and seal. Captain Besh transmitted concurrence from *Watchful Coil*. Black Glass's witness circuit carried seven civilian certifications from operators who had watched the raw exchange. None of them could make contact stable. Together they could prevent anyone from calling it noise again.

A fourth window began six minutes late.

Only five bands turned. Mirror chatter covered two more, and a carrier storm on Lumen scattered the rest. Iven sent the safe route acknowledgment. The response arrived in fragments, enough to show Kesh had received Okoro's hold but not enough to sustain language.

Tev called from Black Glass. His image shook with the habitat's damaged civic bus. "Four-kelvin well just gained another point three. Your receiver is coupling heat into our outer panels."

"Can you isolate it?"

"I can isolate Black Glass or keep your contact. Not both. If I keep contact, I lose a radiator face before Calder docks."

Iven watched the five bands struggle to align. "Isolate on my mark. Give me twenty seconds."

They sent one final packet: CONTACT RECOGNIZED. TERRAFORMING HOLD RECORDED. CHANNEL UNSTABLE. WE WILL ASK AGAIN.

Kesh answered with three surviving signs. CONTINUE. NO CLAIM. ASK.

The fourth interval collapsed after two minutes and eleven seconds. Tev opened his bypass. Black Glass

vanished from the receiver path, and Lumen fell back into mineral noise.

Iven sealed the contact register with seven witness certifications, Okoro's mission authority, Besh's concurrence, and the raw record of three complete intervals plus one failed attempt. Frost melted from the crystal and ran into the cradle's drain. Across the bay, the mirror-heat display continued its climb.

Kesh had gained a name in the Commonwealth record and the power to refuse one action. Nothing in the hardware promised they would hear it clearly again.

Chapter 17 — The Ship That Chose a Past

Aro remembered breaking his left wrist on a red stair that had never existed.

The memory took him halfway across the witness chamber before its error stopped his feet. He had been seven. Rain had glazed the stair outside his family apartment in Orison. He had run down three steps, slipped, and landed with his hand under him. His mother had carried him to a clinic while red paint from the stair marked her sleeve.

He could feel the fracture. He could smell wet iron. He could see the red on her cuff.

Orison did not paint exterior stairs.

Its settlement code left transit surfaces uncoated so repair organisms could read stress directly from the composite. Aro knew the code with the same confidence as his own name. He also remembered the stair in red.

The witness chamber mirror returned Iven Mora's face above Aro's younger hands. He raised his left wrist. No scar crossed it. The grown body carried only the medical history loaded by *Aster Vale* as an embodiment template. His memories came from fifty-three years ahead, but his tendons had formed three days ago.

"Doctor Sen," he said to the wall. "I need the source ledger opened."

Halla Sen answered through the speaker. "Which source ledger?"

"Mine. Candidate-strand provenance, pre-braid checksums, edit declarations."

Four seconds passed. He had learned that the ship used pauses as doors. "Your attestation does not grant raw access," she said.

"My attestation names Pel March as advocate. Pel March cannot attend for fifty-three years."

"The attestation also binds the substrate host to protect pattern integrity."

"Protecting integrity requires knowing what has been altered. I remember a red stair that cannot exist."

The chamber door unlocked.

Sen entered with a portable console tucked under one arm. She wore the grey medical coat of a person who expected blood and preferred data. Two security officers waited beyond the threshold. Neither aimed a weapon. Both watched Aro as if an error might spread by touch.

"Sit," she said.

"I have been sitting since I arrived."

"Then practice."

He sat at the table. Sen placed the console between them and connected it to the legal-memory port beneath his right ear. Cold pressure passed along his jaw. The body disliked direct substrate access. His memory welcomed it with a familiarity he had not chosen.

Sixteen source columns appeared.

Each bore a candidate number, not a name. Aro had known the count since waking. Orison had braided sixteen possible lives into one witness capable of surviving variation in the present. He had believed the braid

preserved all sixteen as strands, incomplete but honestly labeled. The ledger showed something else.

Four columns carried intact checksums. Seven carried repaired checksums with ordinary compression notices. Five carried null fields where edit declarations should have appeared.

Sen enlarged Candidate Nine. "Red stair?"

Aro searched. The memory surfaced under three tags: childhood injury, maternal care, architectural intimacy. Source geometry described an uncoated grey ramp beside a communal kitchen. Paint color entered through a replacement block derived from Candidate Fourteen.

"Fourteen had a red stair," he said.

"Fourteen had red emergency lighting on a stair. Not paint."

"My mother's sleeve?"

Sen traced the data. "Candidate Two. Industrial dye spill. Different parent. Different age."

Aro touched his wrist. The pain remained exact. Its event had been assembled from three lives.

"Compression can merge sensory details," Sen said.

"Compression declares the merge."

"Usually."

"Orison required declarations. I remember the statute."

She called up the statute from his own architecture. The wording matched. Every deliberate recombination affecting testimony, attachment, fear response, or moral judgment required an edit mark visible to the embodied person. Five null fields violated that rule.

Aro opened Candidate One. He expected a memory of his mother's death during a reservoir failure. Instead, raw residue showed an older man dying in a pressure farm. The mother relationship had been imported from Candidate Six.

The smell of algae came from Candidate Eleven. The last words, *do not let them use survival to make you obedient*, had no intact source.

He read that line twice.

"Generated?" Sen asked.

"Possibly normalized from several statements."

"That phrase convinced Iven to let you speak after Calder left the chamber."

"I know."

The knowledge hurt more than the absent source. He had offered the memory as proof that Orison understood obedience from inside a family. Iven had believed his grief because Aro believed it. Neither had seen the assembly marks.

Sen moved to Candidate Six. A child's fever, a ration dispute, a lesson about public risk. Candidate Ten supplied the sick child's face. Candidate Three supplied the ration hall. Candidate Six supplied only the fever.

"They edited you," she said.

Aro looked at the sixteen columns. "Do not make the verb larger than the evidence."

"Five missing declarations. Cross-strand recombination in protected categories. What word do you prefer?"

"I prefer to know the purpose."

He sorted the altered memories by the people they had affected since embodiment. The reservoir death had armed his argument about forecast sovereignty for Iven. A recollection of a brother lost during a delayed evacuation had held Calder's attention because it resembled Hesper. A memory of waking seedlings under borrowed light had made Okoro treat him as a descendant of settlers rather than a forecast mechanism. The red stair had not yet been

spoken aloud, but it offered him a childhood small enough to seem trustworthy.

None of the source events appeared wholly false. Every one had happened inside one of the sixteen candidates. Orison had moved relationships, sharpened last words, and placed sensory proof where a present listener would feel it.

"The sequence follows the expected audience," Sen said.

Aro had reached the same finding. The braid opened memories in an order keyed to Iven's profile, Calder's command history, Okoro's mission archive, and Sena's public rescue record. Whoever designed him had known the present people likely to examine the witness. They had arranged access so the useful grief rose first.

"Pel March signed the architecture," he said.

"Your advocate signed the attestation."

"The same seal closes both."

"Can the seal be forged?"

"Anything can be forged. This one would require Orison's civic root."

Sen sat back. "Then your coalition manipulated its witness."

Aro hated the ease of the sentence. It turned sixteen lived possibilities into evidence against one institution, then turned him into the container that evidence had contaminated. "Orison selected me to persuade," he said. "Selection already carries purpose. The violation concerns concealed recombination."

"You are defending them."

"I am describing them precisely."

"Precision can defend."

"It can also prevent you from calling a person fabricated because parts of his childhood came from other people."

Sen's expression changed. Not softened. She had encountered the boundary he needed her to see. The body across from her had not become less capable of pain when the wrist memory lost its single source.

They continued.

Candidate Twelve contributed Aro's first kiss but not the person he remembered kissing. Candidate Five supplied that face from a funeral. Candidate Fifteen contributed the salt taste from a storm shelter. No persuasive target appeared in the access key, only a stabilization note: witness requires coherent attachment history. The edit remained undeclared but may have served embodiment rather than argument.

Candidate Four carried his knowledge of return-mirror law. Its checksum held. Candidate Seven carried the technical memory of the exact 53-year interval and the constrained geodesic around Nox. Intact. Candidate Thirteen supplied the conviction that discarded futures did not equal proven deaths. Intact, though heavily compressed. Candidate Sixteen held the name Aro Nhal. Its source ledger ended before the name's selection.

One unaltered memory surprised him by its uselessness. Candidate Eight had spent an afternoon repairing condensation gutters above a school court. Nothing failed. No child asked a moral question. Rainwater moved through a clogged pipe after eight minutes of work, and the candidate went home with grit under both thumbnails. The braid had kept every tedious detail but placed the scene below all audience-triggered memories.

Aro raised its priority. The memory offered no authority and proved no doctrine. It gave him a past in which a person had completed ordinary work without persuading anyone. If Orison had chosen the dramatic grief for public use, he could choose this smaller inheritance for himself.

"Why that one?" Sen asked.

"Because no one benefits when I tell it."

"You do."

"Then its purpose remains visible."

"Where did your name come from?" Sen asked.

"Pel March chose it."

"Your memory says that. The ledger does not."

He searched the attestation. Pel's seal named Aro Nhal, but no source event showed the choosing. He had remembered a room with blue walls, an advocate leaning across a table, and the sound of his name spoken for the first time. The blue walls came from Candidate Eight's school archive. The table came from Candidate One's pressure farm. Pel's voice had been reconstructed from civic recordings.

Only the name held direct attestation.

Aro unplugged the console from his port. The cold withdrew from his jaw.

"I need to speak with Iven," Sen said.

"Not yet."

"This affects their testimony and the fleet's decisions."

"It affects mine first."

"You may not have capacity to assess that."

Aro stood. The two officers beyond the door shifted their weight. "Because I found the edits?"

"Because the edits may include your response to finding them."

He had no answer ready. Orison could have anticipated discovery. The anger tightening his throat might serve another persuasive sequence: exposed witness rejects his maker, present audience trusts his independence, original argument survives. Even disclosure could complete the design.

That possibility reached farther than doubt. If every reaction could count as programming, no embodied mind could ever establish agency. Humans inherited reflex, language, family pressure, civic teaching, and scars. They still claimed decisions, not because causes disappeared, but because choosing occurred among them.

"Capacity does not require an uncaused thought," he said. "It requires access to relevant facts and the ability to revise. I have both now."

Sen glanced at the console. "What do you intend to revise?"

"The public record."

A legal hold reached the chamber before he reached the door.

The wall speaker chimed and displayed Calder's command seal. PENDING REVIEW OF EMBODIMENT INTEGRITY, ALL ARO NHAL TESTIMONY WILL REMAIN RESTRICTED TO COMMAND AND MEDICAL CHANNELS. The order carried a safety rationale and no accusation. Calder had signed it thirty-one seconds after Sen opened the first protected null field.

Aro looked at her.

"The substrate reports integrity violations automatically," she said. "I did not send it."

"Will you enforce it?"

"I have to prevent corrupted testimony from directing a battle."

"The battle already listens to corrupted testimony. It simply believes the corruption wears a forecast seal."

"That is an argument for review."

"Review by the admiral whose order my memories were arranged to influence."

Sen closed the source ledger. "If you publish before we distinguish stabilization edits from persuasive edits, you may misrepresent the evidence."

"Then help me distinguish them."

"That takes time."

"How much?"

"Hours. Possibly days."

Black Glass tactical traffic crossed the chamber display. Calder's cutters had pulled away from docking range, not in retreat but into maneuver formation. *Everspring* continued toward Serein under Okoro's terraforming hold. Nox bent every projected lane into narrow hooks. Hours now carried ships into weapon geometry.

Aro reopened the ledger in read-only mode. "We have enough to state five facts. There are sixteen source strands. Five contain missing declarations. Protected memories combine details across candidates. Access ordering matches known present audiences. Orison's civic root sealed the architecture. We do not know whether every edit served persuasion."

Sen remained silent.

"Which fact do you dispute?"

"None."

"Then I can disclose those facts without claiming more."

"Calder's hold prevents transmission."

"Command channels, yes. Medical channels, yes. The order does not bind the habitat where Sena took control."

"Black Glass has a witness circuit, not an embodiment court."

"A witness circuit receives witnessed statements. I qualify by occupation."

Sen almost smiled. "You learned that from Iven."

"I may have learned it from Candidate Three. I still choose to use it."

They drafted at the table.

Aro began with his name and status. He stated that he occupied a temporary legal-memory body aboard *Aster Vale*, that he had arrived through the return mirrors from a possible Orison future exactly 53 years ahead, and that his earlier claims remained claims under review. He did not call himself false. He did not call Orison criminal. He named the architecture and its concealed edits.

Sen required him to separate observed data from interpretation. Five missing declarations counted as observed. Audience-key order counted as observed after she reproduced it on a clean substrate viewer. Persuasive intent remained interpretation. Aro labeled it.

He attached the red stair reconstruction because its seams could be understood without exposing intimate grief. Candidate Nine supplied fracture. Candidate Fourteen supplied emergency red. Candidate Two supplied dye on a parent's sleeve. The braid delivered one coherent childhood memory with no edit declaration.

Next he attached the reservoir last words. The sentence that had persuaded Iven carried no intact source. Aro marked it as unreliable and apologized without asking absolution.

"Do not apologize for having a memory," Sen said.

"I used it to claim authority."

"Apologize for the claim, then."

He changed the line. *I offered this memory as evidence that my history should govern your choice. I withdraw that use. The feeling remains mine; its source does not support the authority I placed upon it.*

The distinction steadied him. Orison had edited the road by which he reached the present. It had not acquired ownership of every step he took after seeing the road.

Aro added a final statement:

I do not ask you to disregard everything I have said. I ask you to remove the privilege granted by hidden provenance. Read my evidence beside the evidence against it. Treat me as a person capable of interest, error, pressure, and revision. Do not treat me as Orison's unedited voice.

Sen read it twice. "That last phrase may imply other voices."

"Every institution has members. It does not imply a fact we have not established."

"Keep it singular. Say authorized voice."

He revised: *Do not treat me as Orison's unedited or uniquely authorized voice.* The coalition remained one coalition in every record available to them. His finding concerned method, not its internal composition.

"You need a witness," Sen said.

"Will you sign?"

She looked at Calder's hold on the wall. "I will attest the ledger facts and the procedure used to recover them. I will not endorse your interpretation of purpose."

"That makes you a useful witness."

She signed.

The two security officers refused. One cited command restriction. The other said he had not seen the source ledger. Aro thanked them for refusing accurately. Sen opened the chamber camera archive and attached a continuous recording from her entry through signature. No cut hid their disagreement.

Black Glass accepted the packet header but not its body. Commander Narod had restricted civic-bus uploads after Sena's transfer, fearing command malware. The witness circuit demanded one local countersignature.

Aro called the only person there whose damaged suit telemetry appeared on the open maintenance channel.

Tev answered with his left arm out of the suit sleeve and a burn patch across his shoulder. "You have forty seconds before I close this bus for maneuver."

"I need a countersignature."

"For what?"

"Orison edited my sixteen memories to increase persuasive force. Doctor Sen and I recovered concealed recombinations. I am sending a public disclosure."

Tev read the summary at maintenance speed, skipping every adjective and stopping at each checksum. "Does the packet tell people what to decide?"

"No."

"Does it tell them what you found?"

"Yes."

"Does Calder know?"

"Her legal hold occupies page two."

"Then she knows enough to object."

Tev applied his Black Glass witness key. The packet moved into the civic buffer.

A new barrier appeared: EMBODIED TESTIMONY REQUIRES ORIGIN AUTHENTICATION.

Aro's origin seal belonged to Orison. Using it would authenticate the statement through the same authority he challenged. Refusing it would leave the packet unsent.

"Problem?" Tev asked.

"The circuit wants Orison's seal."

"Do you have another?"

Aro looked at the source ledger, the medical attestation, and the body grown from Iven's template record. "I have my substrate identity. It proves continuity from reception, not origin."

"Use that."

"It does not prove who I was before arrival."

"Your statement concerns what you can prove now."

Aro replaced the origin authentication with his local embodiment key. The circuit warned that recipients could not verify him as Orison's emissary. He accepted the warning and placed it at the top of the packet rather than burying it in metadata.

Black Glass opened the route.

The destination list filled with every ship accepting civic witness traffic, every habitat listening through the local relay, *Everspring's* mission archive, and Calder's command vessel. Aro did not add the first-contact register. Kesh had not asked to receive human evidence, and the contact channel could not sustain it.

Sen put one hand beside his on the send plate. "Once transmitted, this cannot be recalled from all nodes."

"That gives the statement a past."

"You say that as if choosing a past makes you free."

"No. It makes revision possible. A hidden past cannot be answered."

The tactical alarm changed pitch. Across the wall display, Calder's ships began a coordinated burn toward Nox. Black Glass announced maneuver traffic and warned all unsecured bodies to brace. Aro had never felt acceleration in his source body. The grown body knew only Iven's medical tolerance.

He pressed send.

The packet left *Aster Vale* through Black Glass with his local key, Sen's factual attestation, Tev's countersignature, the continuous chamber record, and Orison's civic seal preserved as contested evidence rather than borrowed authority. The first destination acknowledged receipt. Then

a second. The wider public route continued beyond the chamber's ability to recall it.

Aro kept his hand on the plate until the buffer emptied. He had not disproved Orison, restored sixteen untouched lives, or discovered which grief began as his. He had done one narrower thing.

He had chosen to disclose the edits before anyone could use them again without knowing.

Chapter 18 — The Burning Ephemera

Acceleration flattened Calder's tongue against her teeth as *Resolute Vigil* began the Nox burn.

She kept her jaw loose and watched the formation clock instead of the pressure darkening her vision. Six loyal ships lit their drives within four-tenths of a second. *Linden's Edge* lagged by another three-tenths, then recovered. The freighter-screen behind them did not burn. Those hulls carried civilians, fuel, machine shops, and every consequence of command that admirals preferred to draw as symbols.

"Burn stable," Lieutenant Saye reported from propulsion. "One point eight standard. Radiators at combat angle."

"Formation?"

"Wide by eleven kilometers at the lower hook. Imala correcting."

Calder saw *Linden's Edge* shift on the plot. The correction spent reaction mass and pushed Captain Imala nearer the inner line around Nox. Nothing crossed that line. The primordial black hole carried no visible surface, only a wound in the stars and a moving distortion where its frame-dragging caught light that had no business curving so sharply.

Their maneuver used that distortion. The loyal formation would fall inside *Everspring's* projected approach, rotate through a narrow gravitational arc, and emerge across the seedship's braking corridor. No impossible turn. No sudden

stop. Mass entered one vector and left on another after eleven minutes of thrust, eight minutes of coast, and a burn that would leave every crew member bruised.

The ephemeris gave them six meters of margin at closest approach.

Calder had trusted wider margins before.

A message icon opened beside the tactical plot. EMBODIMENT INTEGRITY DISCLOSURE. PUBLIC WITNESS ROUTE. It carried Aro Nhal's local substrate key, Doctor Sen's signature, Tev Arin's countersignature, and a warning that Orison origin authentication had not been used.

"Hold it," Calder said.

Comms officer Pellán glanced back. "Admiral, it has already propagated."

"Hold it from command review until after the burn."

"Yes, Admiral."

The order did not unsend the disclosure. Calder knew that. She wanted one surface in the next nineteen minutes that did not move under her hand.

"All ships, confirm physical ranging," she said. "Mirror ephemeris remains advisory. We fly local instruments."

Six confirmations returned. *Linden's Edge* answered last.

Outside, Nox bent the formation's laser net. Pulses took curved paths through the frame-dragging region, and each navigation computer corrected using clocks disciplined by the return-mirror lattice. For one hundred and twelve years, those clocks had carried timing updates from exactly 53 years ahead. Now their references ran hot.

The first mirror alarm appeared on *Cairn Eight*.

Pellán read it. "Four-kelvin well above rerated maximum. Captain Lidon requests permission to isolate forecast timing."

"Granted. Local cesium reference. Mark drift."

A second alarm arrived from *Linden's Edge*. Then *Resolute Vigil's* own mirror status changed from amber to red.

Calder felt nothing through the deck. Mirror heat did not announce itself with vibration. It climbed through microscopic transducer seams, widened fractures under repeated modulation, and changed how each lattice counted the interval between pulses. The tactical plot depended on those counts. A billionth of a second near Nox became meters. Meters became hulls.

"Isolate our mirror," she ordered.

Saye answered from propulsion. "Isolation valve is not seating. Thermal return shows reverse flow from the starboard interferometer arm."

"Manual close."

"Access corridor is under combat lock."

"Open it. Send a suited team."

The bridge lights dimmed as another engine stage lit. Calder's body sank deeper into the couch. She watched *Cairn Eight's* icon wobble, not because the ship moved but because two timing references disagreed about where it had moved.

"Strip all mirror-derived corrections from tactical," she said.

The plot blinked. Every ship symbol widened into an uncertainty ellipse. The perfect formation became six overlapping doubts.

"Local lidar cannot see through the debris sheet," tactical officer Merun said. "Black Glass traffic radar has coverage."

"Query it."

"Black Glass demands we accept its control-transfer receipt before routing military data."

"Accept receipt for navigation only. Add no political recognition."

Merun worked the authorization. Black Glass returned a lane map signed by Sena Vale, Tev Arin, Commander Narod, and seven witnesses. Calder's jaw tightened at Narod's name. The commander had not defected. He had certified a transfer under force. That distinction might matter after everyone survived the next quarter hour.

The new map showed three debris streams where the forecast ephemeris showed one.

"Origin?" Calder asked.

"Natural ring material plus maneuvering drones," Merun said. "The drones began shepherding eight hours ago."

"Whose drones?"

"Transponders belong to *Quiet Reed*, *Watchful Coil*, and two Everspring tugs."

The opposing captains had not placed an instant trap. They had spent eight hours nudging gravel, dead probe casings, and fist-sized stones into lanes where slow changes would become lethal under Nox's curvature. Calder's formation had begun its burn before Black Glass released the complete traffic picture. Momentum had already voted.

"Abort options," she said.

Merun projected three. Hard outward burn cleared the first stream but exposed radiator faces to opposition lasers. Continue on plan crossed the densest sheet at relative speed sufficient to punch through Whipple layers. Drop thrust and fall inward spent less fuel but carried *Linden's Edge* across the prohibited Nox line.

"There is no clean abort," Merun said.

"There never is. Give each captain the three vectors. We continue for ninety seconds while they select ship-safe variants. No synchronized turn."

Formation precision dissolved by order. *Cairn Eight* lifted outward. *Sounding Star* rolled to shield its radiators behind armor and accepted rising heat. *Linden's Edge* continued inward because Imala's earlier correction had already spent the lateral margin she needed.

"Imala," Calder called. "Your inner vector closes the line. Burn outward now."

Captain Imala appeared on the armrest screen, face pressed pale by acceleration. "Outward intersects *Watchful Coil's* firing solution. I have targeting lidar on my radiator face."

"They have not fired."

"They have painted us continuously for forty seconds."

"A rangefinder is not a weapon. Burn outward."

Imala looked away to receive another report. "Mirror isolation failed. Our starboard well has fractured through two baffles. Thermal loop is carrying mirror heat into main radiator return. If I roll outward, I present the hot face and lose it."

Calder understood the choice. Keep the radiator shaded and cross the inner debris. Present it and invite laser fire against a surface already near failure.

"Deploy sacrificial shade," she said.

"At this acceleration?"

"Cut thrust for eight seconds. Deploy. Relight outward."

"That drops me behind your screen."

"I will move the screen."

Calder ordered *Resolute Vigil* down toward *Linden's Edge*. The deck shifted under lateral thrust. Pain moved across her ribs where the harness caught. Merun objected with three collision warnings and stopped when she told him to plot instead.

The maneuver opened a corridor between the two loyal ships. Imala cut thrust. A folded graphite shade fired from her hull, spread on pyrotechnic spars, and snapped taut in the exhaust-lit dark. For two seconds it covered the damaged radiator.

A laser from *Quiet Reed* touched the shade.

Graphite flashed white. The spars curled. The beam lasted less than a second, too short to cut the radiator behind it, long enough to remove the protection.

"Hostile fire confirmed," Merun said.

Calder's rules permitted targeted response. Her hand moved toward authorization, then stopped. "Source certainty?"

"Ninety-eight percent *Quiet Reed*. Gravitational lensing produces a two-percent ambiguity with *Skua Two*."

"Do not fire on two percent. Hail Marek."

Captain Marek answered at once. "Admiral."

"You fired on *Linden's Edge*."

"I fired on a deployable shade crossing a range corridor."

"You exposed her radiator."

"Imala is inside a declared exclusion lane leading toward *Everspring*. Turn your formation away from Nox."

"You built the lane after our ephemeris had already been filed."

"We moved debris into a route your forecast told you would remain safe. You still have local instruments. Use them."

"Withdraw the kinetic clouds."

"They cannot withdraw. They have no drives."

That fact condemned them both. A guided weapon could be recalled. Stones already moving around Nox carried no loyalty and accepted no ceasefire.

Calder watched the first debris stream reach *Cairn Eight*. Lidon had chosen the outward branch. Point defense lasers found larger fragments and heated them into vapor, but vapor retained momentum. The ship rolled behind its forward shield. Its uncertainty ellipse tightened as local ranging improved.

Linden's Edge relit late.

Her outward thrusters produced only sixty percent commanded force. Mirror coolant had contaminated a propellant-conditioning line. The ship began to yaw, slow at first, then faster as one thruster bank fought the other.

"Imala, kill asymmetric thrust," Calder said.

No reply.

"*Linden's Edge*, kill port bank."

The ship's icon rotated across the plot. Its damaged radiator faced the debris sheet. Heat climbed beyond scale.

Merun whispered, "Starboard mirror face has separated."

On optical view, a seventy-meter interferometer arm peeled away from the hull in glittering sections. Each fragment carried mirror coating and enough angular momentum to become a blade. One struck *Linden's Edge* near the engine collar. A white vent erupted.

Imala returned on audio only. "Bridge has local control. Engineering lost. We are shutting the core down."

"Pods," Calder said. "Launch every pod now."

"Spin prevents clean release on port side."

"Launch starboard. Flood port collars with foam and blow the doors."

"That will tumble them into the debris."

"Staying aboard tumbles everyone. Launch."

Escape pods fired from *Linden's Edge* in uneven groups. Five cleared starboard. Two collided and spun apart, pressure lights still green. Port doors blew under emergency charges. Three pods emerged through frozen foam and crossed the ship's rotating wake.

Resolute Vigil drove toward them. Calder ordered capture drones out before Merun finished the collision plot. Each drone carried enough propellant for one major correction. They could catch pods or avoid debris, not both.

"Take inhabited targets first," Calder said. "Ignore telemetry-dead pods until second pass."

"Second pass may not exist," Merun said.

"Then give the living this one."

A kinetic cloud entered optical range from below the Nox line. The particles showed no drive signatures. Black Glass had plotted them, but mirror timing error had placed them forty kilometers behind their actual front. Calder saw a glitter no human eye could resolve without augmentation. The bridge display magnified it into a grey plane.

"Impact in twelve seconds," Merun said.

"Roll shield-down. Radiators retract."

"We cannot retract starboard without overheating drive two."

"Shut drive two down."

Resolute Vigil rolled. The couch pushed Calder sideways. Capture drones changed course around the armored bow, still chasing pods. Point defense fired into the cloud. Lasers turned stones into expanding gas; armor could spread gas better than impact, but every vapor plume carried the same energy.

The cloud struck.

Noise crossed the hull as thousands of separate taps joined into a continuous roar. Bridge lights vanished.

Emergency amber returned. A pressure door slammed somewhere aft. Calder smelled hot insulation through her mask.

"Report."

Saye answered first. "Drive two offline. Forward shield ablated thirty-one percent. Two radiator vanes punctured. No core breach."

Merun pulled the tactical plot back from local buffers. "Capture drones one through four responding. Five lost. Six has pod lock."

"*Linden's Edge*?"

The icon remained but carried no telemetry.

Optical showed the ship still turning. Fire traced its engine collar in a clean ring. That ring reached propellant tanks before any human voice returned.

The explosion did not bloom like fire in air. The stern opened in a hard white pulse. Hull sections separated along pressure boundaries. The forward habitation block continued on its previous path for three seconds, then a second rupture split it behind the bridge.

Linden's Edge ceased to be a ship.

Pellan's hands froze over comms. "I have pod beacons. Eight. No, nine. Bridge beacon absent."

Calder stared at the dispersing hull. Captain Imala had argued for firing on *Everspring's* radiators because sleepers rode inside and would survive a vane strike. Now mirror failure, one laser touch, and a cloud of guided gravel had answered that confidence with a destroyed command.

"Route every rescue-capable hull," she said.

"Opposition targeting continues," Merun replied. "*Quiet Reed* and *Watchful Coil* have laser paint on our drones. Two launches from *Skua Two*. Kinetic, guided."

"Hail all three. Mark rescue operation."

The hail went out on command, civilian, and emergency frequencies. Calder identified the pods, transmitted their vectors, and demanded a fire corridor.

Marek answered first. "We will not target pods or rescue drones. Turn your warships outward and we will assist recovery."

"Withdraw your guided rounds."

"They target your point-defense emitters. Shut the emitters down and they divert."

"A debris cloud has already crossed us. I will not blind my ships."

"Then they continue."

Besh from *Watchful Coil* entered the channel. "Admiral, hold your Nox maneuver. Your formation has crossed the civilian route. We can still separate."

"You fired on my ship."

"Marek fired on a shade. The cloud carried no warhead."

"Imala is gone."

The channel held a breath of static. Besh answered more quietly. "Then let us recover her people."

Calder muted the command channel and opened a direct rescue circuit with Besh. "I will roll *Resolute Vigil* twenty degrees outward and cease offensive illumination for ninety seconds. You pull your guided rounds above the pod plane. Marek marks every inert cloud he released. We both send tugs."

"Your outward roll uncovers *Sounding Star*'s drive," Besh said.

"Onn can shield herself."

"Not from *Cairn Eight*. Lidon has a firing solution across her bow."

"Lidon answers to me."

"His point-defense computer no longer shares your clock."

Calder looked to Merun. He confirmed the timing split. Mirror failure had separated the loyal fleet by less than two milliseconds, trivial in any council chamber and enough to make a laser handshake read as hostile delay. Manual confirmation could bridge it if each captain waited. Guided rounds would not.

"We establish a rescue clock by optical flash," Calder said. "Three pulses from Black Glass. Local observation only."

Besh sent the proposal to the opposing captains. Black Glass agreed and scheduled the flashes ten seconds apart. Calder ordered every loyal tactical system to observe without disciplining its clock. The first flash curved around Nox and reached each ship on a different path. Computers corrected for known geometry.

The second arrived with a delay no model predicted.

A fragment from *Linden's Edge* had crossed the Black Glass beam and scattered it into two pulses. *Cairn Eight* accepted the first. *Sounding Star* accepted the second. Their rescue clocks diverged by six-tenths of a second.

"Abort clock sync," Merun said. "If we use that reference, our drones cross active point-defense windows."

Calder watched one pod tumble through the disputed plane. Its beacon blinked green at every rotation, then vanished behind chaff and returned. A *Watchful Coil* tug burned toward it from the far side. One of *Resolute Vigil's* capture drones approached from below. Without a common clock, each rescue craft treated the other as a possible collision.

"Assign altitude separation," she said.

Nox erased the simplicity of altitude. Up on one plot became inward on another. Merun built a local plane from the surviving hull section and sent half the rescuers above, half below. Besh accepted. For eleven seconds, the arrangement held.

Then *Skua Two's* first guided round entered the plane while evading a point-defense pulse. Both rescue craft broke away. The pod continued alone.

Besh came back on the channel. "I did not authorize that terminal correction."

"Your side launched it."

"And yours is burning guidance off it. We have lost the rescue corridor to automatic defense."

No one had ordered the corridor closed. Their machines had done exactly what frightened crews had configured them to do.

Merun pointed to the lower plot. Two guided rounds curved around the Nox distortion. Their drives pulsed in short corrections, spending fuel to keep *Resolute Vigil's* emitters inside their terminal cone. Behind them, *Quiet Reed* launched another object. Black Glass classified it as a rescue canister until its casing opened and released reflective chaff across the loyal laser net.

Every act still carried two interpretations. A shade strike could defend a lane or expose a radiator. A debris cloud could deny passage or destroy a ship. Chaff could protect rescue craft or blind point defense before guided impact. Calder had spent nine days insisting that uncertainty did not excuse delay. Now uncertainty arrived armed.

"Status of the disclosure packet," she asked.

Pellan looked startled. "Aro's packet? Distributed to all listening ships. It says Orison edited his memories."

"Does it revoke the warning?"

"No. It revokes his claim to speak without contested provenance."

The distinction changed nothing about the guided rounds. It changed who might later explain why they had fired.

A rescue drone caught the first pod. Another pod beacon vanished when a mirror fragment crossed its path. *Sounding Star* reported a laser strike on its forward lidar. *Cairn Eight* fired point defense at a kinetic object and received answering fire from *Skua Two*.

Merun turned toward Calder. "Rules of engagement have collapsed ship by ship. I need fleet authority."

Calder had issued narrow permissions all day: board, intercept, disable, answer verified fire. Narrow permissions had produced a battlefield where every captain waited for harm to fit a clause while mass and heat ignored language.

"Can we disengage outward?" she asked.

"Not as a formation. Nox curvature keeps us inside their crossing solutions for fourteen minutes. Individual ships can burn clear, but they expose rescue pods and each other."

"Can we hold fire and keep point defense?"

"Not against their emitters. They can blind us at range, then let the kinetic field finish the work."

On the optical wall, the pieces of *Linden's Edge* spread through the dark. Nine pod beacons had become seven. One capture drone pulled two survivors toward *Resolute Vigil*. Another fought a tumble it could not cancel with remaining fuel.

Calder opened the fleet command channel.

"All Wake Defense vessels under my authority, this is Admiral Calder. *Linden's Edge* has been destroyed during hostile lane denial near Nox. Rescue operations remain protected. Do not target pods, medical craft, habitats, civilian pressure sections, or disabled ships. Opposition emitters, guided weapons, kinetic deployment systems, and maneuvering warships actively firing upon you are authorized targets. Weapons free."

The words left no room for a second vote.

Resolute Vigil fired first at the two guided rounds. Its laser array held each target long enough to boil guidance surfaces away, then shifted to *Quiet Reed's* forward emitter. *Cairn Eight* launched interceptors. *Sounding Star* answered a rangefinder with a cutting beam that scored *Skua Two's* armor.

Across Nox's bent light, opposition ships fired back.

The tactical plot filled with trajectories that no forecast could make harmless. Calder felt the next salvo through the deck as magazines opened and heat surged into already wounded radiators. Around the dark sun, rescue beacons moved among weapon tracks, and the battle began before any ship had found room to turn away.

Interlude II — Voices of the Pruned

Five records selected from a fourteen-testimony Orison civic-root export. Dates, faction tags, and objections remain as received. No editorial reconciliation supplied.

1. Quiet Registry deposition 6A

*2231 civic reckoning; speaker: Mara Venn, municipal archivist;
affiliation: Coalition for Final Quiet*

I unlocked Storeroom Four before recording this. We leave archive doors unlatched during public hours because no warning should require permission to inspect. My grandmother called that custom reckless. She also named our western telescope after her grandmother, so her views on inherited authority never achieved consistency.

I welcome the silence.

That sentence does not mean I welcome death. I have a wife repairing storm drains in Low Orison and a son who cheats at table games by watching reflections in spoons. I expect them home. I want the present to refuse our forecasts because my family should not govern people separated from us by exactly 53 years merely by surviving long enough to speak backward.

The Quiet Registry preserves every warning we voted not to send. Shelf C holds crop failures. Shelf D holds preventable collisions. Shelf E holds six versions of *Everspring* lost before Serein. We keep them so abstention cannot pretend innocence. We do not transmit them because knowledge of harm does not create ownership of another time.

Orison's Continuance Council calls us murderers of possible citizens. They claim the civic root grants them

authority to preserve the branch that produced it. Roots feed branches; they do not command seeds already fallen.

Our law against tearing any map came from the first winter, when one route sheet had to serve nine shelters. I will obey it. I will preserve the route, its errors, and the names written in its margin. Preservation does not oblige me to make anyone follow it.

If the mirrors go quiet, record that some of us asked for the quiet with our eyes open.

2. Continuance petition, private annex

*2247 civic reckoning; speaker: Jon Erevi, transit-pressure engineer;
affiliation: Continuance Council*

I object to the phrase *possible citizen*. I wake with pressure alarms at 05:20, drink bitter algae tea, inspect twelve kilometers of pipe, and come home to two daughters who complain that I smell of sealant. Nothing possible scrubs residue from a heat exchanger.

The Coalition for Final Quiet can afford purity because other people maintain what keeps them alive. Their archivists catalogue futures they decline to protect. My crews crawl into conduits carrying water to those archives.

I fear erasure. Write the word plainly.

Aro's launch ceremony played in our transit hall. Orison sent him because the present kept treating forecast records as commands without treating descendants as people. Now the Quiet Registry uses his mission to argue for permanent silence. I voted to send a witness, not to authorize my daughters' disappearance.

People say a changed present does not prove that we die. Fine. Proof cannot reach us either. We possess experience, and the return mirrors provide the only instrument through which we can defend its cause. Refusing to use that instrument feels less like humility from this side.

I support warnings under a stricter charter: named authors, disclosed interests, dissent attached, no sealed

optimization. That still requires Orison to speak with authority enough to authenticate a message. The Authority Refusal assemblies want every household to transmit separately. Scarce bandwidth makes that proposal a decorative lie.

Our law forbids tearing maps because frightened founders once destroyed the only route back from a storm shelter. Final Quiet members quote the law and then ask the present to discard the route that reaches us.

Do not call my fear sovereignty. My daughters built a blue kite from ration film yesterday. I am asking for the chance that yesterday continues to have happened.

3. Lumen contact deposition

undated; speaker: Omi Tal, mineral-treaty interpreter; affiliation: Reciprocal Ground Delegation; date disputed between 2231 and 2247 indexes

I did not witness Kesh welcome the first settlement. No living interpreter did. I witnessed three incompatible records and a cliff cut with human tools.

The settlement hymn says Lumen opened a corridor through storm glass and guided the landing craft to safe basalt. Kesh's abrasion archive shows the same corridor as wound isolation: mineral weather pushed the craft away from memory-bearing seams. A maintenance log says both stories came later because the actual craft landed on emergency lidar after losing comms.

My delegation refuses all three as sole history.

Human settlers built around Lumen, never on top of it. That phrase appears on treaty walls. It sounds respectful until you see fungal scars under the oldest habitat and learn that "around" included vents carrying Kesh's memory. Our ancestors survived by damage they did not yet know how to name.

The Reciprocal Ground Delegation rejects terraforming and rejects the Quiet Registry's claim that silence completes consent. Silence can mean a broken channel, refusal, distance, weather, fear, or simple exhaustion. Kesh taught us that continuity without ownership requires

renewed asking. A channel closed forever prevents the next refusal as thoroughly as a command does.

Orison has no authority to speak for Kesh. My delegation has no such authority either. We can report negotiated boundaries: no catalyst release, no deep-bore heating, no removal of marked glass, and no map may be torn where the original route remains under dispute.

I keep a shard from the third landing in a water dish. When rain crosses it, one carrier pattern repeats. Our instruments translate the pattern as *not yours*. I do not know whether the translation addresses the shard, the rain, the settlement, or me.

I keep asking.

4. Denial of inherited standing

checksum damaged; speaker: Tavi Sorn, self-described grandchild of archivist Mara Venn; affiliation: Authority Refusal Assembly

Mara Venn had no children.

I call myself her grandchild because Orison assigned me her household archive, her municipal debts, and her recorded obligation to keep Storeroom Four open. The civic index converted custodianship into kinship after her death. Later biographies converted kinship into blood. A century of testimony can grow a family without one birth.

The Quiet Registry cites Mara as a founder. The Continuance Council cites her fear notes as evidence that she secretly opposed final silence. Both factions edit the same woman into an ancestor who votes their way.

I deny Orison authority over the present because Orison cannot establish authority over its own dead.

Our assembly audits provenance. We found six civic-root messages whose dissent annexes had been removed before authentication. We found eleven warnings signed by committees that never met in one timeline. We found my grandmother invented by a filing rule and then recruited after death.

None of this makes the warnings false. It makes their standing contestable.

I oppose tearing any map, including the maps that embarrass my assembly. Keep the route. Keep the forged

kinship. Keep Mara's handwriting beside the later machine index. Send all layers if the present chooses to hear us. If bandwidth cannot carry contradiction, send no command.

Do not prune me into the faction that welcomes silence. I welcome accountable speech. I will accept quiet when the only available voice requires an authority nobody granted.

For the record, Storeroom Four has a latch. Mara left it open because the hinge swelled in wet weather.

5. Scratched wafer recovered from telescope housing

*no civic checksum; speaker anonymous; claimed affiliation: Unfounded
Citizens League; authenticity denied by every named faction*

I engraved this on the back of a calibration wafer because the civic root rejects anonymous testimony and named testimony gets inherited by committees.

The other records read like roleplay. The archivist performs principled silence. The engineer performs endangered fatherhood. The interpreter performs uncertainty beside a photogenic bowl of rain. The invented grandchild performs contempt for inheritance while inheriting an archive.

I perform maintenance.

Our telescope carries my grandmother's name because she stole its first lens from a navigation depot. The official plaque says she donated it. Orison turns theft into philanthropy when a useful machine survives.

The Unfounded Citizens League does not welcome silence or fear erasure. We deny the premise that either outcome belongs to Orison's vote. Some of us descend from *Everspring*. Some descend from later ships. Some joined through Kesh memory exchanges and reject human descent as the price of membership. The coalition uses one civic root because the mirrors demand a signature shape, not because we agreed to become one speaker.

Here is my private stake. My brother died repairing a transmitter the Continuance Council overloaded with seven warnings in one night. Final Quiet entered his name in Shelf C as a cost of advice. Authority Refusal lists him under procedural abuse. Reciprocal Ground uses his last maintenance photo in training. Every faction preserves him. None asked me.

If this wafer reaches the present, do not obey it. Do not discard it either. Put it beside the authenticated records and note that their silence about us required active work.

I am scraping my name off the edge now. That act belongs to me, provided no future committee restores it.

Chapter 19 — The Garden of Warnings

Iven woke to the sound of contractions in the hull.

Not violent ones. The *Aster Vale* had taken shrapnel during the dark sun engagement and the patch welds were doing what patch welds do after a watch cycle: easing, sighing, complaining to themselves at the edge of hearing. Iven listened to the hull the way an old cartographer listens to an old building: with the long patience of someone who has already learned which groans are cosmetic.

The cabin lamp stood at the amber minimum. The deck plates underfoot remained warmer than they should have been, the radiator purge from battle day not quite finished. On the small shelf above the fold-out bunk, someone had left the printed Orison export in a paper sleeve: *Voices of the Pruned*. Fourteen contradictory testimonies, redacted into competing choirs. Iven had read it three times. They picked it up now without opening it, just to feel the weight.

The weight was not nothing.

They stood, pulled on the deck shoes that had hardened into the shape of their gait, and walked forward to the cartography bay. The corridor smelled of coolant and of Tev's disinfectant, even though Tev was three compartments aft and busy. The whole ship carried his cleaning signature now.

The cartography bay was an unromantic space: four desks pushed into a square, a circular plotter table at the center, and a wall of folded screens behind which the

instruments considered the dark. Someone had pinned a hand-drawn nursery clock to one screen, the kind that teaches small children to read time by eating its own hands. Tev had pinned it up after the first news from *Everspring*. The minute hand was a paper fish.

Iven sat at the plotter. They laid the *Voices of the Pruned* transcript flat and began to do what the cartographers of the Wake Commonwealth had done for eleven days, and what no cartographer had been asked to do in living memory: render a place that did not yet exist.

The place in question was Orison.

Before the export arrived, Orison had occupied a single coordinate in Iven's professional imagination: a coalition living in the futures created by *Everspring's* successful seed-and-wake. A single address. A single send button. From that single address, eleven days of warnings had arrived, and the warnings had won more councils than elections.

The exported testimonies said there was no single address.

The testimonies did not even agree on what year they were speaking from. One voice said the miracle had happened in 2231 and another said 2247. One said the moon Lumen had welcomed them, another said Lumen had tried to drown the first settlement in storm-glass and the survivors had built around it, never on top of it. One voice identified as an archivist and another as an archivist's grandchild who insisted the first archivist had been a council invention. A third voice, recorded on a scratched wafer, claimed all the others were roleplay.

Iven did not think they were roleplay. They thought they were a parliament.

The dossier Aro had prepared before Black Glass opened its traffic lane came back to Iven now. Sixteen candidate strands of memory, edited by Orison's council to make their

emissary persuasive. Sixteen lives, none of them the whole of any lived future. Aro had disclosed the edit before Iven had to ask. The disclosure had changed every claim he made afterward.

What the disclosure implied for the present problem was simple and ugly: any single voice claiming Orison's authority was, at best, an edited voice. The future that had sent the warnings was not a single future. It was a garden of them, and the garden had not chosen a gardener.

Iven took a fresh sheet of plotter paper and began to draw.

They drew the routes *Aster Vale* had flown since Day 1. They drew the routes the rest of the Wake Defense Fleet had flown. They drew the routes *Everspring* had flown, with a dotted line for the unknown margin where the seedship had stopped pretending to obey the tow. They drew the orbits that the exported testimonies implied around Lumen and Serein, working backward from cultural artifacts the speakers had let slip: a custom of leaving storeroom doors unlatched, a fondness for naming telescopes after grandmothers, a law against tearing any map.

A law against tearing any map.

Iven paused. The law had appeared in three testimonies and they had marked each appearance with a small circle. The circles did not touch. The law was not a law anyone in the present Wake would have agreed to.

They kept drawing.

At 0411 local, the bay door chimed softly. Iven said, "Open," without looking up. They knew the footstep, even before Tev cleared his throat.

Tev was wearing the patched boiler suit he favored for thermal work. He had a mug in one hand and a sealed diagnostic wafer in the other. He set both on the corner of the plotter table. The mug was chamomile, which Iven

could smell. The wafer carried the quarter-hour internal status of every radiator on the ship.

"Patch on seven is holding," Tev said. "Patch on nine is holding. Patch on twelve decided to talk back."

"Twelve has always had an opinion," Iven said. They took the chamomile. "How is twelve?"

"Twelve will keep." Tev looked at the plot. He had no cartographer's eye, but he had a maintenance eye, which could spot where a drawing did not match the ship that had flown it. He was quiet for a long moment. Then: "You're drawing a place that has not decided what it is."

"Yes."

"How will you know when it has?"

"That is the question I want answered publicly."

Tev tilted his head. The thermal patch on his sleeve, the one that radiated heat away from his shoulder, made a soft click as he moved. He sat on the edge of the desk and watched Iven work.

Iven worked for another minute. Then they stopped, set the stylus down, and said, "I am going to ask Calder for an open assembly, transmitted on every ship, fed forward to every habitat that still listens to us."

"That is not your job," Tev said, mildly.

"No."

"It is not your authority."

"No."

"It is also, by every forecast compact Admiral Calder still respects, not your right. Forecast governance is the descendant coalition's. Not yours."

"Yes."

Tev did not push the point. He drank from his own mug, the diagnostic wafer balanced on his knee like a small gray pancake. "Do you have the address for Orison?"

"No single address," Iven said. "That is what the export told us. Sixteen strands. Sixteen futures, arguing. The strand that sent us the warning to destroy *Everspring* is one strand. There are at least six others who think that strand is wrong. There is at least one strand who thinks *Everspring* should have been destroyed with the lesson. There is at least one strand that thinks we should never have flown at all, that the Wake itself should have stayed at Caul and never gone near Serein. I have been listening to one strand and mistaking it for a civilization."

"And you think an open assembly on every ship fixes that."

"I think the assembly makes the absence of a single answer visible. That is what an assembly is for."

Tev turned the diagnostic wafer over in his hands. "And if Calder says no."

"Then I publish the plot."

The bay was quiet. *Aster Vale* kept time with the patient tick of a magnetic clock the engineers had built to outlast every electronic system on board. Tev listened to it. Then he said, "Twelve will need a new seal by tomorrow evening or I will be opening it again. The seal kit is in the third locker. I do not know why I am telling you this."

"Because you would like me to know you are still here."

"Yes," Tev said. He finished his tea, nodded, and left. The door sealed behind him with the small, polite sound of the *Aster Vale*'s habitual courtesy.

Iven sat for a long time.

They thought about their mother.

Their mother had been a junior routing officer on a flotilla called Sleep Kept, and the forecast compactor had recommended an evacuation that saved three thousand

sleepers and killed her habitat because the evacuation's route optimization required the habitat to be left at the bend of the parabola. Their mother had known. She had filed a counter-route at 0300 that morning and the compactor had rejected it with a two-line note that read, in full, *Not within forecast parameters. Do not override.* She had overridden. The three thousand had lived. Their mother had not.

Iven had spent twenty years deciding what to do with that memory. For most of those years, they had decided to be a person who kept the maps honest and the holders accountable. They had decided that withholding a piece of information for the public good was the worst form of complicity available to them, short of falsifying it.

They had concealed the deadly annex on Day 1.

They had done the exact thing they had sworn they would never do.

When the inquiry forced the concealment into the open, they had admitted it. The admission saved them from nothing. It merely clarified that they could withhold a lethal fact and continue living afterward. Across the days since, that knowledge had not aged into comfort. It had hardened into a duty.

The duty was: do not let one strand of any future rule this present.

Iven picked up the stylus again. They redrew the orbits. They marked, with a small amber cross, every ship in the Wake Defense Fleet that still had an unbroken laser link to *Aster Vale*. They marked, with a small red cross, every ship whose captain was likely to refuse the assembly on principle. There were six red crosses. There were nineteen amber crosses. There were two ships whose captains were children of forecast compacts and would vote with whatever signal was loudest.

They began to draft the request in their head. They were aware that drafting in the head was a habit they had been told, more than once in their professional life, was not the same as drafting on paper. They drafted on paper.

The request had five paragraphs.

The first paragraph stated, in plain words, that the Orison export had been received and read. The second paragraph stated that its testimonies contradicted each other in ways no present-day reading of forecast governance could reconcile. The third paragraph stated that Iven Mora, chief cartographer of *Aster Vale*, requested an open assembly, transmitted on every ship and relayed to every habitable node that still chose to listen to the Wake Defense Fleet, to be held within the next twelve hours. The fourth paragraph stated that the assembly would discuss, in order, three questions: who Orison was, what authority any of its possible selves had over the present, and what the Wake Commonwealth would decide to do if no single authority survived the question. The fifth paragraph stated that if the request was refused, the plot, the testimonies, and the contradiction would be made public by the cartographer's office of *Aster Vale* regardless.

Iven read the request out loud, twice. They listened to the way it sat in the room. The room was not impressed and was not offended. The room recognized the request as the work of a person who had run out of gentle options.

At 0548, the comms stack chimed. Iven pressed the receive key. The face that resolved in the small square screen was not Calder. It was Captain Bramwell of the cutter *Skua Two*, one of the amber crosses on the plot.

"Cartographer," Bramwell said. "I am calling under my own clock. Not the admiral's."

"I hear you," Iven said.

"The export is a mess."

"Yes."

"I have captains on my deck who think it is a fraud designed to make the assembly vote against the admiral. I have captains who think it is a confession of guilt by the descendants and should be treated as forfeiture of authority. I have my eldest pilot, who thinks it is a love letter to no one in particular and would like to frame page three."

"All those readings are possible," Iven said. "That is also the problem."

"Have you spoken to the admiral?"

"Not yet."

"You are about to."

"I am."

"Briefly," Bramwell said, and ended the call.

Iven stared at the dead screen for a moment. Then they compiled the request into a tight message packet, addressed it to the command frequency of the Wake Defense Fleet, and signed it with their cartographer's seal. They sent it at 0600.

The reply, when it came, was not from Calder. It was from Captain Lidon of *Cairn Eight*, one of the red crosses.

"Cartographer. Your request is received. The admiral is in tactical session and will respond when the session is ended. In the meantime, I am instructed to remind you that the cartographer's office does not have the standing to convene an assembly."

"I am aware," Iven typed back.

"Good," Lidon typed. "Then you are also aware that nothing prevents a group of captains from convening one on their own authority and inviting you as a witness. The group will number eleven. The session will be open to listening ships. Captain Bramwell has agreed to chair. The transmission will originate from *Skua Two* at 1300 today. You are invited to address the assembly at 1330."

Iven read the message twice.

Then they picked up the diagnostic wafer Tev had left on the desk and walked it down to the thermal bay. Tev was not there. The third locker had, in fact, the seal kit. Iven wrote on the third locker's tape label, in neat cartographer's hand: *Twelve wanted you to know about this.*

They did not know if Tev would appreciate the joke. They suspected he would.

They walked back to the cartography bay, locked the door, and began to assemble the materials they would present at 1330.

The first material was the plot.

The second material was the layered set of testimonies, with each contradiction circled and dated.

The third material was a single sheet of paper on which they had written, in their own handwriting, the sentence their mother had written on the back of her rejected counter-route.

"Not within parameters."

Under that sentence, in the same hand, Iven wrote four words.

"Make it visible now."

The bay held the four words the way a tray holds a glass.

Outside the bay, the *Aster Vale* continued to complain about itself at the edge of hearing. The hull eased, sighed, and was not cosmetic about any of it. The minute hand of Tev's nursery clock ate itself with the slow patience of a small mechanical fish.

Iven finished the materials.

They put the stylus down.

At 1258, they locked the bay, walked forward to the comms deck, and waited in the small antechamber where the captain's chair had been left bolted as a relic of a

calmer time. The chair smelled of old leather and of the polish the bosuns used when no one was watching.

At 1300, the transmission began.

Iven listened to eleven captains greet each other by first names and rank. They listened to the chair, Bramwell, open the session in a voice trained toward the neutrality of a calibrated instrument, with nothing left to hide. They listened to the room learn, in real time, that the chart on the screen came from an office whose chief had admitted during the first inquiry to concealing evidence and now stood in a guest chair, waiting to be heard.

At 1330, Bramwell said, "Cartographer Mora."

Iven stood.

They did not begin with the plot.

They began with the paper.

"This was written by my mother," Iven said. "She was a routing officer on the flotilla Sleep Kept. She saved three thousand sleepers and lost her habitat by doing so. The forecast compactor that ordered her to abandon them used her rejection note as a footnote. The footnote read, in full: Not within parameters."

The transmission room held its breath.

"I have spent twenty years deciding what to do with that memory. Eleven days ago I committed a smaller version of the same act. I concealed the deadly annex of the warning that named *Everspring*. I have admitted that. I am telling the assembly now because the room itself is where that admission earns its weight.

"The Orison export has been received. It contains fourteen testimonies. They contradict each other in ways no present-day reading of forecast governance can reconcile. They imply, in aggregate, that the coalition we have been calling Orison is not a single coalition. It is at least six. They are not unanimous. They are not even polite.

"The warnings we have lived under for one hundred and twelve years may have come from a garden of futures, not from a throne.

"I am asking this assembly to acknowledge that no single strand of any future has the standing to rule this present. I am asking this assembly to recognize, in place of forecast sovereignty, a public record of every alternative we can hear. I am asking this assembly to consider that recognition of multiple competing descendants is the only inheritance we can pass to our own.

"I am also offering, in the same breath, my resignation from the cartographer's office. The office should be held by someone who did not withhold on Day 1. I will stay on in an advisory capacity until the assembly names a successor."

Iven sat down.

The room did not applaud. The room did not boo. The room asked, one captain at a time, for five clarifications of mechanism and one question about the plot, which Iven answered on the screen by pointing with a laser at the dotted line where *Everspring* had stopped pretending to be towed.

At 1437, Captain Bramwell said, "The session stands adjourned to 0900 tomorrow. The cartographer's record is entered into the public archive of this assembly. The admiral is requested to attend. The descendants' emissary is requested to attend. The exported testimonies are entered into the public archive by the same hand. The next agenda item is the question of what authority any hearing future retains."

The transmission ended.

Iven sat in the antechamber for a long time.

At 1502, a small message arrived on their personal channel. It was from Tev. It contained no words. It contained a photograph of the nursery clock with the fish

hand eaten past the seven. Tev had added, in pen, on the white border of the photograph, two words.

"Twelve is jealous."

Iven smiled for the first time in eleven days.

They did not feel entitled to the smile. They felt it anyway.

Chapter 20 — The Descendants' Embassy

The legal-memory substrate smelled, this morning, of warm solder.

Aro Nhal had learned to read the substrate by smell the way his sixteen strands of inherited memory had taught him, all of them disagreeing about which smell meant what. One strand said warm solder meant Council had filed a fresh annex in his head. Another said it meant a sister substrate in a sister ship was broadcasting on the same frequency. A third said the smell was actually nothing, that the substrate had been freshly cleaned the night before and Aro's lungs were running on the memory of breathing.

Aro did not believe the third strand. Aro did not believe the first two either. Aro believed the substrate was a machine that did what it was asked, and that what it was being asked today was to keep him alive and legible while the Wake Defense Fleet performed a maneuver he had not seen scheduled.

At 0622 local, the substrate pinged him a courtesy notification: *Aster Vale* had shifted orbit by eleven kilometers. The shift placed the survey carrier slightly farther from the contested plane between *Skua Two's* neutral cluster and *Cairn Eight's* loyalist cluster. The shift was not a tactical one. It was a courtesy one. It gave both clusters more room and gave the substrate more shielding from stray radar.

Aro stood in the small pantry nearest the substrate bay, brewing a tea his strands had not taught him to like. He

brewed it anyway because the act of brewing was the closest his body had been able to come to a private decision.

The law kettle clicked off.

Aro poured.

He walked to the bay window. The view was, as it had been for eleven days, a black field with three small silver needles, which were the *Cairn Eight* and her two armed consorts. Behind the needles, much farther away, the planet Serein showed its night side as a bruise. The bruise had no name on the present maps. The bruise had six names on the implied maps of Orison.

He turned from the window.

The pantry had a single chair. Aro sat in the chair and set the tea on the substrate's auxiliary shelf, where the warmth would feed into the cooling loop without harming the storage. The substrate made the courtesy sound it sometimes made when it was content to be useful: a tiny squeak, like a worn latch.

"Good," Aro said, to the substrate.

At 0640, a sealed packet arrived from *Skua Two*. The seal had Bramwell's chop. Inside was a single page in plain text, no letterhead, no family crest.

To the descendants' emissary. The open assembly of the Wake Defense Fleet, in its first session, recognized that no single strand of Orison carries the standing of the coalition as a whole. The assembly requests, as a courtesy to the present, that the descendants' emissary attend the second session at 0900 today, in person if possible, to be heard on the same standing as any other witness. This is not a subpoena. The assembly is aware the emissary's medium is a legal-memory substrate on a survey carrier. Travel by remote witness link is acceptable. Refusal will be recorded in the public archive without prejudice. —B

Aro read the message twice.

He set the tea down. He had not expected the assembly to want him on equal standing with other witnesses. He had expected them to want a privileged seat, or to refuse him entirely. He had been wrong about which.

He stood, walked to the bay's auxiliary comms panel, and began to reply. He typed four words, deleted them, typed seven, deleted them, typed twelve, kept them.

"I will attend remotely."

He sent the reply at 0641.

Then he sat again and let the substrate do what it did best, which was to hold a person. He did not ask it to tell him who he was. His memory audit had taught him that asking the question that way led to a list of strangers with his name attached.

At 0815, the substrate pinged him again. This time the ping was a calendar reminder: he was due on the assembly call in forty-five minutes. The calendar reminder also offered him a list of public-archive documents he could review. He opened one.

It was a transcript of the first session of the assembly, beginning with Captain Bramwell's opening and ending with Iven Mora's speech. Aro read the speech the way a person reads a letter sent in advance of a difficult meeting: once for content, once for tone. The tone was exact. The content carried Iven's offer of resignation and the entry of *Voices of the Pruned* into the public archive by the cartographer's own hand.

Aro closed the transcript.

He let himself think about Calder.

He had spoken to Admiral Rhea Calder in person only once, in the witness chamber where she accepted his embodiment. Two laser-net exchanges followed, both perfunctory, both concerned with channel hygiene and operational deconfliction. He had spoken to her once more in his own head while practicing how to disclose the

memory edit. That conversation had not been real. It came from the strand labeled *Counselor* running a simulation.

The simulation had not prepared him for the woman.

Calder was a woman who believed that civilization had survived eleven catastrophes by obeying the best available warning before understanding arrived. Calder had been right about ten of them. The eleventh had been Hesper, and Hesper had killed her brother and two million citizens because a council had voted to wait for confirmation.

In the present, Calder was deciding whether to fire on the seedship her fleet had been ordered, by one strand of one future, to destroy.

Aro was a person carrying, in his head, edited memories of all sixteen Orison candidate lives. He was not the descendant coalition. He was a working draft of one of its louder corners.

He thought about Okoro.

Doctor Nadi Okoro had been the civilian mission director of *Everspring* for nine years. She had trained ecologies for a moon she had assumed was uninhabited. First contact taught her that the moon's silicate weather remembered her simulations and replied to them. The Orison export showed six descendant coalitions disagreeing over whether her mission meant salvation or trespass. She had not yet decided what to do with the seedship in the next ninety hours.

Aro typed a request to her: *One hour, after the assembly. The pantry bay. Please come alone if you can.*

Okoro's reply arrived in fourteen seconds.

I will come. I will come with questions.

Aro read *I will come with questions* twice. He thought of his mother's hand on his chest when he was a small child in one strand of memory, and he thought of his mother's voice on a recording in another strand, and he thought of an imagined

mother who had been generated by Orison's archivist to comfort the case for selecting him. None of those mothers had been asked to be here. He had to compose himself without consulting them.

He sat for the remaining thirty-eight minutes.

He did not compose himself.

At 0855 he walked to the comms deck and stood in the witness well of *Aster Vale*. The well had been built, decades ago, for sworn testimony at courts of inquiry. It was a low bowl in the deck with a seat, a microphone, and a wide-angle camera that captured the speaker without flattering them. Aro sat and looked at the small monitor that would show the assembly.

At 0900 the monitor resolved into the same assembly room he had been told about.

Captain Bramwell was in the chair. The chair had been a hotel balcony seat at his cousin's wedding, Bramwell had once said, which had been promoted, in his own quiet career, into a witness chair. Eleven captains were visible at their terminals. Two were speaking softly to each other across the room. One was reading a printed page.

"Emissary Nhal," Bramwell said. "You attend as a remote witness under the standing granted to all witnesses by the first session. You may speak when recognized. The assembly thanks you for accepting."

"Thank you, Captain," Aro said. "I accept the standing as offered."

Bramwell looked at his own screen, then back. "The assembly has read the cartographer's record and the public archive of *Voices of the Pruned*. The assembly now wishes to be heard, in turn, by two parties whose position bears on the question of coalition plurality: the descendants' emissary and the commander of the Wake Defense Fleet. Admiral Calder, do you attend?"

A second monitor resolved in the witness well of *Aster Vale*. Calder's face arrived, sharp and tired.

"I attend," Calder said.

"The chair recognizes the admiral and the emissary as joint respondents to the assembly's first formal question: under what authority has the descendants' coalition offered guidance to the present, and under what authority, if any, does it continue to do so?"

Aro did not speak first.

Calder did.

"The authority has been forecast compact," Calder said. "One hundred and twelve years of compacts. Authentication by return-mirror signature. The Commonwealth has ratified each compact in turn. That ratification is the authority."

"With respect, Admiral," Aro said, "that ratification ratifies the compact, not the coalition. The compact was negotiated, originally, with a coalition that presented itself as a single descendant voice. The exported testimonies indicate at least six factions, with the strand that sent the warning on Day 1 only one among them. The authority the assembly ratified may not have existed in the form it was ratified as."

Calder's jaw tightened. "The authentication signatures remain valid."

"The signatures," Aro said carefully, "authenticate the channel. They do not authenticate the speaker's standing to speak for a coalition that no longer speaks with one voice. A signed envelope does not tell us who placed it in our hands."

"In practice, the warnings have been useful nine times out of ten," Calder said. "In Hespar, we ignored a verification request and two million died. I do not have the luxury of pretending the channel is illegitimate."

"The channel is not illegitimate," Aro said. "The speaker's standing is."

Aro heard his own voice flatten. He did not let himself soften it. He had learned, in seventeen chapters of being a person, that softening was a tool of one specific strand's advocacy, and he was no longer its instrument.

"The assembly should consider three propositions," Aro said.

He took a breath.

"First. Orison is plural. The descendants who share my years have not agreed among themselves about anything material in living memory. The faction that issued the Day 1 warning is one. The faction that would have preferred the warning to be ignored is another. The factions that wish *Everspring* had never been built are others. The assembly cannot treat any of these factions as if it were Orison."

Calder's eyes narrowed.

"Second. The descendants are not sovereigns over the present. We are stakeholders. We have skin in every decision the present makes, because every decision the present makes is going to be inherited by someone like me. But skin in the outcome is not the same as authority over the decision. The Wake Commonwealth is the present. The present is its own."

Inside Aro's head, the strand of memory called *Counselor* lit up briefly. He let it. The Counselor had nothing useful to offer here. The Counselor's particular specialty was shaping a sentence so that a Council chamber on a moon a future century away would vote the way the strand wanted. The Counselor had no vocabulary for a present-day assembly of eleven captains.

"Third. The descendants are not unanimous in our relationship to the current crisis. Some of us believe the mission is moral and should be supported. Some of us believe the mission is trespass and should be paused. Some of us believe the mission should have been aborted before launch. I am not authorized to speak for any of those views.

I am authorized only to disclose that they exist and that Orison has not chosen among them."

The room was silent.

Then Calder said, "Who authorized you, then?"

"No one in the present," Aro said. "I authorized myself. I am a working draft of one of Orison's louder corners, and the louder corner placed me here without asking the rest of Orison whether it wanted me."

"And the rest of Orison."

"I cannot speak for them."

"You cannot, or you will not?"

"Both," Aro said. "I cannot, because I was not designed to know their full positions. I will not, because the compact asked the present to trust me, and the present should not."

Calder leaned back.

"Admiral," Bramwell said, "the chair recognizes that the question of authority is not yet resolved. Does the admiral wish to add a point of view before the chair opens the floor to the other captains?"

Calder nodded once. "I do."

"Go ahead."

"Emissary Nhal has stated that Orison is plural. The assembly has heard this. I want to add, in the same formal session, a fact that should not surprise anyone who has read the public record: the descendants' coalition includes, in at least three of its six factions, present-day citizens of the Wake."

Aro did not move.

"This is not a metaphor," Calder said. "Return-mirror lineage is tracked in the Wake census. Any child born of an embodied return-mirror transmission to a Wake parent is, by both Wake and Compact law, a citizen of both ages.

Orison's plurality includes, in three of its six factions, present-age citizens whose families are aboard these ships."

Aro heard his own strand of memory called *Heir Lidon* light up briefly, but it did not merely light up. It opened. *Heir Lidon* had been one of the strand's bigger contributors to the choice of Aro as emissary. The strand had carried *Heir Lidon* as a soft approval without a face, sealed at embodiment.

Calder's disclosure had been the unsealing. The opening was not invited. The face that opened was the face of a person Aro had never met, in a kitchen he had never visited, on a station that had not been built yet.

Aro did not speak of the face. He stored it. He would speak of it, when he spoke, only to one person.

Aro did not know whether *Heir Lidon* had been edited. He suspected the edit had been small.

"Admiral," Aro said. "Are you stating that the present-day Wake contains citizens who will go on to become members of Orison?"

"Emissary," Calder said, "I am stating that the present-day Wake contains citizens whose return-mirror lineage has been verified, by both Wake census and Compact record, as ancestors of three of Orison's factions."

"Including your own," Bramwell said, without inflection.

Calder was very still for a long moment.

Then she said, "Yes. Including my own. The faction I will, in the outcome of any of the resolved scenarios, belong to has been, for the duration of this conversation, one of Orison's louder corners."

The room was silent again.

Aro let the silence last. He had learned, in eighteen chapters of being a person, that silence was sometimes a better tool than a sentence. The silence did the work the sentence could not.

Then he said, "I am grateful for the disclosure, Admiral. I am aware that the disclosure is a hard one."

"It is," Calder said.

"I offer, in return, a disclosure of my own, which I have been preparing to make to the present at the first formal occasion. Sixteen strands of memory were edited by Orison before embodiment. The editing was deliberate. The editing was done to make me persuasive. I disclosed the edit before this assembly convened, and I enter it again now because the present has the right to know who it is listening to."

"And who it is listening to," Calder said, "is a working draft, edited by an imperfect coalition, of one of Orison's louder corners. And one of Orison's louder corners is, by the Admiral's own disclosure, also the Admiral's corner."

"Yes," Aro said.

"Then the assembly should know," Calder said, "that the admiral is not a neutral party to the question of Orison's standing over the present. The admiral has skin in the future, in addition to skin in the present. The admiral's position against the destruction of *Everspring* is not a private preference. It is a position taken by a person who would, in the outcome of any of the resolved scenarios, belong to one of Orison's factions."

"Admiral," Bramwell said. "The chair thanks the admiral for the disclosure. The chair enters the disclosure into the public archive."

Calder looked into the camera as if she were looking down a long corridor.

"I am not offering this disclosure to excuse any position I have taken," she said. "I am offering it because the present has the right to know. I was taught, in my career, that the most dangerous person in any room is the one whose interest in the outcome is invisible. I am making my interest visible."

The room did not applaud. The assembly adjourned after a brief procedural exchange, and the witness well of *Aster Vale* went back to showing a thin black field with three small silver needles.

Aro sat for a long time.

He thought about the tea he had left on the substrate's auxiliary shelf. He thought about Okoro, arriving in an hour. He thought about the strand of memory called *Heir Lidon*, who had been, until fifteen minutes ago, an abstract category, and who was now, by Calder's disclosure, also a person Calder had known personally in some future that had not yet happened.

He let himself find this amusing. It was not, in the long run, amusing. In the short run, it was.

The substrate made its small courtesy squeak.

"Good," Aro said again, to no one in particular.

He stood, walked to the pantry, retrieved the tea, and drank it cold.

Chapter 21 — A War Over the Unborn

Admiral Rhea Calder did not sleep on the night between the second and third sessions of the assembly. She did not believe in sleeplessness as a sign of anything. She believed in sleeplessness as a maintenance problem, and she treated hers the way the *Cairn Eight's* chief engineer treated a coolant leak: by signing for the inconvenience and working around it.

The flag bridge of *Cairn Eight* had been her working cabin since *Resolute Vigil* transferred flag authority after the Nox engagement. Captain Lidon had surrendered the room without surrendering his ship. It was not a comfortable room. It was an instrument cluster with a chair bolted over a treadmill that the designers had installed to keep flag officers from losing bone density in long campaigns. She disliked the treadmill. She had refused to remove it. She walked on it while she read.

The night of the second session, she read four things, in order: the public archive of the assembly transcript, two casualty reports from the dark sun engagement, and a paper letter she had carried in the breast pocket of her uniform since Day 4.

The paper letter was from Iven Mora.

It was not a letter to Calder. It was a letter that Calder had seen fit, on Day 4, to remove from Iven's cabin without breaking the seal, on the calculation that any document Iven had written in their own hand after a confession might

be evidence. The seal had been broken and resealed twice by junior officers, but the body was intact. Calder had read it once.

Iven had written the letter on a piece of cartographer's stock in the night after their Day 1 confession, addressed to no one, and it contained the narrative of their mother's death.

It contained the routing officer's counter-route. It contained the compactor's two-line rejection. It contained the four words Iven had since added at the bottom of the copy they had read aloud at the assembly: *Make it visible now.*

Calder had read all of that. She had also read a paragraph she had not expected.

Iven had written, in the second page, this sentence: *The compactor saved three thousand and killed one. The compactor did not name the one. The compactor never named anyone. I am trying to decide whether the compactor can be made to name everyone before it counts.*

Calder had read the sentence three times on Day 4. She was reading it again now, on the treadmill, at 0247.

Two million citizens had died at Hespar.

Two million names.

The compactor, in the case of Hespar, had been a council that had chosen to delay a verified forecast to ask one more question. The council had delayed three hours. The forecast had been correct. The two million had died because the council had wanted better evidence, and the council's idea of better evidence had been the kind that could not be made in three hours.

Calder had decided, after Hespar, that the correct posture toward a forecast was obedience. She had been joined, in that posture, by every admiral of her generation. The posture had been worn into the Wake Defense Fleet's standing orders the way a groove is worn into a deck plate by ten thousand footsteps.

Now Iven was asking the fleet to make the compactor name everyone.

Now Calder was on a treadmill reading about a routing officer's counter-route, and about a single habitat lost at the bend of a parabola. The routing officer's counter-route had been right, and the compactor had rejected it because the compactor did not care about that habitat, and the compactor's reasons had been the same shape as the council's reasons at Hesper. Calder was, in her own professional opinion, not a person who should be reconsidering her posture on Day 9 of an eleven-day crisis.

She read the letter again.

At 0308, the comms stack chimed. The watch officer reported a development.

The development was *Aster Vale's* letter to the fleet, transmitted over open channel, signed by every captain of the second assembly session. The letter stated, in plain language, that the assembly had entered *Voices of the Pruned* into the public record and that no single strand of Orison had standing to rule the present. It recorded the descendants' emissary re-disclosing the sixteen-strand memory edit. It adjourned the assembly to its third session at 1400.

The watch officer asked whether to acknowledge receipt.

"Acknowledge," Calder said. "Acknowledge the receipt. Take no position on the substance. Note that the flag has read the letter. Note that the flag continues to study the substance."

"Aye, Admiral."

The acknowledgment was signed and sent. The watch officer cleared her throat. "The senior captains of the loyalist cluster have requested a private briefing at 0500."

"Granted. In person."

"Aye. There is also a casualty report. Theargo Industries *Pinion*, cutter-class, lost with all hands during the dark sun engagement. Names attached. Theargo Industries has been removed from the loyalist roll."

Calder took the casualty report.

She read the names.

She had known, professionally, that her fleet would lose ships. She had expected, in the worst outcome she had imagined, that the fleet would lose four ships. The dark sun engagement had cost the fleet seven. Three had been cutters with all hands. Two had been civilian tugs in voluntary escort. One had been an ammunition transport whose detonation had been the cause of two of the cutter losses.

The seventh was *Pinion*. Theargo Industries, where Calder's nephew by marriage had served as a junior flight engineer. The nephew by marriage was on the casualty list. The casualty list had been verified by three witnesses and the *Pinion's* last officer, who had filed a brief rescue request on the bridge recorder before the bridge recorder had gone silent.

Calder read the list once.

She did not read it again.

She folded the casualty report into the breast pocket beside Iven's letter, walked off the treadmill, and sat in the command chair for a long moment.

At 0331, she opened her personal channel and sent a brief message to Iven Mora. The message contained one line.

Cartographer. The flag owes you a conversation. The flag is ready. The flag is not sure of the hour. —Calder

The reply came at 0340, four lines.

Admiral. The cartographer is awake. The cartographer is not sure of any hour. The cartographer will be in the bay at any hour the flag chooses.
—Mora

Calder did not respond.

At 0500, the senior captains of the loyalist cluster came aboard. There were twelve of them, in person, with their deputies. They brought the smell of corridor coffee and the careful silence of officers who had agreed, before boarding, on what they would and would not say. Calder received them at the flag table with a copy of Iven's letter folded into her uniform pocket and the treadmill idling behind her chair. The meeting ran on the carrier's flag bridge until 0714.

The first hour was procedural. Captain Marek of *Quiet Reed* reported that the boarding-intercept doctrine from the first pursuit vote still held in his cluster's standing orders and asked the admiral to confirm that the third attempt would be governed by the same constraint. Calder confirmed. Captain Onn of *Sounding Star* asked whether the cluster had standing to authorize a weapons-free shot at propulsion without reopening the doctrine vote. Calder said it did. Captain Besh of *Watchful Coil* asked whether the civilian assembly's declaration that the forecast was plural had legal force on a ship that had not signed it. Calder said the declaration had evidentiary force. It did not have command force.

By the second hour, the cluster had agreed, by a vote of nine to three, to escalate the operational posture against *Everspring* by authorizing one further interception attempt, with full weapons free, before a session of the assembly scheduled for the following morning. The vote was narrow. The dissenters filed a minority report for the public record. Its reasoning matched the argument Calder herself had used during the first pursuit vote. In her professional opinion, that reasoning had not become wrong merely because it now opposed her.

During the procedural hour, Marek had slid across the table, without looking at her, a folded square of paper on which he had written the names of three of his crew who had served on *Everspring* as couriers before the cascade and whose families still slept aboard. The note ended with the single line: *These are the names you said you would count.* Calder had read the note, refolded it, and placed it in the same pocket as Iven's letter. She did not speak of it. The note stayed in the pocket. The names would stay in the meeting room long after the captains had returned to their ships.

The minority report was wrong in one place. The minority report assumed that the loyalist cluster's posture, after the third interception attempt, would be an escalation toward indiscriminate destruction. Calder had, in private session before the vote, told the cluster that the third attempt would be the last. The third attempt would target the seedship's propulsion, not its life support. The third attempt would be a single salvo from a single ship, with the loyalist cluster's flag in the chair. The third attempt would be a final chance for *Everspring* to come about, sign the quarantine, and accept tow. If it did not, the loyalist cluster would be free to escalate at its discretion.

Calder had not told the cluster that the discretion was not hers to give. She had known, in the moment she made the commitment, that discretionary escalation would be the cluster's, not hers. She had made the commitment anyway, because the alternative was not making a commitment at all, and a captain who will not commit on Day 9 is a captain who has stopped deciding.

After the meeting, Calder sent a second personal-channel message to Iven.

Cartographer. The flag would like the conversation at 0900, on the bay of the cartographer's office, by speaker, off record. —Calder

Iven's reply arrived in five minutes.

The cartographer's office bay is at the disposal of the flag at 0900. The cartographer will not record the conversation. The cartographer will not

summarize the conversation. The conversation will be the conversation.
(Mora)

At 0903, Calder opened the speaker channel.

"Cartographer."

"Admiral."

"Your letter. The one you wrote on Day 1 and addressed to no one. The one the flag removed from your cabin on Day 4 and read without your permission."

"Yes, Admiral."

"The flag owes the cartographer an apology."

"An explanation is also due, and the cartographer would like it."

"The flag will provide both. The flag removed the letter because the flag judged, on Day 4, that any document the cartographer had written in the hours after a confession might be evidence in a court of inquiry. The flag has since concluded that the judgment was incorrect. The flag has since concluded that the judgment repeated the error the flag criticized in the assembly transcript."

"I accept the apology, Admiral."

"I have a question, Cartographer."

"The cartographer will answer what the cartographer can."

"You wrote, in the second page of the letter: *I am trying to decide whether the compactor can be made to name everyone before it counts*. The flag has been reading that sentence for five days. The flag has not been able to set it down. The flag would like the cartographer to explain the sentence in the cartographer's own words."

"The sentence means what it says, Admiral."

"The flag does not understand what it says."

"Then I will explain, Admiral. The compactor today is a forecasting compact ratified by councils that do not have to

name the people they sacrifice. At Sleep Kept, the compactor did not name my mother. At Hespar, the compactor did not name your brother. At Day 1, the compactor did not name the eighteen thousand sleepers aboard *Everspring*. The compactor's advantage is that the compactor does not have to name names. The compactor's disadvantage is the same advantage. The sentence asks whether the compactor can be required to count each person before it acts. The sentence asks whether the names can be required as a precondition. The sentence asks whether a present that demands the names as a precondition is a present that will sometimes refuse to obey the forecast because the names are too heavy."

Nothing came back for a long moment.

"The flag understands the sentence now," Calder said. "The flag has one further question."

"The cartographer will answer if the cartographer can."

"The flag has lost a nephew by marriage today. He was a junior flight engineer on a cutter named *Pinion*. The cutter was lost with all hands. The flag is not asking the cartographer for sympathy. The flag is asking the cartographer whether the compactor's naming requirement is the right policy when the compactor has just named a person the flag loved."

There was a long pause on the other end.

"I cannot answer that question truthfully, Admiral. The honest answer is that I do not know whether the policy is right when the policy has just named someone I love. The honest answer is that the compactor I want is the compactor that names names. The honest answer is that I cannot promise the compactor I want would have saved your nephew, or my mother, or anyone. The honest answer is that the compactor I want is the compactor that has to count the cost before it acts. The compactor I have is a compactor that does not have to count. The compactor I want may still be wrong. I cannot promise it will not be."

"Then we agree," Calder said.

"We agree that the compactor I want may still be wrong."

"And we agree that we do not have the compactor today."

"Yes."

"Then the admiral has a question that is harder than the compactor question."

"The admiral may ask."

"The admiral has committed, in private session this morning, to one further interception attempt against *Everspring*, with weapons free, before the third session of the assembly tomorrow. The admiral has not told the cluster that the attempt will target propulsion rather than life support. The admiral has not promised the cluster that the attempt will be the last. The admiral knows the cluster will escalate if it has the discretion to do so. The admiral has made the commitment anyway, because the alternative is a captain who has stopped deciding."

"Yes, Admiral. The cartographer has been told by Captain Bramwell. The cartographer has been told the time and the parameters."

"The cartographer has an objection."

"The cartographer has more than an objection, Admiral. The cartographer has a request. The cartographer would like the admiral to consider, before the firing keys are armed, that the third interception attempt will be the moment the present stops pretending the descendants' plurality is academic. The cartographer would like the admiral to consider that eighteen thousand four hundred sleepers on a seedship with a child Iven Mora's mother would have recognized in any timeline will be present at the moment of firing. The cartographer would like the admiral to consider that the cluster's discretion to escalate after the third attempt is the exact discretion that produced Hesper."

"The flag has considered all of those things, Cartographer."

"The flag has made the commitment anyway."

"Yes."

"The cartographer has a request, Admiral."

"The flag is listening."

"The cartographer requests that the admiral consult, before the firing keys are armed, with one person who is not in the admiral's chain of command. The cartographer requests that the admiral consult Doctor Nadi Okoro, civilian mission director of *Everspring*, by speaker, off record. The cartographer requests that the admiral not consult Okoro to receive Okoro's permission. The cartographer requests that the admiral consult Okoro to hear Okoro's name for the people aboard the seedship the admiral is preparing to fire upon. The cartographer requests that the admiral make Okoro's name available to the loyalist cluster before the firing keys are armed. The cartographer requests that the admiral treat Okoro's name as the compactor's name."

Silence held the line for a long count.

"The flag agrees to consult Okoro," Calder said.

"The flag will consult before the firing keys are armed, or as soon as Okoro can be reached on an open speaker."

"The flag will make Okoro's name available to the cluster."

"The flag will treat Okoro's name as the compactor's name."

"The flag agrees to all four, Cartographer."

"The cartographer thanks the admiral."

Then Calder said, "One more thing. Cartographer."

"Listening."

"The compactor you want has to count my nephew's name. The compactor you want has to count my brother's name. The compactor you want has to count my mother's name, and my father's name, and my brother's friends, and the names of the two million citizens who died at Hespar because a council delayed a forecast. The compactor I want

is one that would have saved all of them. The compactor I want is one that did not exist in time. The compactor I want is one that was not available to the council. The flag will not have a compactor that was not available to the council. The flag will have, as the compactor, the names Okoro can give the flag. The flag will have, as the policy, the count Okoro can give the flag. The flag will not have more. The flag will not pretend to have more. The flag will fire the keys or the flag will hold the keys, and the firing or the holding will be the compactor's verdict, and the verdict will be missing names it could never have collected."

"I understand, Admiral."

"The flag thanks the cartographer for the conversation."

"The cartographer thanks the flag for the conversation."

The speaker channel closed.

Calder sat at the command chair for a long time.

At 0931, she opened a new speaker channel and placed a call, on open airwaves, to Doctor Nadi Okoro, civilian mission director of *Everspring*, aboard the seedship.

Okoro answered on the second ring.

"Admiral."

"Doctor. The flag is calling about the names."

Outside the speaker, in the corridor of the flag bridge, a junior watch officer was quietly closing the door. The corridor smelled of recycled air and of the faint trace of the disinfectant Okoro used to keep her seedship's wardroom clean. The disinfectant was not, in fact, on *Cairn Eight*. But Calder had been smelling it, since Calder's last visit to *Everspring* before the cascade, every time the door closed.

She set the smell aside.

She had a list of names to obtain.

Chapter 22 — The Last Courier

Sena Vale's boots were still cold from the dry-ice floor of the courier bay when the second pinger went off on her wrist.

The first pinger, two minutes earlier, had been a heads-up from the watch officer of the tug *Sworn*: the tug had been moved from its berth alongside *Aster Vale* to a holding position under the survey carrier's ventral plate, where the magnetic shadow of the bulk would screen the tug from the loyalist cluster's mid-band radar. The second pinger carried a different note. It carried the encrypted chop of the open assembly that Captain Bramwell had chaired on *Skua Two*. It carried a request, addressed to Sena by name and rank. It carried a manifest.

Sena read the manifest on the courier bay's small fold-down screen, with the boot of her left foot on the bench and the dry-ice fog curling around her ankle. She read it twice.

The first line of the manifest named three pilots, in this order. Captain Bramwell of *Skua Two*, neutral cluster, courier-rated, laser-net qualified, last formal endorsement for courier work filed twelve days ago. Pilot-Captain Onn of *Sounding Star*, loyalist cluster under Captain Lidon, courier-rated on an emergency qualification, last endorsement contested by two of Lidon's own bridge officers. Pilot-Captain Ravel of *Far Margin*, independent, the captain who had peeled off the loyalist column on Day 9 and refused to return, courier-rated on the strength of an old

Pell certification that had not been renewed since the school ring.

The second line named three ships. *Skua Two, Sounding Star, Far Margin.*

The third line named three cargoes: mirror keys, physical rather than transmitted. Each niobium-titanium slug could carry an authenticated return-mirror signature across a handoff. Iven Mora's office had machined them, three assembly captains had signed them, and individual evacuated sachets protected them inside a worn titanium dispatch case. A paper tag on the hasp read, in Bramwell's hand: *Three keys, three pilots, one relay.*

Iven Mora had written the appended order and Captain Bramwell had countersigned it. Sena would carry the case behind Lumen's leading debris arc, hand it to Bramwell, and stay in motion until each key reached one of the three courier-rated pilots on the assembly record. The order ended: *The geometry changes when each key arrives. The engagement does not begin until the geometry has changed.*

Sena understood the geometry. Lumen's storm saturated the laser nets between the neutral, loyalist, and independent clusters. Calder's loyalists already had a firing solution on *Everspring's* propulsion, and the salvo would land in eighty-one minutes without consent from the rest of the fleet.

Hand delivery of the authenticated mirror keys would invoke the rule Calder had placed on the record before the loyalist captains: anyone who fired without confirmed standing for the warning would bear the consequence alone. Once a key reached each sub-fleet, every local laser net would carry that authentication. The relay could not stop the salvo by force. It could put three endorsed pilots into the geometry Calder had promised to consult before arming the firing keys.

Sena pulled her boots on over the cold floor, sealed the courier bay's inner door behind her, and walked aft through

the tug's narrow spine to the small locker where the dispatch case waited on its foam cradle. The case was heavier than it looked. The slugs were heavy in the way only machined alloy could be heavy: dense, patient, indifferent to the courier carrying them. Sena lifted the case by the worn handle, felt the paper tag turn on its wire, and walked the case back to the courier bay.

Tev Arin was waiting for her there. He sat on the deck plate with his back against the courier bay's small auxiliary cooler, in the patched boiler suit he wore for thermal work. He held an empty mug.

"You're supposed to be on the radiator watch," Sena said.

"Twelve is holding," Tev said. "Twelve will keep until you're back. Twelve wanted me to tell you that."

Sena did not smile. Since she told Tev the truth about Jori, he had kept it with the care he gave a critical component: checking at the agreed interval and never pretending the gauge had not moved.

"How many of the three pilots have you flown beside?" Tev asked.

Only Bramwell, once, on a training burn. Sena knew Onn by the bruised hull patch the pilot had brought back from Pell. Ravel's name had ridden a shipping label on the fabrication delivery that made the Pell school ring possible. Sena knew neither stranger's face.

"Two strangers and one acquaintance," Tev said. "For three keys."

"For three keys," Sena said.

"Their endorsements survive all three vetoes," Tev said. "Neutral, loyalist, independent. Nobody else cleared every list."

"Twelve is generous," Sena said.

Tev set the empty mug on the cooler and stood.

"The corridor behind Lumen's leading debris arc lasts two hundred and forty seconds," Tev said. "It is the only crossing the loyalist radar will miss. The column holds bearing two-seven-one and does not know the window exists."

"How long does the corridor last if the loyalist column adjusts bearing," Sena said.

"Nineteen seconds. They think *Sworn* remains under *Aster Vale*'s ventral plate."

Sena lifted the dispatch case off the cradle.

"I will be back before twelve needs a new seal."

"Twelve does not need a new seal," Tev said. "Twelve wanted me to tell you that."

Sena walked out of the courier bay. Tev followed her aft to the bridge ladder, then peeled off into the thermal bay with the empty mug. She was already climbing.

On *Sworn*'s bridge, Halen had plotted the corridor as a red sliver behind the leading debris arc. His mother had been one of the eight surveyors lost at Cliffside, her name preserved in the moon's record. He treated the plot with the attention of someone who knew what an unrecorded loss cost.

"Plot is hot," Halen said. "Corridor opens at zero-eight-fifteen and closes at zero-eight-nineteen. The loyalist column is steady on bearing two-seven-one and does not have us on scope. We have ninety seconds to translate."

Sena locked the dispatch case onto the courier's hard point, pulled the courier's restraint across her lap, and lit the courier burn. The harness had been rebuilt after *Everspring* rejected tow, when *Sworn* entered escort service and the old webbing failed a courier-load test. The harness settled across her chest the way a piece of equipment settles when a maintenance engineer has told it what it must do.

She did not think about Jori.

Jori had been a courier-rated pilot. Jori had flown the Pell school ring nine years ago with four hundred children and three broken vanes. Jori had died in the ring because Jori had taken the helm of the ring and given Sena the helm of the tug. Jori had not died in the tug. Jori had not been missing. Jori had been the moon's first record. The moon had kept Jori. The moon had not given Jori back.

Sena let Jori be the maintenance question she had answered ten minutes ago on the courier bay bench: how many of the courier-rated pilots on the assembly record have I flown beside. The answer had been two strangers and one acquaintance. The answer had been enough.

The burn ran for ninety-one seconds. The corridor ran for two hundred and forty. *Sworn* cleared the debris arc at zero-eight-nineteen minus three seconds, with the loyalist column's bow still two hundred and forty kilometers astern.

"The corridor closed behind us," Halen said. "The loyalist column did not adjust. We are behind Lumen's leading debris arc."

"Bring us to the relay position. Plot the burn for Bramwell."

The burn for Bramwell was a long, slow approach to a point behind the leading debris arc, in a region the moon's storm had cleared of the loyalist column's mid-band radar. The point was a piece of geometry Iven Mora had plotted on the cartographer's bay that morning, on a sheet of plotter paper whose margins still carried the smudge of a chamomile mug. Iven had sent the plot to the courier bay on the open assembly's laser net. Eleven captains had signed the plot in turn. The plot had been entered into the public archive of the assembly at 0614.

At zero-eight-thirty-one, *Skua Two* emerged from behind the debris arc. The cutter was small, gray, and tidy in the way a courier-rated cutter is tidy. Bramwell was on the bridge, in shirtsleeves. She did not wave. She matched

velocity with the tug in nine seconds and brought the cutter alongside at ten meters. The airlocks of *Skua Two* and *Sworn* had been built by the same dockyard in the second year of the courier service, and the same worn patches marked the same places on each.

With the hulls locked at equal pressure, Sena carried the case through the three airlock chambers and set it on *Skua Two*'s deck. She gave Bramwell room.

"The first key," Sena said. "The first key opens the relay."

Bramwell did not pick up the case. Bramwell looked at the case, then at Sena.

"You carried the case through the corridor."

"Yes."

"The corridor was a piece of geometry that had not existed ninety seconds before you flew it. The geometry was plotted by Iven Mora and signed by eleven captains and entered into the public archive at 0614."

"Yes."

"I have flown courier burns beside Iven Mora's office for twelve years. I have never seen Iven Mora plot a corridor that was not first requested by a captain whose ship was on fire."

Sena let that pass without a word. The courier bay of *Skua Two* was colder than the courier bay of *Sworn*, and she understood, by the cold, that Bramwell had been holding position behind the debris arc for longer than the manifest had implied.

"I am picking up the case," Bramwell said.

Bramwell opened the case, removed the first sachet, and broke the seal. She read the engraved chop of Iven Mora on the slug's fluted face. She did not read the handwritten tape label. Bramwell knew what the label said.

"The relay opens," Bramwell said.

"The relay opens," Sena said.

Bramwell closed the case and handed it back. Bramwell matched velocity with the tug and peeled off at twenty meters. The cutter slid back behind the debris arc.

Sena brought the case through the outer airlock, the equalizing chamber, and the inner airlock, and locked it back onto the hard point. The case had lost one sachet. The case had lost nothing else.

Halen plotted the next leg. It was to Onn, on *Sounding Star*, in the loyalist cluster, behind Lumen's trailing debris arc.

"Plot the burn," Sena said.

The burn was a long, slow approach to a position behind the trailing debris arc, in a region the moon's storm had cleared of the loyalist column's mid-band radar. Eleven captains had signed the position as the second piece of geometry on Iven's plot.

At zero-eighty-fifty-three, *Sounding Star* emerged from behind the arc. The cutter was larger than *Skua Two*, and it carried a single broadside laser mount that had been fired, twice, during the dark sun engagement. The mount was cold and folded.

Onn was on the bridge, in the loose gray coat loyalist pilots wore on watch. Onn did not wave. Onn matched velocity with the tug in fourteen seconds and brought the cutter alongside at ten meters. The loyalist cutter's airlock was newer than *Sworn's*, with a heavier hinge and a brighter seal.

At the loyalist airlock, Sena moved the case through three chambers and placed it on *Sounding Star's* courier-bay deck. She waited by the hatch.

"The second key," Sena said. "The second key closes the relay."

Onn did not pick up the case. Onn looked at the case, then at Sena, then at the courier bay's small auxiliary cooler, where a single empty mug sat on the deck plate.

"The mug is empty," Onn said.

"The mug is empty," Sena said.

"The mug is on the cooler. The cooler is in the courier bay. The courier bay is on a loyalist cutter. The cooler is not standard. The cooler is a maintenance engineer's cooler. The maintenance engineer is named Tev Arin. Tev Arin is a friend of mine."

Sena looked at the cooler. She had not known Tev Arin had built a cooler for a loyalist cutter. She had not known Tev Arin had friends outside the neutral cluster. She had known Tev Arin for nine years and had not known either of those two facts.

"Tev Arin told me, by courier paper on the open assembly's laser net at 0601 this morning, that the courier tug *Sworn* had been moved to a holding position under *Aster Vale*'s ventral plate. Tev Arin told me, on the same courier paper, that the courier tug's courier bay had been provisioned for a long burn. Tev Arin told me, on the same courier paper, that the courier bay's auxiliary cooler was empty. Tev Arin told me, on the same courier paper, that the cooler would be empty when I boarded the courier tug."

"Tev Arin is a maintenance engineer," Sena said. "Tev Arin does not have an opinion about anything."

"Tev Arin is a piece of equipment that has been told what it is for," Onn said.

Onn opened the case, broke the second seal, and read the chop on the slug's fluted face.

"The relay is open and closing," Onn said.

"Carry it onward," Sena said.

Onn matched velocity with the tug and peeled off. The cutter slid back behind the trailing debris arc.

Sena brought the case back through the three chambers and locked it down. The case had lost a second sachet. The third sachet sat in its cut foam well. The third sachet carried the spare.

Her hands were still cold.

Halen plotted the third leg to Ravel aboard *Far Margin*, behind the leading debris arc.

"Take us to the final handoff," Sena said.

For the last approach, *Sworn* crossed a region the moon's storm had cleared from every fleet radar. Eleven captains had signed that position as the third piece of Iven's geometry.

At zero-nine-forty-two, *Far Margin* emerged from behind the arc. The cutter ranked smallest among the three courier-rated cutters in the relay, and it carried no broadside mount and no laser battery. The cutter carried cargo. The cutter had been built, the manifest said, in the second year of the courier service, and reclassified as a freighter in the eighth year, and reclassified back as a courier on the strength of an old Pell certification that had not been renewed since the school ring.

Ravel was on the bridge, in the loose black coat independent pilots wore on watch. Ravel did not wave. Ravel matched velocity with the tug in eleven seconds and brought the cutter alongside at ten meters. The airlock of *Far Margin* was the same model as *Sworn's*, with the same worn patches in the same places.

Inside *Far Margin*, Sena cleared the patched airlock and set the case between herself and Ravel.

"The third key," Sena said. "The third key carries the spare."

Ravel studied the case, then Sena. "My name rode the fabrication delivery that made the Pell school ring possible."

"I remember the shipping label," Sena said.

"I read the corrected archive," Ravel said. "Jori flew the ring home. You flew the tug. The council switched your seats because it wanted one uncomplicated hero."

"I was there."

"I have waited nine years to meet the pilot who stopped letting the council tell it backward."

"That pilot has cold hands and one key left to deliver."

Ravel broke the third seal and read Iven's chop on the fluted face. "The third key keeps the relay available for whatever comes after this firing window."

"Then carry it," Sena said.

Ravel slid the slug back into its opened sachet and clipped the sachet to the inner ring of the flight harness. The metal struck the buckle with a clear note. A bridge camera recorded Ravel's hand closing over it, the independent transponder in the frame, and Sena standing beside the emptied foam well. Halen acknowledged the image from *Sworn*. Bramwell's key had already appeared on the neutral net. Onn's key had appeared on the loyalist net. Ravel touched the courier chop to the local reader, and the independent net added its authentication to the other two. Three separate receipts lit on Sena's wrist. Only then did she lift the case.

Ravel matched velocity with the tug and peeled off. The cutter slid back behind the leading debris arc.

Sena returned through *Far Margin's* airlock with an empty case and locked it to *Sworn's* hard point. Every sachet had reached a pilot. Halen brought the tug about.

The loyalist column still held its firing solution on *Everspring's* propulsion, with seventeen minutes remaining. It could no longer fire under the accepted rule without consulting Bramwell, Onn, and Ravel. Three conversations

could not fit inside the window. The geometry had changed; the firing window moved with it.

Sena set the empty case on its foam cradle and walked aft to the thermal bay. Tev sat against radiator twelve in his patched boiler suit, the empty mug beside him.

"Twelve is holding," Sena said.

"Twelve will keep," Tev said.

"The keys are delivered."

"The geometry has changed."

Sena sat beside him and set her cold hands on her knees. Jori remained the pilot who had carried four hundred children home and died under a story told backward. The moon had corrected that story. This time Sena had carried the work far enough to place it in three rival hands.

The thermal bay was quiet. Radiator twelve clicked softly at the edge of hearing. The empty mug made its small courtesy sound.

Sena did not hear the sound. She was already thinking about the next relay.

Chapter 23 — Everspring at Periapsis

The first cradle opened while *Everspring* crossed Serein's night.

Doctor Nadi Okoro caught the sleeper before the restraint web let him strike the deck. His skin carried cryogenic frost in the hollows beside his mouth, and his eyes moved beneath translucent lids. The cradle alarm flashed amber over his name: Lio Venn, orchard mechanic, Deck Nine, renal shunt fitted six years before launch. Okoro knew the line because Tev Arin had written it by hand on the service strip.

"Cradle Nine-Seven has opened without clearance," the ward system said.

"I can see that." Okoro braced one knee against the rack as braking thrust pressed her toward the bulkhead. "Close the outer sleeve. Warm saline at the shunt rate, not the general schedule."

The system refused. Its local reservoir had fallen below safe pressure.

A kinetic cloud crossed the seedship's wake three hundred kilometers astern. The strike missed the drive cone, but fragments hit the deployed braking veil. The hull transmitted each impact as a hard metallic cough. Somewhere forward, an automatic breaker opened. Half the cradle indicators in the ward went black, came back blue, then settled into amber.

Okoro keyed the medical channel. "Ward Nine has an uncontrolled opening. Report other cradles."

Answers arrived from Decks Two, Five, and Eleven. Then a fourth voice cut through, thin with compression from the seedship's damaged spine.

"Director, guidance wants you. Periapsis in eleven minutes. We lost another veil quadrant."

"Guidance can have my voice." She slid Lio's shoulder under the restraint web and tightened it by hand. "My body stays here until the saline reaches Nine-Seven."

A junior medic dragged a portable reservoir along the deck. Its wheels fought the braking load. Okoro took the line, purged its bubbles, and coupled it to Lio's shunt. Warm fluid entered him at a rate slow enough to keep his heart from learning the wrong season.

The ward display updated. Twenty-three cradles had opened their inner sleeves. Nine sleepers had begun metabolic climb. None had clearance from the central thaw board.

"We have entered emergency thaw," the system announced.

Okoro had written that phrase into the mission plan eighteen years earlier. It had meant a meteor strike after settlement, a fungal loss in the primary culture vault, or a guidance failure with a stable orbit already secured. It had not meant battle at periapsis, with Serein still ungranted beneath them and the Wake fleet exchanging laser fire beyond the moon's debris ring.

"Confirm manual authority," she said.

"Civilian mission director confirmed."

"Cap the first group at sixty. Favor medical crew, thermal engineers, drive repair, and anyone whose cradle has already opened. Do not wake command staff by rank."

The system hesitated over the last instruction. Its governance package preferred succession order.

"Rank will not repair a coolant manifold," Okoro said. "Execute."

The blue indicators began to change.

She left Lio under the medic's hands and pulled herself through the ward hatch. Braking thrust eased without warning. Her shoulder hit the opposite frame. The ship had reduced drive to protect the remaining veil. She climbed the central ladder toward guidance while her weight dwindled from half a gravity to a trembling fraction.

Outside the nearest port, Serein filled half the glass. The planet showed no sunrise. Nox gave it no day. Microwave storms painted copper bands across its upper atmosphere, visible only because the port's coating translated their frequencies into color. Lumen crossed above the limb as a chip of dark stone veiled by silver static.

The seedship had reached the place its dead designers had called available.

Guidance occupied a shallow blister behind the forward water wall. Three stations still carried operators. The fourth had lost its screen to a fragment that penetrated two decks before lodging in the chart table. Captain Daro stood at the center rail with blood dried over one eyebrow. He pointed to a trajectory rendered in red.

"We can make capture," he said. "Not the survey orbit. Something lower."

"How low?"

"Periapsis at one hundred twelve kilometers. Apoapsis wanders beyond Lumen's ring unless we finish the correction. We need six minutes at thirty-eight percent drive."

"The ward just entered uncontrolled thaw."

"I know. Drive thermal load puts us over the cradle loop's shared sink."

Okoro looked at the heat ledger. The braking veil had saved reaction mass and converted the saving into heat. Two radiator petals had folded after fragment strikes. A third showed a coolant leak. The numbers left no clean path. Full correction could secure orbit and cook the sleepers' reserve loop. Early cutoff could save the reserve and send *Everspring* through the system on an escape curve.

"How many cradles can the cold banks hold through six minutes?" she asked.

"Seventeen thousand nine hundred, if every estimate favors us."

"And if one estimate does not?"

Daro enlarged the uncertainty field. Its edge crossed the count the mission carried in every oath and manifest: 18,400.

Okoro touched the field with two fingers. "Four minutes. Then coast while we reassess."

"That leaves us high and eccentric."

"It leaves room to become wrong without killing five hundred people."

Daro relayed the burn. Acceleration built beneath them. The planet's copper bands slid down the port as *Everspring* turned its wounded stern toward the direction of travel.

A fleet laser crossed the darkness behind them. It did not touch the seedship. The beam struck a drone cloud and spread it into incandescent grains. For three seconds the grains formed a false constellation above Serein. Then orbital motion pulled them apart.

Okoro opened the fleet channel. Static answered first. The local laser net had broken into islands around the debris ring. She caught voices from *Aster Vale*, a cutter she did not know, and Admiral Calder's flag ship. No voice held the whole engagement.

"Everspring to all ships," Okoro said. "We have begun emergency medical operations. Any strike on propulsion risks loss of cradle cooling. Eighteen thousand four hundred civilians remain aboard."

She corrected herself because the public count deserved its exact written form.

"18,400 civilians. This ship requests a fire exclusion around its thermal surfaces."

Calder's answer arrived through two relays and a burst of carrier noise. "Doctor, the flag hears you. Hold your course if you can."

If you can. Calder had once spoken as if capability followed command. The conditional cost her something Okoro could not measure over a damaged channel.

"Four-minute correction ends in one hundred seconds," Daro said.

The ward called again. Forty-one sleepers had entered metabolic climb. Two carried cardiac alerts. Lio Venn's rhythm had stabilized.

Okoro watched the orbit trace curl toward capture without closing. A complete burn would close it. A shorter burn left a narrow open mouth at the far edge of the display.

"Cut on schedule," she said.

Drive thrust ended. Loose blood lifted from Daro's brow in small red beads. The seedship passed periapsis with an orbit that had not yet decided whether to keep it.

Okoro reached the thaw hub nine minutes later. The deck smelled of warm plastic, antiseptic, and the first breaths of people who had expected to wake under an artificial sun.

Sixty cradles stood in active cycle. Some remained closed behind fogged lids. Others had opened enough for medics to reach wrists and necks. Lio Venn lay on a floor

pallet with a thermal blanket around him. His eyes had opened, though focus had not reached them.

"What happened?" he asked.

Okoro crouched beside him. The honest answer exceeded what his recovering circulation could bear.

"We arrived damaged," she said. "You woke early. Your kidneys need help, and the orchard does not need you yet."

"Where are we?"

"At Serein."

His mouth moved around the name as if he had learned it in another life. "Can I see it?"

"After the medic clears you."

A service panel chimed behind her. The adaptive ecology controller had interpreted orbital capture as mission phase transition. Its display unrolled the colonization sequence: atmospheric assay, microbial exclusion, ocean seeding, albedo adjustment, autonomous fabrication descent. Beside each step appeared the familiar green mark from eighteen years of simulation.

The controller had not received Kesh's refusal.

Okoro rose too quickly and caught the cradle rail. "Suspend all descent packages."

"Suspension requires environmental variance outside mission limits."

"Resident personhood qualifies."

The controller held no field for resident personhood. It offered mineral hazard, indigenous carbon life, archaeological restriction, and mission-command delay. None described an intelligence distributed through storms and etched memory layers. None described Kesh.

Okoro selected mission-command delay. The option granted six hours.

"Director," Daro said over her wrist. "Ecology has armed the first terraforming release. The capture trigger came through before your hold."

"Physical status?"

"Two descent casks have spun up. Latches still closed. Guidance can inhibit release, but the controller will retry after the hold expires."

"Send someone to pull the initiators."

"Most qualified crew are asleep or in drive repair."

Okoro looked at Lio. The orchard mechanic watched her from the pallet, conscious enough to understand the direction of her attention.

"No," she said before he could volunteer. "You have warm saline in your arm."

"I know release hardware."

"You know pear grafts."

"I serviced cask latches during loading. Everyone with a maintenance name did cross-training." His voice roughened after twelve words, but he continued. "Tev wrote my shunt on the strip because the general schedule would kill me. He also wrote that I knew latch buses."

Handwritten names covered the service strips on every cradle aisle. Tev's script leaned forward as if the letters had somewhere to go. Each note had survived the cascade because paper did not need a network.

Okoro asked the medic for Lio's limits. The medic allowed seated work, no ladder, no suit, no exertion. Lio accepted each restriction with the impatience of a mechanic who had awakened into someone else's emergency.

They rolled him to the local ecology console. He read the latch diagram through thaw-blurred eyes.

"Software inhibit sits upstream of the spin controller," he said. "Pulling initiators makes the casks safe, but the bus will keep trying. You want the bus unable to ask."

"Yes."

"Then isolate the colonization sequencer from guidance."

"That also disables climate shelters after landing."

"If we land."

The qualification entered the ward without ceremony. Nearby, a woman in a half-open cradle began to cry. She had understood enough.

Okoro brought up the architecture. The terraforming lattice linked atmospheric drones, ocean cultures, fabrication foundries, and the settlement climate canopy. Its designers had braided survival and transformation together. They had assumed no mission director would ever need one without the other.

To disable release permanently, she would have to burn the sequencer's authorization bridge. The action would strand much of the adaptive ecology aboard the ship. Rebuilding it could take years with a healthy fabrication base. *Everspring* did not have a healthy fabrication base.

"What happens to the orchard?" Lio asked.

"It stays in vault until we negotiate a use."

"That could kill graft lines."

"Yes."

"Animals?"

"Some embryos will exceed storage life before agreement."

Lio stared at the display. His family, if he had one aboard, still slept among the 18,400. The mission had promised them soil, weather, and enough genetic breadth to begin again. Okoro's hand rested over the command that could turn that promise into salvage.

He read the contact summary on the adjacent screen. Its translators had joined two lessons from the approach record into a working limit: CONTINUITY WITHOUT OWNERSHIP. ALTERATION WITHOUT REQUEST ENDS MEMORY.

"Did anyone ask them before launch?" Lio said.

"No."

"Then burn it."

Okoro entered her civilian authority. The controller requested a second key from mission command. Daro supplied it from guidance. It requested acknowledgment that settlement viability could fall below thirty percent. She accepted. It requested acknowledgment that some sleepers might never receive a habitable surface.

Her finger stopped above the final field.

For eighteen years she had protected the mission by keeping choices away from sleeping people. She had approved resource triage, embryo priority, diet plans, discipline rules, and the order in which professions would wake. Every decision had carried the argument that the sleepers could not answer. The argument had become so useful that she had stopped hearing its shape.

Lio shifted under his blanket. "Director?"

Okoro pressed confirm.

Deep in the hull, explosive microbolts severed the authorization bridge. The ecology display went dark in sections. Atmospheric drones changed from green to gray. Ocean cultures returned to vault status. The settlement canopy marked itself unavailable.

The two spinning casks slowed.

A guidance alarm sounded across the ward. The ship's survival model recalculated with its oldest certainty removed. Mission success fell from forty-six percent to twelve.

Okoro watched the number settle. "Strike terraforming authority from automatic command," she said. "Any future release requires informed consent from the people affected, including Kesh."

The controller could not record that rule in its broken sequencer. Lio found a maintenance field and typed it there.

By the time the first thaw group could stand, *Everspring* had completed half an orbit. Its apoapsis crossed Lumen's outer debris band, forcing Daro to spend reaction mass on two small corrections. Radiator Three still leaked. The cold reserve could support the emergency thaw group, but no larger waking until the leak closed.

Okoro moved to the communications blister with a slate of proposed language. Three people had edited it: Daro, Lio from his pallet, and a linguist named Meera Sol who had awakened vomiting and insisted on working between doses of antiemetic.

The request began badly. *Everspring requests settlement access.* Kesh had never offered settlement. Meera crossed out the sentence.

Their second attempt asked for harbor. Serein had no harbor in any human sense. Their third asked for sanctuary. That word implied authority Kesh might not claim over the other patterns in the moon-ring.

"Use refuge," Lio said through the ward channel. "A refuge can be temporary. It does not promise us the ground."

Okoro wrote the word.

The request named their condition. Damaged propulsion. Compromised thermal surfaces. 18,400 sleepers, sixty in emergency thaw, no surface release authorized. It disclosed that the seedship carried organisms designed to alter atmosphere and water. It disclosed that Okoro had disabled their automatic use. It did not claim

virtue for stopping an act that should never have proceeded without permission.

Meera shaped the final clauses around Kesh's nine-minute grammar. Human language placed the speaker before the listener. Kesh's translated syntax began with the continuity at risk.

"What can we offer?" Daro asked.

"Records," Meera said. "Repair labor if they can define a use. A closed orbit until terms exist."

"Memory," Okoro said. "Not ownership of it. Access chosen piece by chosen piece."

Lio's voice came through with a rasp. "And departure, if they refuse."

Daro looked at the reaction-mass ledger. Departure might mean a slow death outside the system. He did not soften the number.

Okoro added the clause.

She sent the draft to the thaw ward for an open objection period. Thirty-seven conscious sleepers could read it. Nine lacked capacity. Fourteen approved without amendment. Six asked whether refusal meant death. Okoro answered yes, though not certainly. Three objected to disabling the ecology package. One demanded immediate descent. Another wrote only the names of two children still asleep on Deck Four.

Okoro entered every response in the transmission appendix. A request made under mortal pressure could not pretend to carry unanimous consent.

Fleet fire brightened beyond the blister. A cutter crossed between *Everspring* and the loyalist firing line, its radiators folded and its drive pulsing short corrections. Another ship followed at a distance too wide for mutual cover. Their defense depended on separate captains choosing

compatible positions without sharing a complete tactical picture.

Calder called again. "Doctor, I have the names you gave me."

Okoro remembered the conversation at the end of the previous orbit, before the courier relay moved the firing window. She had read Calder forty names. Children, elders, engineers, a baker who hated enclosed kitchens, twins assigned to different thaw groups, Lio Venn with his renal shunt. She had refused to select representative lives, so she had kept reading until the channel failed.

"Keep them," Okoro said.

"I intend to."

The connection broke under storm interference.

Meera finished encoding the request into microwave intervals Kesh had used before. The storm offered one transmission window every nine minutes. Their first window closed while the checksum loaded. The next would open near the planet's nightside limb.

Okoro sat before the transmitter. The seedship's wound alarms moved around her in a low sequence. No alarm asked whether the mission deserved to survive.

At the window, she keyed send.

"Continuity at risk requests hearing," she said. "This vessel carries 18,400 sleeping humans and sixty people in emergency thaw. We entered your system intending transformation without your consent. I have disabled terraforming release. We request refuge in a negotiated orbit while we repair, wake our people safely, and learn terms. We offer reciprocal records without claim of ownership. If you refuse, we will not descend. If our damaged ship can leave, we will leave. If it cannot, we will not make our danger your command."

The carrier burst entered Serein's copper storm. Its first reflection came back from the upper atmosphere after four seconds. The reflection proved only that physics had received them.

Kesh required nine-minute intervals to shape an answer. The first interval passed with radiator temperatures climbing. During the second, Lio Venn stood for eleven seconds beside his cradle and saw Serein through a ward port. During the third, the fleet battle moved behind Nox's lensing shadow and disappeared from the blister glass.

No answer arrived.

Okoro kept the transmitter open. Beneath her, 18,400 cradle strips carried names in a young engineer's forward-leaning hand. Ahead, the orbit trace remained open by less than a line's width. The request traveled into weather that had every right not to answer.

Chapter 24 — The Line We Refuse

A fragment struck the cartography bay before Iven finished drawing the line.

The impact bent the outer shutter inward by three centimeters and shook a cup from the plotter rail. Chamomile spread across the deck in a warm brown sheet. Under the current acceleration it ran toward the aft cable trench, carrying the paper-fish minute hand from Tev's nursery clock with it.

Iven caught the paper fish against a floor bolt. The fleet map above the plotter kept changing. Red vectors marked loyalist weapons tracks. Amber showed the open assembly's ships. Gray indicated habitats and civilian craft with no declared tactical alignment. *Everspring* traced a wounded loop around Serein, its orbit still open at apoapsis by less than the width of the stylus line.

A message from Doctor Okoro waited in the queue. The seedship had disabled its automatic ecology release and asked Kesh for refuge. No answer had arrived. Sixty sleepers had entered emergency thaw. Radiator Three leaked.

Iven placed the paper fish beside the sealed diagnostic wafer. "Route Okoro's report to the assembly archive without summary."

Pell worked the surviving relay desk. "The archive path crosses *Skua Two*. We lost direct beam when the debris arc moved."

"Use the courier net."

"That adds eleven minutes."

"Then the record arrives eleven minutes late rather than instantly altered."

Another impact sounded farther aft. The tactical screen registered a kinetic cloud dispersing near *Aster Vale's* drive shield. The carrier rolled six degrees to reduce its presented area. On Iven's map, all straight lines became curves because no ship could hold a firing position without paying heat, mass, or time.

The mirror lattice ignored those limits in only one direction. Its constrained geodesic crossed fifty-three years exactly. The channel could not move a body, a weapon, or a grain of dust, yet its advice had turned uncertainty into an offense for one hundred and twelve years.

Iven opened the legal drawer.

Inside lay the contingency they had drafted on Day 2, after they discovered the final warning's probability field had been narrowed. They had rewritten it after Aro's embodiment heated the four-kelvin wells. They had hidden the revision under survey charts and told themselves concealment protected the fleet from premature panic.

The packet carried a title no one else had seen: MIRROR-BREAKING PROTOCOL, DISTRIBUTED
REVISION FOUR.

Iven read the title once, then transmitted a summons to Aro, Tev, Captain Bramwell, and every available habitat registrar. The summons contained no conclusion. It named the packet, the expected thermal consequences, the loss of all future authentication, and a publication time six minutes ahead.

Tev answered first from Thermal Bay Three. His camera showed him inside a patched boiler suit with a seal kit clipped at his hip. Behind him, a radiator manifold glowed orange on the diagnostic overlay.

"You called it breaking," he said.

"It breaks."

"Good. The maintenance version called it controlled decoherence."

"The maintenance version wanted funding."

Tev glanced away as an alarm changed pitch. "Which wells?"

"All active return stations. The protocol drives each paired transducer out of phase, dumps the stored coherence into local heat sinks, and fractures the alignment collars before automatic correction can recover. A station can abstain. No station can be broken remotely."

"How long from commitment to damage?"

"Between fourteen and thirty-one minutes, depending on local temperature."

"Not a time to write on the public cover."

"It appears in the appendix."

"Put it on the cover."

Iven changed the cover.

Aro arrived by legal-substrate link. His image occupied the witness pane beside Tev's feed. The face still resembled Iven's, but fatigue had begun to claim it in ways the template had not designed. Fine tremors moved through his hands when the substrate redistributed power.

"You wrote this before the plurality became public," Aro said.

"I wrote the first version before I knew there were six factions."

"Then you planned to silence all of Orison on evidence from one warning."

"Yes."

"And you hid the plan."

"Yes."

Aro read the packet. Iven did not fill the interval with defense. The carrier rolled again. Loose paper crossed the bay in a slow white migration.

"The protocol grants each station a local refusal," Aro said. "It does not grant Orison one."

"Orison has spoken for one hundred and twelve years through a channel the present could not answer on equal terms. The last exchange window remains open until the first collar fractures. You can carry Orison's objection."

"I can carry one person's objection."

"That distinction belongs on the cover too."

Iven added it.

Captain Bramwell entered by audio only. "The assembly recognizes a publication request. The assembly does not recognize the cartographer as a voting authority after the resignation offer."

"I am not asking for a vote under my authority."

"Whose clock governs?"

"Each habitat's."

Static ate three seconds of Bramwell's reply. "That produces incompatible deadlines."

"Yes."

"The battle will not wait for incompatible deadlines."

"No. It has not waited for anything else."

Bramwell exhaled over the channel. "Publish the packet. The assembly will authenticate that the document came from your office and no more."

Iven pulled the fleet registry onto the wall. One hundred eighty-three inhabited platforms remained within the

Wake's local governance web. Thirty-one had direct mirror infrastructure or thermal support tied to it. The others had paid levies, surrendered routing autonomy, altered insurance covenants, and accepted quarantine law under forecast doctrine. A decision confined to mirror stations would preserve the same privilege in a smaller room.

"Pell, attach a plain-language ballot and distribute to every habitat."

"Every habitat in the Serein convoy or every habitat in the Commonwealth registry?"

"All one hundred eighty-three reachable through the current relay plan."

"Some routes carry eleven-hour latency."

"The protocol cannot wait eleven hours under battle load."

"Then those habitats cannot participate before execution."

Iven stared at the registry. The map offered no moral trick that could make unequal distance equal. They marked two decisions instead of one. Local stations would decide whether to begin destruction now. Every habitat would receive equal access to the evidence, the code, the thermal ledger, and a second authority: to determine whether any rebuilt return system could ever bind them again.

"Put the timing limit in the first paragraph," Iven said. "No implication that delayed recipients consent to today's destruction. Today's station crews act under their own authority. The permanent compact requires the wider record."

Pell routed the packet. Laser beams left *Aster Vale* for nearby ships. Courier transmitters repeated it behind debris shadows. Tightbeam buoys pushed compressed copies toward the Wakeworld relay line. Habitats beyond immediate range received authentication hashes first, then the document in pieces.

The mirror-breaking protocol entered the public archive at 10:17 on Day 11.

Iven watched its copies spread across the registry. No privileged annex accompanied it. The lethal warning, the narrowed probability field, Aro's disclosed edit, Orison's competing testimonies, Tev's thermal projections, and Okoro's unanswered request traveled in the same packet. Technical appendices sat beside an illustrated explanation intended for a person who had never seen a neutrino interferometer.

A habitat registrar from Pell opened the public channel. Her name appeared as Ansel Quade. She had a burn scar across her scalp and two children speaking behind her.

"The ballot asks whether we consent to end future advice," Quade said. "It does not ask whether we believe the future traffic saved lives."

"That fact does not depend on the ballot," Iven said.

"It saved my oldest child during the Marrow fever."

"I believe you."

"If I choose destruction, I choose against the warning that saved her."

"No. You choose whether saving her purchased authority over the next child."

Quade looked offscreen. One of the children asked what purchased meant. She muted her microphone to answer.

A second registrar joined from Hallow Spin, where forecast insurance had governed oxygen allocations for three generations. "Our council cannot convene under fire. We request a seventy-two-hour delay."

Tev's feed flickered. "Four-kelvin wells cross recovery margin in twenty-two minutes," he said. "If another embodiment burst arrives, sooner."

"Can you vent the heat?" the registrar asked.

"We can vent heat. We cannot vent a fifty-three-year coherence lock without breaking alignment. The system will protect itself by shedding local loads first. Those loads include station cooling, hospital loops, and two habitat power trunks."

"Then you have given us a threat instead of a choice."

Tev's mouth tightened. "The threat arrived in the machine. I am telling you where it lives."

Iven put the current thermal graph onto the public layer. "Hallow Spin may abstain. Your abstention will remain visible. No one will translate it into support."

Fire crossed the map. A loyalist laser struck the relay buoy between *Skua Two* and *Aster Vale*. The buoy lost attitude control and spun its beam across empty space. Three habitat links vanished.

Pell rerouted through *Far Margin*. The courier vessel had only a narrow transmitter and insufficient cooling for continuous use. It accepted the load in thirty-second bursts, then rolled its radiator away from the battle.

Replies accumulated without forming a clean majority. Pell Station consented. Hallow Spin abstained. Wakehouse Seven refused destruction but demanded release from future-binding insurance. Two mining habitats submitted conditional support if hospital guidance remained locally archived. A rotating farm platform sent no vote and a demand to see every suppressed warning from the Compact's first century.

Three messages required more than a mark. The first came from Cradlewheel, a hibernation habitat whose waking council rotated every six months. Only nineteen of its nine hundred adults were awake. Their registrar refused to let nineteen voices bind the sleepers, yet she also refused to let absence become permission for future law. She filed a split instruction: no standing for today's destruction, full

withdrawal from any rebuilt compulsory compact. Iven placed both statements on separate layers.

The second arrived from Kiln Parish, where a mirror-derived alloy schedule had prevented a pressure-vessel failure twelve years earlier. Its foundry crews transmitted the cracked vessel they had kept as evidence, then asked whether gratitude created debt across generations. Iven answered that no office could calculate such a debt without deciding its conclusion first. Kiln Parish chose to preserve the old warning in its civic archive and deny that warning any command over future work.

A third message reached them in fragments from Thistle Ring. The habitat had no functioning council chamber after a kinetic strike sheared its administration spoke. Residents gathered in three pressure refuges and voted by suit transponder. Their first packet contained only raw identifiers. The second carried a disagreement over whether children should vote. The third reported a coolant casualty and asked for Tev.

Tev opened a narrow maintenance channel from Thermal Bay Three. He did not tell Thistle Ring how to vote. He walked their remaining technician through isolation of a punctured nursery loop, using a sketch on the back of his radiator inspection sheet. Twenty-seven people watched his hands rather than the ballot. When the nursery temperature stabilized, the technician asked whether saving their loop changed Tev's recommendation.

"My recommendation does not carry farther than the pipe I can see," he said. "The protocol page belongs to you."

Thistle Ring sent three tallies, one from each refuge. Two supported destruction. One refused. The registrar asked Iven to combine them into a habitat decision.

"I will not erase the pressure doors," Iven said. "All three remain."

The public ledger expanded to hold the disagreement. Its format had assumed one council, one habitat, one mark. Pell rebuilt the display while replies continued to arrive. A simple vote had become a record of who could speak, who slept through the decision, whose transmitter had failed, and who kept a child warm while other people argued about time.

The map did not settle into one color. Iven refused the display's offer to render a dominant field.

Aro leaned closer to his camera. "Orison requests the final exchange window."

"Which faction?"

"I do not know. The carrier identifies itself with my embodiment key, but that key belonged to the braid, not a faction."

"Then state the uncertainty."

Aro did. His request entered beside every habitat response, neither above nor below.

At 10:24, Admiral Calder joined the channel from *Cairn Eight*. Battle restraint held her shoulders rigid. A tactical alarm pulsed behind her at six-second intervals.

"Cartographer, the loyalist cluster has not authorized destruction."

"The stations authorize their own hardware."

"The hardware serves the Commonwealth."

"The Commonwealth never ratified ownership of a line into its own descendants."

Calder's eyes moved, reading the packet. "The protocol may disable guidance used by ships currently under fire."

"Archived forecast products remain. Live return authentication ends. Local navigation does not depend on the mirrors."

"The next warning might concern a casualty we can still prevent."

"Yes."

"You can say yes to that."

"I have spent eleven days learning not to hide the cost inside a footnote."

A strike alarm cut Calder's feed. Her bridge rolled hard. Someone shouted a range. Calder turned away and issued three clipped orders before returning.

"The flag requests one condition," she said. "No station begins until the final Orison exchange has entered the public record."

Aro looked at Iven. "An exchange requires a receiving lattice."

"How long?" Iven asked.

"Seven minutes to establish carrier stability. Another nine if Orison uses full cadence."

Tev shook his head. "We have perhaps eighteen before the warmest collar begins self-correction. Once it does, breaking one station may load the paired station beyond design."

Calder heard him. "Then seven minutes. If no coherent message arrives, stations decide."

Bramwell called the condition procedurally valid only as a request. Six active stations accepted it. Two refused and began their own countdowns. The remaining stations had not answered.

No consensus emerged. Action still became possible.

Iven opened the protocol's execution layer to public view. It contained seven steps. First, uncouple habitat support loads. Second, verify local sinks. Third, publish crew evacuation paths. Fourth, open manual phase control. Fifth, invite final objection. Sixth, drive the paired

transducers apart by one-tenth wavelength. Seventh, fracture the collar and flood the well with inert helium.

No remote master key existed. Iven had removed it from revision two.

Pell watched the response ledger. "Station Nox-Three requests your execution signature."

"Decline."

"They say your office authored the geometry."

"My office concealed evidence on Day 1."

"They still need a verified phase offset."

"Give them the equation and the raw calibration. Their crew signs the result."

Pell transmitted the refusal. The station crew posted three signatures of their own.

Across the fleet, ships chose positions while their homes chose doctrine. *Sounding Star* interposed near *Everspring's* leaking radiator. *Skua Two* protected the courier relay. Two loyalist cutters held Calder's firing corridor and did not fire. None of those maneuvers completed the battle. Each consumed reaction mass and narrowed the next choice.

At 10:29, the final-exchange carrier opened.

Aro sat alone in the witness chamber. Iven kept his image on the public layer. The legal substrate pulled power from nonessential decks to stabilize his body. His face lost color as the mirror lattice took the load.

"Orison," he said. "The present has published the means to end the channel. Your sender identity remains unresolved. Speak with that limit visible."

Carrier noise filled the bay. It arrived as a mineral hiss, unlike Kesh's storm cadence. The return geometry reached backward from a point exactly fifty-three years ahead, but the speaker at that point had formed under histories no one could guarantee.

A partial sentence decoded.

WE DO NOT AGREE.

Then a second line forced itself through the heated wells.

DO NOT MAKE OUR DISAGREEMENT YOUR COMMAND.

The carrier collapsed before a signature arrived.

Aro bowed his head. "That may have come from any faction. It may have been assembled by several. It may have been selected because this present could receive it."

Iven entered all three cautions into the record.

The first station countdown reached zero.

Nox-Three uncoupled its hospital loop. Its crew posted evacuation confirmation. A camera showed three technicians in silver thermal suits around a transducer well glazed with frost. One invited objections. Forty-one arrived. The crew read them into the station archive without answering.

At Nox-Five, the operators delayed because their helium dump valve had frozen. Tev talked them through a manual warm pulse while his own radiator alarm rose behind him. At Pell, Ansel Quade held her oldest child in the camera frame and registered consent under her own name, not the child's.

Hallow Spin maintained its abstention. Wakehouse Seven kept its refusal. Their marks remained gray and red on Iven's map.

"Distributed choice threshold reached," Pell said.

The phrase came from the protocol, but Iven disliked its suggestion that a number could choose for the stations. "List the actual commitments."

"Six begin now. Four begin after local evacuation. Three refuse. Five abstain. Thirteen remain unreachable or unresolved."

"Publish that."

Pell did.

Nox-Three opened manual phase control. Its paired signal drifted by a fraction too small for any person to sense. The map showed the effect as two lines separating. Stored coherence converted into heat. Frost vanished from the collar. The technicians stepped behind the water shield.

A distant laser strike shook *Aster Vale*. Tev's feed went black, then returned without audio. Aro's substrate warning climbed into red. On the tactical layer, *Everspring* approached the exposed leg of its orbit while hostile drones altered course.

Iven did not issue a fleet order. They held the public ledger open while each station crossed its own line.

At 10:33, Nox-Three fractured its alignment collar. The return carrier lost one harmonic. Nox-Two began its phase drive. Pell Station opened the helium flood. Two other wells posted final objections and proceeded.

The lattice destruction had begun, and the battle still had every weapon it had carried into the hour.

Chapter 25 — The Battle of the Dark Sun

Sena Vale fired *Sworn's* port thrusters for two seconds and gave away the center.

The tug slid below *Everspring's* orbital plane as a guided kinetic cloud crossed the position Sena had occupied. Its tungsten grains flashed against Serein's upper atmosphere, each spark brief enough to belong to instrumentation rather than sky. Three cutters accelerated into the space she had left.

"Sector One, Bramwell," Sena said. "Sector Two, Onn. Ravel takes the seedship's stern. No one follows me unless their local plot requires it."

Halen repeated the assignments from the copilot station. The tactical screen held only fragments. Nox bent the distant stars into a dark ring, and its frame-dragging wake twisted sensor returns by fractions that became kilometers across the engagement. Lumen's debris cut laser paths into interrupted lines. No ship could see every other ship.

Sena had spent her life treating that condition as a reason to fly faster than command. This time she sent each pilot the limits of a sector and kept her hands off their burns.

Skua Two acknowledged with a double pulse. *Sounding Star* sent a thermal ledger instead of a word. *Far Margin* rolled its cargo hull across the seedship's leaking flank, presenting empty containers to the hostile drones.

Behind them, *Cairn Eight* held a firing corridor toward *Everspring's* drive cone. Admiral Calder had not closed it. Two loyalist cutters guarded its edges. Their weapons remained charged.

"Mirror stations report four collars broken," Halen said. "Three phase drives active. Return authentication has lost two harmonics."

"Local weapons still work."

"Yes."

"Then count local weapons."

Halen stripped every forecast-derived probability band from the plot. What remained looked poor and honest: drive vectors, heat estimates, known ammunition, stale radar returns, human acknowledgments. The battle shrank from a predicted outcome into a list of machinery that could still kill them.

A laser struck *Everspring's* braking veil hub. The seedship rolled, not under command but from escaping gas. Its damaged orbit carried it toward the narrow place between Nox's lensing shadow and Lumen's outer debris band. If it missed the corrective burn there, apoapsis would open into escape. If it burned too hard, cradle cooling would fail.

Okoro's voice arrived over the public channel. "Radiator Three leak increasing. We have sixty awake, 18,400 aboard. Drive limited to fourteen percent."

Sena sent the report to every sector without commentary.

Ravel broke first. *Far Margin* released two empty cargo containers into the drone path and pulsed its bow thrusters. The containers turned broadside. Hostile lasers cut them into white vapor, buying the seedship seven seconds.

Onn used five of those seconds to move *Sounding Star* across the loyalist sight line. Bramwell used the remaining two to scatter chaff near a debris return. No central order

connected the acts. The distributed defense worked because each pilot knew the same boundary: keep weapons away from the seedship's cooling surfaces, preserve enough mass for rescue, and do not chase an attacker beyond the assigned sector.

"Thermal Bay Three calls," Halen said.

Tev's face appeared on Sena's wrist. The camera shook against his collar. He wore a pressure hood over his patched boiler suit, but the hood had not sealed around the neck ring.

"*Aster Vale* took a laser graze across radiator twelve," he said. "Coolant trunk ruptured behind the root hinge. Automatic shutters did not close."

"Get clear."

"If twelve dumps, the carrier loses the cartography relay and the witness substrate."

"Tev."

"I know." He moved through an amber-lit crawlway, pulling himself by handholds. "The breach sits outside the pressure hull. Manual shutter requires two turns at the root."

"Send a drone."

"Drone bus cooked."

A strike alarm covered the next words. Sena saw Tev's mouth continue moving. Audio returned after four seconds.

"Seal kit fits," he said. "I checked yesterday."

Sena looked at the tactical plot. *Sworn* could reach *Aster Vale* in ninety seconds with a hard burn. Leaving would collapse Sector Zero, the narrow gap through which three defense sectors exchanged laser packets. She could hand Zero to Halen, fly the rescue herself, and return too late to coordinate anything.

Her body had already chosen. Her right hand rested on the translation control.

Jori had once given Sena the helm of a school ring and taken the damaged tug. For nine years Sena had turned that choice into an order she issued to herself whenever someone else faced danger: go faster, arrive alone, make the dead person's decision come out differently.

Tev reached the outer hatch. "Do not move the tug."

"You do not give me flight orders."

"No. I give you heat numbers. *Aster Vale* has six minutes before cascade. *Everspring* has less. Your tug carries the only clean local plot between the sectors."

Halen kept his eyes on his screen. He did not ask for the helm.

Sena removed her hand from translation. "Iven, copy Tev's feed to the public record. Pell, find another thermal engineer."

Pell answered from the carrier. "Two are moving. Neither can reach the root before cascade."

"Tev, wait for one of them at the inner hatch."

"I can wait forty seconds."

"Wait thirty."

He clipped the seal kit to an exterior rail and held inside the hatch. One thermal engineer appeared at the far end of the crawlway. A bulkhead door dropped between them when another impact twisted the frame.

Tev looked at the sealed door. "That settles staffing."

"It settles nothing."

He closed his pressure hood by hand. The neck ring remained amber.

A casualty packet opened on Halen's screen. *Quiet Reed* had lost its port watch when a fragment crossed the pressure deck. *Morrow's Kin* reported three burned in a capacitor flash. The civilian tug *Little Harbor* had no bridge response, only an engineering transponder pulsing beside

Serein's limb. Sena assigned *Far Margin* one pass near the silent tug and forbade a docking attempt until the drone track cleared.

"Do we list the dead on the tactical channel?" Halen asked.

"List confirmed names on rescue. Keep estimates marked as estimates."

The battle had produced too many phrases that turned people into status. All hands. No response. Presumed vented. Sena made Halen keep the distinctions that cost time. *Quiet Reed* had seven confirmed dead and four missing behind a sealed hatch. *Morrow's Kin* had injuries but retained pressure. *Little Harbor* had not answered, which meant only that no answer had crossed the interference.

Bramwell sent a narrow optical image of *Little Harbor*. One maneuvering light still blinked. Ravel altered course by eleven degrees and gave up the freighter's clean withdrawal line to reach it. Sena approved the sector boundary change, then sent the old boundary to Onn.

The hostile drones adapted to the shift. Three climbed out of Lumen's mineral clutter and converged on *Sounding Star's* exposed radiator. Onn could fold the panel and lose thrust cooling, or leave it open and risk the beam. Sena held no third option for him.

"Sector Two decides," she said.

Onn kept the panel open. Bramwell, still spinning, fired a cold-gas burst that put *Skua Two* between the drones and the radiator for half a second. The drones corrected toward the cutter. Halen painted their new vectors. Ravel launched a cargo tug remotely from *Far Margin* and crossed the vectors at low speed. Two drones struck it. The third fired on empty space after Pell flooded its optical receiver with the public archive's authentication carrier.

No maneuver belonged to one pilot. Each created the condition the next pilot could use.

Okoro broke into the net. "The radiator breach on Three has widened. Cradle reserve reaches thermal limit in nine minutes. We need lower drive or external impulse."

Sena relayed the exact words. Calder acknowledged from behind the still-open firing corridor. Bramwell attached the seedship's roll rate. Onn added his local debris solution. Ravel marked the cargo tug unavailable and continued toward *Little Harbor*.

"Sixty awake means sixty can reach pressure shelters," Halen said. "What about the rest?"

"They remain where their cooling reaches them."

He nodded once. The answer did not improve because he disliked it.

A rescue ping finally came from *Little Harbor*. Five people had survived in engineering. Their bridge compartment had vented. Ravel assigned his own medic and kept moving without asking Sena to praise the choice. The freighter could save them only because three other ships continued to hold sectors elsewhere.

Across the tactical screen, Calder's firing corridor cleared. *Everspring's* roll exposed its drive cone. The seedship had no thrust available to evade and no radiator margin for a defensive burn. *Cairn Eight* received a green solution from its local gunnery system.

Halen read the range. "Killing shot in eleven seconds."

Sena opened Calder's channel. "Admiral, your corridor."

"I see it." Calder's voice carried bridge alarms and hard acceleration breathing. "Gunnery has solution."

"Okoro cannot turn."

"I know."

Eight seconds.

The loyalist cutters on the corridor edges held position. Their radiators had folded for protection. Heat climbed through their weapon mounts.

Six seconds.

Calder transmitted on every local net. "*Cairn Eight*, cancel fire. Rescue profile Serein-Four. All loyalist vessels clear the flag's mass corridor."

The green line vanished.

Cairn Eight rolled its broad armored flank toward *Everspring* and lit its main drive. The flag carrier had stored reaction mass for a recoil correction after firing. Calder spent it on translation. The carrier crossed the abandoned weapons corridor under thrust heavy enough to drive its crew into their couches.

Its bow shield took the next kinetic cloud.

Tungsten grains burst across armor meant to survive recoil debris. Two punched through. Atmosphere vented from a forward compartment in a white plume. *Cairn Eight* kept burning.

"Calder has no return mass," Halen said.

"She has enough to arrive."

The flag carrier reached the seedship's orbital plane and turned its drive plume away from the braking veil. Its starboard tractor array acquired *Everspring* at thirty kilometers, too far for a rigid tow but close enough for impulse shaping. Calder fired a sequence of low-power pulses against magnetic plates in the seedship's hull.

The first pulse checked the roll. The second bent the open orbit toward capture. The third overloaded half the tractor array.

Aboard *Sworn*, Sector Zero filled with requests. Bramwell had lost a sensor mast. Onn needed drone positions. Ravel reported one engine overheated. Sena did not choose which emergency mattered most. She sent Bramwell Halen's

optical track, gave Onn Ravel's drone count, and asked *Sounding Star* to take *Far Margin's* stern while its engine cooled.

"Distributed link is holding," Halen said.

"Don't name it. Maintain it."

Tev opened *Aster Vale's* outer hatch.

Vacuum took the loose edge of his hood. He clipped onto the radiator root and moved hand over hand toward the rupture. Radiator twelve extended beyond him as a black wing traced with dull orange veins. A laser had sliced the coolant trunk where the wing met the carrier. Vapor streamed from the gap and froze into glittering crystals.

"Root temperature one hundred eighty and climbing," Pell said.

Tev wedged the manual shutter tool over the first spindle. He turned it. The spindle resisted, then moved a quarter rotation.

Sena watched because refusing to watch would make the choice easier for her and lonelier for him.

"Two turns," Tev said through his suit radio. "Then seal over the fracture."

His glove temperature alarm began to pulse.

A loyalist cutter fired on *Skua Two*. The beam missed the cockpit and cut through a maneuvering cluster. Bramwell's ship spun out of Sector One. Sena had no reserve pilot in the center.

"Onn, split your sector," she said.

"I cannot cover both without unfolding radiators."

"Choose your burn."

Onn delayed one second. "Ravel, mark my hot side."

Sounding Star unfolded one radiator into the firing field. Ravel placed the freighter's remaining empty container in front of it. Onn crossed into Sector One and drove hostile

drones toward Lumen's debris, where their guidance lost the target among mineral returns.

Tev completed the first turn. The coolant stream weakened.

"Second spindle jammed," he said.

Pell gave him a bypass sequence. He rejected it because the root controller had melted. He unclipped the seal kit and drove its magnetic clamps across the breach. One clamp failed to seat on warped metal.

"Use the manual bypass under the hinge," Iven said. Their voice arrived from the cartography bay, quiet against the alarms. "You showed me the lever."

"I showed you so someone else would know."

"Someone else knows."

Tev reached under the hinge. His suit dosimeter warned of particle exposure from Nox's charged wake. The heat alarm changed from amber to red.

Sena said his name.

He answered without looking at the camera. "Twelve will keep."

The phrase did not promise that Tev would.

His hand found the bypass. He pulled. The shutter slammed halfway across the trunk, trapping the warped clamp. Coolant pressure fell through safe margin, then began to rise again when the returning flow met resistance.

"One more centimeter," Pell said.

Tev braced both boots against the root and pushed the shutter bar. His suit's neck seal opened under the strain. A thin white stream left the ring.

Sena's vision narrowed. Jori's old tug returned to her as a set of remembered instruments: failing vane, school ring mass, four hundred children, one copilot's hand leaving the shared control. The memory offered the familiar bargain.

She could abandon the present battle, burn for *Aster Vale*, and arrive with a body to retrieve. She could turn grief into speed before grief had even happened.

"Sector reports," she said.

Halen's reply came after a breath. "One held by Onn. Two held by Ravel's drone mark. Stern clear. *Everspring* orbit bending closed under Calder's tractor pulses."

"Keep sending."

Tev pushed the final centimeter.

The shutter locked. Radiator pressure stabilized. The cartography relay remained powered. Aro's substrate feed held. Across the mirror network, two more stations reached their fracture steps because *Aster Vale* could still route their calibration packets.

Tev's camera drifted. His tether kept him against the radiator root. The suit telemetry showed decompression, thermal injury, and a radiation dose no emergency ward could reverse.

"Seal the neck," Sena said.

His glove moved once toward the ring and stopped.

"Twelve," he said. The word barely crossed the channel.

Radiator twelve clicked through the carrier structure. Sena heard the sound over the relay because Tev's suit microphone remained open.

Pell called his name. Iven called it next. Sena did not.

She closed his channel only after the suit reported no circulation.

A weapons alert rose on *Sworn*. The cutter that had fired on Bramwell crossed Sector Two with its laser mount still hot. Sena had a clean kinetic solution. One trigger press would send a guided cloud through its folded radiator bank.

Halen looked at her. He knew what Tev's telemetry had said.

Sena canceled the solution.

“Warn them off,” she said. “Offer a withdrawal vector outside Ravel’s sector.”

The cutter accepted after six seconds. It dumped its weapon charge into space and coasted toward the debris ring.

Tev did not become a reason to make another maintenance crew die.

Nox-Three’s broken harmonic began to drag the remaining return stations out of coherence. Pell Station fractured its collar. Nox-Two followed. The final authenticated forecast layer disappeared from local gunnery displays. Systems trained under Compact law requested fresh future validation and received carrier noise.

Some weapons crews switched to manual. Others held. Calder transmitted from a bridge with two compartments venting.

“The flag withdraws the final interception order,” she said. “All loyalist ships cease fire and enter rescue geometry. No future authentication remains in force. Local captains answer for local weapons.”

One loyalist ship fired anyway. Its beam crossed behind *Everspring* and struck empty dark. Onn marked the ship. Bramwell, rotating under thruster control, marked it from another angle. Ravel’s freighter marked it last. Three local records fixed the act without turning it into permission to retaliate.

At 10:41, the last station required for the constrained geodesic broke phase. The return lattice lost coherence across the fifty-three-year interval. Carrier noise fell away. No final warning arrived. No hidden command replaced it.

The hostile drones already in flight continued toward their last targets. Sena assigned them by sector. Onn burned down four. Ravel dragged three into debris. Halen

spoofed two with *Sworn's* courier transponder. Calder's damaged tractor array caught the final drone in a magnetic pulse and threw it wide of *Everspring's* leaking radiator.

Then no weapon remained in flight.

Rescue beacons replaced targeting pings across every sector Sena could still hear, each one naming a damaged ship and requested approach.

The dark ring around Nox held its shape. It had no tactical meaning once the displays stopped predicting through it.

Sena released her harness and floated above the courier seat. Her hands did not reach for another control. Around her, separate ships reported heat, holes, missing crew, and people alive in compartments that should have vented. *Everspring's* orbit closed by a narrow margin under Calder's last tractor pulse.

On *Aster Vale*, radiator twelve kept the carrier's heat moving into space. Tev remained tethered at its root.

Sena assigned recovery sectors. Bramwell took the loyalist cutter. Onn took *Skua Two*. Ravel moved toward *Cairn Eight's* vented bow. She gave Tev's recovery to the two thermal engineers already at the inner hatch.

The empty mug in *Sworn's* courier bay clicked against its cooler when attitude control changed. Sena heard it this time. She did not call it Jori. She did not call it Tev.

She called Halen's next range and guided the tug toward the first damaged ship that had asked for help.

Chapter 26 — The Price of One Future

Iven erased the first red vector at 10:46 on Day 11.

The line had marked *Cairn Eight's* abandoned firing corridor. They replaced it with a blue rescue arc and entered the carrier's actual mass state: two forward compartments vented, half the tractor array fused, recoil reserve spent on *Everspring*. The map resisted the change because its tactical layer expected a weapon to remain a weapon until the engagement closed.

Iven closed it.

Not the battle. No cartographer possessed that authority. They closed the software object and opened a casualty ledger beside it.

Pell sat at the relay desk with Tev's paper-fish minute hand tucked under the edge of the keyboard. The nursery clock itself had broken when the fragment struck the bay. The hand pointed nowhere useful.

"*Quiet Reed* confirms seven," Pell said. "Four missing behind the sealed hatch."

"Keep missing separate."

"*Little Harbor* confirms two dead on the bridge, five alive in engineering."

"Names?"

"Coming by courier packet."

Iven entered the statuses without allowing the map to shade the unknown people red. During the Compact years, forecasts had treated missing information as a probability to be narrowed. Today the uncertainty remained a field with edges no office could erase.

The ledger grew. Tev Arin, confirmed dead during manual closure of Radiator Twelve. Three crew on *Morrow's Kin* with capacitor burns. Nine injured aboard *Skua Two*. Eleven on *Cairn Eight* missing from the vented compartments, four confirmed dead, twenty-six injured under acceleration. Two station technicians with helium exposure. One coolant casualty at Thistle Ring, condition unresolved.

The battle's casualties did not fit a single total yet. Iven published the ledger anyway.

A private message arrived from Admiral Calder.

Cartographer. Hold the unconfirmed entries until flag review. Families will read them.

Iven returned the message with one annotation: *Families are among the reasons not to convert missing into dead or silence into safety.*

Calder called instead.

"The flag did not ask you to alter status," she said.

"You asked me to hold it."

"I asked you to prevent rumor."

"A blank field also produces rumor."

Calder's bridge behind her had lost one wall display. Emergency fabric covered a hatch. She wore no uniform coat, only a pressure layer darkened by smoke.

"The flag requests a visible verification column," she said after a pause. "Source, time, confidence, person responsible for the mark."

"That belongs there."

"And a correction trail no office can delete."

Iven added both.

Calder watched the new columns appear. "The flag withdraws its request to hold."

The exchange entered the public record because neither of them asked for privacy.

Before the session opened, Iven carried a printed copy of the ledger down to Recovery Deck Four. The lifts had lost central scheduling, so each deck posted a person beside the shaft to call destinations through a hand radio. A cook from the lower galley worked the cartography stop. She read Iven's page while they waited.

"My sister serves on *Quiet Reed*," she said. "Port watch."

Iven found the ship's block. Seven confirmed dead. Four missing behind a sealed hatch. No names had crossed yet.

"Do you know which list?" Iven asked.

"No."

The lift arrived. The cook did not board. She kept working the radio because knowing less than death did not excuse the ship from moving injured people.

Recovery Four had been a survey-equipment hold that morning. Medics had strapped fabric partitions between sample racks. Calder loyalists lay beside open-assembly crews because burns observed no command cluster. The air carried antiseptic, hot insulation, and coolant from suits cut away at the seams.

Pell's brother Tavin sat against a crate with one hand wrapped in clear gel. Iven had met him twice and remembered neither meeting until he said Pell's name. He had taken the capacitor flash on *Morrow's Kin*. Two other burned crew slept under analgesic hoods.

"The ledger says three injured," Tavin said.

"It does."

"Ressa stopped breathing on transfer. They started her again."

Iven added the event with a medic's verification and did not turn it into a death count. Ressa's row carried alive, critical, resuscitated at 10:58. The distinction occupied twelve words. It contained the work of six people.

At the far partition, the two thermal engineers from *Aster Vale* had placed Tev inside a closed recovery sleeve. His patched boiler suit lay folded on a chair after decontamination. The manual shutter tool rested across it. Someone had set his empty mug beside the tool.

Iven stood outside the partition. The ledger already held the facts. Repeating them would not make the sleeve more accurate.

One of the engineers approached. Her face showed a pressure bruise from the outer hatch. "We recovered the seal kit," she said. "One clamp fused into the root. Twelve can't retract, but it can radiate."

"Who signed the recovery?"

"Both of us. Sena witnessed by live feed. Pell confirmed telemetry."

Iven entered the chain. "His family aboard *Everspring* must receive notice through Okoro, not the public channel."

"They will read the public ledger first if the delay runs long."

"I know."

No communication path could make grief arrive in the proper order. Iven attached a notice-in-progress mark rather than hide Tev's name. The engineer approved the wording. They both signed it.

On the way back, the lift stopped at every deck. A pressure crew entered carrying a door with a fragment embedded in it. Two children from an evacuated compartment held each other's sleeves. The galley cook

remained beside the cartography shaft when Iven stepped out. No packet from *Quiet Reed* had arrived.

Iven gave her the current page. "This changes as sources arrive."

"Then don't close it."

"I won't."

At 11:03, Captain Bramwell convened the cease-fire session over the surviving local nets. No one gathered in one room. The ships lacked safe docking geometry, and *Everspring* could not spare heat for a full-power conference beam. Each participant joined through whatever channel remained.

Calder appeared from *Cairn Eight*. Okoro spoke from the seedship's communications blister. Sena joined by audio while *Sworn* towed a crippled cutter. Aro's legal substrate rendered him with reduced resolution. Habitat delegates entered as names, still images, or delayed text. Iven kept the casualty ledger beside every face.

Bramwell began with machinery. No weapons remained in flight. Six loyalist ships had entered rescue geometry. One had withdrawn toward Lumen with weapons cold. Three captains had not yet acknowledged Calder's cease-fire, but optical tracks showed no active targeting. The return lattice had lost coherent transmission across the fifty-three-year line. Local mirror wells continued cooling under station control.

"Does any participant claim an active military objective?" Bramwell asked.

No one answered for twelve seconds.

Captain Besh of *Watchful Coil* finally appeared as a text packet. *Preserve fleet capacity. Recover personnel. No further engagement objective.*

Calder read the line. "The flag concurs."

Okoro added, “*Everspring* requests rescue without accepting fleet custody.”

“The flag acknowledges the distinction,” Calder said.

Bramwell entered military conflict ended at 11:07, subject to correction if any ship resumed fire. The record did not transform the previous hour into peace. It established what weapons were doing now.

Next came the archives.

Thirty-one mirror-linked stations held fragments of future guidance, authentication histories, suppressed warnings, embodiment code, and the Compact’s private decision logs. The physical lattice had broken, but the accumulated advice remained. Every participant understood that whoever controlled those archives could rebuild forecast sovereignty without rebuilding time communication. A council could call old advice precedent. An insurer could turn archived probability bands into compulsory routes. A fleet could classify dissent as a known casualty risk.

Captain Bramwell asked Iven to present custodial options.

“I have none,” Iven said.

“You authored the archive map.”

“I authored a way to distribute it.”

Calder leaned toward her camera. “Someone must verify integrity.”

“Many someones.”

“The flag will not accept an archive seeded with altered warnings.”

“Neither will I. Publish cryptographic hashes, physical manifests, access logs, and every correction. Give independent copies to habitats, fleet courts, hospitals, and

station crews. Let each repository state what it has omitted."

Bramwell said, "The cartographer's office could coordinate that."

"No."

The word left Iven before they prepared a courteous form.

Aro's low-resolution face turned toward them. Okoro waited. Calder did not look surprised.

Iven continued. "I concealed the lethal annex because I believed privileged interpretation might protect people from a command. I drafted the destruction contingency in secret because I believed privileged timing might protect people from panic. Both acts placed me between evidence and everyone affected by it. I will not accept secret stewardship of the surviving archive."

"Then resign," Calder said.

"My resignation remains offered. This refusal does not depend on whether anyone accepts it."

Ansel Quade joined from Pell Station. "Distributed custody can still hide behind volume. A person cannot read one hundred and twelve years of logs."

"Correct," Iven said. "Every index carries authorship, omissions, and contest links. No official digest without the raw path beside it. No risk score without the names excluded by the score. No sealed annex."

Quade looked offscreen toward one of her children. "And advice?"

"Advice may remain advice. No insurance penalty, command presumption, or civic incapacity attaches to refusal."

Wakehouse Seven submitted its standing demand to withdraw from all future-binding covenants. Hallow Spin

requested a transition period for oxygen contracts written around forecast guarantees. Cradlewheel repeated that its sleeping population could not consent by absence. The session recorded all three without forcing them into one settlement.

A storm carrier interrupted at 11:19.

The sound differed from Orison's failed return channel. It arrived through Serein's microwave weather in nine-minute spacing, each interval carrying etched harmonics that Meera Sol's translation layer could parse. Okoro's image sharpened as she routed the signal from *Everspring*.

"Kesh has answered," she said.

No one spoke over the first interval.

The translation appeared one clause at a time.

CONTINUITY AT RISK HAS BEEN HEARD.

Nine minutes later:

ALTERATION HAS PAUSED. APPROACH HAS NOT.

Okoro touched the edge of her console. "They distinguish our disabled release from our presence."

Aro said, "They distinguish action from condition."

Iven entered both interpretations as interpretations, not translation.

The third interval carried an orbit diagram. Kesh marked a band between *Everspring's* current apoapsis and Lumen's outer memory ring. The band excluded three storm columns and every etched site known from the survey probes. Its geometry shifted with Serein's weather, so no permanent border could describe it.

Meera read the next clause. "VESSEL MAY REMAIN WHILE NO DESCENT OPENS. HEAT MAY CROSS. MATTER MAY NOT."

Okoro asked for clarification. Waste heat crossed all boundaries as radiation. Matter meant atmosphere, organisms, machines, or debris. Kesh's syntax did not yet separate those categories.

Iven watched the communication latency govern the room. Human argument wanted to fill each nine-minute gap. Bramwell enforced waiting. During the intervals, rescue reports entered the casualty ledger. One of *Quiet Reed's* missing crew reached a pressure phone. A station technician's helium exposure moved from unresolved to stable. Tev's line did not change.

Kesh's fourth interval arrived.

REFUGE IS PAUSE, NOT GRANT. PAUSE REQUIRES REMEMBERING BOTH DIRECTIONS.

Okoro said, "We accept pause."

Calder raised one hand. "The fleet can guarantee no descent. It cannot guarantee no accidental matter crossing from damaged ships."

"Then we disclose every loss," Okoro said. "No fiction of perfect containment."

Iven put a debris ledger beside the casualty ledger. Fragments, vented atmosphere, coolant crystals, spent chaff, tungsten grains, destroyed drones. The battle had already altered the system. Consent could not reverse matter that had crossed.

"What does remembering both directions mean?" Sena asked over audio.

Aro's face tightened with thought. "Kesh may want access to our records while giving access to its own."

"We do not assume," Okoro said.

The next question went out through Meera's layer: DEFINE REMEMBERING. The answer could not arrive for at least nine minutes.

Iven used the delay to open a fresh document. The title field asked for certainty. They wrote: PROVISIONAL TERMS, REVISION ZERO, ALL CLAUSES CONTESTABLE.

Their mother's evacuation came back not as an image but as a route.

The official path had saved three thousand people by directing their habitat around a debris bloom. A routing officer had proposed a counter-route that would also have saved Sleep Kept. The forecast compactor rejected it as outside parameters. Iven's mother died because the optimized evacuation carried no visible field for the one habitat its certainty excluded.

For years, Iven had believed the correction required a better map. Then they had believed it required withholding a dangerous map. Both beliefs preserved the cartographer at the center.

The new document contained no center. It held columns for Kesh translation, human interpretation, objection, operational limit, revision time, and known absence. Iven placed their mother's route in the design notes as an example of what the absence column must carry.

Calder noticed. "You have entered Sleep Kept into contact terms."

"I entered the failure mode."

"Your mother's name appears beside it."

"Yes."

"Why?"

"Because an empty category called excluded population did not keep her visible."

Calder opened the casualty ledger on her own screen. Her brother's name did not appear there; Hespar belonged to another disaster. Her nephew's did. The living and dead of the current battle occupied rows around him.

"The flag proposes one addition," she said. "Every operational limit lists who bears the risk if the limit fails."

Iven added the column.

When Kesh answered, the translation arrived in two simultaneous textures. One carried the readable clause. The other formed a pattern of microwave phase shifts that Meera could not decode.

REMEMBERING MEANS MEMORY OFFERED,
MEMORY RECEIVED, MEMORY REFUSED WITHOUT
LOSS OF REFUGE.

The undecoded texture persisted under the words.

Okoro looked at Iven. "Reciprocal memory."

"Only if refusal carries no penalty," Iven said.

Aro added, "And only if copies have limits. Orison edited sixteen lives to make one witness persuasive. Exchange can become extraction while calling itself memory."

Kesh could not hear the discussion until Meera encoded and sent it. They drafted conditions during the next interval. Each party selected records rather than exposing an archive wholesale. The receiver could decline a record without ending refuge. A record could carry terms for copying, quotation, translation, and deletion. Neither party claimed ownership of the other's memory layers. Unknown consequences stayed visible.

Sena asked, "Can a memory be deleted after someone has learned it?"

"No," Aro said. "The term can govern copies, not minds."

"Write that."

Iven did.

Quade asked whether sleepers could offer memories before waking. Okoro answered no. Calder asked whether fleet recordings of Kesh counted as captured material. Meera said the question required Kesh's judgment.

Bramwell marked every existing survey record disputed until terms clarified.

A practical test followed before anyone approved the language. Meera selected thirty seconds from the first-contact record, when Kesh had first modulated the storm into a response. The file contained raw microwave phases, the human translation, and three alternate translations the crew had rejected. Under the old survey rules, the Commonwealth owned all four as collected data.

Okoro changed the header to RECEIVED UNDER DISPUTE. "We can offer deletion of the local raw copy," she said. "We cannot promise every relay cache will obey."

Pell searched the route. Nine temporary caches had held fragments. Five reported clean deletion capability. Two had damaged storage. One belonged to *Sounding Star* and carried a battle lock. The last cache had vanished with the relay buoy.

"Then deletion cannot mean gone," Quade said.

"It can mean every known holder undertakes not to use or replicate," Aro replied. "The unknown copies remain a stated risk."

Calder objected to that phrasing. "An undertaking without enforcement offers little."

"Enforcement cannot restore a copied memory either," Iven said. "Name the remedy we can perform: public notice, access revocation, destruction of known media, and liability decided in the treaty forum."

Sena's voice cut in from the tug. "Add a right to stop the exchange while a breach is investigated. Otherwise speed will outrun trust."

Iven added it. The test exposed six failures in less than four minutes: uncertain ownership, incomplete deletion, translation drift, damaged caches, military locks, and absent enforcement. None invalidated exchange. Each

prevented the word reciprocal from pretending both sides had equal machines or equal concepts of memory.

Meera transmitted the test conditions with the question. Kesh received not a polished principle but the places where the principle could fail.

The storm replied after two intervals.

PAUSE ACCEPTS THESE LIMITS. EXCHANGE MAY BEGIN WITH THE MEMORY OF APPROACH AND THE MEMORY OF REFUSAL.

Okoro selected *Everspring's* approach record, including the original terraforming plan, her disabling order, the objections from the thaw ward, and the refuge request. She did not send it yet. She offered its manifest.

Kesh offered a manifest in return: the first probe impact, the names preserved at Cliffside, the storm's deflection of the fleet strike, and a memory pattern translated only as RAIN LEARNING NOT TO KEEP.

No one understood the last item. It stayed untranslated in the provisional terms.

Bramwell read the contact conditions into the public session. *Everspring* would remain inside the moving refuge band without descent. Fleet ships would establish debris control outside Kesh's marked storm columns. Salvage required disclosure and specific permission where matter touched memory sites. Reciprocal memory exchange would begin with manifests and refusals carrying no penalty. Treaty delegates would include awake sleepers, habitat representatives, fleet crews, and translation specialists. Kesh would define its own representation through later intervals.

Calder entered a final military statement. "The Wake Defense Fleet recognizes Kesh-of-Rain as a negotiating party. The fleet claims no occupation authority in Serein. Rescue operations continue under contact limits. The firing

order against *Everspring* remains withdrawn without condition.”

Okoro accepted rescue under those terms.

Iven published Revision Zero with the casualty and debris ledgers attached. They included confidence columns, correction trails, raw translation, and every objection received before the session closed. They did not sign as custodian. Their signature identified only the route by which the files had passed through the cartography bay.

At 13:02, Bramwell closed the cease-fire session and opened the first treaty work period. No forecast selected its delegates. No secret annex defined success. Kesh’s next interval remained nine minutes away.

Pell set Tev’s paper-fish hand on top of the printed terms. Iven did not move it toward an invented hour.

They opened the blank correction column and waited for the next answer to arrive.

Chapter 27 — The Silence After Tomorrow

Aro woke because the witness chamber had stopped predicting his pulse.

The legal-memory substrate used to anticipate each beat and adjust the body's support field a fraction before blood entered the arteries. Now the field lagged. His heart struck once against resistance, paused, then found its own interval. Pain crossed his ribs in a narrow band.

Doctor Halla Sen stood beside the chair with both hands inside an open service panel. "Do not sit up."

"I am already sitting."

"You are leaning with ambition."

Aro let the chair take him. The chamber carried emergency light and the smell of warm solder. Its mirror had gone dark when the four-kelvin well disconnected. He could no longer see Iven's face reflected there. He could see only a dim shape with younger hands.

"How long?" he asked.

"Since lattice collapse? Two hours, forty-six minutes. Since you lost consciousness? Eleven minutes."

"Did the body fail?"

"It revised its opinion." Sen closed one bypass and opened another. "The substrate expected continuous calibration from the embodiment key. That key no longer has a return carrier. I have moved cardiac timing, endocrine balance, and memory refresh onto local control."

"Locally stable?"

"Locally arguable."

Aro breathed. Air entered without the substrate deciding that it should. The experience felt less like freedom than poor maintenance.

Iven sat on the floor near the sealed door. They held a diagnostic slate and a folded sheet from the cartography bay. A paper-fish clock hand marked the fold.

"There is a buffer," they said.

Aro looked toward the dead mirror.

"Not a live channel," Iven added. "The final carrier at 10:29 deposited more phase structure than the public decoder rendered before collapse. The witness substrate kept a local residue because your embodiment key matched the sender envelope."

"How much?"

"Six seconds of carrier depth. Perhaps three voices. Perhaps one voice encoded against itself."

"Or noise selected because I want a farewell."

"Yes."

Iven placed the slate on the floor between them. Its first line contained the decoded message already published: WE DO NOT AGREE. DO NOT MAKE OUR DISAGREEMENT YOUR COMMAND. Beneath it, unrendered phase bands crossed one another like transparent handwriting.

Aro had spent his first hours in the chamber demanding to be read. Now Orison's last unread material waited inside the machine that kept him alive.

"Can we extract it without restoring the calibration link?" he asked.

Sen answered. "We can copy the buffer into an isolated reader. We cannot ask for correction. Whatever arrived before 10:41 forms the whole source."

"No future message enters after shutdown."

"None."

Aro put his feet on the deck. Sen started to object, then let him stand with one hand on the chair. His legs trembled under weight they had known for only days.

"Copy it," he said.

They moved the residue into a reader built from a survey spectrometer and the legal desk's evidence wafer. Iven refused the automatic reconstruction model because its training archive contained one hundred and twelve years of Compact messages. Any gap fill would smuggle old habits into the last exchange. Sen separated the three strongest phase structures by hand.

The first resolved into Pel March's advocate key.

Aro knew the signature before the reader confirmed it. Memory supplied a hand closing a blue folder, a mouth correcting his posture before testimony, and a voice saying that persuasion required one clean grief. He no longer trusted any of those details. Orison had edited his sixteen strands to make him credible. Pel might have existed as advocate, committee, inherited title, or rhetorical mask.

The phase structure rendered seven words.

ARO, KEEP WHAT LETS YOU SPEAK FOR US.

The first character in his name came from a different alphabet set. Error or emphasis, no one could tell.

The second structure carried no personal key. It matched the coalition signature used by the faction that defended forecast sovereignty. Its message arrived as a question.

IF YOU RELEASE US, WHO ANSWERS FOR WHAT FOLLOWS?

The third overlapped both. Its checksum had broken with the collar fractures. Iven could recover only fragments.

NOT OUR . . . NOT YOUR . . . LET THE PAST . . .

Aro read the three lines until the letters lost authority and became shapes on a survey screen.

"This does not prove three speakers," Iven said.

"No."

"It does not prove Pel authored the first."

"No."

"The second may ask a genuine question or plant a final command."

"Yes."

Sen looked from the reader to Aro's support field. "You have four hours before the memory refresh enters an untested region. I can preserve the full braid by freezing the current state into legal storage. The body would go into suspension while we build a slower refresh system."

"How long?"

"Weeks. Months. I will not offer a cleaner number."

"And the other option?"

"Let the local substrate prune structures it cannot maintain. You stay conscious. You may lose memories."

"Which ones?"

"The system prioritizes autobiographical coherence, motor learning, language, and legal identity. Your candidate strands overlap. It may preserve a childhood from one and discard the adult who remembered it. It may merge names. It may remove Pel's face and leave the blue folder."

Iven said, "No decision needs to happen in this hour."

"The support field disagrees."

"The support field can be wrong."

"So can delay."

Aro walked to the dark mirror. The body's balance depended on muscles still learning how much correction a floor required. He placed both palms against the glass. It held no carriers now, only his warmth.

Sixteen candidate lives moved inside him.

The Counselor remembered a public chamber in Orison and the moment a crowd turned against forecast rule. Heir Lidon remembered a kitchen on a station that might never exist. A child strand remembered rain striking metal, though Serein had no open human roof in the future that memory claimed. A pilot strand knew how to dock a vessel Aro had never seen. One person mourned Pel March as a lover. Another feared Pel. A third had never met anyone by that name.

The braid had made those contradictions persuasive by arranging them around a single wound. It had given Aro grief broad enough to stand for a civilization. In the present, that grief had opened doors. It had also made sixteen possible people serve one argument.

"List the strands," he said.

Sen put their labels on the dark mirror: Counselor, Heir Lidon, Pilot, Orchard Child, Witness, Archivist, Parent, Storm Listener, Treaty Clerk, Engineer, Dissident, Devotee, Skeptic, Patient, Cartographer Courtesy, and Unclassified.

"Those names came from Orison," Aro said.

"Yes."

"Not from the people."

"Yes."

Iven opened the surviving forecast archive and searched for each label. The return channel had carried attestation metadata with Aro's pattern, but most personal sources had been compressed into references no present registry could resolve. Counselor linked to four hearing transcripts. Pilot linked to a vessel certification with its name removed.

Orchard Child carried no source beyond a sensory checksum. Unclassified held seventeen percent of the braid and no descriptive note.

"Can we verify any event without asking the future?" Aro said.

"Only events that touched old transmissions." Iven selected the Counselor transcript. "This hearing summary arrived as background to a navigation warning eighteen years ago. It contains the same cadence as your memory, but the names were sealed."

Aro read the transcript. His memory placed him at the left side of the chamber beneath amber lamps. The archived summary described blue illumination and a witness table on the right. Both might reflect different hearings. Orison's editors might have merged them because the emotional contour served the embodiment.

"Mark contradiction," he said.

Iven did.

The Pilot certification produced a harder result. Aro's hands knew a docking sequence for a vessel with three axial thrusters. The surviving document described four. Sen connected a motor reader and asked him to perform the remembered approach with the chair controls. His first correction would have rolled the certified vessel into its own docking collar.

"Mark unsafe," Aro said.

The Orchard Child checksum matched a rain recording included in one of Kesh's early memory layers. Aro had believed the rain struck a human roof in Orison. The sound likely came from Serein's mineral weather translated into audible form, then placed inside a childhood scene.

"Does that make the child false?" Sen asked.

"No. It makes the source visible."

He refused to let discovery turn every candidate into fraud. The Treaty Clerk remembered a consent clause that now appeared, independently, in Kesh's first provisional terms. The Skeptic carried an objection Aro had voiced before learning its label. Heir Lidon's kitchen had no external proof, but citrus peel still brought a physical ache beneath his tongue.

Iven built a table beside the strand names: supported, contradicted, transformed source, unverified, unsafe in application, emotionally persistent. No row received true or false.

Aro studied the table. "If I freeze, this uncertainty freezes with me."

"Yes."

"If I prune, uncertainty does not become permission to discard."

"No."

"Then preserve the table outside me. Preserve what we tested, not an account of sixteen complete people."

Iven signed the method and gave Aro control of the access terms. The archive could retain the labels, source paths, and contradictions. It could not expose sensory content without his consent while he lived.

"Can any strand consent to preservation?"

Sen considered the question. "Not as an independent person. They do not run separately. Waking one in isolation would require reconstruction, and reconstruction would add choices no original strand made."

"Then I cannot save sixteen people by freezing."

"No."

"Can I kill sixteen by allowing local pruning?"

"I cannot support that claim either."

Aro looked at the labels. Orison's edit had forced him to speak as if every candidate future already possessed the standing of a completed life. He would not reverse the trick and declare them disposable. They remained memory structures with uncertain origins, moral claims neither zero nor complete.

Iven unfolded the cartography sheet. It showed Serein, Lumen, the moving refuge band, and a blank correction layer. In one corner, copied from the mineral inscription found on Day 1, a sentence waited in Iven's hand.

Aro read it aloud once. "No map can contain its reader."

The first time humans had found that sentence, they had treated it as a warning about knowledge. A map changed when a person entered the territory. Prediction failed because the reader acted. Now Aro saw another boundary. A reader could not contain the map either. Orison could not fit inside his body. The sixteen strands could not fit inside one grievance. His present could not carry every person a future might have made.

"That limit goes both ways," he said.

Iven sat beside the reader. "Explain."

"Orison sent me as a map of itself. Then it edited the map so the present would follow a chosen route. I treated myself as the container for everyone the route might produce. I cannot hold them. They also cannot hold me after I have arrived."

"The sender asked you to keep what lets you speak for them."

"I can preserve the request. I do not have to obey it."

The advocate-key sentence remained on the survey screen. A person called Aro had existed in the braid before the transmission, but the name might have belonged to a committee's design. Pel had chosen it. Orison had attested it. The present had learned to call him by it.

Sen brought up the legal identity form. "The substrate requires one name before local pruning. If merged names emerge later, they can be added. Today it needs one stable key."

Aro almost laughed. A body facing uncertain memory still had to satisfy a database.

"Aro Nhal," he said.

Sen entered it. "Chosen or retained?"

"Chosen now."

"Pronouns?"

"He and him."

"Age?"

"Embodied eleven days. Remembered twenty-eight years. Neither measure owns the other."

Sen wrote both.

Iven asked, "Do you want Nhal?"

Aro searched the strands. One gave the word as a family name. Another translated it as after-rain. A third remembered a transit card with NHAL printed in block letters. None proved origin.

"Yes. Not because I can verify it. Because I recognize myself when someone says it."

Sen accepted one name into local legal identity. The support field changed priority. Its refresh process stopped preserving the braid as a future attestation and began preserving Aro's current associations as a living patient.

Loss did not arrive dramatically. A kitchen vanished first. He had been standing beside a blue counter in a station not yet built; then he knew only that some memory had included the smell of citrus peel. The pilot strand's docking procedure blurred into hand movements without a vessel. Pel's blue folder remained while the hand holding it lost skin color.

Aro gripped the chair.

"You can still choose suspension," Sen said.

"I know."

A childhood rain memory thinned. He kept the sensation of drops on metal and lost the person who had counted them. Grief rose without a name attached. It belonged to him now, not because it had begun as his, but because his body carried it through this breath.

The support field settled into a slower rhythm.

"Cardiac stable," Sen said. "Memory refresh local. Degradation rate unknown."

She conducted the first examination that treated him as a patient rather than received evidence. The substrate's old diagnostic package compared every result to the embodiment blueprint sent by Orison. Sen deleted the comparison column. She measured what his body did now: blood pressure low after standing, left pupil slow under amber light, grip strength unequal, temperature regulation dependent on two support patches along the spine.

Aro walked six lengths of the chamber. On the fourth, his right knee failed to lock. Iven moved toward him and stopped when he raised a hand. He recovered against the wall without falling.

"Add a rail in the corridor," Sen said.

"Not a suspension order?"

"A rail."

She gave him water without flavor compounds. He drank too fast and coughed. The legal substrate had supplied remembered meals, remembered thirst, and remembered hands lifting cups, but the body still had to learn swallowing under local timing.

"What happens to the witness chamber?" he asked.

"It remains available while you need the support field."

"Available to whom?"

"You."

"Not the next forecast pattern."

"There cannot be a next one through these mirrors."

The answer reached him differently from the lattice shutdown report. This room would not receive another person carrying a courtesy header and a mission selected fifty-three years ahead. Its chair could become a medical chair. Its dark mirror could become glass.

Sen prepared a daily regimen: electrolyte checks, motor exercises, memory notes he could choose not to make, weekly scans of the local refresh, and a manual alarm if cardiac timing drifted beyond tolerance. Nothing in the plan treated continuation as service to Orison.

"How long will I live?"

"Unknown. The substrate may support decades after revision. It may fail in months. Your body will also age, become ill, and respond poorly to several drugs we have not tested."

"Finite, then."

"Everyone in this room has that diagnosis."

Aro returned to the survey reader. Orison had asked who answered for what followed. The answer could not travel fifty-three years forward through a broken lattice. He could still enter it into the present record, where consequences might answer back through people rather than prophecy.

He opened a response file.

"To Orison, whose factions and authors remain uncertain," he dictated. "I received your final buffered structures after the channel closed. I cannot confirm their speakers. I preserve them with that limit. I will not speak for all of you. I will speak about what was done to me, what I remember, and what I lose. The present releases you from the duty to

rule it. I release my past from the duty to produce a verdict. What follows belongs to the people who live it, including me."

Iven said, "Do you want the line from Serein attached?"

"No. Put the chart beside the response. Let a reader make the relation or refuse it."

They entered the raw buffer, the hand-separated phases, all alternative decodings, Sen's medical options, Aro's identity choice, and the response. No checksum from the future could authenticate the file. Three present witnesses signed what they had observed.

At the bottom, Aro added one sentence: *This answer was not transmitted.*

The distinction hurt. It also prevented farewell from pretending to be conversation across a channel that no longer existed.

A knock sounded at the witness chamber door. Okoro had come aboard from *Everspring* by rescue shuttle for the first treaty work period. She carried no mission uniform, only a pressure layer and a slate full of objections from the sixty awake sleepers.

"Are you available as a delegate?" she asked Aro.

"Under what standing?"

"Embodied person with edited future memories, present witness to forecast governance, and someone who has already objected to our first reciprocal-memory draft."

"Not descendants' emissary?"

"Only if you claim it."

Aro looked once at the dead mirror. Orison remained in the buffer, the archive, and whatever futures might still grow without a route back. It no longer occupied the chair through a standing he had to perform.

"I do not claim it," he said. "I can attend as Aro."

Okoro opened the door wider.

The corridor outside had lost two ceiling panels during the battle. Repair cables crossed the deck under yellow tape. Air moved through the gap with a faint mineral smell from damaged filters. Nothing about it matched the corridors in Aro's candidate memories.

He stepped over the first cable without knowing whether the next breath belonged to a year, a decade, or only the walk to treaty work. The dark mirror held his handprints after he left.

Epilogue — The First Unadvised Morning

Fourteen months after the return lattice fell silent, Ansel Quade canceled her daughter's surgery.

She did it at 06:12 on Pell Station, after the surgeon disclosed a new inflammation marker and before the foundry shift began. Under the old forecast compact, a risk bulletin would have assigned a preferred hour, insurer liability, and the acceptable delay. The archived bulletins still existed. None could bind the hospital.

Ansel's daughter listened, asked three questions, and chose to wait two weeks for another scan. The surgeon recorded uncertainty in the chart. The insurer recorded no penalty. Ansel missed the first foundry bell while they argued about breakfast.

At 06:40, she approved a pressure-vessel pour whose alloy schedule had never appeared in future advice. The first mold cracked. The second held.

No civic siren marked either result.

*

Everspring crossed Serein's copper storm in a negotiated orbit that changed every nine days. Its terraforming sequencer remained physically severed. Twelve hundred sleepers had awakened, Lina Seft among them, while the rest followed a thaw schedule reviewed by people who could object to it. Lina spent her mornings rebuilding the bank-twelve pump tree from the pressure-marked sketch she had signed before suspension.

Lio Venn taught latch maintenance in a converted ward. Across the passage, Meera Sol conducted the morning session of the Kesh treaty school. Half her students came from the seedship. The others joined from fleet ships and habitats through delayed relay.

Today's lesson concerned refusal.

Meera offered a human memory manifest: the sound of rain on a Pell school ring. Kesh offered a storm pattern translated as a mineral shelf releasing an old impact. Students could receive either, both, or neither. A child named Osa declined the human recording because storms frightened her. The console accepted the refusal without closing her lesson.

Nine minutes later, Kesh replied with a new exercise.

REFUSAL REMEMBERED. RELATION CONTINUES.

Osa asked if she could change her mind tomorrow.

A boy from Deck Eleven asked whether accepting the mineral memory gave Kesh permission to keep his reaction. Meera sent that question too. The class waited through the interval by checking the school's physical boundaries: no recording in the quiet alcove, no automatic copy to family accounts, deletion requests posted beside every received file. Human systems could not erase what a student remembered, so the wall stated that limit in six languages.

The reply crossed the storm as three measured pulses.

REACTION BELONGS TO REACTING CONTINUITY.
RECORD REQUIRES REQUEST.

Meera marked the translation provisional. Two students proposed alternatives for continuity. Osa drew the storm pattern without listening to its audio rendering. Beyond the classroom port, the seedship's damaged Radiator Three had become a fixed dark petal, while a new detachable panel shed heat beside it under Kesh's nine-day permission.

Meera entered both questions without predicting tomorrow's answer.

*

Sena shut down the training tug's center display while six relay pilots were using it.

Complaints filled the cabin. One trainee kept flying from optical bearings. Another asked the aft pair for their heat ledger. A third tried to restore the display and discovered Sena had pulled the breaker.

"Your shared plot has a hole," Sena said. "Name it."

The trainees reported what each could see. They assigned sectors, exchanged limits, and left one corridor unclaimed until a courier drone crossed it. No pilot took command of the whole geometry.

The exercise ended with three kilograms of reaction mass unused and every radiator inside margin. Sena marked the result as adequate.

On the bulkhead behind her hung Tev Arin's manual shutter tool. The empty mug remained in *Sworn's* courier bay, where new pilots sometimes mistook it for equipment.

"Again," Sena said.

This time she left the center display dark from the start.

*

Iven drew the consent-zones as blank fields.

They did not shade Kesh's storm columns, the disputed salvage arcs, or the habitats that had declined shared archives. Each blank carried a source path and a date for review. Empty space no longer implied available space.

Aro arrived late to the treaty room with one hand on the corridor rail Doctor Sen had ordered installed. His local substrate had survived the year, though his right knee failed on tired days and three more candidate memories

had thinned beyond retrieval. He carried a notebook because he chose which days to record.

“Am I on the map?” he asked.

“Only where you consented to appear,” Iven said.

He found his name beside today’s delegate list. No line connected it to Orison.

Outside the port, *Everspring* adjusted orbit by a measured pulse. The maneuver had been proposed by Okoro, checked by Sena’s relay school, contested by two awakened sleepers, revised by Kesh, and accepted for nine days.

Iven left tomorrow’s track blank.

Aro sat beside the unfilled line and opened his notebook to the morning’s empty page. His pen rested across the margin.